

Feeding the City: How Agricultural Productivity in Villages Shapes Urban Structural Change*

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Job Market Paper (most recent version [here](#))

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Abstract

I ask whether agricultural productivity in rural areas affects outcomes in cities located elsewhere. I use Indonesia's Transmigration Program, which resettled over two million people from Java and Bali to new agricultural villages in the Outer Islands in the 1980s, to destinations that were plausibly as good as randomly assigned. In cities surrounded by more productive villages, the change in sectoral employment is much greater than the change in population: agricultural employment falls and modern-sector employment grows. New arrivals cannot account for the shift, and the settlers themselves largely stay in their villages, so residents already in the city are changing sector. The effects concentrate in cities less connected to global markets. In a spatial general equilibrium model, the reallocation of urban employment out of agriculture still occurs when migration is shut down, but it does not occur when food trade is shut down. Food trade between rural and urban areas is necessary for urban centers to expand their modern-sector employment, especially when they are less connected to global markets. The more cities rely on local food supply, the more rural productivity gains translate into urban growth and structural transformation.

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1 Introduction

Many development programs implemented by governments and often supported by external financing target rural areas: raising farm productivity, expanding cultivation, and improving rural livelihoods. Governments dedicate about 4% of their public spending to agriculture in the countries where farming is nearly 30% of the economy ([World Bank 2007](#)). These policies are typically evaluated using rural outcomes. Yet, when goods and labor markets are linked across space, rural productivity improvements can also reshape the towns and cities nearby. Higher productivity may attract and retain workers in agriculture, while also reducing liquidity constraints for more well-off farmers to finance migration and leave for cities. Higher agricultural output can lower the local cost of living (especially when staple foods are locally sourced), raise rural incomes and demand for non-agricultural goods and services, and alter migration incentives and urban labor supply. The more cities rely on local food supply, the more likely rural spending is also urban policy. Most existing research estimates the effects of agricultural productivity gains within the region that received them, so the effect on cities cannot be separated from the effect on the villages themselves. How trade in goods (food) or in labor (migration) spreads across space and whether these channels translate into urban growth and structural transformation is ultimately an empirical question.

In this paper, I study rural-to-urban spillovers by leveraging a policy that (i) generated a large and spatially dispersed rural shock, (ii) produced plausibly exogenous differences in rural productivity across the newly settled villages, and (iii) can be linked to urban outcomes over several decades. Indonesia’s Transmigration Program provides this setting. Although the program was designed as a rural resettlement and agricultural development policy, it created sharp, localized changes in rural population and agricultural production in the hinterlands of many towns. Transmigrant households did not choose their destination villages, and farming households who ended up in destination villages more similar to their places of origin are more productive. Higher agro-climatic similarity between origin and destination raises village rice productivity by roughly 20% per standard deviation, for reasons unrelated to local economic conditions ([Bazzi et al. 2016](#)), since farming skills transfer better to places that resemble home in geography and climate. These villages supply food to the cities around them.

I estimate rural-to-urban spillovers comparing cities exposed to a more or less productive rural hinterland, measured as the average agro-climatic similarity of the transmigration villages within a five-hour journey of the city. I use three decades of Indonesian census microdata and village and household surveys along with the satellite-derived urban footprints of the Global Human Settlement Layer to construct a consistent-boundary sample of urban centers on the Outer Islands. I measure long-run changes in outcomes, holding baseline levels of population, employment, and urban structure in cities in 1980 constant, showing that agro-climatic similarity exposure is uncorrelated with baseline city outcomes. Cities surrounded by more productive farm villages change their sectoral composition more than their size: employment moves out of urban agriculture into manufacturing and services, urban population does not grow detectably, while the farmers who generated the productivity gains stay in their villages and keep farming. Comparing the growth in population to the growth in modern

employment, this is not a trivial reallocation of newly arrived migrants: back-of-envelope calculations show that new arrivals cannot account for the structural shift. Most of the fall in the agricultural share comes from workers of a given birthplace exiting agriculture and going into modern employment.

Cities transform through the trade in food, as lower spending on agricultural goods among urban residents frees income for other sectors, and the response concentrates in the closed cities, those less connected to global markets, where food availability and food trade are a binding constraint. Food expenditures of urban residents go down while they do not reduce food consumption, and demand for non-agricultural goods and services rises, pulling labor into the modern sector. The countryside moves the opposite way. Villages with higher agro-climatic similarity keep at least as large a share of their workers in agriculture and produce more food, while the cities' trade proxy, food consumption minus own production, rises with exposure and indicates that cities are importing more food from rural hinterlands. The farmers who generated the productivity gains stay in their villages and keep farming, and the migration that does occur comes from other rural areas in the same province, which now have to compete with the more productive villages, while the share of urban residents born on Java or Bali falls.

Cities that are further away from trading routes represent more closed economies that mostly trade with their hinterland, while a city that can already import cheap food gains little from a better local harvest. I measure openness by each city's travel time over land and sea to the Thai and Vietnamese hubs that ship the bulk of Indonesia's rice imports. The sectoral response is larger in the closed cities, by about 40% on the agricultural employment share and by more than double on modern-sector employment, and city size does not grow detectably in either group.

I then use a spatial general equilibrium model to quantify the contribution of agricultural productivity to urban sectoral employment change, and to decompose the trade and migration channels behind the fall in the urban agricultural employment share. I do not observe trade flows directly in any Indonesian data source at this level; in the data both trade and migration operate at the same time and only their general equilibrium effects are observed, so the model decomposes them: I shut down trade, raising trade costs, and shut down migration separately. The two sectors are agriculture, producing food, and combined manufacturing and services, the same two sectors as in the empirical analysis; workers choose a location, with migration costly, and within each location a sector; and preferences are Stone-Geary, with a subsistence level of food consumption that has to be satisfied, which generates the sectoral shift. With the counterfactual exercise in the model, I show that even in the absence of migration the sectoral reallocation can be achieved, whereas in the absence of trade it cannot.

I make two contributions. First, my finding that a village productivity gain transforms the sectoral composition of the cities it feeds adds a supply-side spillover to the literature on spillovers of local shocks across space, which has followed such shocks through demand and through capital, and adds the hinterland's productivity to the explanations of urbanization without growth. Local shocks

have been followed through demand,¹ from cash transfers that raise consumption in nearby non-recipient villages (Egger et al. 2022) to resource booms that raise income and employment in surrounding counties and new jobs in one urban sector that create jobs in others, and agro-processing plants that raise employment in the villages around them, and through capital, as agricultural windfalls flow through the banking system into industrial growth elsewhere (Bustos et al. 2020); the shock here reaches the cities through the supply of the food their villages sell them, and the companion paper on India studies the reverse direction, urban growth reaching villages (Asher et al. 2022). Studies of local productivity shocks will therefore need to incorporate spillovers on both population and sectoral employment among the places that buy goods from the affected villages. Where the work on urbanization and structural transformation (Gollin et al. 2016; Jedwab & Vollrath 2015; Gollin 2023; Henderson & Turner 2020) documents cities that add people without industrializing, and the spatial accounts of structural change follow the transformation with the people who move (Eckert & Peters 2025), cities surrounded by more productive villages grow more intensively than they grow in size, and the change in sectoral employment is much greater than the change in population. More broadly, the finding speaks to the macro-development literature on agricultural productivity and structural transformation (Gollin et al. 2007; Restuccia et al. 2008; Gollin et al. 2014; Herrendorf et al. 2014; McArthur & McCord 2017; Gollin et al. 2021), which works at the level of countries, where the countryside that receives a productivity gain and the cities it reaches enter the same aggregate. Closest to this paper, Bazzi et al. (2016) establish the productivity variation itself. Settlers assigned to agro-climatically familiar destinations farm more productively, raising village rice yields and nightlights within the settlements. I take that village-level variation as the shock and follow it out of the settlements, asking what the productivity of a city’s surrounding villages does to the city itself. Complementary work in progress studies the villages’ own outcomes under the same program’s variation in market access, where better-connected villages shift out of agriculture in place yet their income per capita falls (Mitchell 2026), while I show that hinterland productivity moves the cities those villages feed.

Second, my finding that the response concentrates in the closed cities moves the empirical test of Matsuyama’s openness prediction to the side that buys the food, the city with the villages that feed it, and adds to the literature on internal trade and migration a case in which regions are well integrated through food while barely integrated through people, so migration flows understate integration. Existing studies estimate what an agricultural shock does inside the locality with the source of the shock, where the sign of the non-farm response varies with which markets happen to be open (Bustos et al. 2016; Emerick 2018; Santangelo 2019; Foster & Rosenzweig 2004; Hornbeck & Keskin 2015); across Indian districts, Moscona (2019) finds that the Green Revolution’s productivity gains raised agricultural employment and land use and reduced urbanization and manufacturing employment, most in the districts more involved in trade. The empirical tests of that prediction (Matsuyama 1992; Bustos et al. 2016) and the within-country tests that interact an agricultural shock with regional openness (Moscona 2019; Liu et al. 2023; Marden 2016) measure the response on the shocked side. Matsuyama’s prediction separates closed economies, which shift away from agriculture when agricultural productivity rises,

¹(Feyrer et al. 2017; Allcott & Keniston 2018; Moretti 2010; Edwards 2024; Kraus et al. 2024)

from open economies, which keep specializing in agriculture because they face effectively unlimited demand from world markets, and here the openness that matters is the destination’s, the city’s access to outside food. The literature on internal trade and migration (Tombe & Zhu 2019; Bryan & Morten 2019; Morten & Oliveira 2024; Lagakos et al. 2023; Budi-Ors 2026) quantifies how goods-market and labor-market integration shape aggregate income and infers integration from how people move; here goods trade transmits the shock while labor flows stay small, and in a world where migration is low, trade still accounts for the urban structural change, whereas migration without trade does not enable that transformation. The importance of food trade as the channel has implications for studying trade and migration spatial frictions in spatial models of structural transformation (Tombe 2015; Nath 2025).

The rest of the paper is organized as follows. Section 2 describes the program and the urban setting, Section 3 the data and the construction of the exposure measure, Section 4 the framework and the empirical strategy, Section 5 the reduced-form results, Section 6 the model and its quantification; Section 7 concludes.

2 Setting

In this section, I provide further details on the Transmigration program, as well as descriptive statistics on urbanization and structural transformation on the Outer Islands. I also explain how I construct a consistent sample of urban areas over time to look at the long-run changes.

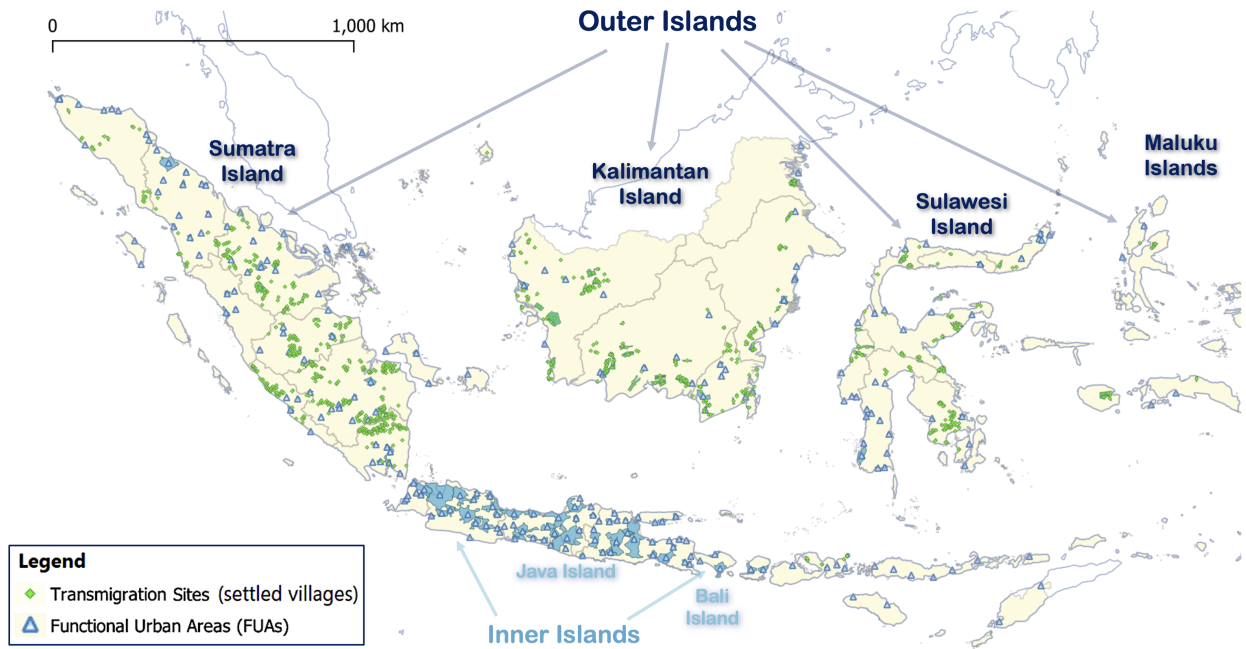
2.1 The Transmigration Program and Quasi-Random Variation in Agricultural Productivity

Indonesia’s Transmigration (TM) Program was one of the largest resettlement initiatives in modern history. The program was aimed at alleviating overpopulation in the densely populated Inner Islands, primarily Java and Bali, by moving people to much less densely populated areas, such as Sumatra, Kalimantan, Sulawesi, and Papua. During the peak period between 1979 and 1988, the policy facilitated the resettlement of some two million people into newly established villages across the Outer Islands (Figure 1 maps the settled villages against the urban agglomerations they surround). Origins were selected on population density, and settlers were drawn predominantly from rural, agricultural districts. Every transmigrant household was provided with a free land lot, a house, and free transportation to the destination in new settlements, mostly on previously unused land. The program was explicitly rural, expanding agricultural production and food security in the receiving regions, not targeting cities. In this paper, I focus on the population shock from the 814 villages settled between 1979 and 1988, the peak of the program, which form the complete set of settled villages in the Bazzi et al. (2016) replication data.²

Two additional features of the program matter for the identification strategy. First, the rollout was large and concentrated in time. The program continued until the late 1980s, when it was abruptly canceled in 1988 due to declining natural resource rents, which used to fund the ballooning cost of

²Agro-climatic similarity is measured over all 814 settled villages.

Figure 1: Transmigration villages and urban agglomerations



Notes: village locations from the Transmigration census (the settled-village sample of [Bazzi et al. 2016](#)), city polygons from the Global Human Settlement Layer ([Moreno-Monroy et al. 2021](#)). “FUA” stands for functional urban area, an urban agglomeration region identified consistently across the globe based on population density, built-up area, other remote sensing data, and commuting zones. Papua is not shown: it lies outside the harmonized census panel, since the 1980 baseline measures the design requires are unavailable there, and it is excluded from the analysis throughout ([Section 2.2.1](#)).

resettling households ([Bazzi et al. 2016](#)); afterward, resettlement continued only at much lower intensity and in a more self-financed form. Second, TM villages were geographically dispersed across the Outer Islands, generating substantial spatial heterogeneity in which towns were exposed to new rural settlements and how close those settlements were.

For this project, the key feature of Indonesia’s Transmigration Program is that, beyond moving people, it generated quasi-exogenous differences in how productive newly settled villages became. Transmigrant households did not choose their destination villages. While the program did not implement an explicit randomization of transmigrants to destinations, the implementation of the resettlement made the match between a household and its destination village as good as random with respect to the household’s origin and characteristics, because which of the newly established villages a household was sent to depended on the specific timing of its arrival in the resettlement centers. Trains and ships would depart to different destination villages from one day to the next, and transmigrants’ final destinations were not matched to any of their personal characteristics or preferences. Previous research has shown that the resulting agro-climatic similarity between settlers and their villages is uncorrelated with the villages’ predetermined characteristics, including their potential crop yields and pre-program development in the surrounding area ([Bazzi et al. 2016](#)), and the same assignment process underlies the as-good-as-random matching of transmigrants from different districts on the main islands to their destination villages that

Bazzi et al. (2019) exploit. Conditional on where the government chose to open a settlement site, the composition of settlers arriving at that site was therefore plausibly as-good-as-random.

The *agro-climatic similarity* measure at the center of this design was developed by Bazzi et al. (2016) as a novel metric for assessing the skill transferability of migrants in agricultural settings. It quantifies the compatibility between the agro-climatic conditions of migrants’ origins and their new settlements, averaging the agro-climatic distance between each origin district and the village over the village’s settlers, weighted by the share of settlers from each origin district³, and provides a proxy for the potential agricultural productivity of transmigrants in their destinations. The measure is grounded in agronomic data on essential land attributes and climate characteristics, which remain largely unchanged over time: soil and terrain characteristics, climate variability (measured by rain and temperature variability), and stable geographic characteristics such as distances to rivers and coasts. Migrants’ farming techniques came from the environments in which they had previously worked. The index quantifies the match between those techniques and the new environment, influencing migrants’ potential productivity and integration success. The agro-similarity measure predicts productivity. Bazzi et al. (2016) show that the index strongly predicts agricultural performance and broader development outcomes in transmigration villages. That pattern is consistent with skill transferability and inconsistent with selective placement of migrants into more promising sites. The assignment process also makes the resulting variation in match quality plausibly orthogonal to unobserved determinants of later growth.

Appendix Figure A5b shows the distribution of the village-level agro-similarity index, with 75% of TM villages reporting positive rice production in the analysis sample; in Bazzi et al. (2016), a one-standard-deviation increase in agro-similarity raises village rice productivity by roughly 20%, the first stage behind the exposure measure. This variation is economically relevant given the importance of rice in urban consumption. In the household survey data, rice accounts for more than 25% of urban household expenditures on agricultural goods (Badan Pusat Statistik 2011).

The program’s geography implies that towns were exposed to different combinations of (i) the scale of nearby TM settlement and (ii) the productivity of those surrounding villages. Some towns are surrounded by TM villages with high agro-climatic match and thus higher agricultural productivity; others are surrounded by villages with low match; and many towns are exposed to few or no TM villages at all (appendix Figure A3a shows the joint variation in these two variables). By comparing cities with variable productivity levels of nearby TM villages, I aim to disentangle the effects of rural productivity on urbanization.

Settlers were a non-trivial share of the population these hinterlands already held. Across the 133 cities I study, the settlers placed in a city’s rural hinterland, meaning the rural subdistricts within 135 km that lie outside any urban center, amount to about 7% of the population that hinterland held at the 1980 census, a baseline that precedes 84% of the placements. The newcomers thus constituted

³For village v with settler shares π_{dv} from origin districts d , the index is $AS_v = -\sum_d \pi_{dv} d(x_d, x_v)$, where $d(x_d, x_v) = \sum_g |x_{dg} - x_{vg}|$ is the sum of absolute differences between origin and destination across the standardized agro-climatic characteristics g , so that a larger distance means lower similarity; Section 3.3 states the construction in full.

a substantial population shock for the receiving economies in their destination regions, large enough to produce urban effects and small enough to leave the rural landscape they entered in place. Settlers complemented the pre-existing rural population and did not replace it. The share varies across cities, with roughly a seventh of them above 15%. The next subsection turns from the program to the cities around it.

2.2 Urbanization and Industrialization in the Outer Islands

Indonesia is administratively divided into provinces, 26 of them in the 1980 census files. Provinces are divided into districts, the first sub-province level, called *kabupaten*, and districts into subdistricts, *kecamatan*, the finer level I use, referred to as subdistricts throughout the analysis; some datasets are available only at the district level, but the analysis runs on data available for subdistricts. In the rest of the analysis, rural subdistricts are aggregates of their villages, and urban centers are aggregates of subdistricts overlaid with the urban-center boundary from the Global Human Settlement (GHS) layer, introduced with the urban sample in [Section 2.2.1](#). The official urban classification operates at the village level. BPS, the national statistical agency, classifies each village as urban or rural from its population density, its share of farm households, and its access to urban facilities, and I use that administrative classification for baseline measurement, the 1980 urban population share below.

The Outer Islands entered the study period overwhelmingly rural. Overall they are less urban than the main islands, and as economies that are not as highly developed, they are substantially more agricultural. In 1980 they held 52 million people, 35% of Indonesia’s population, spread over most of its land area: 67% of outer-island workers were in agriculture, against 51% on Java and Bali, and 18% of the population lived in administratively urban villages, against 25% on the main islands.⁴ Over the following three decades the Outer Islands more than doubled their urban share, similar to the main islands. By 2010 their population had grown 80%, against 50% on Java and Bali, their urban population had more than tripled, from 9.3 to 35.1 million, taking the urban population share to 37%, and their agricultural employment share had fallen by 15 percentage points, to 52%.

Over the period the service share on Java and Bali increased by roughly a third, from 34% to 45%, whereas on the Outer Islands it increased by more than 50% relative to baseline, from 23% to 35%. Manufacturing moves in the opposite direction. On the Outer Islands it is not growing as a share, drifting from 6% to 5% even as it grows in levels, while on Java and Bali it rises from 11% to 15%. So the Outer Islands have largely services replacing agriculture, a different type of urbanization from the manufacturing-led path, and a setting where positive results on structural transformation are not what one expects a priori. The urbanization this paper studies is therefore transformation at an early stage, arriving in cities that begin the period small and substantially agricultural. The next subsections define the urban units the analysis follows through this transformation, and the agriculture inside them.

⁴The aggregates in this subsection are computed from the harmonized census files described in [Section 3](#). Papua is outside the harmonized panel and is excluded throughout, from the 1980 aggregates as well as the 2010 ones ([Section 2.2.1](#)).

2.2.1 The Urban Sample

The main analysis focuses on urban subdistricts on the Outer Islands of Indonesia. I exclude Java and Bali because there are no settled TM villages there in the program sample, and I exclude Papua because baseline (1980) census measures needed for pre-period controls are unavailable and spatial coverage is sparse in early years.

The Global Human Settlement (GHS) layer provides year-specific urban-center boundaries as well as a static functional-urban-area layer, which delimits each city together with its commuting zone, consistently across countries, from population density, built-up area, and commuting patterns (Moreno-Monroy et al. 2021). I use the urban-center boundaries, since they are available over time and finer, to define which subdistricts and neighborhoods go into the definition of urban. I spatially match the Urban Centre (UC) polygons to the subdistrict (ADM3) boundaries and take the urban sample to be the subdistricts that overlap a UC, where an intersection counts, with no minimum-overlap threshold on the intersecting area. Section A.3 details the polygon-to-city mapping and the chain from the satellite layer to the estimation sample.

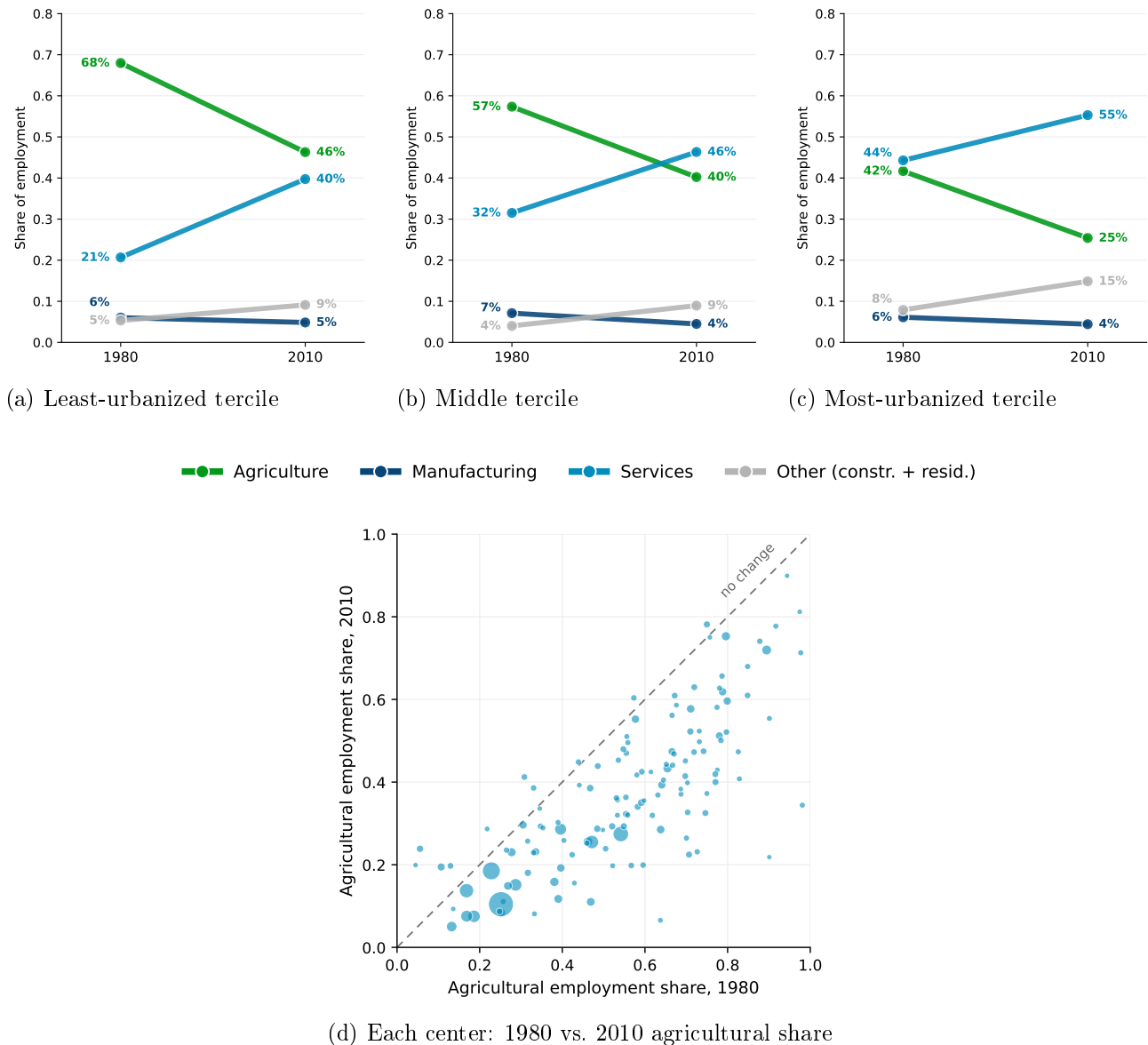
Reclassified urban-center sample. The satellite data have limitations, especially for Indonesia, and Section A.2 measures the gap. I therefore go through an additional reclassification, using the fine subdistrict-level data from 1980 and two key definitions that are meaningful distinctions of cities, density and the share of agricultural employment. Density is meant to capture communities that are more urban in a meaningful way, and there is no better measure for it; the share of agriculture identifies communities that have likely started urbanizing but not fully. Together they provide a simple but meaningful check on which other areas could be viewed as urban but might not be captured by an urban layer with some partially missing historical satellite data. The rule uses only pre-treatment information. A subdistrict is promoted to urban when its 1980 population density exceeds 500 persons per square kilometer and its 1980 agricultural employment share falls below one half, the dense, non-agricultural profile of the satellite-defined centers (Figure A4a in the appendix shows the diagnostics). Section A.3 names the added and re-attached centers and gives the chain from the satellite layer to the estimation sample; the sample rule itself is stated with the travel-time window in Section 3.3. Because both rules use only 1980 observables, the sample does not condition on the treatment.

Urbanization Intensity and Sectoral Composition.

Panels (a)-(c) of Figure 2 trace each sector’s mean employment share from 1980 to 2010, one panel per tercile of urban population % in 1980, for outer-island urban centers whose nearest transmigration village is within 300 minutes. Two patterns stand out. First, in 1980, many “urban” subdistricts in the Outer Islands remain highly agricultural. In the least-urbanized tercile (1980 urban share below 11%), agriculture accounts for 74% of employment against 18% for services. Second, structural transformation is pervasive across all three terciles. In the middle tercile (urban share 11-33%), services rise from 29% to 44% while agriculture falls from 61% to 42%, the two lines crossing; even the most urbanized tercile (urban share of 33% or more), where services already lead agriculture in 1980 (43% versus 40%),

sees agriculture keep falling, to 25% by 2010. The tercile means also average over wide center-level dispersion. Panel (d) plots every center against the no-change diagonal, and more than nine in ten lie below it, with the median center's agricultural share falling by 20 percentage points and a tenth of centers dropping by 37 percentage points or more. These descriptive patterns motivate focusing on both (i) urban population and employment growth and (ii) sectoral transformation in the analysis, and the dispersion in panel (d) is the cross-city variation the identification strategy exploits.

Figure 2: Sectoral employment change across urban centers, 1980 to 2010



Notes: Outer-island urban centers whose nearest transmigration village is within 300 minutes of cost-distance travel time, the headline reduced-form sample. Panels (a)-(c): UCs split into terciles by 1980 urban-population share ($UrbanShare_c$; cutoffs 11% and 33%), held constant across both years; “Other” combines construction with mining, utilities, and undefined. Panel (d): each center's 1980 (horizontal) and 2010 (vertical) agricultural employment share, marker area proportional to 2010 census population; the dashed line is the 45-degree line. Source: Population Census 1980 and 2010, aggregated to the UC level.

Summary Statistics

Table 1 summarizes the main estimation sample of outer-island urban centers (UCs), the unit of analysis: the centers within 300 travel-time minutes of a transmigration village with a 1980 baseline the census recorded directly (Section 2.2.1). Summary statistics for the full outer-island UC universe are in Appendix Table A6.

Table 1: Summary statistics: main estimation sample ($N = 133$ outer-island urban centers)

Panel A: Levels (counts and the travel time to the nearest transmigration village)						
	Mean	SD	p50	Min	Max	N
Total population 1980 (UC, BPS census)	118,928	220,979	67,518	5,785	2,078,951	133
Total population 2010 (UC, BPS census)	229,569	405,501	114,751	28,284	3,796,167	133
Total employed 1980 (UC, BPS census)	32,861	57,605	18,858	2,226	537,303	133
Total employed 2010 (UC, BPS census)	90,333	152,478	47,909	10,952	1,416,355	133
Agricultural employment 1980 (UC, BPS census)	17,328	21,398	10,610	261	148,091	133
Agricultural employment 2010 (UC, BPS census)	27,389	30,024	18,405	1,298	161,274	133
Modern-sector employment 2010 (manufacturing + services)	53,136	110,670	23,173	1,533	1,059,416	133
Distance to nearest TMV (cost-distance, min)	107	85	77	1	289	133

Panel B: Shares, the unemployment rate, and the exposure measures						
	Mean	SD	p50	Min	Max	N
Urban population share 1980 (UC)	0.261	0.229	0.244	0.000	0.807	133
Employment % in agriculture, 1980	0.559	0.219	0.577	0.045	0.981	133
Employment % in agriculture, 2010	0.374	0.184	0.362	0.050	0.899	133
Employment % in the modern sector, 2010	0.517	0.157	0.524	0.094	0.827	133
Unemployment rate 2010	0.123	0.041	0.117	0.038	0.227	133
TMV settler count (IDW within 300 min, '000s)	0.470	0.685	0.133	0.001	4.125	133
TMV AgSim (z, 300min)	-0.012	0.973	-0.123	-2.930	2.692	133

Notes: Main estimation sample: the $N = 133$ outer-island urban centers within 300 travel-time minutes of a transmigration village with a 1980 baseline the census recorded directly and 2010 outcomes present (the estimation sample of every regression in the paper). The rows are the seven headline outcomes, the two exposure measures with the travel time that defines the window, and the four 1980 baselines the main specification conditions on; every row is measured on the whole sample. The modern sector is manufacturing plus services, and every employment share uses the broad employment count as its denominator. Full urban-center universe and the wider set of descriptives: Appendix Table A6. UC = urban center; TMV = Transmigration Village; IDW = inverse-distance weighted.

The city-level exposure measure is built from all 814 program villages (Section 2); Appendix Table A7 summarizes the transmigrant villages whose agro-climatic similarity to the migrants' origin regions builds it. They are small (around 1,900 settlers placed at founding) and overwhelmingly agricultural. Appendix Table A8 shows how the three samples entering the structural-transformation regressions (cities, rural subdistricts without a transmigrant village, and rural subdistricts hosting one) compare. Cities are far larger and markedly less agricultural, yet agro-similarity exposure is balanced across the three groups (mean ≈ 0.67 in each), consistent with the quasi-random allocation of settlers underlying the identification.

2.2.2 Urban Agriculture

Urban agriculture is a non-trivial part of the economy in many developing-country cities, especially cities beyond the capital region, and it is something that many poor urban workers still engage in as their primary or secondary activity. A city is not equivalent to the modern sector of manufacturing and

services. These cities have substantial agriculture still, and non-agriculture. This is partially because of food availability, and some of these cities are retaining agriculture because there is not enough food. The mixed zones where rice cultivation and non-agricultural activity occur side by side at urban densities have a name of their own in Southeast Asia, *desakota*, from the Indonesian words for village and city (McGee 1991), and agricultural employment inside cities declines with development and with city size in the cross-country record (Gollin et al. 2016; Jedwab et al. 2025).

The boundaries of the cities studied here come from the global harmonized urban layer (Section 2.2.1), and outcomes are measured on the subdistricts that overlap the satellite footprint (Section 3.2). A city’s totals therefore include the cultivation in its ring. In 1980 the average city in the estimation sample had 55.9% of its workers in agriculture; in 2010 the share is 37.4% when every member subdistrict is counted whole, rural edges included, 18% when only the villages that touch the city’s satellite footprint are counted, and 14% when, in addition, only the villages the census itself classifies as urban are kept; the share falls at each step because each definition strips away more of the farmland at the edge of the city (Appendix Table D10). The pattern also holds when I subset to density, which means that within regions of comparable density urban and peri-urban agriculture survives and is important. The level of urban agriculture depends on how the city is defined, but the ranking of cities does not, since the alternative definitions rank the sample’s cities nearly identically (correlations 0.68 to 0.94). The margin also matters for what the paper can observe. I define structural transformation within a region, whereas classical models of structural transformation do not separate city from village: when people move to the city, and the city is assumed to produce manufacturing and services, that is structural transformation. Because I separate the two regions and look at cities only, if I assumed that there is no agriculture in cities, I could not observe structural transformation at all. As cities get access to cheaper food, fewer people within the urban areas will be working in the remaining peri-urban agriculture. Figure 2 shows the reallocation in progress across the sample, and Section 5 shows it responds to the productivity of the hinterland.

3 Data and Outcomes

3.1 Data Sources

In this project, I merge several distinct datasets to create a unique panel and cross-sectional dataset of Indonesia’s urban areas and transmigration villages: administrative population census microdata (1980, 2000, 2010), Transmigration program village records and replication files from prior work on Indonesia’s Transmigration program (Bazzi et al. 2016, 2019), and satellite-derived remote sensing layers used to define urban areas and to construct travel-time (cost-distance) measures between villages and cities.⁵

Population census (1980, 2000, 2010). I use the population census individual-level microdata (full count in 2000 and 2010 and the 5% population sample for 1980) to measure population, employment, sectoral employment composition, and migration patterns. This includes data on each person’s current

⁵Further details can be found in Section A.1.

subdistrict and district of birth, which is crucial for identifying migrants from the Inner Islands to the Outer Islands. The 1980 data identify location down to the subdistrict level (level 3), and the 2000 and 2010 data down to the village/community level (level 4). For 2000 and 2010, village-level boundary shapefiles (level 4) are also available, and I use them to define city boundaries and village catchments (Section 3.4). The 2000-to-2010 subdistrict link itself is administrative rather than spatial, and is described below. I use 1980 outcomes as baseline controls and construct post-policy outcomes primarily in 2010, with decade growth measured over 1980-2000 and 2000-2010.

Transmigration Census and replication files. Data on settled transmigration villages come from replication files for two previous papers on the transmigration program, based on the Transmigration Census data. Information on the established year, the village locations, the number of migrants initially settled, and the village-level agro-climatic similarity index of Bazzi et al. (2016) is sourced from the Transmigration Census and previous research replication files.

Administrative boundary crosswalks. To harmonize subdistrict identifiers across census waves, I build on Olken (2009), which links historical BPS subdistrict codes (1976-2002), to match 1980 baseline measures to 2000 subdistrict identifiers. That source omits several provinces by construction, including Aceh and Maluku, so I extend it with additional passes built from the 1980 census codes themselves and from the BPS village panel. The extension adds 182 subdistricts holding 4.1 million people in 1980, 7.9% of the outer-island total, of which 120 are in Aceh and 52 in Maluku. To align 2000 and 2010 geography I work at the village level, because Indonesia’s subdistrict boundaries were extensively redrawn over the decade and the village is the finer unit both censuses record.⁶ Each 2010 census village is matched to its year-2000 village through the BPS village-history file, which records every village’s administrative code in each vintage from 1998 to 2013, and the 2010 village is then counted in that year-2000 village’s subdistrict. This is an administrative lineage published by the statistical office rather than a geometric assignment: a village that was renumbered, or whose subdistrict was split, is followed by its own code history. Where the file does not list a village, or where a 2010 village descends from year-2000 villages in two different subdistricts, the assignment is left to the boundary overlay and flagged. Section A.2 gives the construction, the share of villages each route places, and the checks the assignment passes.

Geospatial remote-sensing data layers. I use the Global Human Settlement (GHS) Urban Centre polygon layer to define urban agglomerations. GHS provides consistent urban agglomeration boundaries based on population, built-up area, and density. I focus the primary analysis on this satellite-based sample since it is based on a globally consistent and comparable definition; Section 2.2.1 details the

⁶Indonesia redrew subdistrict boundaries repeatedly between the two censuses, so a village’s 2010 subdistrict is frequently not the one whose territory it sat in during 2000. Matching village polygons to the year-2000 polygon they most overlap therefore reads boundary change as relocation, and does so most where boundaries moved most, which is in and around growing cities. The administrative lineage follows each village’s own code through every intermediate vintage, so a subdistrict split, merge or renumbering is recorded rather than inferred, and the assignment does not depend on how closely two boundary vintages happen to coincide.

construction.⁷ I also use the gridded raster friction surface created by Weiss et al. (2018) and combine it with maritime shipping routes to construct an Indonesia-specific friction surface for travel over both land and water, to more precisely account for Indonesia’s complex geography. This global raster is the study’s single travel-cost input; its vintage, and the limited role it has in the design, are set out in Section A.2.

3.2 Urban Outcomes

Aggregation to urban centers. Outcomes are measured on the subdistricts that overlap an urban center (Section 2.2.1) and dissolved into one urban-center area, the unit that corresponds to the meaningful urban labor market; Section A.3 sets out the levels the data come at and why the subdistrict is the finest comparable unit. I aggregate census outcomes (population, employment, sectoral composition, migration) and the city-level exposure measures to the urban center level by summing total counts for extensive variables (population, employment) and taking appropriately weighted averages for shares and price indices.

I construct urban outcomes from census data at the subdistrict level. Outcomes fall into three groups: population and employment growth, sectoral composition, and migration-based measures of demographic change, aggregated for each city c from individual microdata.

Population and employment. I use total population and total employment levels from the 2000 and 2010 censuses. I analyze both (i) endline levels, the logs of population and of total employment in 2010, and (ii) growth, the log change in each over 2000-2010. Employment is measured as total employed persons (constructed from counts), consistent with the reduced-form urban growth outcomes in the main results. The 2010 census supports two employment counts, a narrow one restricted to respondents recorded as economically active and a broader one covering everyone with a valid industry code; every employment share in the paper uses the broader count as its denominator, so the sectoral shares add up within a city. **Sectoral employment and shares.** Using census sector classifications, I construct sectoral employment in levels and shares. In particular, I measure: (i) employment in agriculture, manufacturing, and services; and (ii) service-sector subcomponents used in the main tables (e.g., trade, transportation, and other services), expressed as shares of total employment. I report results both in levels (log counts) and in shares, depending on the specification.

Migration outcomes. Migration measures are constructed from individual-level census microdata (primarily 2010). For each individual observed in 2010, I observe the current subdistrict of residence, while place of birth and residence five years prior are recorded only at the coarser district level. All lifetime and five-year migration measures are therefore defined as district-to-district or province-to-province moves. For each urban center, I construct the share of 2010 residents who: (i) were born in a different province (or a different district), (ii) were born in the same province (or the same district), (iii) were born in Java/Bali, (iv) lived in a different district five years earlier, (v) lived in a different

⁷The urban agglomerations shown in the descriptive maps are the GHS functional urban area (FUA) polygons, the larger agglomerations from an alternative GHS layer that adds commuting zones around the urban centers.

province five years earlier, (vi) lived in Java/Bali five years earlier. These outcomes allow me to quantify whether urban growth is accompanied by in-migration and to characterize the origins of migrants.

Prices, consumption, wages, and production (Susenas 2002/2011; PODES 2003). From Susenas, I construct subdistrict-level food/crop price indices using unit values, a housing/rent price index as a proxy for non-tradable/service prices, consumption expenditures and expenditure shares, and sectoral wages (agriculture vs. non-agriculture). These measures have no 1980 baseline, so they identify differences in end-of-period levels across cities. I also construct “real” wages by deflating nominal wages with the relevant local price indices. From PODES, I aggregate agricultural production (especially rice) to the subdistrict level to measure peri-urban agricultural production within urban subdistricts. For a subset of outcomes, I additionally construct an indicative “rice surplus” measure (production minus consumption) combining PODES production with Susenas consumption aggregates.⁸

3.3 Exposure: Agro-Climatic Similarity, Settler Counts, and Cost Distance

The main explanatory variable captures quasi-exogenous variation in the agricultural productivity of Transmigration villages arising from differences in the match between migrants’ origin environments and destination environments. I use the agro-climatic similarity index developed in [Bazzi et al. \(2016\)](#), which was designed to measure skill transferability of agricultural migrants across space.

Village-Level Agro-Climatic Similarity. Following [Bazzi et al. \(2016\)](#), agro-climatic similarity in a TM village v aggregates the agro-climatic “distance” between each migrant’s origin district and the destination village. Let x_d denote a vector of agro-climatic characteristics in origin district d (e.g., soil and terrain, rainfall and temperature averages, and stable geographic attributes such as distance to rivers/coast), and let x_v denote the corresponding vector for destination village v . Let π_{dv} be the share of migrants in village v born in district d . The village-level index is:

$$\text{Agroclimatic Similarity}_v = AS_v = (-1) \times \sum_{d=1}^I \pi_{dv} d(x_d, x_v), \quad (1)$$

where $d(\cdot)$ is an agro-climatic distance metric, the sum of absolute differences between origin and destination across the standardized characteristics, as in [Bazzi et al. \(2016\)](#). The leading (-1) converts a dissimilarity measure into a similarity measure. I use the index in the min-max rescaled form stored in the replication data, so $AS_v \in [0, 1]$ across the 814 program villages. [Bazzi et al. \(2016\)](#) show that this index predicts realized agricultural success of TM settlements (including higher productivity outcomes), motivating its use here as a proxy for the intensity of the rural productivity shock. Appendix [Figure A5b](#) shows the variation in the agro-similarity across settled Transmigration villages.

⁸The empirical analysis uses two Susenas waves jointly: Susenas 2002 for food consumption quantities (1-year aligned with PODES 2003 production, giving a clean consumption-production surplus measure) and Susenas 2011 for prices, wages, food share, housing rent, and consumption-by-food-category aggregates (1-year aligned with the 2010 Census outcome year, supporting the structural-transformation explanation). See Appendix [Table A1](#) for a per-variable summary of source wave, sample size, and construction.

The city-level exposure measure. To map the village-level productivity variation into an urban shock, I construct a city-level exposure measure that aggregates AS_v across the transmigration villages inside each city’s travel-time catchment. The variation in the resulting aggregated index of agro-similarity is shown in [Figure 3](#).

I take the village-level agro-similarity index AS_v (see [Section 3.3](#)) and average it across the villages that fall within a fixed travel time of the city, giving each of them equal weight. [Figure 4](#) illustrates the spatial matching of a sample urban agglomeration to nearby villages; proximity is measured throughout by cost-distance travel time in minutes. For each city, the exposure measure is therefore

$$AgSimilarity_c = \frac{1}{|V_c|} \sum_{v \in V_c} AS_v, \quad V_c = \{v : \tau_{cv} \leq 300 \text{ minutes}\}, \quad (2)$$

where τ_{cv} is the least-cost travel time in minutes between city c and village v (the cost-distance metric described in [Section 3.3](#)), V_c is the set of transmigration villages inside city c ’s 300-minute catchment, and $|V_c|$ is the number of villages in that set.

The settler count. To separate the role of productivity from that of scale, I construct a second, purely quantitative exposure measure: the number of individuals initially placed in nearby TM villages. Village-level initial placement counts N_v come from the Transmigration Census. I then aggregate N_v to the urban center as an inverse-travel-time-weighted sum over the same villages inside the city’s 300-minute catchment, a different aggregation from the exposure measure’s unweighted catchment mean (see [Section 4](#)). This controls for the possibility that cities surrounded by more TM settlements experience different population and demand dynamics independent of the productivity shock.

To control for the overall number of migrants resettled around a given city c under the Transmigration Policy (*TransMigrants*), I calculate the inverse-distance-weighted number of transmigrants across villages around each city. Unlike the agro-similarity exposure, which is an unweighted average over the catchment, this scale control is an inverse-distance-weighted sum of the initial settler counts N_v across the villages v in the catchment V_c (so it rises with both the number and the size of nearby settlements), with proximity measured by cost-distance travel time d_{cv} :

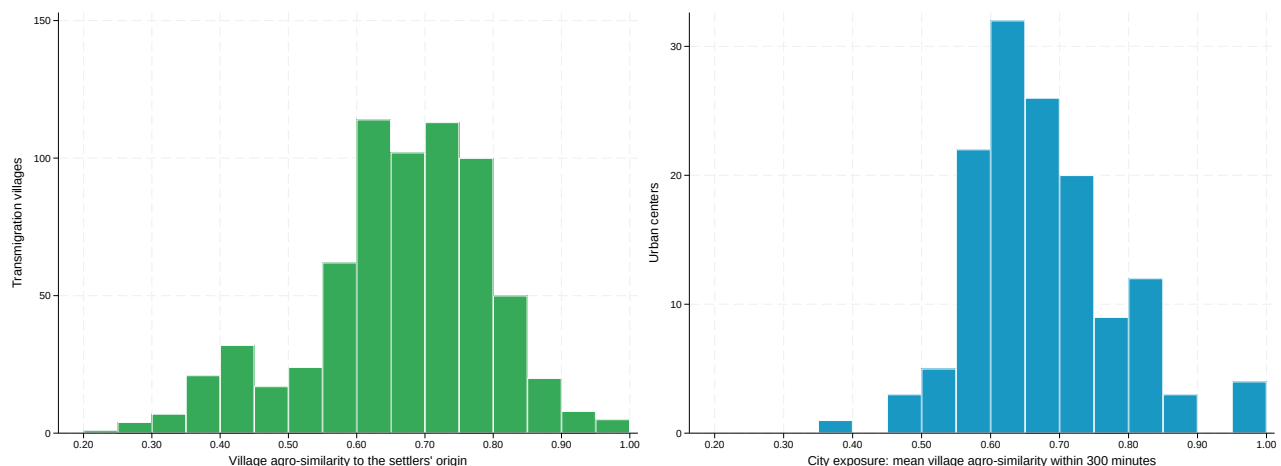
$$TransMigrants_c^{IDW} = \sum_{v \in V_c} \frac{1}{d_{cv}} N_v, \quad (3)$$

where $TransMigrants_c$ is the aggregated scale of TM settlement around city c : the sum, in thousands, of settler counts N_v across the villages in city c ’s catchment V_c , each inversely weighted by d_{cv} , the least-cost travel time in minutes between city c and village v .

Weighting of the similarity index and the settler count. Two measures enter from the village level: the agro-similarity index AS_v of [Bazzi et al. \(2016\)](#), an index of similarity from 0 to 1, and

the number of individuals initially placed in each village, which accounts for the size of the villages. The settler count enters inverse-distance weighted, as in (3). Villages that are further away contribute fewer migrants than the ones that are closer, and the number of migrants affects productivity but also generates demand for urban services, so distance is salient in that measure. The agro-similarity measure is first and foremost about agricultural productivity, and goods can be transported over distance more easily than people can travel to access services, so the city index is the average similarity of the surrounding villages without an inverse-distance weight. A productive village that is a little further away is not the same as a low-productivity village. The higher trade costs to a more remote village are not equivalent to that village being less productive, the two are different channels, and the distance dimension is in some ways accounted for by the weighted total number of migrants placed, which is a scale effect. Rescaling the similarity index by distance would not make sense under this interpretation, and different distance weights do not affect the estimates much (Section D.3). Because AS_v is rescaled to the unit interval, so is the city aggregate; in the regressions I enter it as a z-score computed over every urban center with a measured exposure.

Figure 3: Transmigration-village agro-similarity and the city exposure built from it

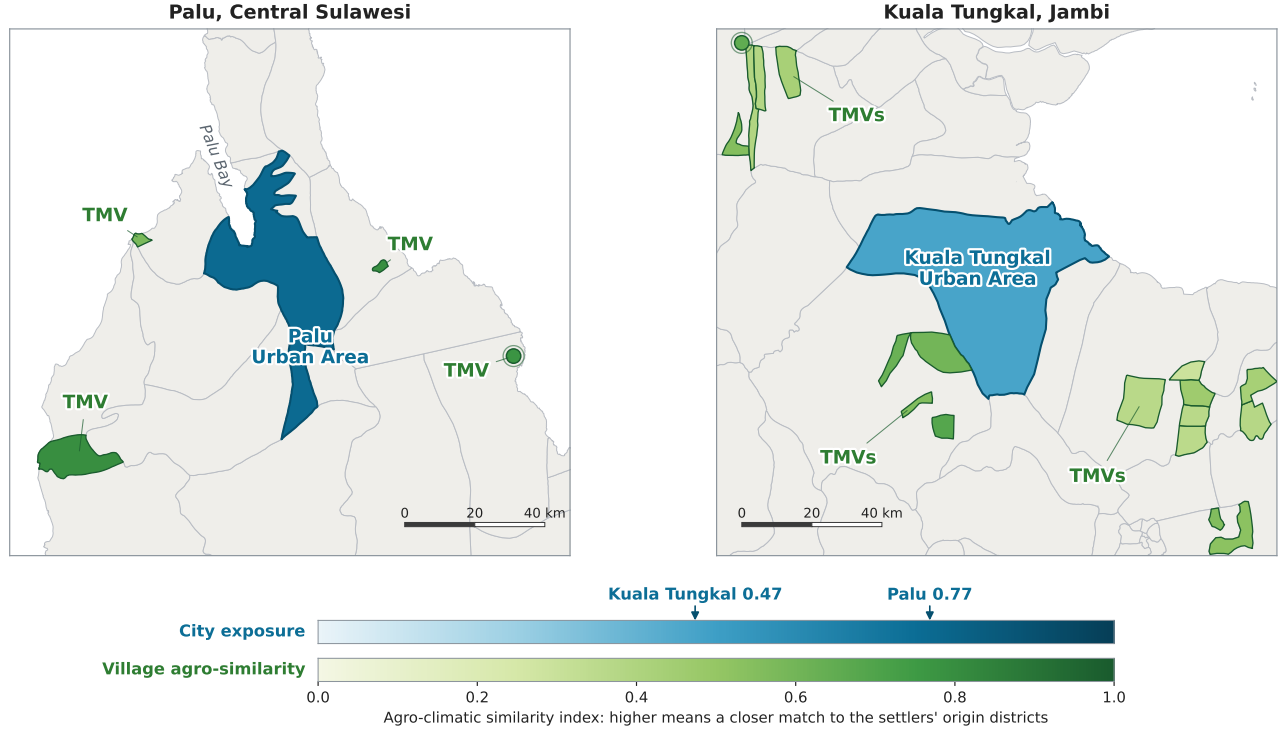


Notes: Left panel: agro-similarity to the settlers' region of origin for the 680 transmigration villages that lie within 300 cost-distance minutes of an outer-island urban center. The program has 814 villages in all and every one of them has a measured similarity; the other 134 sit beyond 300 minutes of any urban center and so enter no city's exposure. Right panel: the city-level exposure measure, the unweighted mean of village agro-similarity across the villages inside that window, over the 137 urban centers with at least one village in it. Averaging is why the city distribution is the narrower of the two: it runs from 0.38 where the village index runs from 0.22. Bin width 0.05 in both panels, drawn on one axis so the two are read against each other.

Maps in Figure 4 show two sample regions. Both cities are surrounded by a number of Transmigration villages in their vicinity, but one receives a larger agricultural productivity shock (better match quality in the villages), and the other gets a smaller productivity shock as, on average, less productive villages surround it.

Cost-distance travel time. I measure proximity using cost-distance travel time. Combining the satellite-derived layers with the administrative data, I construct a full travel-time matrix between villages and cities that is representative of the actual time it takes to travel from one place to the next:

Figure 4: Sample Areas of Urban Agglomerations & Nearby Transmigration Villages



Notes: Urban subdistricts and TM villages, each shaded by productivity: urban productivity exposure in the urban center vs. transmigration village productivity in the villages around it. The two regions are Palu, Central Sulawesi (left) and Kuala Tungkal, Jambi (right), drawn at one common scale. Exposure is the unweighted mean of village agro-climatic similarity over the transmigration villages within 300 travel minutes, so a city's shade is the average of the village shades in its catchment. Palu averages 0.77 over 9 villages and Kuala Tungkal 0.47 over 14, most of them beyond the mapped window, so each panel shows the part of its city's catchment that falls in the frame. A village whose boundary is absent from the village polygon layer is drawn as a circle at its recorded location. The map highlights the variability in the number, size and proximity of villages to cities in the two sample regions.

for all city-village pairs, the minimum travel time on a 1km-grid friction raster. In an island setting travel time is the better measure of proximity. The friction raster is the [Weiss et al. \(2018\)](#) travel-speed surface, and I add marine travel links so that travel over water between the islands is part of the estimates. This produces a full city-village travel-time matrix τ_{cv} , which selects the villages inside each city's exposure catchment and with it the main estimation sample (nearest transmigration village within 300 minutes), and which supplies the inverse-travel-time weights for the settler-count control.

Main estimation sample and proximity cutoffs. After constructing the city-village travel-time matrix, I restrict the estimation sample to urban centers whose nearest transmigration village is within a meaningful, achievable transportation distance of 300 travel-time minutes, $\min_v \tau_{cv} \leq 300$, where the minimum is taken over the subdistricts that make up the urban center, so a center enters the sample when its closest subdistrict has a program village within the cutoff. The 300-minute window keeps 134 of the 186 centers, of which 130 have a 1980 baseline the census recorded directly. [Figure A3b](#) shows the full distribution of travel time to the nearest transmigration village and the percentile the 300-minute cutoff falls at. The restriction does change the estimand. The exposure measure is a mean over the transmigration villages inside the window, so a city with no village inside it has no exposure

value at all, and the dropped centers have no measurement, not weak exposure. The coefficients are therefore identified off variation in the composition of nearby villages among cities that all have some transmigration exposure, and are local to that comparison.

3.4 Baseline Controls

Controls aggregated from villages. I include a rich set of controls capturing agro-climatic and geographic features of TM villages to ensure that identifying variation in the agro-similarity exposure is not mechanically driven by underlying land quality or geography. These controls include the components used to construct the agro-climatic distance metric in [Bazzi et al. \(2016\)](#) (e.g., soil and terrain attributes; rainfall and temperature patterns; and stable geographic characteristics such as distances to rivers and coasts). I aggregate each village covariate X_v to the urban center as an unweighted mean over the same villages as the exposure measure, $X_c = \frac{1}{|V_c|} \sum_{v \in V_c} X_v$. I refer to this set as “agro-climatic TMV controls” in the tables.

City-level baseline controls. Finally, I include city-level baseline controls to account for heterogeneity in initial conditions and to improve precision. One important set of controls includes **1980 baseline levels** of population and employment for each urban subdistrict, constructed from the 1980 census and matched forward to consistent subdistrict units. Some villages were settled in 1979 and 1980, before the 1980 census that provides these baselines, so the baseline precedes most placements and not all of them. The 1980 agricultural employment level enters the same way and in every specification. Where relevant, I also include baseline migration measures to address the possibility that some cities were already on different trajectories prior to exposure to nearby TM settlements.

The main specification (the Main Spec) conditions on four 1980 baselines: log population, log employment, the urban population share, and log agricultural employment, the last two being the baseline characteristics the balance test of [Section 4.3](#) flags.

The 1980 urban population share. Urban centers differ in how much of their population is administratively urban, and I summarize this with the 1980 baseline urban-population share, $UrbanShare_c$. For urban center c , the share is the 1980 census population living in administratively urban villages, summed across the center’s constituent subdistricts, divided by the center’s total 1980 census population:

$$UrbanShare_c = \frac{Pop_c^{urban}}{Pop_c^{total}} = \frac{\sum_{k \in c} Pop_k^{urban}}{\sum_{k \in c} Pop_k},$$

where k indexes the subdistricts that aggregate into center c , and Pop_k^{urban} counts residents of the villages the 1980 census classifies as urban (the BPS administrative *village status*). The split is administrative. The GHS Urban Centre determines which subdistricts belong to c , and the urban-rural division within each subdistrict comes from the census.

3.5 Trade and Openness Measures

Three measures summarize each city’s access to food trade beyond its own hinterland. The route measure is introduced here; [Section B.1](#) constructs all three, compares them, and validates the route measure against the 2004 rice-import ban.

Route openness. The preferred measure uses predetermined trade geography only. For each urban center I compute the route-aware least-cost travel time to its nearest customs port of entry, over the same land-and-sea friction surface as every travel time in the paper, plus the onward sea leg from that port to the nearer of Bangkok and Ho Chi Minh City, the hubs from which the bulk of Indonesia’s rice imports ship, with both legs converted to hours at the surface’s shipping-lane speed (adapting the distance-to-the-rice-routes construction of [Bazzi 2017](#)). Smaller totals mean closer alignment with the rice-import routes, hence higher baseline openness. The anchor set is the full customs port-of-entry list, so no realized trade quantity enters the measure. Route openness is the openness split variable in [Section 5.3](#).

The two other measures, gravity market access over realized rice-import volumes and the distance to the nearest customs port of entry, are constructed in [Section B.1](#); the first is the alternative split of [Table B1](#), and the second enters the balance tests of [Section 4.3](#) as a predetermined trade-access covariate.

4 Conceptual Framework and Empirical Strategy

This section sets out how a productivity gain in a city’s hinterland can reach the city ([Section 4.1](#)), the equation that estimates its effect ([Section 4.2](#)), and the identification argument behind it ([Section 4.3](#)).

4.1 Framework: Mechanisms Linking Rural Productivity to Urban Outcomes

The productivity shock first and foremost affects food production, directly. The villages that have a better match of people produce more food, and everything else follows from that. The region is the city and its rural hinterland, and the framework captures how a shock to some parts of that hinterland affects the urban center. Some areas get a food productivity shock, which raises agricultural production and reduces prices, and that has spillover effects on the city through the trade and migration channels, through price and wage effects. Many classical accounts treat this as a reallocation of people. Excess labor gets released from agriculture and goes to the cities.⁹ That is not the only thing at play. A priori it is not clear which channels or flows dominate, but they all shape growth versus employment and the structural transformation in cities. This simpler framework covers a single region and captures those effects within it; the spatial equilibrium model of [Section 6](#) adds the interactions across urban regions, through trade and migration between cities. The mechanism map ([Figure 5](#)) shows the effects, the price

⁹The closed-economy account descends from the dual-economy and food-problem literatures, in which an economy has to feed itself before it can industrialize ([Lewis 1954](#); [Schultz 1953](#); [Gollin et al. 2007](#); [Matsuyama 1992](#)); in the open-economy version the prediction reverses, and a more productive place keeps producing food ([Matsuyama 1992](#); [Bustos et al. 2016](#)). [Section F.15](#) sets out both accounts model by model.

channel through the city and the labor channel through the countryside, which the two paragraphs that follow take in turn.

As rural productivity rises, food production goes up, meaning that there is more agricultural output available at lower prices. This affects cities first through trade, as cities have access to cheaper food. That means that, on the one hand, cities themselves do not need to produce as much food within, and the remaining peripheral urban agriculture can decline as workers exit urban agriculture, a labor supply effect in the cities. At the same time, on the goods market, households spend less money on food as it is cheaper, and they can reallocate additional disposable income to services and other non-agricultural items, which creates a labor demand effect in the modern sector, as residents demand more services. The price channels work through the cities, and everything that sits behind cheaper food is a price effect. None of it requires anyone to move, and the pass-through into the price a city’s households face is larger the more of its food the city has to source from its own hinterland.

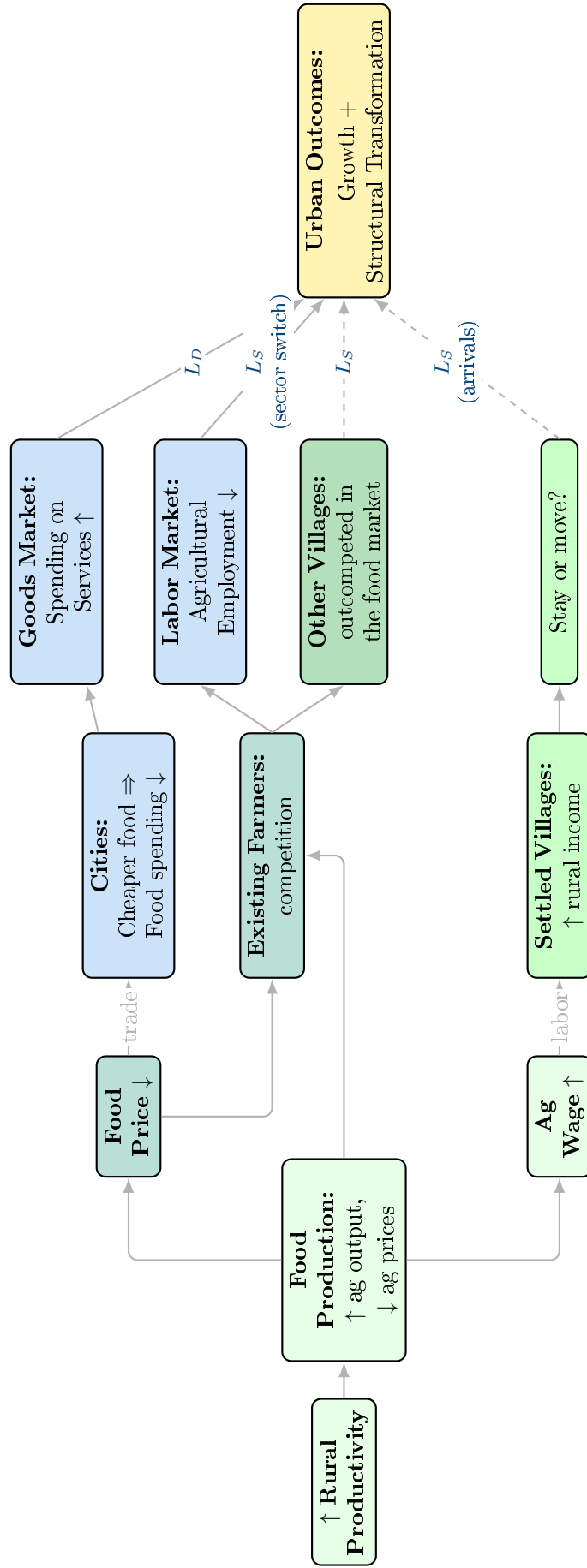
The wage effects work in the rural areas, as the workers adjust. Places that experience the positive productivity shock have higher incomes coming from agriculture, and that might have an ambiguous effect on out-migration. As they earn enough money, they might be more inclined to stay, whereas at the same time, as they earn enough money, they might be more easily able to afford the cost of moving to the city, as the liquidity constraints are a salient barrier to many rural-urban migrants (Bryan et al. 2014). At the same time, other rural areas that do not experience the shock are now at a disadvantage, so they are facing the increased competition from places that have the shock, while at the same time also being negatively affected by lower prices. In the model the shock enters as the productivity of the villages that received it, holding other things constant, so the rise in their income is the direct effect of higher productivity, and every other adjustment is part of the general equilibrium response.

Each region with the outer-island city and its hinterland together approximates the closed economy from Matsuyama’s theory (Matsuyama 1992), and how closed it is varies with the city’s access to outside food markets. Comparing more open and more closed urban regions therefore provides the variation to test the theoretical predictions within one country. Composition against size and who moves separate the two channels. The price channel moves a city’s employment composition without moving anyone, while the labor channel moves composition and population together and empties the villages that became better at farming (Section 5.1, Section 5.2 and Section 5.4). Where the response concentrates is the price channel’s signature, a large response in the cities poorly connected to outside food markets and a small one in the well-connected ones (Section 5.3). In the model I have sector switching and location choice, and it measures how much of the transformation each channel accounts for (Section 6).

4.2 Estimating Equation

I look at urban outcomes in 2010 as a function of the productivity induced by the Transmigration Program, measured by agro-climatic similarity, where β captures the causal effect of interest. I control

Figure 5: Mechanism Map



Notes: The figure sets out the routes by which a productivity gain in the transmigration villages can reach a nearby city; the text follows each route in turn, and the model in [Section 6](#) formalizes the same mechanisms. Each arrow into the outcome box is named for the labor-market side it acts on: L_S where the city's labor supply changes, by residents switching sector or by people arriving, and L_D where labor demand changes because spending has moved toward services. Green marks the agricultural side and blue the urban side, while blue-green marks the competition channel, which acts on both. Solid arrows are the goods routes the framework signs; dashed arrows are the migration routes it does not.

for the urban baseline in 1980, the earliest year for which I have comprehensive data, so the specification estimates a long-run difference:

$$y_c = \beta AgSimilarity_c + \gamma TransMigrants_c^{IDW} + C_{c,1980} \eta + X_c \nu + \varepsilon_c \quad (4)$$

where y_c is an endline urban outcome for urban center c , aggregated from its subdistricts (e.g., 2010 population and employment, or decade growth over 1980-2000 and 2000-2010), $TransMigrants_c$ is the inverse-distance-weighted sum of settlers across the Transmigration villages inside city c 's 300-minute catchment, in thousands (a scale control), $AgSimilarity_c$ is the *unweighted mean* of agro-similarity across the transmigration villages inside city c 's 300-minute catchment (2), entered as a z-score computed over every urban center with a measured exposure, $C_{c,1980}$ is the city's baseline characteristics (log population, log employment, the urban-population share, and log agricultural employment, all in 1980), and X_c contains unweighted means, over the same villages, of their agro-climatic suitability characteristics (climate, soil, terrain, and geography), including every characteristic used to construct the similarity measure.

For population and employment the same outcome measured in 1980 enters $C_{c,1980}$ directly, so the comparison runs between cities that started the period at the same size. For the compositional outcomes, the 1980 sectoral *shares* are not among the controls, so β there compares cities of the same baseline size, urban share, and 1980 agricultural employment level. The level enters and the share stays out, because the share is the ratio the compositional outcomes are measured in, so conditioning on it would hold the baseline composition fixed by construction. The specification for the outcomes without a baseline relies on the same identifying assumption, but has a slightly different interpretation. Prices, wages, and expenditure have no 1980 measurement to condition on, so those regressions compare levels across cities at the end of the period. This is why the price and wage results are suggestive of the mechanism.

To account for the differential distribution of villages around cities, as some cities might have received a differential number of villages in their immediate proximity around them depending on the greenfield land availability during the site selection, I include the overall exposure measure for cities, the settler-count control $TransMigrants_c^{IDW}$, to allow for the effects of agro-similarity to be conditional on overall village population, identifying its causal effect more precisely. The city-level policy exposure accounts for the variable (and not necessarily exogenous) placement of villages relative to cities, such that the coefficient on this variable in the main regression does not capture a causal effect, but controls for the overall number of migrants. Holding the count fixed, β is identified off the composition of that exposure, whether the villages a city happened to draw were well or poorly matched. The two terms are aggregated differently over the same villages. The count is an inverse-distance-weighted sum, as in (3), which assigns higher weights to the closer villages, with the assumption that similarly sized villages will have a stronger impact on the city if located closer, while the similarity measure is an unweighted

mean, as in (2), so it captures composition only and is invariant to how many villages a city happens to have. See a sample region of higher- and lower-productivity villages and exposed cities in Figure 4.

The Main Spec is estimated without fixed effects, and spatial correlation in the errors is handled by Conley standard errors with a 200-kilometer cutoff. The inclusion of island fixed effects is tested in one of the alternative specifications (Appendix Figure D3), and the robustness analysis varies the inference choice as well.

The estimated β captures how urban outcomes change in response to being surrounded by, on average, one standard deviation more productive villages within the 300-minute travel-time window of Section 2.2.1. There are different crops (Appendix Table E8 reports staples, rice, and legumes separately), and different ways to measure output: harvest or yield, per person or per hectare. I treat the estimates as a reduced form, without taking a stance on which of these measures is the relevant one, and Section 6 puts their magnitudes into economic units.

4.3 Identification

While the assignment to destinations was random, the total number of Transmigration villages and their locations around cities were not random, adding a layer of complexity to the analysis. I therefore control for the number of settlers, because I am not claiming that all village placements were as good as random, only the agro-similarity averaged across the settlers in each village. That measure is the match quality between the places settlers came from and the places they were sent. I also control for the characteristics of the destination villages. Among cities with equally large nearby settlements on similar land, some drew cohorts whose origin conditions resembled their new home and some did not, and that difference is the treatment.

The exposure measure is based on Bazzi et al. (2016), who show that at the village level it is a predictor and a proxy for rural productivity. One standard deviation higher agro-climatic similarity is associated with 20% higher rice productivity. I show a validation consistent with that within the estimation sample, where the measure does predict agricultural yields. Moving across the similarity range raises village agricultural productivity by about half a ton per hectare against a median near 0.43, a gradient estimated at conventional significance in levels (Appendix Figure C5 and Table C8).

The footprints that define the sample come from the 2020 GHS satellite classification, so the sample geography is endline geography. Three features keep this from contaminating the design. The sample is held fixed across all census years, so every within-unit comparison uses constant boundaries rather than boundaries that respond to the shock. Measured in 1980, exposure does not predict which of these units were already large (Appendix Table C5). And the growth margin is intensive rather than extensive. Exposure transforms cities that already exist, so the shock is not creating the units that a 2020-defined sample would then select (Section D.2).

The random component of the treatment is established at the village level. The program did not match migrants to destinations on agro-climatic similarity, so where each cohort ended up was as

good as random (Bazzi et al. 2016), and within a cohort it is plausible that people could have been assigned to another village among those that opened up around the same time, rather than a village that opened in 1979 being compared to one that appeared in 1987.

The quasi-random variation therefore sits at the village level, but I average it across villages to construct the city’s exposure, and since the village locations were chosen by the government, I do not argue that individual locations or village size are random. Once I am averaging the shock from each village to the city, a non-random component appears inside that averaging. The function incorporates the centrality of the city and how many villages happened to be selected around it, and averaging a random shock in a structure that has a non-random component means the aggregated shock itself now has non-randomness in it. In my case, a city surrounded by transmigration sites with land more similar to the Java and Bali origin mix might be bound to have higher exposure regardless of how settlers are reshuffled across the as-good-as-randomly assigned destinations, because of its pre-existing characteristics.

I condition on those characteristics separately, and that helps with identification; to do it more systematically, I run the test of Borusyak & Hull (2023). Whichever weights I choose to average the shocks, whether they read the distance or the population, the exposure structure they produce is predetermined, and that structure is the non-random component. Holding that structure constant and reshuffling people isolates the part of the exposure left once the expected exposure embedded in it is removed. The expected exposure is the average a given city would have received across all counterfactual settler assignments, and I subtract it from the actual exposure using the fixed weights, so the remaining variation is purged of the expected exposure.

The permutations behind that counterfactual take two forms that bracket the design. One reshuffles the village match within province-by-settlement-wave cells, the set of villages that opened around the same time, and a more demanding one holds each site’s destination fixed and re-draws only which settler origins arrived. I detail the construction in Section C.1.

Placebo and Balance Tests at Baseline

The placebo and balance checks use the 1980 baseline data to check whether urban outcomes at baseline were already correlated with the agro-similarity exposure, that is, whether there was selection at baseline; if there were, it would pose more serious concerns about the exogeneity of the later productivity exposure. The 1980 census is a baseline rather than a strictly pre-program year. Settlement began in 1979, and 127 of the 814 villages, accounting for 19.5% of the settlers, had been placed by the time of the count. A separate appendix check excludes the villages already settled by the time of the count from the construction of the exposure, and the balance is unchanged (Appendix Table C5). In addition, for village effects to trickle down to cities it is plausible that some time is needed, so there is less of a concern with this. Figure 6 plots each 1980 baseline characteristic’s standardized association with exposure, conditional on the settler count and the agro-climatic controls, the margins the design itself holds fixed. Baseline levels and shares are uncorrelated with the agro-similarity exposure. Twelve of

the fourteen rows are statistically indistinguishable from zero, among them city size, the unemployment rate, modern employment in levels and in shares, every birthplace row,¹⁰ and all three trade-access measures. Log agricultural employment is significant at the 10% level and the 1980 urban population share at 5%. Both are 1980 baselines the estimating equation conditions on, so the estimates below already hold both of them fixed. Every estimate below conditions on the 1980 urban population share and on 1980 log agricultural employment, the two baseline characteristics this test flags, so the imbalance is held fixed. Adding the 1980 birthplace shares to the control set leaves the compositional results unchanged (Appendix Table C7), and an exposure rebuilt from the villages settled after the census returns the same picture (Appendix Table C5).

Appendix Table C5 runs the same test in levels for the fuller set of baseline outcomes, urban center-level aggregates of 1980 individual census records regressed on average agro-similarity across villages, under two constructions of the exposure and two village sets, and reports it in Section C.1. Every weighting variant inside the 300-minute identification window behaves like the headline specification, and restricting the exposure to the villages settled after the count leaves that unchanged.

¹⁰The 1980 census records place of birth at the province level, so it cannot support the finer district-level split that is used in the 2010 decomposition.

Figure 6: Baseline balance: 1980 city characteristics and exposure



Notes: Each row plots the standardized association between a 1980 baseline characteristic and exposure, from a regression of the (standardized) characteristic on agro-similarity, the transmigrant settler count, and the agro-climatic control set, the same conditioning as the levels test in Appendix Table C5, with Conley 200 km spatial standard errors and 90% confidence intervals. The upper block (teal circles) collects 1980 baseline city characteristics: the ten outcomes of the placebo table, in the same order, and the 1980 urban population share; the estimating equation conditions on that share, and on log agricultural employment, as baseline controls. The birthplace rows split the 1980 population by province of birth, since SP1980 records place of birth at province level only; the two log counts are parts of one total and move independently, and the share row is the born-elsewhere count over 1980 population. Row N varies with the outcome, from 137 on the trade-access rows to 134 on the city rows and 131 on the unemployment rate, and the placebo table reports each one. The lower block (navy diamonds) collects predetermined trade openness: capacity-free port access, route openness to the Thai and Vietnamese rice hubs, and distance to the nearest customs port of entry. Urban-center sample whose nearest transmigration village lies within 300 minutes.

5 Reduced-Form Results

This section establishes the reduced-form case in four steps. Cities exposed to more productive transmigration villages move employment out of agriculture and into the modern sector, by much more than they grow in size (Section 5.1). The mechanism runs through the goods market. The effective cost of food falls, the exposed city stops producing its own staples and buys them from its hinterland, and household spending tilts toward non-food, while the settlers who received the productivity gain stay in their villages (Section 5.2). The transformation is concentrated exactly where a trade mechanism predicts, in cities weakly connected to outside food markets, and is offset in cities that can already import cheap food (Section 5.3). The villages that received the gain keep their workers and their settlers, and the village side and the city side of the same shock move in opposite directions (Section 5.4). These

facts identify the shock that the structural model in [Section 6](#) then decomposes into its food-trade and migration components.

5.1 Urban Structural Transformation

Table 2: Urban growth and structural transformation: the shift from agriculture to the modern sector (Modern = Manufacturing + Services)

	Population + Employment			Agric Empl		Modern Empl	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	ln pop	ln emp	unemp rate	ln L agric	agric %	ln L modern	modern %
TMV AgSim (z, 300min)	0.060 (0.059)	0.033 (0.056)	0.009 (0.007)	-0.167* (0.096)	-0.053*** (0.018)	0.169*** (0.061)	0.068*** (0.015)
Transmigrant count (IDW)	✓	✓	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓	✓	✓
Mean dep var	11.839	10.930	0.123	9.816	0.374	10.229	0.517
Std dev dep var	0.858	0.843	0.041	0.898	0.184	0.979	0.157
Observations	133	133	133	133	133	133	133

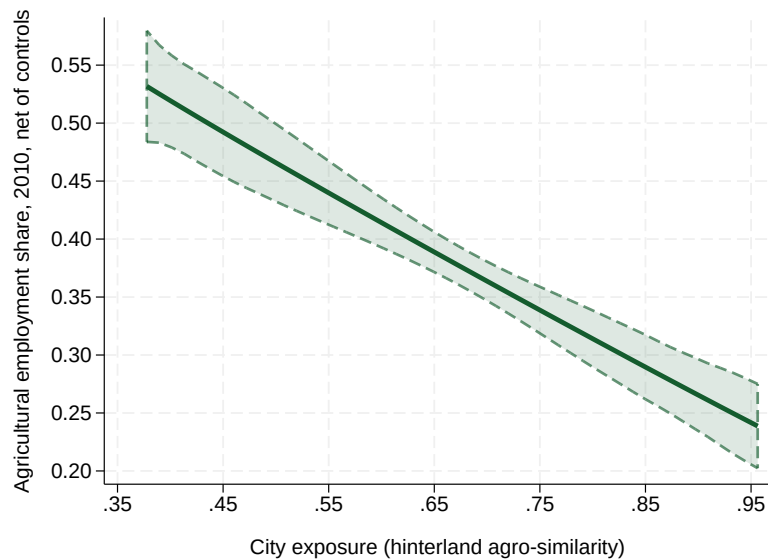
Notes: UC-level regressions, 300min cost-dist sample. Modern sector = manufacturing + services, per the Matsuyama 1992 framing. Cols 1-3 = pop, total empl (ln), and unemployment rate; cols 4-7 = ln agric employment, agric share, ln modern employment, modern share (all 2010). Main Spec controls, with the transmigrant settler count entering in thousands (shown as ✓; polynomial variants are reported in the appendix as a robustness exercise). Spatial Conley SEs (dist cutoff 200km), no prov FE. Stars are from the analytic Conley p-value. [Table C4](#) reports the randomization-inference p-value for every column under each permutation scheme, and [Table D12](#) runs the same columns across spatial-inference radii. *** 1%, ** 5%, * 10% significant.

[Table 2](#) reports the headline estimates across the main sample of urban centers, those whose nearest transmigration village lies within 300 minutes of travel time. The point estimates on population and total employment are positive but not distinguishable from zero at the headline radius. There is no effect on the unemployment rate, so cities are not growing at the expense of people being jobless. Agricultural employment is declining and modern-sector employment is growing, in both levels and shares. The agricultural employment share falls by 5.3 percentage points, the agricultural level falls with it, at the 10% level, and modern-sector employment (combined manufacturing and services) rises by 0.169 log points in level and 6.8 percentage points in share. The analysis works with the modern sector as a whole throughout; its manufacturing and services components are reported separately in Appendix [Table E6](#). Every column includes the Main Spec controls (the nearby transmigrant count, the 1980 baselines for population, employment, urban structure, and agricultural employment, and the agro-climatic control set), with Conley spatial standard errors at a 200-kilometer cutoff.¹¹

The change in sectoral employment is much greater than the change in population. [Figure 7](#) shows the relationship behind the share result: the agricultural employment share against exposure, with the Main Spec controls partialled out, and the fall is close to linear over the whole range.

¹¹Appendix [Table D12](#) re-runs the same columns at 150, 450, and 500 kilometers and with clustering on the catchment components; the three compositional columns reject at the 10% level under every variant that returns a standard error, and the agricultural level does not survive the component clustering.

Figure 7: The agricultural employment share against city exposure, semiparametric fit



Notes: Partially linear regression of the 2010 agricultural employment share on city exposure, with the Main Spec controls entering linearly (the settler count weighted by inverse distance within 300 minutes, the 1980 baselines, and the agro-climatic controls) and exposure entering nonparametrically. The curve is a local linear fit (Epanechnikov kernel, bandwidth 0.30) of the share net of the controls' linear part, with a 90% confidence band. Exposure is the mean agro-climatic similarity of the transmigration villages within 300 minutes of the city, in index units; the same relationship per standard deviation of exposure is column 5 of [Table 2](#). Sample: the urban centers of the estimation sample.

In 1980, as the villages were being settled, exposure is uncorrelated with city population and employment structure ([Appendix Table C5](#)). The two 1980 baselines it does correlate with, the urban population share and log agricultural employment, are controls in every column, and adding the 1980 birthplace shares leaves every headline estimate unchanged ([Appendix Table C7](#)). The placebo tests are in [Section 4.3](#). The natural concern is that the result is trivial in the share. Migrants who move to cities most of the time do not move for agriculture, so more migrants coming in would move the share by themselves. That is why the level matters. Modern-sector employment is growing, and growing more than the population itself, which means there must be sectoral switching. The data reject that account. Arrivals farm too. Workers born outside the city's district farm at a rate of 27.1% against 40.4% for the locally born, so producing the estimated fall through arrivals and nothing else would require an inflow of 105.6% of a city's workforce, against the 3.4% the employment response implies, and a one-sided test rejects that account at the 10% level ($p = 0.036$).¹² The rejection holds at every arrival farming rate the 2010 census measures ([Table 3](#); [Section E.2](#)).

Crossing each city's 2010 workforce by birthplace and by industry splits the fall into two components: a composition component, from the rising migrant share of the workforce, and a within

¹²[Hamory et al. \(2021\)](#) use the IFLS panel to study sectoral switching and find that individual fixed effects remove over 80% of the estimated agricultural productivity gap, so the gap largely reflects selection into sectors. The IFLS sample with sectoral histories in the outer-island cities that I study is too small, covering too few people and too few unique cities, the level at which I define treatment.

Table 3: Could arrivals alone have produced the fall in the agricultural share?

	Estimate
<i>Panel A. What the 2010 census measures</i>	
Agricultural share of workers born in this district	0.404
Agricultural share of workers born outside it	0.271
Comparison city's agricultural employment share	0.374
<i>Panel B. The two coefficients on exposure the accounting compares</i>	
Agricultural employment share	-0.053*** (0.018)
Log total employment	0.033 (0.056)
<i>Panel C. The accounting</i>	
Inflow required to produce the fall through arrivals alone	105.6%
Inflow the estimated employment response implies	3.4%
Gap in logs, implied less required	-0.688 (0.382)
One-sided p against the arrivals-only account	0.036
<i>Panel D. What the fall in the agricultural share decomposes into</i>	
Total fall, 1980 to 2010	-0.038** (0.017)
Within, workers switch sector (91% of the fall)	-0.034** (0.016)
Composition, the migrant share rises (16%)	-0.006 (0.004)
Interaction (-8%)	+0.003 (0.005)
Within component on the arrival-recency cut	97%
Urban centers	133

Notes: 133 urban centers, Main Spec, Conley spatial standard errors at a 200-kilometer cutoff. Panel A reports unweighted means over the estimation sample, built from the individual records of the 2010 census; a migrant is a worker born outside the district of residence. Panel B reports the two coefficients on exposure, estimated jointly as one stacked system so that Panel C can use their cross-equation covariance, with standard errors in parentheses and stars taken off that same inference. Panel C applies the accounting identity of [Section E.2](#): a city of L workers with an agricultural share s that takes in a gross inflow g , a share a of whom also farm and with no incumbent changing sector, ends with its share lower by $g(s - a)/(1 + g)$, so producing the estimated fall requires an inflow of 105.6% of the workforce against the 3.4% the employment response implies. On a city of the sample's mean employment, 91,821 workers, that is 96,979 arrivals required against 3,102 estimated. The p-value is one-sided against the null that the employment response is at least as large as the identity requires, computed by the delta method on the stacked covariance, and its cell is shaded by evidence strength. The rejection holds at the 10% level at every arrival farming rate the 2010 census measures, from 0.211 to 0.281 across the four migrant definitions ([Table E16](#)). Panel D crosses each city's 2010 workforce by birthplace and by industry and splits the fall into a within component, incumbents changing sector, a composition component, the migrant share of the workforce rising, and their interaction; shares are each component over the total fall and need not sum to 100 because the interaction offsets. The last row repeats the within component on a second cut that splits the workforce by arrival recency instead of by birthplace, and the two cuts agree. The construction and the two-panel figure are in [Section E.3](#).

component, from workers of a given origin leaving agriculture. Table 3, Panel D, puts 91% of the fall on the within component and 16% on composition, the two adding to more than the whole because a small interaction term offsets them; the within share is above the 80% that Eckert & Peters (2025) attribute to within-region change in the 1880-1920 United States (estimates and bands in Appendix Figure E3a). The census is a repeated cross-section, so the within component still mixes two mechanisms: workers changing sector, and one generation replacing another with nobody changing jobs, which Budi-Ors (2026) puts at 68% of Indonesia’s aggregate agricultural-share fall.

Two censuses ten years apart separate the two mechanisms, because ten-year age bands let a birth cohort be followed one band forward. Among native-born cohorts, which in-migration between the waves cannot reshape, about half of the effect of exposure on the 2010 agricultural share was already in place by 2000, a third comes from workers within a cohort leaving agriculture between 2000 and 2010, and the remainder from older, more agricultural cohorts being replaced by younger ones (Figure E3b; the shares and their intervals are in Section E.3). Incumbent urban workers are switching sectors, an adjustment that an inflow of newcomers cannot supply, and Section 5.2 traces that switching to the goods market.

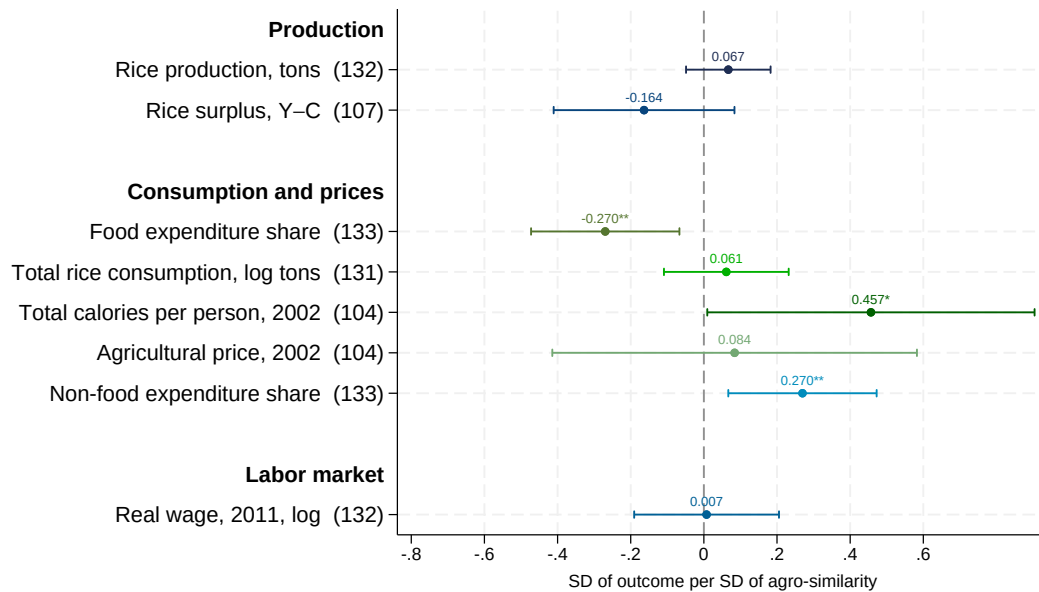
These estimates also survive the design-based test of Section 4.3 in its double-robust form. With the exposure a city would expect under counterfactual settler assignments entered as a control, the agricultural employment share falls by 0.042 and the modern employment share rises by 0.051, the first at the 5% level and the second at the 1%, against headline values of -0.053 and $+0.068$ (Appendix Table C2). The scalar form subtracts that expected exposure from the treatment and leaves little variation to identify anything, so its coefficients attenuate toward zero and no outcome rejects under randomization inference (Figure C2; Appendix Table C4). The design-based support for these results therefore comes from the double-robust column, and the precision comes from the Conley standard errors in Table 2.

Design-level robustness. Section D.3 re-estimates the three headline outcomes across 85 specifications that vary the control set, island fixed effects, the construction of the exposure measure, and the travel-time cutoff. Appendix Figure D2 shows that evidence grouped by design choice, Table D9 defines each variant, and Appendix Figure D3 extends the grid to every outcome. No specification is both wrong-signed and distinguishable from zero for any of the three outcomes. Modern-sector employment and the agricultural employment share each clear the 10% level in 49 of the 85, log population in 13. The tighter 180-minute window, on a much smaller sample, is the one family that misbehaves, and the specification space excludes it for that reason. The estimates are insensitive to whether the settler-count control enters linearly, in logs, or with a quadratic term (Section D.6), and to controlling in addition for the settlers’ size relative to the pre-existing rural population of the city’s hinterland (Appendix Table D1). The extended robustness material, including the price-measurement checks behind Section 5.2, is collected in Section C.1.

The proximity window and the sample construction are varied the same way. Appendix Figure C3 re-estimates the effects at every proximity window from 200 to 500 minutes, with the exposure itself reconstructed at each radius. The estimates keep their sign throughout, the sectoral responses hold at most windows of 300 minutes and wider, and the tighter windows are imprecise. Appendix Table A5 re-estimates the seven main outcomes without the reclassified urban areas and then under stricter urban criteria, on 81 centers, and the sectoral results hold in both. The modern-employment result is the same in the cities that entered 1980 more and less agricultural (Appendix Table D11), while splitting at the median 1980 population separates the cities. Modern-sector employment rises by 0.145 log points more in the larger half and population by 0.176 more, and the agricultural share moves alike in both halves (Panel B of Appendix Table D11).

5.2 The Goods-Market Mechanism: Food Trade, Not Migration

Figure 8: The goods-market mechanism: standardized responses to exposure (90% CIs)



Notes: Standardized response of the exposed city's goods and labor markets to a one standard deviation increase in agro-similarity, each outcome z-scored over the urban centers inside the 300-minute window, from the Main Spec reduced form with Conley 200 km spatial standard errors; 90% confidence intervals. Rows are grouped by mechanism into the city's own production, household consumption and prices, and the labor market. Sample sizes are in parentheses next to each row: coverage is the full estimation sample for the census-based rows, with the Susenas- and PODES-based consumption and rice rows limited by survey coverage. The calorie row sums every item the Susenas food module records and is measured in the 2002 wave, the wave the rice-surplus row's consumption input also uses; the other household rows are Susenas 2011. The agricultural price is the urban center's own raw-food Paasche index, read at that same 2002 wave. The real-wage row is the log of the urban center's 2011 median wage over its own local price index, on the full estimation sample; it is the wage response Table E2 reports. The number printed above each marker is that row's coefficient. Sample restricted to urban centers whose nearest transmigration village lies within 300 minutes. *** 1%, ** 5%, * 10% significant.

Figure 8 collects the responses of the city's goods and labor markets in one picture. The price response is not distinguishable from zero in either direction, so the figure shows a quantity adjustment. Urban residents buy more of their food from the rural hinterland and produce less of it themselves (the agricultural price in the 2002 wave is +0.08 of a standard deviation with a standard error of 0.30, the

agricultural-price row of the figure). The rural food production has more comparative advantage in these higher-productivity villages, and so urban dwellers are now buying the food, which is why their consumption is not reducing, but their own production minus consumption is reducing (the rice-surplus row; the city’s own production row is flat). Importantly, the food expenditure share is going down without the city lowering the total quantity of rice it consumes or households lowering the calories they consume (the rice row is indistinguishable from zero, and calories, if anything, rise), which is consistent with the idea of subsistence in food consumption, that once basic food consumption needs are satisfied, it will not grow as much. And so now the residents, through this potentially cheaper food, have access to satisfy those basic needs and allocate and spend more money on non-agricultural items, therefore raising demand for non-agricultural items including services and manufacturing products, which then drives the demand for labor in that market up as well. The model in [Section 6](#) formalizes this demand structure, non-homothetic preferences with a subsistence term in food.

[Table E2](#) reports these budget and price outcomes in their own units, and adds the wage response. The non-food expenditure share rises by 1.3 percentage points, distinguishable from zero at the 5% level, while housing rents and all three wage measures, the non-agricultural wage, the overall wage, and the real wage, are statistically indistinguishable from zero in this 2011 wave. Rents do not move detectably in the 1996 wave either ([Appendix Table E12](#)). Flat wages alongside expanding modern-sector employment are the pattern the model must produce through its offsetting price and land terms, since the headline calibration sets the agglomeration elasticity to zero ([Section F.7](#)), and they are why the reallocation shows up in employment composition.

The price effects, and the fact that the unit-value point estimates in the 2011 wave are small, positive, and insignificant, have several causes. First, prices are measured imprecisely. Second, they combine two effects. Some of the food is cheaper from the rural hinterland. At the same time, the appendix shows evidence of a quality improvement. One of the household consumption modules had data broken down by type of rice, and that is also consistent with more households buying higher-quality rice, although those estimates are also imprecise.¹³ The 2002-2011 window also brackets the national rice-price inflation that followed Indonesia’s 2004 import restrictions, so the small positive measured coefficient is not evidence against a locally cheaper delivered price. The size of the delivered-price decline is model-implied.

[Table 4](#) splits the 2010 residents aged 20 to 55 by birthplace into four groups: born in the same district as the city, born elsewhere in the same province, born in another outer-island province, and born on Java or Bali; the column labels abbreviate these four groups.

The policy moved two million people, so a mechanical effect is possible. Settlers could live in these villages for a few years and then move to the cities. The data show otherwise. Cities surrounded by more productive villages have fewer residents born in Java or Bali at the end of the period. That

¹³The appendix reports the price result across seven price measures, all small and imprecise, together with the decomposition behind the quality-upgrading explanation. Households in exposed cities buy less rice per capita while paying more per kilogram, and the city-level rice-price growth is absorbed by the city’s overall CPI growth ([Figure D4](#)).

Table 4: Where urban growth comes from: 2010 residents by birthplace

	Birthplace decomposition (% of 2010 cohort)			
	(1) Same district	(2) Same province	(3) Other outer-island province	(4) Java or Bali
TMV AgSim (z, 300min)	-0.043* (0.023)	0.043** (0.018)	0.018 (0.021)	-0.018* (0.011)
Transmigrant count (IDW)	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓
Mean dep var	0.667	0.130	0.116	0.087
Std dev dep var	0.187	0.092	0.129	0.084
Observations	133	133	133	133

Notes: Each column is a separate urban-center-level regression on the main city sample: the outer-island urban centers whose nearest transmigration village (TMV) lies within 300 travel minutes and whose 1980 baseline the census recorded directly. The regressor, TMV AgSim (z, 300min), is agro-similarity: the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, standardized over every urban center with a measured exposure. Every column includes the Main Spec controls: the inverse-distance-weighted (IDW) count of settlers placed within 300 minutes, the 1980 log population, log employment, urban population share, and log agricultural employment, and the sixteen agro-climatic covariates averaged over the same window. Standard errors in parentheses are Conley spatial standard errors with a 200-kilometer cutoff. Outcomes are the four mutually exclusive shares of the 1955 to 1990 birth cohort of 2010 residents (aged 20 to 55 at the census) by birthplace, in column order: born in the same district as the city, born elsewhere in the same province, born in another outer-island province, and born on Java or Bali; the four shares sum to one within each urban center. *** 1%, ** 5%, * 10%.

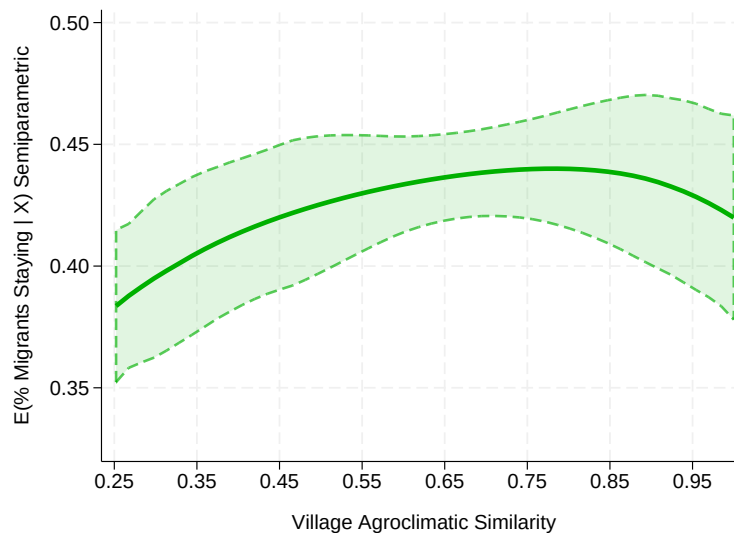
share is the best proxy for transmigrants I have in the city, so if anything, the cities surrounded by more productive villages do not have more of those transmigrants. Other rural migrants are moving in from elsewhere in the same province. They are likely primarily residents of other villages, some of whom could be transmigrants, and children already born on the Outer Islands.

The settlers do not leave the villages where they are well matched. The villagers are, if anything, slightly more likely to stay in villages that are more productive, and the relationship is nearly flat (Figure 9), which is also saying that if cities are experiencing more migrants, they are not coming from the more productive villages, if anything.

Migration is a secondary force. The model’s counterfactuals confirm that the sectoral reallocation can occur without migration and cannot occur without trade. The people best placed to transmit the shock into town are the ones who stay, and the in-migration that does occur is a regional labor-market response to modern-sector opportunity, with no farm labor being released. The model in Section 6 captures this with a low location-choice elasticity, consistent with the high internal mobility costs that Bryan & Morten (2019) document for Indonesia, and a companion analysis of urban emergence finds that exposure transforms existing cities near the productive villages (Section D.2).

Three rival explanations of the same shock. A rural productivity gain can reach a city through more than the food market, and the theory itself names the rival explanations, set out in full in Section F.15.5: workers released off the farm, capital moving out through the banking system (Bustos et al. 2020), and a factor-biased gain that pulls labor into farming at a fixed outside price (Bustos et al.

Figure 9: Productive villages retain their settlers: settler stay-share vs. agro-climatic similarity



Notes: Semiparametric (local linear, Epanechnikov kernel, bandwidth 0.30) fit of the village settler stay-share against the village's agro-similarity, with a 90% confidence band, partialing out the same village geographic controls and estimated on the common village sample with both employment and migration outcomes non-missing ($N = 808$). Plotted over the 0.25 to 1 similarity range. The settler stay-share named in the text divides a village's 2000-census residents born in Java or Bali, excluding residents who reported living in Java or Bali five years before the census and residents born after the village was settled, by the settlers the program placed there; the plotted fit uses the broader numerator without the two exclusions. The linear gradient under each definition, with island-clustered, wild-bootstrap, and Conley inference, is in Appendix Table C9.

2016); outside the theory sits a fourth possibility, that the state, having settled the villages, built the roads and facilities that transformed the cities. Each is tested. The labor release is a prediction about the villages that received the gain, and those villages retain their workers and their settlers (Figure 10, Figure 9), so there is no release for the classical account to send. A capital mechanism and a factor-bias mechanism do not give a city's food-import connections any role in whether its labor market responds to the hinterland, so both fail to predict the concentration of the response in trade-closed cities that Section 5.3 establishes. The state-provision explanation is tested directly. Exposure does not predict road-quality improvement over 1990 to 2007, the facilities a city acquired, or the placement of the INPRES school-building program (Section E.4), so the transformation does not track public investment.

5.3 Heterogeneity: The Open-Economy Test

The trade mechanism makes one further prediction. A city that can already import food cheaply gains little from a more productive hinterland. Its delivered food price barely depends on the local harvest, so its labor allocation does not respond either. A city weakly connected to outside food markets depends on its own hinterland, so the same shock has its largest effect there. This is the delivered-price mechanism's own prediction, in the spirit of the openness result in Matsuyama (1992), and it turns trade openness into a testable dimension of heterogeneity. The structural transformation is expected to concentrate in

Table 5: Agro-similarity $\times$ trade openness: exogenous route geography

High openness = below-median route distance to the Thailand/Vietnam rice routes (exogenous geography)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	ln pop	ln emp	unemp rate	ln L agric	agric %	ln L modern	modern %
AgSim, more closed cities: β_1	0.130	0.113	0.003	-0.152	-0.065***	0.283***	0.093***
	(0.086)	(0.082)	(0.009)	(0.132)	(0.024)	(0.090)	(0.022)
AgSim, more open cities: $\beta_1 + \beta_2$	0.020	-0.011	0.012*	-0.177*	-0.046**	0.107	0.054***
	(0.057)	(0.059)	(0.007)	(0.096)	(0.021)	(0.067)	(0.016)
High openness (=1)	0.023	0.005	-0.005	0.044	0.003	-0.012	-0.013
	(0.067)	(0.082)	(0.011)	(0.150)	(0.028)	(0.084)	(0.019)
Mean dep var	11.839	10.930	0.123	9.816	0.374	10.229	0.517
Std dev dep var	0.858	0.843	0.041	0.898	0.184	0.979	0.157
Observations	133	133	133	133	133	133	133

Note: High openness = below-median route distance to the Thai and Vietnamese rice hubs. Main Spec throughout: settler count, 1980 city baselines and the agro-climatic controls. Conley 200 km

Notes: Sample: the headline-sample outer-island urban centers (nearest Transmigration village within 300 minutes of cost-distance travel time), every one of which has a measured route-openness position. The route measure is defined in Section 3.5; the gravity market-access split is Table B1. The four sector columns, two agricultural and two modern, are set in bold. Measurement caveat: the international leg is a great-circle approximation, which understates distances from western ports whose actual sea route rounds the Strait of Malacca; this compresses the open end of the gradient but preserves the ranking. *** 1%, ** 5%, * 10%.

trade-closed cities and to be muted in trade-open ones. Table 5 runs the test on predetermined trade geography, each city's route-aware cost-distance to the Thai and Vietnamese hubs that ship the bulk of Indonesia's rice imports (Dawe 2008), in the spirit of Bazzi (2017); because this is fixed baseline geography, identification relies on it. A second measure, gravity market access built from realized rice-import volumes and in part an equilibrium outcome, is reported as additional evidence in Table B1. The specification interacts exposure with an indicator for high openness, so β_1 is the exposure effect in trade-closed cities, β_2 the high-openness offset, and $\beta_1 + \beta_2$ the net effect in trade-open ones.

The causal claim rests on the exogenous route geography. In trade-closed cities a one standard deviation increase in exposure lowers the agricultural employment share by 6.5 percentage points and raises modern-sector employment by 0.283 log points, both precisely estimated, while city size does not respond detectably in either regime. In the more open cities the agricultural share falls by 4.6 percentage points, and modern-sector employment rises by 0.107 log points, an estimate not distinguishable from zero, so the closed-city response is about 1.4 times the open-city response on the agricultural share and about 2.6 times on modern-sector employment. The offset itself, β_2 , is -0.176 log points on modern-sector employment (s.e. 0.093), distinguishable from zero at the 10% level, as is the offset on the modern share, and 0.019 on the agricultural share (s.e. 0.026), which is not. The agricultural level falls at the 10% level in the open cities and imprecisely in the closed ones. The offset is attenuation, not reversal. The open-city effects keep the closed-city sign, as a delivered-price mechanism implies, where specialization along comparative advantage would flip it. Agricultural productivity of the surrounding hinterland does not play as much of a role in shaping urban outcomes in cities with easier access to open markets, which may be because food supply is much less of a binding constraint in these cities at baseline; there are no baseline data on prices and food availability, so the paper cannot say much

about that directly. The split is difficult to rationalize outside the goods market. A migration or amenity mechanism, generic income growth, land conversion at the urban fringe, or public provision gives a city's rice-import connections no role in whether its labor market responds to the hinterland, while a food-trade mechanism requires exactly that switch, so each rival would predict effects in open and closed cities alike, and public provision also fails the direct tests of [Section E.4](#). Of the paper's reduced-form results, this one points most directly to the delivered cost of food as the operative channel. Construction and validation of both openness measures, including a validation of the route measure against the price response to Indonesia's 2004 rice-import ban, are in [Section B.1](#).

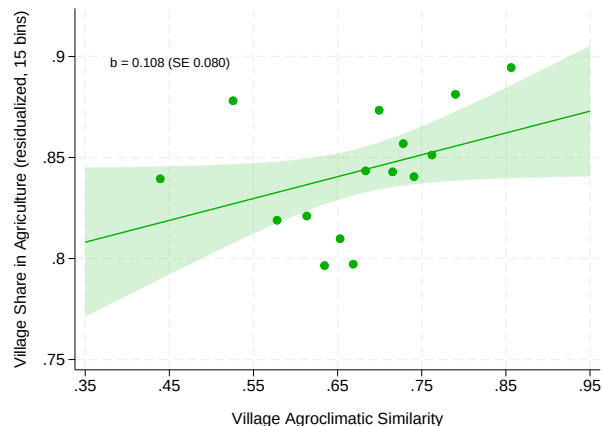
5.4 The Village Side

If I look at the villages themselves, the results here are across villages, based on the village agro-climatic similarity.

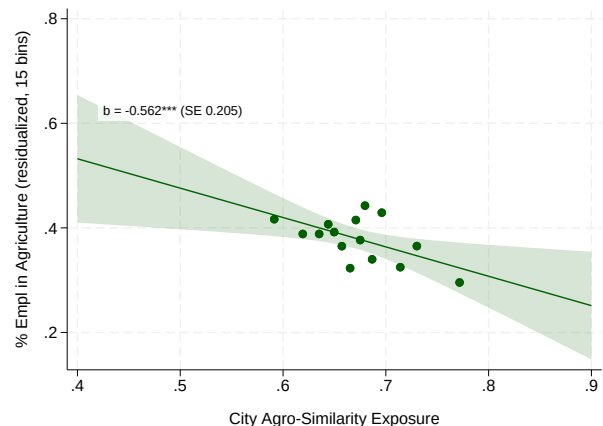
The villages that received the productivity gain respond in the opposite direction from the cities they feed.

Figure 10: The two sides of the food-trade relationship: agricultural employment share vs. agro-similarity

(a) Transmigration villages: share rises



(b) Nearby cities: share falls



Notes: Binned scatter plots: dots are 15 quantile-bin means of the outcome and the exposure, each residualized on the panel's controls with means added back, and the linear fit and its 95% band are computed on the residualized coordinates; the in-plot coefficient is the linear gradient with its Conley 200 km standard error. **(a)** The share of a village's workers in agriculture against the village's own agro-climatic similarity, residualized on village terrain, soil, geographic, and climate controls, plotted over the 0.35 to 0.95 similarity range. The gradient on the common village sample with both employment and migration outcomes non-missing ($N = 808$) is 0.066 (island-clustered SE 0.083; wild cluster bootstrap over the four island clusters, Webb weights, $p = 0.83$; Conley 200 km $p = 0.32$). **(b)** The 2010 agricultural employment share of urban centers whose nearest transmigration village lies within 300 minutes, residualized on the Main Spec controls, plotted over the 0.4 to 0.9 exposure range; the same relationship in standardized units is column (5) of [Table 2](#).

[Figure 10](#) plots the agricultural employment share against agro-climatic similarity in the transmigration villages (panel a) and against hinterland exposure in the nearby cities (panel b). In the transmigration villages, panel (a), the agricultural employment share is flat to mildly increasing in agro-similarity, with no sign of the negative gradient a release of labor requires. Well-matched villages

keep farming. In the nearby cities, panel (b), the same variation lowers the agricultural employment share (Table 2, column 5). The result within cities is almost the exact mirror of that in the villages themselves, where the more productive villages are retaining more workers in agriculture. In response to the positive productivity shock, the villages focus on agriculture and trade this food with the cities, and the cities then reduce agricultural employment, freeing up workers who no longer need to produce food within the city given the higher food supply from the villages.¹⁴ These are the two trade regimes of Matsuyama (1992) visible in the same setting, the village specializing in its comparative advantage while the city it feeds, with cheaper food, moves its employment out of agriculture and into the modern sector, and a test confined to the shocked villages misses the pairing.

The symmetrically opposite responses in the villages and in the cities are additional evidence for the open-against-closed hypothesis of Matsuyama (1992). These predictions hold exactly. And it is the city with less of a comparative advantage in agriculture now, and sufficient food availability, that is enabling the economies to transform, whereas the directly shocked villages are perhaps more consistent with the initial hypothesis, continuing to focus on agriculture and not leaving differentially for cities.

Winners and losers of rural productivity growth. A national average of agricultural contraction and modern-sector expansion conceals within-country variation in how the gains to agricultural productivity accrue, differing between the regions that experience the growth and the regions that benefit from it indirectly through trade. Figure 11 separates the two. It estimates the same specification in three areas: the urban centers, the rural subdistricts that one of the 814 settlement villages reaches, and the rural subdistricts that the settlement villages do not reach, standardizing every outcome within its own area so the three are read on one scale.

The cities gain the most from the rural productivity shock. They show the strongest evidence of structural transformation, and modern-sector employment growth outpaces population growth. The modern employment share rises by 0.43 of a standard deviation and log modern employment by 0.17, against 0.07 on population, which is not distinguishable from zero. There is no population growth here, even though the point estimate on it is on the large side.

The rural subdistricts with no transmigration village match the cities to an extent, in a similar sectoral reallocation of employment. The agricultural employment share falls by 0.27 there and the modern share rises by 0.26, both precisely estimated. Yet the modern-sector growth is more contained there, at 0.11 in logs against 0.17 in the cities, and the point estimate on overall employment is negative, at -0.06 , so this does not look like overall growth in those areas.

The areas that experienced the positive productivity shock are effectively not changing. The point estimates on population and on overall employment are positive, at 0.06 each, which is consistent

¹⁴The village side is suggestive. The conditional gradient of the village agricultural share on agro-similarity is positive under every specification (0.066 on the common sample of $N = 808$), and Conley standard errors and a wild cluster bootstrap both leave it indistinguishable from zero, with or without a quadratic; the city side, panel (b), is the headline estimate and is precise.

with the historical accounts, and on 191 subdistricts both are indistinguishable from zero. There is no reallocation out of agriculture into the modern sector. This one is a measured zero, not a matter of statistical power. The standard errors on the three composition outcomes are smaller here than in the larger group with no settlement, and the intervals exclude the reallocation the other two areas show. The pattern is consistent with the same hypothesis, in which village economies are open economies. They trade with the cities, and facing more demand coming from the cities they continue in agriculture and stay there.

Figure 11b continues the comparison into prices and wages. On these outcomes the gain accrues to the rural areas that host the settlements. Where the transmigration villages are more productive, the subdistricts around them face lower food prices and higher real wages, with the 2002 agricultural price falling by 0.23 of a standard deviation and the real wage rising by 0.27. The nominal wage there rises by 0.16 and is not distinguishable from zero, so the real gain is a price effect, not a wage effect.

There is no real wage growth in the cities. The sectoral reallocation there therefore comes without as clear a gain in what a worker earns, and the effect is confined to the movement of workers out of agriculture and into the modern sector. In the rural subdistricts with no settlement the real wage effect is, if anything, negative, at -0.14 , which the data do not separate from zero.

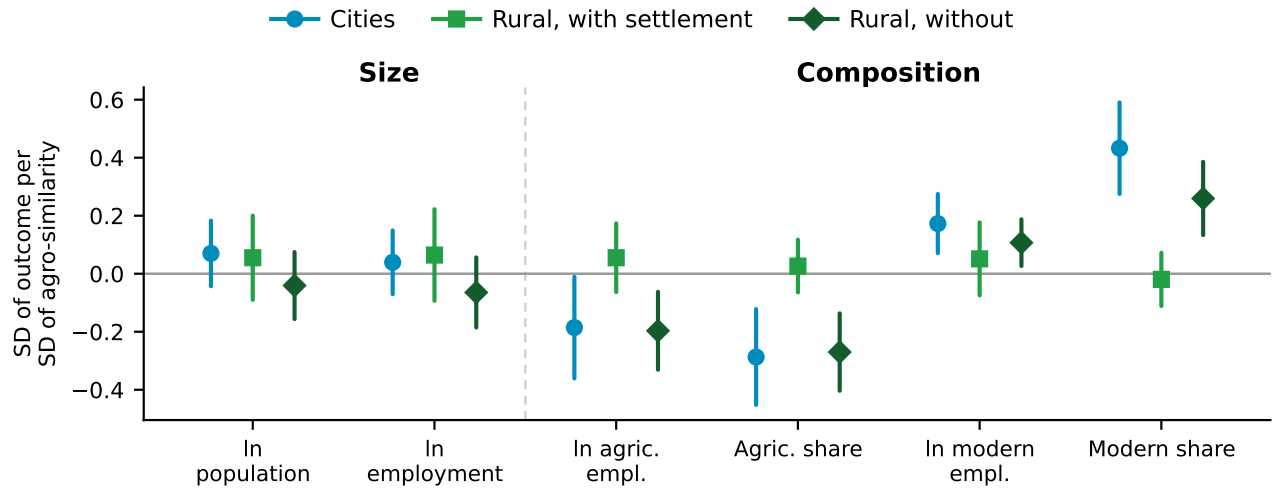
Household spending moves with the prices. The food share of spending falls by 0.26 in the cities and by 0.22 in the hosting subdistricts, and rice calories per person rise by 0.45 in the cities. There is no unemployment response in any of the three areas.

The two panels place the productivity gain and its consequences in different subdistricts. Where the settlements are, the gain appears in the price of food and in the real wage, and employment does not reallocate. The cities and the subdistricts around them are the regions surrounded by more productive hinterlands, which benefit from the higher availability of food and expand their modern sector once the subsistence constraint is satisfied.

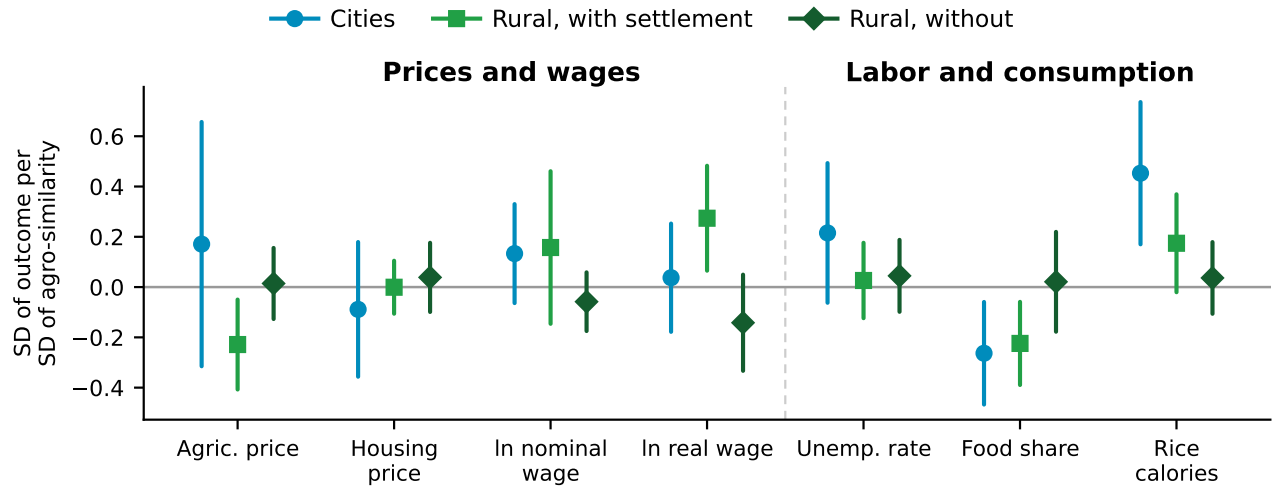
For comparison, a back-of-the-envelope aggregation gives the scale. Across the estimation sample the average city's agricultural employment share fell 18.5 percentage points between 1980 and 2010, from 55.9% to 37.4%, and the headline estimate lowers that share by 5.3 percentage points per standard deviation of hinterland exposure. Cities at the 90th percentile of exposure sit 2.2 standard deviations above cities at the 10th, so the spread in exposure across the sample accounts for something between a quarter and most of that fall, a width set by the standard error on the estimate. Here I compare more exposed cities against less exposed ones. Zero exposure lies far outside the range the estimate is identified on.

Figure 11: The response by location type: cities, rural hinterland, and the settlement hosts

(a) Structural transformation



(b) Prices, wages, and consumption



Notes: Response to a one-standard-deviation increase in agro-climatic similarity, with every outcome standardized within its own area, so the three areas are read on one scale. Agro-similarity is standardized within each area as well, at standard deviations of 0.103 and 0.104, so one standard deviation is the same exposure in both. Bars are 90% confidence intervals from Conley standard errors at a 200-kilometer cutoff. Cities are estimated at the urban-center level and are the main estimation sample; the two rural groups are subdistricts outside any urban center, split by whether a transmigration village overlaps them. “Rural, no settlement” are the other pre-existing rural areas that do not have transmigration villages and are not part of the program; every one of them is within 300 minutes of a village, so its exposure comes from the villages nearby. Main Spec controls throughout. The city regressions add the settler count inverse-distance weighted inside the 300-minute window; the subdistrict level has no such measure, so the rural regressions use the uncut inverse-distance count. (a) 133 cities, 191 subdistricts hosting a settlement and 631 with none, on every row. (b) The two price rows are the 2002 Susenas wave, before Indonesia closed the border to rice imports in 2004, since a 2011 price falls after that break while the program was settled in the 1980s. The subdistrict-level 2002 agricultural price is merged from the Susenas build and reaches 538 subdistricts. Both wage rows are the 2011 wave, because the subdistrict level has no 2002 wage, and wages respond to agricultural prices, so the same caution applies. The nominal wage is the median earner’s wage and the real wage deflates it by the local price index, so the distance between the two rows is the price effect. Each row uses the households its own wave and survey module reach, so the sample sizes vary by row, from 104 to 133 cities, 82 to 191 hosting subdistricts, and 231 to 631 subdistricts with no settlement. Every outcome with its standard error is in Appendix Table E1.

6 A Spatial Equilibrium Model with Trade and Migration

The preceding sections show the results: food, prices, food and non-food expenditures, supply and demand, services, migration. The model quantifies and separates the trade effect from the migration effect. I test the model on the empirical facts. The mapping from each fact to what the model produces for it, with each fact's coefficient and source, is tabulated in Appendix [Section F.2](#).

The model is a two-sector spatial equilibrium model of multiple locations with cities and rural areas, in the spirit of [Matsuyama \(1992\)](#). There are two sectors: agriculture (A), which produces food, and a modern sector (S) combining manufacturing and services, matching the empirics; food is tradable and the modern good is not. Workers choose a location and a sector: where they want to live, which is the migration elasticity, and within each location whether they work in agriculture or non-agriculture. Migration is costly. Preferences are Stone-Geary, with a subsistence parameter. There is a subsistence level of food that needs to be satisfied, but food consumption does not rise without limit as income rises, and that is exactly the Matsuyama mechanism. Manufacturing is folded into the modern composite because both activities respond to the shock in the same direction ([Section 6.1](#) states the fuller case), and the model has no unemployment. Exposure moves population and total employment one for one, a precisely estimated null.

The model organizes the facts through two economic mechanisms and one estimated friction. First, a *food-trade channel*: the shock to rural productivity changes the food availability in nearby cities and thus the consumption basket, the expenditure shares for households, creating additional demand for modern-sector goods and services and moving spending away from agriculture. Second, a *sectoral reallocation channel*: with growing demand for non-agricultural goods and services, their prices go up and there is more labor demand in that market, so workers sort between the sectors within each location; even in the absence of migration, the trade channel changes household expenditures, creates more labor demand in the modern-sector market, and increases the labor supply in that market. Third, *migration*: people do not only switch sectors of work, they can also switch locations, so people in all locations can make different choices. High migration frictions (a low location-choice elasticity) keep the population response modest. The quantitative decomposition ([Section 6.3](#) below) reports how much of the model's shift out of agriculture the food-trade channel is sufficient to reproduce. City size moves through migration, because there is no natural population increase in the model.

6.1 Setup and Equilibrium

This is a multi-location model with J locations and two sectors, agriculture (A), which produces food, and the modern sector (S), which combines manufacturing and services. The shock comes through agricultural productivity in the transmigration-village locations and produces the food spillover. Manufacturing and services are folded into the single modern composite because the two activities respond in the same direction to the shock, so combining them does not change any of the outcomes the model is asked to match. The modern sector admits an agglomeration externality, an external term N_j^η on the exogenous modern productivity scale $\bar{A}_j^S$ in (15), with $\eta \geq 0$; the term does not improve the model's fit,

and the headline sets $\eta = 0$, so the headline model has no agglomeration force (a sensitivity arm uses $\eta = 0.10$, [Section F.7](#)). Some locations are primarily urban (cities), others primarily rural (villages, including transmigration-village hosts). Workers of total measure $\bar{L}$ choose a location j and a sector $s \in \{A, S\}$, supplying one unit of labor inelastically. Location population is $N_j = L_{jA} + L_{jS}$. Production is set up in a traditional way, with land and labor in agriculture (T_{jA}) and housing land and labor in the modern sector (T_{jH}), under perfect competition. Each location's land endowment $\bar{T}_j$ is fixed and is split between farming and housing by a single land market (the von Thünen mechanism). The land constraint enters both sectors with one rent. Land is owned locally, and rents are equally distributed among residents. The rebate is R_j per person, which closes each location's budget. Local ownership is also the realistic case for this setting. In the household consumption surveys people report whether they own or rent, and ownership rates are quite high; in the treated villages people own because that was the whole point of the program. They were given the land. This closed rebate is the baseline variant, and opening the rebate to a national pool is the second land-market closure the counterfactuals bracket ([Section 6.3](#)). The notation, the production functions, the land-value bids, and the endowment measurement are in [Appendix Section F.1](#), and the derivations in [Appendix F](#).

Preferences. Following [Kongsamut et al. \(2001\)](#), preferences are non-homothetic through a subsistence food requirement:¹⁵

$$U(c_A, c_S) = (c_A - \bar{a})^{\psi_A} \cdot c_S^{\psi_S} \quad (5)$$

where $\bar{a} > 0$ is the minimum food consumption, $\psi_A + \psi_S = 1$ are the discretionary-expenditure weights, and $c_A \geq \bar{a}$ (participation). The modern good is a luxury (income elasticity above one) because of the subsistence term. Once subsistence consumption is satisfied, a household with higher income does not spend it proportionally on food. The subsistence term is the idea that the whole release-of-labor assumption needs. A productivity shock releases labor from a sector only under some constraint on how much the economy would produce in that sector, and the subsistence requirement is that constraint. This single non-homotheticity enables the model to generate the agricultural contraction in the cities in response to the positive productivity shock in the villages.

A household in location j and sector s earns a wage, may earn additional rental income through the per-capita land-rent rebate R_j ([Appendix Section F.1](#)), and pays for its subsistence consumption; the remainder is supernumerary income, $\tilde{w}_{js} \equiv w_{js} + R_j - p_A^j \bar{a}$. Utility maximization subject to the budget ([Appendix Section F.3](#)) gives demands in which the household first buys the subsistence bundle $\bar{a}$ and then splits what remains in fixed shares $\psi_A : \psi_S$. The implied expenditure shares are stated in [Appendix Section F.1](#).

¹⁵Stone-Geary matches [Matsuyama \(1992\)](#), whose mechanism this paper speaks to throughout, and the follow-up development structural-change work that used the same utility function. A non-homothetic CES alternative specifies the same non-homotheticity through an additional power on the terms without changing the mechanism the counterfactuals isolate; related open-economy food-problem work uses the same subsistence form ([Nath 2025](#)).

Key properties. The subsistence term generates the prediction that as income rises the food share s_A^j falls and the modern share s_S^j rises, two of the facts I find (**F15** and **F8** in Appendix Table F2); under Cobb-Douglas preferences the two shares are fixed. Substituting the demands back into (5) gives indirect utility proportional to $\tilde{w}_{js}/P_j$, with $P_j = (p_A^j)^{\psi_A} (p_S^j)^{\psi_S}$ the price index over the supernumerary basket (Appendix Section F.3). The sector choice within a location ranks the sectoral wages, and the location choice ranks this indirect utility.

Location and sector choice. Workers make a nested location-sector choice with idiosyncratic Fréchet shocks (Allen & Arkolakis 2014; Redding 2016; Tombe & Zhu 2019). Because the shocks are Fréchet, the final allocations of people by location and sector are pinned down by the choice probabilities, and the sectoral reallocation responds to the wages. The sector stage gives the probability of working in a specific sector given the location, with shape θ_S , and the location stage gives the probabilities of being in each location overall, with shape θ_L (the two displays are in Appendix Section F.1). The location stage ranks the Stone-Geary indirect utility evaluated at the location’s expected income and scaled by the measured 1980 amenity Φ_L^j (standardized built-up area, population density, and PODES-1983 infrastructure). Within location j , the share of people in agriculture is captured by the wage ratios with the elasticity θ_S ; θ_L comes from the choices of people across locations, so the location choice determines the overall population distribution and the sector choice determines the sectoral labor allocation within each location. θ_L introduces the migration friction. A low θ_L produces a modest population response, consistent with the small but positive **F1/F2**.

Trade and prices. The agricultural good is tradable with iceberg costs. To model the trade costs, I use the cost-distance estimates, which capture the realistic travel time between each city and village (Section 3.3). Shipping food from k to j melts part of it away, at an iceberg factor that is a power law in route-aware cost-distance in minutes, $\tau_{kj} = \max(d_{kj}, 1)^\delta$ with $\tau_{jj} = 1$; the unit floor only ever binds own-location shipping, since the smallest off-diagonal distance in the sample is about five minutes. Each origin posts a farm-gate marginal cost obtained by inverting the labor marginal product in (14), $\text{mc}_k = L_{kA}^{1-\beta} / [\beta (A_k^A T_{kA})^{1-\beta}]$, and the delivered food price at j is a CES (Armington) aggregate over all origins:

$$p_A^j = \left[\sum_k (\tau_{kj} \cdot \text{mc}_k)^{1-\sigma} \right]^{1/(1-\sigma)}, \quad \sigma = 4. \quad (6)$$

The cost-distance elasticity δ is reported over the band $\delta \in [0.5, 0.8]$, the range the quantitative-trade literature uses for cost-distance-based iceberg costs (Allen & Arkolakis 2014; Tombe & Zhu 2019), and no headline result turns on the point chosen; the band’s basis, the implied cost levels, and the solved import shares are in Appendix Section F.5. This channel transmits the agro-similarity shock to cities. A productivity gain at transmigration-village hosts lowers their mc_k , which lowers the delivered price p_A^j at every location within shipping reach.

Modern good: non-traded, so p_S^j is determined by local market clearing (equilibrium condition (5) below). The numeraire is the reference city's modern wage (the median-population urban unit), $w_{j^*S} = 1$.

Equilibrium. A spatial equilibrium is a set of prices $\{p_A^j, p_S^j\}$, wages $\{w_{jA}, w_{jS}\}$, land splits $\{T_{jA}\}$, rebates $\{R_j\}$, and labor allocations $\{L_{jA}, L_{jS}, N_j\}$ such that: (1) each sector's wage equals its marginal revenue product of labor (Appendix Section F.4, Section F.6); (2) food prices satisfy the CES sourcing (6); (3) the von Thünen land split and the rebate hold (Appendix Section F.8); (4) sector employment satisfies (16) with $L_{jA} + L_{jS} = N_j$; (5) modern markets clear location by location, which sets p_S^j so that local modern spending equals local modern revenue; (6) population satisfies the location choice (17) with $\sum_j N_j = \bar{L}$; and (7) the numeraire is $w_{j^*S} = 1$. The full system, its summary table, and the mapping from these conditions to the paper's estimating equations are in Appendix Section F.10 and Section F.11.

The closed rebate has a local-financing property, stated in Appendix Section F.13, which the channel decomposition of Section 6.3 relaxes.

The comparative static. An increase in A_v^A (village agricultural TFP) lowers mc_v and hence, through trade, the delivered price p_A^j for nearby cities. Lower p_A^j raises supernumerary income $\tilde{w}_j$ (increasing modern demand and the local modern price, matching the 1996 rent rise in F9, with no detectable movement in 2011) and lowers the relative agricultural wage, so the city's agricultural employment share declines (matching F3, F4) while modern employment expands (matching F5, F6) and the food spending share falls (matching F15). To first order, holding the city's own inputs fixed, the composition response is

$$\frac{\partial \pi_{A|j}}{\partial \ln A_v^A} = -\theta_S \pi_{A|j} (1 - \pi_{A|j}) (1 - \beta) s_{vj}, \quad s_{vj} = \frac{(\tau_{vj} mc_v)^{1-\sigma}}{\sum_k (\tau_{kj} mc_k)^{1-\sigma}}, \quad (7)$$

where s_{vj} is the share of city j 's food sourcing held by the shocked origin (derived in Appendix Section F.10). The response scales with the sector-choice elasticity θ_S and with the sourcing weight s_{vj} , which falls with the trade cost τ_{vj} . Villages trade with cities. Cities that need food buy it from nearby villages, and the more productive the villages are, the cheaper the food is. Cities that are close to ports, close to global trade centers, barely respond, because the shocked origin holds little of their sourcing. That gradient in trade position is the openness heterogeneity of Section 5.3, and it is P3 in general equilibrium; the full response, with population, land, and prices all adjusting, is the object Section 6.3 quantifies.

Beyond the reduced-form results, the model disentangles and identifies the two channels separately. It first rationalizes the effects I observe, and it then estimates the contribution of the trade and migration channels to the main outcomes.

6.2 Taking the Model to the Data

The model is built from relatively standard elements: Stone-Geary utility with its subsistence term, Fréchet location and sector choices, and costly food trade. I calibrate it by choosing two data moments to identify its two parameters, check how well it predicts outcomes that were not used in the calibration, and run two counterfactuals: the channel decomposition and the program’s contribution to aggregate urban change. Some parameters are taken as values from the literature, while others are calibrated to the data; the two choice elasticities, of location and of sector, are estimated from the reduced form. And I cannot estimate them with a first-stage instrumental-variables setup on wages and prices, because the price effect of exposure is null and the wage effect is similarly null, so I cannot get that variation to identify the elasticities in the long-run changes. The null wage effect is partially driven by two forces in the non-agricultural market that offset each other. As people switch out of agriculture and into non-agriculture, labor supply in that market increases; at the same time there is more demand for non-agricultural services and manufacturing, so labor demand in that market is higher as well, and higher labor demand together with higher labor supply results in no effect on real wages. The elasticities are therefore identified by solving the model and matching it to the reduced-form coefficients of [Section 5](#). I compare the data moments and choose two of them to identify the two parameters, the population growth and the agricultural employment decline: population growth pins down the location elasticity, because as people migrate that shapes the population allocation across places, and the agricultural-share decline pins down the sectoral elasticity, people choosing sectors within each location ([Appendix Section F.12](#)). Everything else is either set outside the model, from the household surveys and from production shares, or recovered from the cross-section each city shows before any shock is applied.

Preferences and technology. Preferences and technology are set from what I observe in the data, the food shares and the sectoral shares among them, before the elasticities are estimated, at the values in [Appendix Section F.1](#). The Stone-Geary demand system takes its subsistence food requirement and its food expenditure share from the Susenas consumption modules. The subsistence term in food demand is the main modeling choice, and it is measured in the household data. The subsistence parameter comes from the relationship between the food share of household consumption and income. Production shares come from the census wage bill and from the land and labor inputs of the agricultural block. The food trade-cost elasticity is set within the literature band and reported as a band (the trade and prices block of [Section 6.1](#)). The traded-services block takes its expenditure share from the Susenas 2011 traded-services share of household spending and its distance decay from IFLS household service-flow gravity ([Appendix Section F.13](#)).

City fundamentals. Each city’s agricultural productivity, modern productivity, and amenity are three fundamentals I do not observe directly. I recover them by inverting the model, so that the model reproduces each city as it stood before any shock: agricultural productivity from agricultural labor per hectare of crop land, modern productivity from the census modern wage bill, and the amenity

from the 1980 index of built-up area, population density, and infrastructure. A city missing any of these inputs is excluded. The detail behind each input is in Appendix [Section F.12](#).

The two choice elasticities. The two Fréchet dispersions $\theta = (\theta_L, \theta_S)$ are estimated by indirect inference. I run the exact specification of [Section 4.2](#), with the same controls, on the model panel, and choose the two parameters so that the coefficients estimated on the model panel match those estimated in the data.¹⁶ The specification is identical on the two sides, controls included, because the method requires it. The coefficient matched on the model side has to be the same conditional coefficient as the one estimated in the data. The estimator’s mechanics, the model panel, the headline estimate, and its robustness on the land specification and closure are in Appendix [Section F.12](#).

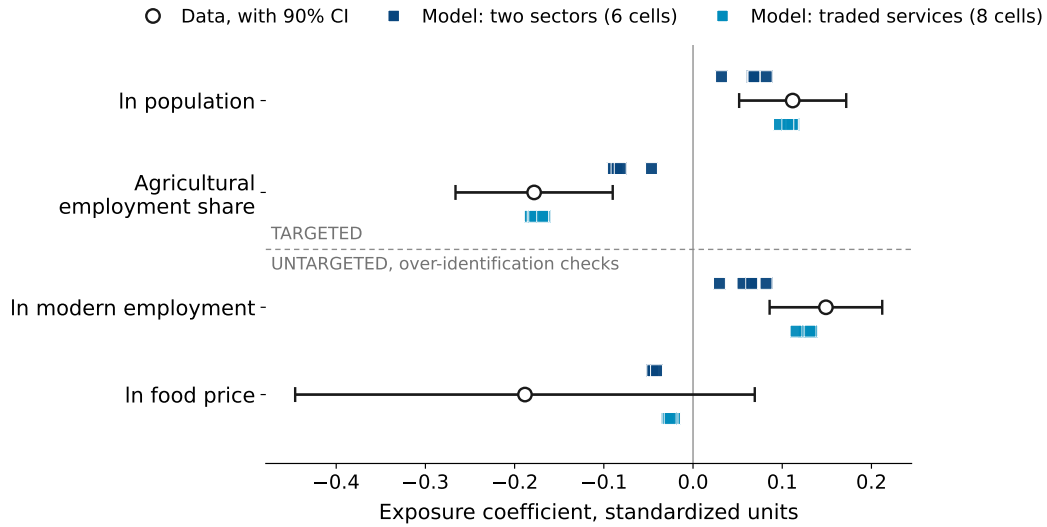
Untargeted validation. With the parameters calibrated, next I check how well the model predicts outcomes that were not used in the calibration ([Figure 12](#)). The growth in modern employment is also well predicted, and the two targeted moments nest this slope. The two migration facts the reduced form establishes, the birthplace composition of urban in-migrants and the IFLS never-mover share of 0.533, enter as over-identifying moments and stay unfitted ([Appendix Section F.14](#)). The rural labor moments provide the same test on the village side. In the model, villages move labor toward agriculture where they do in the data. And the von Thünen logic implies an untargeted price test, the price of food against the agricultural density of a city’s surroundings. In the data the food price falls in the cropland share of a center’s surroundings, at -0.248 with a standard error of 0.099 across 102 cities, and the closed-rebate specification reproduces that gradient at every food trade cost in the band, while the open-rebate specification misses it at the two trade costs at which θ_L reaches the upper bound of the search ([Appendix Figure F1](#)). So the model fits the data on outcomes not directly targeted in the calibration.

The full set of targeted and over-identifying moments, in the same z -on- z units, is in Appendix [Table B2](#).

Migration as an estimated friction. Migration enters as idiosyncratic shocks with a migration cost, estimated using the IFLS household panel data; the location choice is generalized to an origin-destination allocation over the 1980 origin population, with the friction pinned by the subdistrict-to-subdistrict migration gravity estimated in the IFLS ([Appendix Section F.14](#)). The estimated friction is large. The IFLS never-mover share is 0.533, reported as an untargeted over-identifying moment, because coverage limits make its level a lower bound on true mobility. The costly-migration model, in which both the baseline and the shock bear the moving cost from 1980 origins, is a robustness check reported in the same appendix; the migration cost is a friction that dampens over-mobility, and the channel diagnosis is unchanged under it.

¹⁶I do not simulate data. The Fréchet shocks enter through closed-form choice probabilities, so given θ the model’s outcomes, and the coefficients estimated on them, are computed exactly by re-solving the equilibrium. The estimator is therefore classical minimum distance on the reduced-form coefficients, a GMM estimator ([Hansen 1982](#)) whose moments are the coefficients of an auxiliary regression, the design known as indirect inference ([Gourieroux et al. 1993](#); [Smith 1993](#)); with two coefficients and two parameters it is exactly identified, the weighting is immaterial at the estimate, and the criterion is zero when both targets are reached.

Figure 12: Model fit: targeted and untargeted moments against the data



Notes: Each row is a standardized (z -on- z) exposure response. The hollow circle is the data estimate with its 90% confidence interval; each small square is the fitted response of one specification at one food trade cost, dark for the single-tradable specification and light for the traded-services one, across the trade-cost band. The two moments above the dashed rule are targeted; the moments below it are untargeted over-identification checks.

6.3 Counterfactuals

I use the model for the decomposition. It can tell the channels of trade and migration apart, which the reduced form cannot. In the real world both forces operate, and the data show the overall effect after all of them have adjusted; only in the model can I simulate a hypothetical world where there is only trade and no migration, and a different hypothetical world where there is migration but no trade, and compare those unobserved worlds to see what would have happened. Shutting down trade means the food trade costs rise substantially, so each city clears its own food market and the hinterland's productivity gain does not reach it through the goods market. Shutting down migration means people cannot change locations, so the population allocations across space do not change, only the goods are moving and people are paying for them, and the remaining adjustment is workers switching sector inside the city they already live in. A channel's contribution is the gap between the full response and the response with that channel shut. The two contributions need not sum to the total, because the channels interact, so all four states are solved and the interaction is reported as its own term. Economically the interaction is the covariance between the two channels. Trade enables workers to allocate out of agriculture, migration would push the same reallocation further, and as the prices and wages adjust, the combined reduction comes out slightly below the sum of the two pieces. A full autarky counterfactual, shutting trade down entirely, would require trade-cost estimates this setting does not provide. The decomposition instead uses the variation the reduced form already identifies.

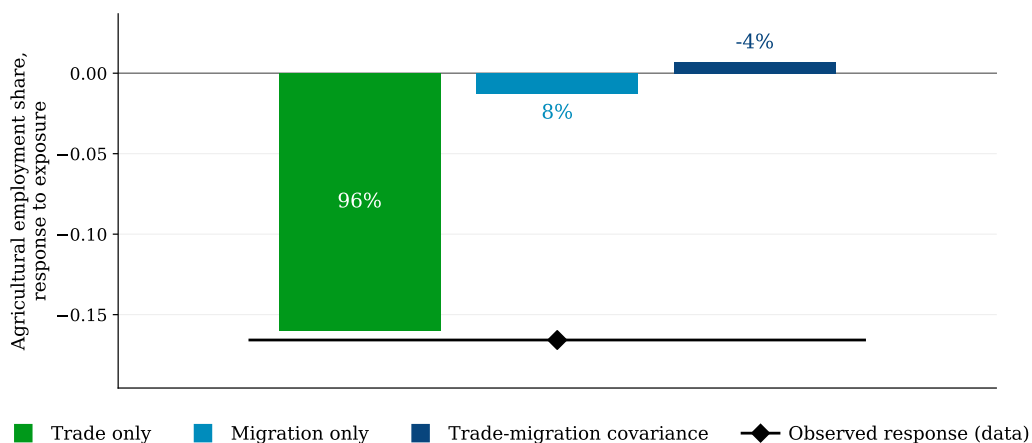
Two predictions come from the reduced form itself. If I shut down trade and leave only migration, the composition response is expected to mostly go away, matching the openness evidence of [Section 5.3](#). Migration enabled without trade is not expected to generate a lower agricultural employment share, and

the decline is expected to be small. If instead I shut down migration and leave trade, that response is expected to stay nearly intact, matching the migration null the birthplace evidence establishes. Neither prediction is fitted by the estimator, so the decomposition can fail them.

The decomposition is run in the traded-services specification, in which a city can export modern output to pay for net food imports. Why the single-tradable baseline cannot produce a trade-only contraction, and the traded block's two parameters, are in Appendix [Section F.13](#).

The decomposition at the estimated parameters. Consider a world where people are not shifting locations. Migration costs are prohibitive, so locations can trade in goods and services but people cannot move. In this world of trade only, the agricultural employment share response is almost fully reproduced ([Figure 13](#), with both outcomes under both closures in [Figure 14](#)). Trade is sufficient to reproduce essentially the entire agricultural-share response, and migration moves city size. The legs of the decomposition under both land-market closures, and the interaction term, are in Appendix [Section F.13](#).

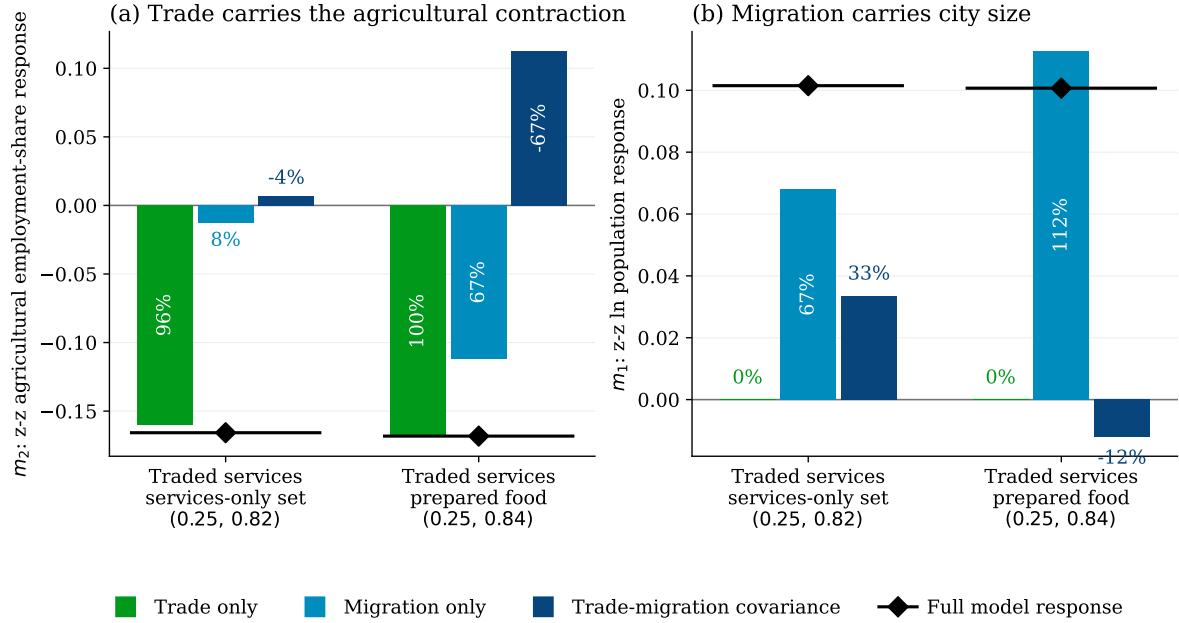
Figure 13: Channel decomposition of the agricultural-share response



Notes: Bars are the model's agricultural-employment-share response to the estimated exposure shock with one channel active at a time (trade only, migration only); the third bar is the residual interaction between the two channels, and the diamond marks the data's observed response. Percentages express each bar relative to the model's full response.

The decomposition is interpreted together with the demand system, because the subsistence term is an essential feature of the model. Preferences are non-homothetic: as a household consumes food, there is a subsistence level that needs to be satisfied before it spends on anything else, and once that level is met the marginal utility of further food is lower. So as people earn more, the income effect does not translate fully into higher food consumption, and the share of food in household spending falls. A lower delivered food price works from the other direction, reducing the cost of meeting the same requirement. Either route releases budget to the modern good, so demand for urban labor rises and workers switch sector. With the subsistence requirement set to zero, preferences are Cobb-Douglas. Expenditure shares are fixed, the food share does not fall with income, and the trade channel does not affect employment composition. Setting $\bar{a}$ to zero and re-solving is the direct test of how much the non-homotheticity matters.

Figure 14: Both outcomes, both traded-services specifications



Notes: Panels show the two targeted outcomes in standardized (z -on- z) units. Within each panel, bars are the trade-only, migration-only, and interaction terms under the two traded-services basket variants, and the diamond marks the full solved response. The population trade-only bar is zero by construction, because the migration switch works by fixing populations.

The aggregate contribution. Alongside the decomposition I want a quantification of how much of the overall agricultural-share decline, the structural transformation itself, the agricultural productivity accounts for. Next, I quantify magnitudes using the model. The model is re-solved with the shock vector set to zero, so that no transmigration village receives its agro-climatic gain, and the resulting urban outcomes are compared against the baseline solution; the difference is the program's contribution to aggregate urban change. I compute it from the aggregate effect, because scaling the reduced-form coefficient up to the program would be inaccurate. There are general-equilibrium forces readjusting that the reduced form does not fully capture. The model can.

I report this as a share of the observed 1980 to 2010 change, in outer-island urban population and in urban sectoral composition, given as a range across plausible values of the two elasticities and both land-market closures. The bracket is tight. The two closures give nearly the same answer, so the result does not depend on which one is imposed. I report both closures as a robustness check. The credibility of the share itself rests on the shock I simulate. The size of the response depends on the size of the shock, so the simulated productivity gain has to correspond to the gain the program generated over the period, and because the effect need not be linear, a coefficient normalized to one standard deviation of exposure does not extrapolate to the program total.

7 Conclusion

This paper studies what rural agricultural productivity does to nearby cities. I leverage the Transmigration Program in Indonesia as a source of exogenous variation in rural productivity: the quasi-random variation in the settlers' agro-climatic skill match across newly settled villages (Bazzi et al. 2016). I construct a city-level exposure to surrounding rural productivity by aggregating village-level agro-climatic similarity over all 814 program villages using travel-time connectivity, while separately controlling for the scale of nearby settlement. The estimation sample is the outer-island urban centers within 300 minutes' travel of a program village.

The reduced-form evidence gives four results. First, cities exposed to more productive transmigration villages change what they do more than how big they are. Agricultural employment is going down while modern-sector employment is going up, in levels and in shares, both precisely estimated, without any change in unemployment. Second, population grows as well, by much less than modern-sector employment. The back-of-envelope calculation says the new arrivals will not account for the structural shift, and most of the fall in the agricultural share comes from workers of a given birthplace exiting agriculture and going into modern employment. The population estimate holds under the spatial-inference corrections, it is the least secure of the four under randomization inference, and the compositional result does not rest on it. The population margin is also intensive: a companion analysis of urban emergence finds no boom of new towns.

Third, the results suggest a goods-market mechanism. Exposed cities reduce agricultural employment while producing less food relative to what they consume, so consumption needs to come from somewhere, and the cities buy more of their food from the hinterland. People spend relatively less on food and buy less rice per person in the 2011 wave, consistent with Stone-Geary preferences, and housing rents rise on a coarse mid-1990s measure as demand shifts toward non-tradables. The timing supports this. In the mid-1990s household survey data, one to two decades after placement, the structural signal is already present while local food prices show little response, so the later food-price movements are a downstream equilibrium object. Fourth, urban growth is driven by in-migration from other outer-island provinces, not by transmigrants relocating out of their settlement villages. The villagers are, if anything, slightly more likely to stay in villages that are more productive.

To interpret and quantify these patterns, I develop a spatial general equilibrium model of multiple locations, villages and cities, with two sectors: agriculture, which produces food, and the modern sector, combining manufacturing and services. Households make two choices, the location and the sector, each governed by its own elasticity, and the main modeling choice is the Stone-Geary utility with a subsistence term in the demand for food, which lets a positive productivity shock in the villages generate agricultural contraction in the cities. Food prices are determined endogenously through costly inter-regional trade, and the modern-sector agglomeration externality is set to zero in the headline calibration. I choose two data moments to identify the two parameters I have, the population growth and the agricultural employment decline, and the model's discrete-choice structure gives estimating

equations that map directly to the reduced-form regressions. I then shut down trade, meaning the trade costs rise substantially, and I shut down migration separately; in the real world both forces are in play and I observe the general-equilibrium effect. Trade is sufficient to reproduce nearly the entire agricultural contraction, and the trade-only response on city size is zero by construction. Migration enters as an estimated friction, disciplined by a gravity-based moving-cost estimate, and it moves the city’s size, not its composition.

These findings speak to a core debate in development and urban economics about whether raising agricultural productivity accelerates or delays urbanization and structural change. Matsuyama argues that closed economies shift away from agriculture in response to an agricultural productivity shock, whereas open economies continue focusing on agriculture, facing effectively unlimited demand from the global markets (Matsuyama 1992); Bustos et al. (2016) test the mechanism in Brazil, and this paper extends it across space. Because island geography only weakly connects regional economies to one another, each urban region with its rural hinterland behaves like a closed economy, and regional structural transformation follows the closed-economy logic. The data show this openness pattern directly. The transformation concentrates in cities weakly connected to outside food markets, and is sharply muted in cities already open to rice imports through major ports. Policies targeting rural areas get translated into urban outcomes. Food trade enables further urbanization in this context, as cheaper food frees up urban workers and spending, without requiring massive rural-to-urban migration in a setting with high internal moving costs.

Several questions stay open. The reduced form measures long differences between census years, and the price and wage side is observed in cross-sections at the endline, so the paper establishes the direction and the timing of the urban response; the path of that response, how quickly the composition of a city moves and how long the food-price signal takes to appear, needs data with a baseline on prices and wages. The counterfactual keeps the settlers where they were placed and asks what a better match would have done, so the paper does not evaluate the program as a whole, and the origin regions in Java and Bali stay outside the accounting. Within the city, the shift out of agricultural employment leaves more people seeking service jobs while the demand for services is also rising, and the higher supply pushes the service wage down as the higher demand pushes it up, so growth need not translate into higher welfare for residents; Figure 11b shows the wage and price responses at the endline. The elasticities are the portable output. The sector and location elasticities govern whether the workers a rural productivity gain displaces switch sectors or switch places, and they can be compared across settings where the gain comes from a different source, a Green Revolution or a cash-crop boom, and across regions more open to food trade than these, where the same gain is exported and the city keeps its composition. Whether the absence of rural productivity growth is part of why some regions urbanize without transforming, as in much of sub-Saharan Africa, cannot be tested in this setting; the elasticities are the input such a test needs.

References

- Acemoglu, D. & Guerrieri, V. (2008). Capital Deepening and Nonbalanced Economic Growth. *Journal of Political Economy*, 116(3), 467–498.
- Allcott, H. & Keniston, D. (2018). Dutch disease or agglomeration? the local economic effects of natural resource booms in modern america. *Review of Economic Studies*, 85(2), 695–731.
- Allen, T. & Arkolakis, C. (2014). Trade and the Topography of the Spatial Economy. *Quarterly Journal of Economics*, 129(3), 1085–1140.
- Asher, S., Chauvin, J. P., Grigoryeva, I., & Novosad, P. (2022). Rural Spillovers of Urban Growth. *Working Paper*.
- Badan Pusat Statistik (2011). Survei Sosial Ekonomi Nasional (Susenas): National Socioeconomic Survey. *Badan Pusat Statistik (Statistics Indonesia), household survey microdata*. Consumption, price, wage, and housing modules. This paper uses the 2002 and 2011 waves; the 2011 wave covers quarters Q1–Q3, the only quarters released.
- Bazzi, S. (2017). Wealth Heterogeneity and the Income Elasticity of Migration. *American Economic Journal: Applied Economics*, 9(2), 219–55.
- Bazzi, S., Gaduh, A., Rothenberg, A. D., & Wong, M. (2016). Skill Transferability, Migration, and Development: Evidence from Population Resettlement in Indonesia. *American Economic Review*, 106(9), 2658–98.
- Bazzi, S., Gaduh, A., Rothenberg, A. D., & Wong, M. (2019). Unity in diversity? How intergroup contact can foster nation building. *American Economic Review*, 109(11), 3978–4025.
- Boppart, T. (2014). Structural Change and the Kaldor Facts in a Growth Model with Relative Price Effects and Non-Gorman Preferences. *Econometrica*, 82(6), 2167–2196.
- Borusyak, K. & Hull, P. (2023). Nonrandom Exposure to Exogenous Shocks. *Econometrica*, 91(6), 2155–2185. Previously NBER WP 27845, “Non-Random Exposure to Exogenous Shocks: Theory and Applications”.
- Bryan, G., Chowdhury, S., & Mobarak, A. M. (2014). Underinvestment in a Profitable Technology: The Case of Seasonal Migration in Bangladesh. *Econometrica*, 82(5), 1671–1748.
- Bryan, G. & Morten, M. (2019). The Aggregate Productivity Effects of Internal Migration: Evidence from Indonesia. *Journal of Political Economy*, 127(5), 2229–2268.
- Budí-Ors, T. (2026). Rural-Urban Migration and Structural Change: A Reinterpretation. Working paper, Bank of Spain, March 2026; formerly CEMFI job market paper.

- Bustos, P., Caprettini, B., & Ponticelli, J. (2016). Agricultural productivity and structural transformation: Evidence from Brazil. *American Economic Review*, 106(6), 1320–1365.
- Bustos, P., Garber, G., & Ponticelli, J. (2020). Capital Accumulation and Structural Transformation. *The Quarterly Journal of Economics*, 135(2), 1037–1094.
- Civelli, A., Gaduh, A., Rothenberg, A. D., & Wang, Y. (2023). Urban Sprawl and Social Capital: Evidence from Indonesian Cities. *The Economic Journal*, 133(654), 2110–2146.
- Comin, D., Lashkari, D., & Mestieri, M. (2021). Structural Change with Long-Run Income and Price Effects. *Econometrica*, 89(1), 311–374.
- Dawe, D. (2008). Can Indonesia Trust the World Rice Market? *Bulletin of Indonesian Economic Studies*, 44(1), 115–132.
- Dijkstra, L., Florczyk, A. J., Freire, S., Kemper, T., Melchiorri, M., Pesaresi, M., & Schiavina, M. (2021). Applying the Degree of Urbanisation to the globe: A new harmonised definition reveals a different picture of global urbanisation. *Journal of Urban Economics*, 125, 103312.
- Donaldson, D. (2018). Railroads of the Raj: Estimating the Impact of Transportation Infrastructure. *American Economic Review*, 108(4-5), 899–934.
- Duflo, E. (2001). Schooling and Labor Market Consequences of School Construction in Indonesia: Evidence from an Unusual Policy Experiment. *American Economic Review*, 91(4), 795–813.
- Eckert, F. & Peters, M. (2025). *Spatial Structural Change*. NBER Working Paper 30489, National Bureau of Economic Research. September 2022, revised July 2025. Under revision at the Review of Economic Studies.
- Edwards, R. B. (2024). Spillovers from Agricultural Processing. Working paper, Australian National University (Arndt-Corden Departmental WP 2024-6, July 2024); first circulated 2019.
- Egger, D., Haushofer, J., Miguel, E., Niehaus, P., & Walker, M. (2022). General equilibrium effects of cash transfers: Experimental evidence from kenya. *Econometrica*, 90(6), 2603–2643.
- Emerick, K. (2018). Agricultural Productivity and the Sectoral Reallocation of Labor in Rural India. *Journal of Development Economics*, 135, 488–503.
- European Commission, Joint Research Centre (2024). *GHSL Data Package 2023*. Technical Report JRC133256, Publications Office of the European Union, Luxembourg.
- Fajgelbaum, P. & Redding, S. J. (2022). Trade, Structural Transformation, and Development: Evidence from Argentina 1869–1914. *Journal of Political Economy*, 130(5), 1249–1318.

- Feyrer, J., Mansur, E. T., & Sacerdote, B. (2017). Geographic dispersion of economic shocks: Evidence from the fracking revolution. *American Economic Review*, 107(4), 1313–1334.
- Foster, A. D. & Rosenzweig, M. R. (2004). Agricultural Productivity Growth, Rural Economic Diversity, and Economic Reforms: India, 1970–2000. *Economic Development and Cultural Change*, 52(3), 509–542.
- Gertler, P. J., Gonzalez-Navarro, M., Gračner, T., & Rothenberg, A. D. (2024). Road Maintenance and Local Economic Development: Evidence from Indonesia’s Highways. *Journal of Urban Economics*, 143, 103687.
- Glaeser, E. L. (2014). A world of cities: The causes and consequences of urbanization in poorer countries. *Journal of the European Economic Association*, 12(5), 1154–1199.
- Gollin, D. (2023). Agricultural Productivity and Structural Transformation: Evidence and Questions for African Development. *Oxford Development Studies*, 51(4), 375–396. First circulated as STEG Pathfinding Paper PP04, April 2021.
- Gollin, D., Hansen, C. W., & Wingender, A. M. (2021). Two Blades of Grass: The Impact of the Green Revolution. *Journal of Political Economy*, 129(8), 2344–2384.
- Gollin, D., Jedwab, R., & Vollrath, D. (2016). Urbanization with and without industrialization. *Journal of Economic Growth*, 21(1), 35–70.
- Gollin, D., Lagakos, D., & Waugh, M. E. (2014). The Agricultural Productivity Gap. *The Quarterly Journal of Economics*, 129(2), 939–993. Verified against Crossref 2026-09-10 (title, journal, volume, issue, pages, authors).
- Gollin, D., Parente, S., & Rogerson, R. (2002). The Role of Agriculture in Development. *American Economic Review*, 92(2), 160–164.
- Gollin, D., Parente, S. L., & Rogerson, R. (2007). The Food Problem and the Evolution of International Income Levels. *Journal of Monetary Economics*, 54(4), 1230–1255.
- Gourieroux, C., Monfort, A., & Renault, E. (1993). Indirect Inference. *Journal of Applied Econometrics*, 8(S1), S85–S118.
- Hamory, J., Kleemans, M., Li, N. Y., & Miguel, E. (2021). Reevaluating Agricultural Productivity Gaps with Longitudinal Microdata. *Journal of the European Economic Association*, 19(3), 1522–1555.
- Hansen, L. P. (1982). Large Sample Properties of Generalized Method of Moments Estimators. *Econometrica*, 50(4).
- Harris, J. R. & Todaro, M. P. (1970). Migration, Unemployment and Development: A Two-Sector Analysis. *American Economic Review*, 60(1), 126–142.

- Henderson, J. V. & Turner, M. A. (2020). Urbanization in the Developing World: Too Early or Too Slow? *Journal of Economic Perspectives*, 34(3), 150–73.
- Herrendorf, B., Rogerson, R., & Valentinyi, Á. (2014). Growth and Structural Transformation. In P. Aghion & S. N. Durlauf (Eds.), *Handbook of Economic Growth*, volume 2 chapter 6, (pp. 855–941). Amsterdam: Elsevier.
- Hornbeck, R. & Keskin, P. (2015). Does Agriculture Generate Local Economic Spillovers? Short-Run and Long-Run Evidence from the Ogallala Aquifer. *American Economic Journal: Economic Policy*, 7(2), 192–213.
- Jedwab, R., Ianchovichina, E., & Haslop, F. (2025). The Employment Profile of Cities around the World: Consumption vs. Production Cities and Economic Development. *World Development*, 188, 106883.
- Jedwab, R. & Vollrath, D. (2015). Urbanization without growth in historical perspective. *Explorations in Economic History*, 58, 1–21.
- Kongsamut, P., Rebelo, S., & Xie, D. (2001). Beyond Balanced Growth. *Review of Economic Studies*, 68(4), 869–882.
- Kraus, S., Heilmayr, R., & Koch, N. (2024). Spillovers to Manufacturing Plants from Multimillion Dollar Plantations: Evidence from the Indonesian Palm Oil Boom. *Journal of the Association of Environmental and Resource Economists*, 11(3), 613–656.
- Lagakos, D., Mobarak, A. M., & Waugh, M. E. (2023). The Welfare Effects of Encouraging Rural–Urban Migration. *Econometrica*, 91(3), 803–837.
- Lewis, W. A. (1954). Economic Development with Unlimited Supplies of Labour. *The Manchester School*, 22(2), 139–191.
- Liu, M., Shamdasani, Y., & Taraz, V. (2023). Climate Change and Labor Reallocation: Evidence from Six Decades of the Indian Census. *American Economic Journal: Economic Policy*, 15(2), 395–423.
- Marden, S. (2016). *The Agricultural Roots of Industrial Development: ‘Forward Linkages’ in Reform Era China*. Working Paper 09/2016, Department of Economics, University of Sussex.
- Matsuyama, K. (1992). Agricultural productivity, comparative advantage, and economic growth. *Journal of Economic Theory*, 58(2), 317–334.
- Matsuyama, K. (2008). Structural Change. In S. N. Durlauf & L. E. Blume (Eds.), *The New Palgrave Dictionary of Economics*. Basingstoke: Palgrave Macmillan, 2nd edition.
- McArthur, J. W. & McCord, G. C. (2017). Fertilizing growth: Agricultural inputs and their effects in economic development. *Journal of Development Economics*, 127, 133–152.

- McGee, T. G. (1991). The emergence of desakota regions in Asia: Expanding a hypothesis. In N. Ginsburg, B. Koppel, & T. G. McGee (Eds.), *The Extended Metropolis: Settlement Transition in Asia*. Honolulu: University of Hawaii Press.
- Mitchell, H. (2026). The curse of connectivity: Evidence from indonesia’s village resettlement program. Job market paper in progress, University of California San Diego.
- Moreno-Monroy, A. I., Schiavina, M., & Veneri, P. (2021). Metropolitan areas in the world. Delineation and population trends. *Journal of Urban Economics*, 125.
- Moretti, E. (2010). Local multipliers. *American Economic Review: Papers & Proceedings*, 100(2), 373–377.
- Morten, M. & Oliveira, J. (2024). The Effects of Roads on Trade and Migration: Evidence from a Planned Capital City. *American Economic Journal: Applied Economics*, 16(2), 389–421. Previously NBER WP 22158, “Paving the Way to Development: Costly Migration and Labor Market Integration”.
- Moscona, J. (2019). Agricultural Development and Structural Change, Within and Across Countries. Working paper, MIT.
- Nagy, D. K. (2023). Hinterlands, City Formation and Growth: Evidence from the U.S. Westward Expansion. *Review of Economic Studies*, 90(6), 3238–3281.
- Nath, I. (2025). Climate Change, the Food Problem, and the Challenge of Adaptation through Sectoral Reallocation. *Journal of Political Economy*, 133(6), 1705–1756.
- Ngai, L. R. & Pissarides, C. A. (2007). Structural Change in a Multisector Model of Growth. *American Economic Review*, 97(1), 429–443.
- Olken, B. A. (2009). Merged Indonesian Kecamatan Codes. *National Bureau of Economic Research Electronic Dataset*.
- Pellegrina, H. S. & Sotelo, S. (2025). Migration, Specialization, and Trade: Evidence from Brazil’s March to the West. *Journal of Political Economy*, 133(12), 3993–4049.
- Redding, S. J. (2016). Goods Trade, Factor Mobility and Welfare. *Journal of International Economics*, 101, 148–167.
- Restuccia, D., Yang, D. T., & Zhu, X. (2008). Agriculture and aggregate productivity: A quantitative cross-country analysis. *Journal of Monetary Economics*, 55(2), 234–250.
- Santangelo, G. (2019). Firms and Farms: The Local Effects of Farm Income on Firms’ Demand. Cambridge Working Papers in Economics No. 1924, University of Cambridge.

- Schultz, T. W. (1953). *The Economic Organization of Agriculture*. New York: McGraw-Hill.
- Smith, A. A. (1993). Estimating Nonlinear Time-Series Models Using Simulated Vector Autoregressions. *Journal of Applied Econometrics*, 8, S63–S84.
- Tombe, T. (2015). The Missing Food Problem: Trade, Agriculture, and International Productivity Differences. *American Economic Journal: Macroeconomics*, 7(3), 226–258.
- Tombe, T. & Zhu, X. (2019). Trade, Migration, and Productivity: A Quantitative Analysis of China. *American Economic Review*, 109(5), 1843–1872.
- Weiss, D. J., Nelson, A., Gibson, H. S., Temperley, W., Peedell, S., Lieber, A., Hancher, M., Poyart, E., Belchior, S., Fullman, N., Mappin, B., Dalrymple, U., Rozier, J., Lucas, T. C., Howes, R. E., Tusting, L. S., Kang, S. Y., Cameron, E., Bisanzio, D., Battle, K. E., Bhatt, S., & Gething, P. W. (2018). A global map of travel time to cities to assess inequalities in accessibility in 2015. *Nature*, 553(7688), 333–336.
- World Bank (2007). *World Development Report 2008: Agriculture for Development*. Washington, DC: The World Bank.

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A Data and Sample

This part documents the sources behind every number in the paper and the rule that defines the estimation sample.

A.1 Data Sources

Data summary. The empirical analysis combines three BPS statistical collections spanning five reference years (1980, 2002, 2003, 2010, 2011), together with the Transmigration Census replication data of [Bazzi et al. \(2016\)](#), which generate the exposure measure, and the geospatial travel-time and raster inputs cataloged in [Table A2](#). The Population Census provides 2010 outcomes (population, employment by sector, migration) and 1980 baseline controls. The Susenas household survey provides 2002 consumption quantities by food item, paired with PODES 2003 production for surplus measures, and 2011 prices, wages, food shares, and housing rent, paired with the 2010 census outcome year. The PODES village survey provides 2003 production quantities by crop. Keeping consumption near the production year gives a cleaner rice surplus = production – consumption measure, while keeping prices and wages near the outcome year supports the structural-transformation interpretation. [Table A1](#) provides a per-outcome summary including the wave / year of measurement, the analysis-level sample size (out of the satellite layer’s 180 Urban Centre polygons, 1,967 subdistricts, or 1,785 split subdistricts in the transmigration-village split), and the construction notes. Availability counts are reported on that satellite layer; the analysis sample has 186 urban centers, every one of which has a travel time to a program village ([Section A.3](#)), and the headline estimation sample is the 130 centers inside the 300-minute travel-time window whose 1980 baseline the census recorded directly.

Table A1: Outcome variables and baseline controls: source wave, sample size, and construction. *Sample N reported as Urban Centre (UC) / subdistrict / subdistrict split into transmigration and surrounding villages.*

Outcome / variable description	Source / wave	Sample N	Construction notes
(A) Population Census: BPS Sensus Penduduk			
Log total population and total employment (2010)	Census 2010	180 / 1,967 / 1,785	Headline urbanization outcomes; constructed from individual-level Sensus Penduduk 2010 records aggregated to the urban-center and subdistrict level.
Log sectoral employment in agriculture, manufacturing, and services (2010)	Census 2010	180 / 1,967 / 1,785	Sectoral employment counts from the 1-digit industry classification on the individual record, aggregated to the geographic unit.
Sectoral employment shares (0–1): agriculture, manufacturing, and services (2010)	Census 2010	180 / 1,967 / 1,785	Sectoral employment divided by total employment. Shares are stored as 0–1 fractions.
Unemployment rate (2010)	Census 2010	180 / 1,967 / 1,785	Unemployed individuals divided by labor-force participants, stored as a 0–1 fraction.

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Outcome / variable description	Source / wave	Sample N	Construction notes
Log change in total population, 1980–2010	Census 1980 + 2010	168 / 1,727 / 1,599	Derived as $\ln(\text{pop}_{2010}) - \ln(\text{pop}_{1980})$. Subdistricts with no 1980 census record drop out. Gaps caused by the crosswalk rather than by the census were closed by the extension described in Section 3.1, so the remaining omissions are places the 1980 census did not reach.
Birthplace decomposition (4 exhaustive categories): share of 2010 residents born in same district / different district same province / different outer-island province / Java-Bali origin	Census 2010	180 / 1,967 / 1,785	Categories are mutually exclusive and exhaustive; they sum to 1. The Java-Bali origin category captures the channel through which the Transmigration Program is hypothesized to redistribute people.
1980 baseline controls: log total population, urban-population share, agricultural employment share	Census 1980	168 / 1,727 / 1,599	Conditioning controls in the headline reduced-form specification. The 1980 aggregates apply the SP1980 per-record inflation weights at the individual level.
(B) Susenas 2011: prices, wages, food share, housing rent (aligned with the 2010 outcome year)			
Household food share of total consumption (2011)	Susenas 2011	175 / 974 / 1,650	Household-weighted mean of food expenditures divided by total consumption expenditures (0–1). Quarters Q1–Q3 are averaged then scaled to annual. Used as a structural moment in the spatial-equilibrium model.
Median monthly wage: all sectors, agricultural workers, non-agricultural workers (2011)	Susenas 2011	175 / 974 / 1,650	Constructed at the subdistrict level from the Susenas 2011 individual labor module by taking the 50th percentile of monthly wage income for currently employed workers. The wage item is net monthly earnings from the main job, in money and goods, and is asked of everyone who worked, so the earner base includes the self-employed and own-account farmers; unpaid family workers, who do not report earnings, drop out of the median.
Log nominal wage and log real (price-deflated) wage (2011)	Susenas 2011	175 / 974 / 1,650	Log transforms of the subdistrict-level median wage. The real (price-deflated) version is constructed for the all-sector wage only, by dividing it by a household-expenditure-weighted price index (the agricultural-goods price index weighted by the food-expenditure share, plus the housing price index weighted by the non-food share).

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Outcome / variable description	Source / wave	Sample N	Construction notes
Composite agricultural goods price index (broad), crops-only price index, and rice-only price index	Susenas 2011	175 / 974 / 1,650	Built from household item-level unit values (expenditure divided by quantity), winsorized at the 5th/95th percentiles within item×province×urban, taken to the subdistrict×item median, and aggregated across items into three index variants: Laspeyres (national expenditure weights), Paasche (local weights), and an equal-weight geometric mean. The broad index covers all agricultural-origin food items; the crops index restricts to plant crops; the rice-only index uses the dominant rice item. As unit values, these indices reflect price × quality (see Section C.1).
Housing price index and median monthly rent (2011)	Susenas 2011	175 / 974 / 1,650	Imputed monthly rent for owner-occupied dwellings combined with median actual rent for renters; it is the urban land-scarcity / non-tradables price proxy in the model.
Non-food price indices: fuels and utilities (2011)	Susenas 2011	175 / 974 / 1,650	Cost-of-living controls; constructed parallel to the agricultural goods price index from household-level unit values.
(C) Susenas 2002: consumption quantities and values (aligned with PODES 2003 production)			
Rice consumption per capita per year: quantity (kg), value (Rp), and calories (kcal)	Susenas 2002	141 / 845 / 746	Built from the Susenas 2002 food-consumption module weekly recall and scaled to annual.
Total dietary calories per person per year (kcal)	Susenas 2002	139 / — / —	Sum of the calorie content of every item the Susenas 2002 food-consumption module records, so it measures the whole diet. Built by pooling households within an urban center, a median of 48 households per center, and excluding household-quarters that record fewer than ten food items, which are partly-recorded modules. A genuine diet of fewer than ten items is implausible. Constructed at the urban-center level only, which is why the subdistrict columns are empty. Plotted as the calorie row of Figure 8.
Subdistrict-total annual rice consumption (kg and tons), winsorized	Susenas 2002	141 / 845 / 746	Per-capita rice consumption multiplied by 2003 subdistrict population, then winsorized at 0.5% to avoid extreme household-level outliers contaminating the subdistrict total. Used in the rice-surplus measure.

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Outcome / variable description	Source / wave	Sample N	Construction notes
Staples consumption per capita per year (kg)	Susenas 2002	141 / 845 / 746	Sum of per-capita consumption (kg) across all staple food items: rice, corn, cassava, sweet potato, taro, and other tubers. The aggregate excludes wheat and sago (not produced in measurable quantities in the PODES sample).
Legumes consumption per capita per year (kg)	Susenas 2002	141 / 845 / 746	Sum of per-capita consumption (kg) across peanut, soybean, and mungbean items.
Coffee & tea consumption value per capita per year (Rp)	Susenas 2002	141 / 845 / 746	Coffee and tea consumption are reported in monetary value because the underlying items use distinct physical units (whole beans, ground beans, leaf).
Spices consumption value per capita per year (Rp)	Susenas 2002	141 / 845 / 746	Sum of pepper, nutmeg, and clove monetary consumption. Units differ across spices, so values are aggregated.
(D) Susenas 2011 consumption by food category (constructed from the quarterly food-module raw data)			
Staples consumption per capita per year (kg), Susenas 2011	Susenas 2011	174 / 1,850 / 1,688	Constructed analogously to the Susenas 2002 staples aggregate using the same item set. Quarters Q1, Q2, and Q3 are averaged then scaled to annual; Q4 was never released by BPS, so the result represents the Q1–Q3-average annualized consumption, with no fourth quarter imputed.
Legumes consumption per capita per year (kg), Susenas 2011	Susenas 2011	174 / 1,850 / 1,688	Same items as the Susenas 2002 legumes aggregate.
Coffee & tea consumption value per capita per year (Rp), Susenas 2011	Susenas 2011	174 / 1,850 / 1,688	Companion to the Susenas 2002 coffee & tea measure, aligned with the 2010 outcome year.
Spices consumption value per capita per year (Rp), Susenas 2011	Susenas 2011	174 / 1,850 / 1,688	Companion to the Susenas 2002 spices measure.
(E) PODES 2003: agricultural production (aligned with the Susenas 2002 consumption year)			
Rice production per capita (kg / person)	PODES 2003	180 / 1,896 / 1,727	Total subdistrict rice production divided by 2000 census population. Headline PODES outcome; available across both village-overlap and non-overlap subdistricts.
Log subdistrict-level rice production (tons)	PODES 2003	180 / 1,896 / 1,727	Log of absolute subdistrict-level rice production in tons; primary moment used in the model calibration.

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Outcome / variable description	Source / wave	Sample N	Construction notes
Rice yield (tons / ha) and rice harvested area (ha)	PODES 2003	180 / 1,896 / 1,727	Yield computed as sum of production divided by sum of harvested area across villages within the subdistrict. Rice harvested area is one of several available land measures (see “Land measure for the agricultural production function” note below).
Staples harvested area (ha)	PODES 2003	180 / 1,896 / 1,735	Aggregate of harvested area across the staples set (rice, corn, cassava, sweet potato, taro, other tubers, sorghum); mirrors the staples production aggregate. Intermediate land measure between rice-only and total arable.
Total arable land area (ha)	PODES 2003	180 / 1,896 / 1,735	Sum of harvested area across all 30 PODES crops: staples (rice, corn, cassava, sweet potato, taro, other tubers, sorghum), legumes (peanut, soybean, mungbean), coffee & tea, spices (pepper, nutmeg, clove), and the remaining plantation, fiber, and tree-crop items (rubber, palm oil, cocoa, sugar palm, sugarcane, cashew, cinnamon, cotton, vanilla, tobacco, jute). Equivalent to the PODES native “total ag area” aggregate; the broadest available land measure.
Staples production (tons)	PODES 2003	180 / 1,896 / 1,735	Aggregate of subdistrict-level production in tons for rice, corn, cassava, sweet potato, taro, and other tubers (mirroring the Susenas 2002 staples consumption aggregate).
Legumes production (tons)	PODES 2003	180 / 1,896 / 1,735	Aggregate of peanut, soybean, and mungbean production in tons.
Coffee & tea production (tons)	PODES 2003	180 / 1,896 / 1,735	Aggregate of coffee and tea production in tons.
Spices production (tons)	PODES 2003	180 / 1,896 / 1,735	Aggregate of pepper, nutmeg, and clove production in tons.
(F) Derived surplus outcomes (PODES 2003 production minus Susenas 2002 / 2011 consumption)			
Rice surplus (tons): subdistrict rice production minus subdistrict rice consumption	PODES 2003 – Sus 2002	141 / 845 / 746	Constructed as PODES 2003 subdistrict rice production minus winsorized Susenas 2002 subdistrict-level rice consumption. Indicates whether the subdistrict is a net rice exporter or net importer.
Staples surplus (tons)	PODES 2003 – Sus 2002	141 / 845 / 746	Production minus consumption for the staples aggregate (rice, corn, cassava, sweet potato, taro, other tubers); built using the matched PODES 2003 production and Susenas 2002 consumption aggregates.

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Table A1, continued from previous page

Outcome / variable description	Source / wave	Sample N	Construction notes
Legumes surplus (tons)	PODES 2003 – Sus 2002	141 / 845 / 746	Production minus consumption for legumes (peanut, soybean, mungbean).
Staples surplus using 2011 consumption (tons) [robustness]	PODES 2003 – Sus 2011	174 / 1,836 / 1,677	Robustness variant: pairs PODES 2003 production with Susenas 2011 consumption, where the main rows above use Susenas 2002. Trades the consumption–production year-gap (now 8 years) for higher consumption coverage and alignment with the 2010 outcome year.
Legumes surplus using 2011 consumption (tons) [robustness]	PODES 2003 – Sus 2011	174 / 1,836 / 1,677	Analog of the staples-2011 surplus for legumes.

Sample-size note. Susenas 2011 outer-island subdistrict-level coverage is 974 of 1,967 subdistricts (49.5% by count) but 66.6% of the outer-island population, with a strong urban-bias by sampling design: subdistricts in the largest population quintile are 3.2 times more likely to be sampled than those in the smallest, and subdistricts with $\geq 47\%$ urban population are essentially fully covered while those with $< 16\%$ urban are only 22.6% covered. The 2011 sample is appropriate for the structural-transformation analysis but describes the urban / kotamadya subset of the subdistrict panel. The 974-subdistrict figure describes the prices-and-wages module; the consumption-by-food-category aggregates in block (D) of Table A1 come from the quarterly food-consumption module, whose coverage is much wider (1,850 of 1,967 subdistricts), which is why that block's sample counts differ from the coverage stated here. Susenas 2002 outer-island subdistrict-level coverage is 845 of 1,967 (43.0% by count, 59.2% by population). Aceh has zero Susenas 2002 coverage and Maluku / North Maluku have zero PODES 2003 coverage, pre-existing structural coverage gaps in the survey-frame design.

Imputation conventions. PODES 2003 subdistrict-level aggregates are produced in two imputation schemes: *strict* (drop missing crop rows) and *imputed-zero* (treat missing crop production as zero). The variables tabulated above use the strict scheme; imputed-zero variants are available for robustness. Susenas 2002 subdistrict-level rice consumption is winsorized at the 0.5% level for the surplus calculation, to avoid extreme household-level outliers contaminating the subdistrict aggregate. The Susenas 2011 consumption-by-food-category aggregates are not winsorized at the variable-construction stage. The 0.5% winsorization applies to the surplus calculation only.

Coffee & tea and spices surplus measures not constructed. PODES 2003 production for coffee, tea, and spices is recorded in tons; Susenas 2002 and 2011 consumption for the same items is recorded in monetary value (Rupiah), because the underlying items use distinct physical units across coffee, tea, pepper, nutmeg, and clove. There is no clean tonnage conversion without auxiliary price imputation, so the corresponding surplus measures are not constructed. Log production and log consumption value are reported as separate outcomes instead.

Land measure for the agricultural production function. Several subdistrict-level land measures are available from PODES 2003: rice harvested area (the historical default), staples harvested area, and total arable land area, plus additional sub-aggregates for legumes (peanut, soybean, mungbean), plantation crops (rubber, palm oil, coffee, tea, cocoa), coffee & tea on their own, and spices (pepper, nutmeg, clove), each summed from the individual-crop harvested-area variables. Individual harvested-area variables are also retained separately for each of the 30 PODES crops (the staples and legumes items listed above, plus coffee, tea, pepper, nutmeg, clove, rubber, palm oil, cocoa, sugar palm, sugarcane, cashew, cinnamon, cotton, vanilla, tobacco, jute). The land endowment that enters the model is the satellite-derived built-up surface $\bar{T}_j$ (GHS-BUILT-S, fixed at the model-unit level; see the model section). It is measured consistently

across rural and urban units and does not depend on crop-mix reporting. The PODES harvested-area measures remain available as agricultural-land alternatives for robustness. The rice-only measure sharply understates productive land in plantation-dominant outer-island districts (especially Sumatra and Kalimantan), the total arable measure is the broadest, and staples is an intermediate option.

A.2 Spatial and Crosswalk Data Sources

Linking the 2010 census to year-2000 subdistricts. Both the 2000 and the 2010 censuses record each person's village, so the link between the two vintages is made at the village level and then aggregated. I use the BPS village-history file (*Histori Master File Desa*, 1998–2013), which has one row per village with that village's administrative code in every vintage, so a 2010 census village and its year-2000 predecessor appear on the same row. Each 2010 village is counted in the year-2000 subdistrict of its year-2000 predecessor.

Three rules govern the matching. First, only rows that are a village in both vintages are used, since the file also includes subdistrict and district heading rows. Second, a 2010 village whose year-2000 predecessors lay in two different subdistricts is left unresolved rather than assigned by a tie-break; this affects five villages nationally. Third, where the file does not list a village, the village keeps the assignment from the boundary overlay, so no village is dropped.

The file reaches 96.3% of the 77,480 year-2010 villages, and places 45,993 of them on the year-2000 outer-island map. Two checks support the construction. First, the number of 2010 villages each year-2000 subdistrict receives should be close to the number the 2000 census recorded there: 41.7% of outer-island subdistricts are within one village of that count and 87.1% are within a factor of 0.8 to 2.0, against 22.1% and 71.5% for a geometric assignment of the same villages. Second, on the Inner Islands, which this paper does not analyze and on which the construction was not chosen, the same comparison is 95.9% against 56.4%, so the improvement is not an artifact of tuning on the estimation sample. The total 2010 population is unchanged: a village changes subdistrict and is never created or dropped.

Spatial sources. The empirical analysis aggregates village- and individual-level data to two spatial units: 2000-vintage subdistricts and outer-island urban centers (UCs). The mapping between the original survey-year administrative codes and the analysis units relies on a suite of crosswalks plus globally harmonized raster products. [Table A2](#) summarizes the spatial and crosswalk inputs.

Table A2: Spatial boundary definitions, administrative crosswalks, and gridded raster inputs.

Spatial input	Source	Description / use in this paper
(A) Spatial boundary definitions		
Transmigration village sites and polygons	Bazzi et al. (2016) ; replication archive	The exposure sample is the complete set of 814 program villages in the Transmigration Census analysis sample, settled during Indonesia's Transmigration Program (1979–1988). Each village has the program-recorded year of settlement, the count of initially placed settlers, and the agro-climatic similarity of the assigned settlers; 716 of the 814 also have a digitized settlement-footprint polygon, and travel times are computed from the replication village centroids for all 814. This set of villages is the spatial reference for constructing the agro-similarity treatment.

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Table A2, continued from previous page

Spatial input	Source	Description / use in this paper
Urban Centre (UC) delineation, outer-island Indonesia	GHS Urban Centre Database, GHS-UCDB R2024A V1.1 (European Commission, Joint Research Centre), multitemporal layer GHSL_UCDB_MTUC_1980_-GLOBE_R2024	Outer-island Urban Centres aggregated from 2000-vintage subdistricts, derived from the GHS-UCDB urban-center polygons; the reclassified analysis sample has 186 urban centers, every one of them with a travel time to a program village (see Section A.3). UCs are the headline analysis unit.
Functional Urban Area (FUA) delineation, outer-island Indonesia	GHS-FUA R2019A (European Commission, Joint Research Centre)	Wider commuting-zone polygons around the urban centers, shown in the descriptive maps.
2000-vintage subdistrict administrative boundaries	BPS (Indonesian Central Statistics Agency) ADM3 shapefiles	2000-vintage subdistrict polygons used as the village-aggregation target and the subdistrict-level analysis unit.
(B) Administrative-code crosswalks across census vintages		
Subdistrict 1976 / 1980 to 2000 / 2002 crosswalk (“Olken crosswalk”)	B. Olken, courtesy of replication / supplementary materials	Hand-built crosswalk linking 1980 Census subdistrict codes to the 2000/2002 administrative vintage. Excludes Aceh, East Timor, and the Malukus by construction (regions where 1976–80 boundary data were not available). The 1980-to-2000 subdistrict mapping is the workhorse for assembling the 1980 baseline-control panel.
Subdistrict 2000 to 2010 crosswalk	Constructed for this paper from BPS administrative-code lists + spatial intersection of 2000 and 2010 subdistrict polygons	1-to-many mapping where post-2000 subdistrict splits (or district splits creating new subdistrict codes) require allocating a 2010 subdistrict to a 2000 subdistrict via spatial-overlap proportions.
Village 2000 to 2010 crosswalk	Constructed for this paper from BPS village-code lists + spatial join	Used to roll up 2010 individual-Census village-level counts into 2000-vintage subdistricts and (where applicable) into the transmigration-village part of each subdistrict overlapping a transmigration village.
Transmigration-village flag	Constructed for this paper from the intersection of the Bazzi et al. (2016) village polygons with 2010-vintage census-village polygons	Identifies which 2010 census-village polygons fall (centroid or area-share threshold) inside a transmigration village; provides the village-level split used for the transmigration-village / surrounding-village decomposition in the split-subdistrict analysis.
(C) Travel-time and great-circle distance matrices		
Cost-distance (travel-time) matrix between UCs and transmigration villages	Constructed for this paper: least-cost travel time over the Indonesia-specific land-and-sea friction surface built from Weiss et al. (2018) combined with maritime shipping routes (vintage stated in Section 2)	Pairwise UC×village travel-time matrix (minutes) used to construct the agro-similarity treatment as the unweighted mean over villages within the headline 300-minute window (weighted variants are robustness). The corresponding subdistrict-level matrix is built analogously for the surrounding-villages extension.

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Table A2, continued from previous page

Spatial input	Source	Description / use in this paper
Great-circle (haversine) distance matrix between UCs and transmigration villages	Constructed for this paper from the centroid coordinates of UC and village polygons	Robustness companion to the cost-distance matrix; a free-space variant of the design-based recentering exercise in Section C.1 .
(D) Gridded raster products		
GHS-POP gridded population estimates (1975, 1980, 1985, ..., 2020)	GHS-POP R2023A (European Commission, Joint Research Centre), 100m resolution	5-year epoch population grids calibrated to the BPS Sensus Penduduk totals at the year-2000 subdistrict level. Used as a back-check on the canonical population time series. Satellite-derived recovery of 1980 population for subdistricts missing from the Olken crosswalk was scoped but not implemented, and no analysis variable in the paper draws on it.
GHS-SMOD gridded settlement-model classification (1975, 1985, 2015, 2020)	GHS-SMOD, R2022A for the 1975, 1985, and 2015 epochs and R2023A for 2020 (European Commission, Joint Research Centre), 1km resolution	Cell-level settlement-density classification (Urban Centre / Urban Cluster / rural). Used as an alternative urban-share measure for robustness. The Urban Centre boundaries themselves come from the GHS-UCDB polygon layer in panel (A).
GHS-BUILT-S built-up-surface raster (5-year epochs, 1975–2020)	GHS-BUILT-S R2023A (European Commission, Joint Research Centre), 100m and 1000m resolution	Built-up-surface fraction per grid cell. It is the land endowment $\bar{T}_j$ in the spatial-equilibrium model (Section 6); the satellite-baseline-recovery exercise it was also scoped for was not implemented.

The census as the 1980 baseline. Every 1980 quantity in this paper comes from the population census, for three reasons. First, there is no observed satellite record for 1980. The GHSL built-up series has observed Landsat anchors at 1975, 1990, 2000 and 2014, and every other epoch is interpolated between them; the 1980 population grid is close to a fixed one-third blend of the 1975 and 1990 anchors, with an implied interpolation weight of 0.28 to 0.33 in every island group. A satellite 1980 baseline would therefore be an interpolation, not a measurement. Second, the gridded population product is not independent of the census in any case, since its counts come from census-based population estimates and the built-up layer only redistributes them within an input unit. At the subdistrict level it tracks the 1980 enumeration at a correlation of 0.57 in levels and 0.62 in logs, with the grid-to-census ratio running from 0.23 at the tenth percentile to 1.87 at the ninetieth. Third, the one observed anchor that straddles the program is the 1990 one, and it already includes the program’s own settlement: within a subdistrict that split between census rounds, a successor’s share of the gridded population rises between 1975 and 1990 by more when that successor hosts transmigrant villages. The census has none of these problems. It is a complete enumeration taken in 1980, on the ground, by the national statistical agency.

A complete-looking raster is not an observed one. [Civelli et al. \(2023\)](#) report sparse 1975 Landsat coverage over much of the archipelago, Sumatra included, in an earlier GHSL release. That gap persists in the release used here, and it can be measured rather than inferred: GHSL publishes an input-count layer, `GHS-SDATA_LDS-QUANT`, giving the number of Landsat measurements behind each cell of each epoch ([European Commission, Joint Research Centre 2024](#)). On that layer **99.3% of Sumatra’s land cells have no 1975 Landsat observation at all**, against 0.7% in 1990, and 98.2% of the island’s 1975 built-up surface and 97.7% of its 1975 gridded population sit on cells with none. Across the outer islands as a whole, 60.7% of 1975 gridded population is unobserved. Where imagery is absent the producers solve the built-up prediction by spatio-temporal interpolation whose dynamic component is drawn from the epochs that *were* observed, which here begin in 1990, after the program. A satellite 1980 baseline for this setting would therefore not

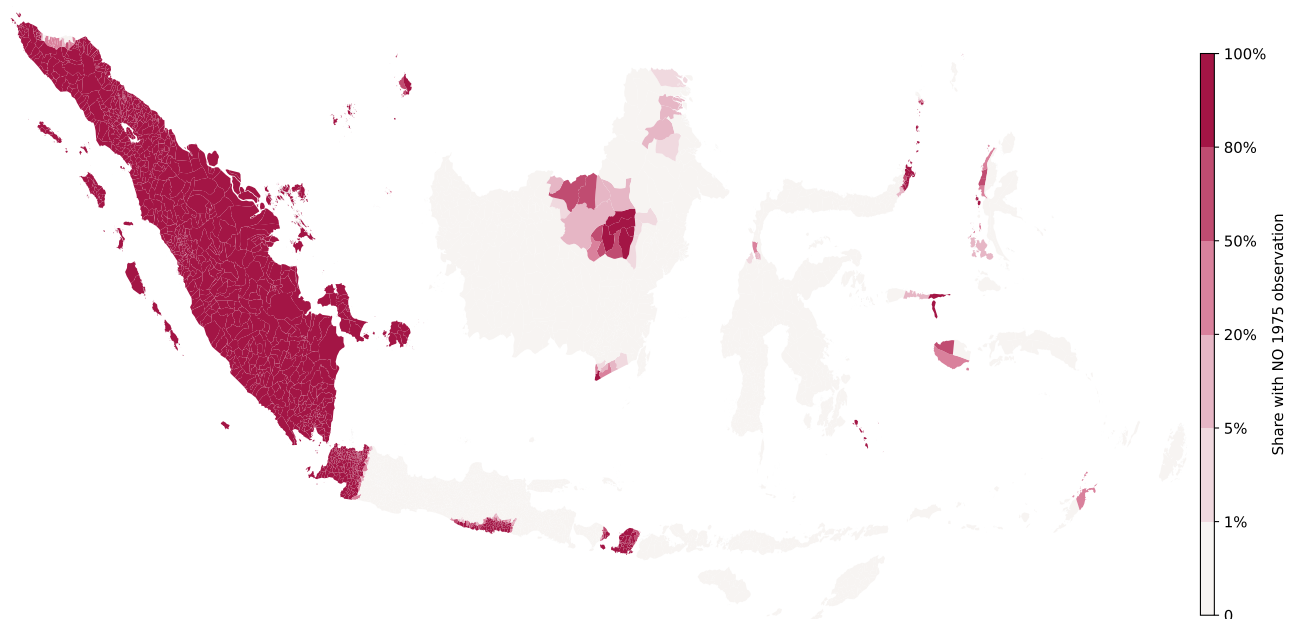
merely interpolate between two anchors; over the treated region it would rest on an anchor that was never measured, informed by settlement recorded after the program had run. The census does not have this problem. It measures 1980 directly.

Table A3: Almost no part of Sumatra has a 1975 satellite observation behind it.

Island group	Land cells	% cells with no 1975 observation	% of 1975 built-up	% of 1975 population
Sumatra	47,743,508	99.3	98.2	97.7
Java and Bali	13,919,709	17.7	26.5	20.4
Nusa Tenggara	6,727,677	6.5	28.6	34.8
Kalimantan	53,835,817	5.5	1.1	0.9
Sulawesi	18,731,243	1.5	3.8	3.6
Maluku and Papua	7,822,973	8.1	16.9	13.9
Outer islands, the estimation region		38.4	53.5	60.7

Notes: Computed from GHS-SDATA_LDS-QUANT R2023A band 1, the producers' own record of how many Landsat measurements support each 100 m cell of the 1975 epoch, summed inside year-2000 subdistrict polygons. A cell with none has 1975 built-up and population assigned by spatio-temporal interpolation. Nationally the mean is 0.83 measurements per land cell for 1975 against 1.19 for 1990, 1.26 for 2000 and 1.43 for 2014.

Figure A1: Where Indonesia's 1975 satellite layer rests on no observation.



Notes: Year-2000 subdistrict boundaries, shaded by the share of the unit's land cells with no Landsat measurement behind the 1975 epoch. Gray units fall outside the raster. 889 of 921 Sumatran subdistricts are entirely unobserved and 909 of 921 fall below the 80% threshold [Civelli et al. \(2023\)](#) use to mark a village as missing.

[Table A3](#) and [Figure A1](#) give the geography of the gap. It is not scattered cloud cover: 61.7% of the variance in the subdistrict-level unobserved share sits between island groups and only 5.8% within a district, so it follows the coverage of the Landsat archive itself. Sumatra, where the program placed most settlers, is the region with almost no record at all, and Indonesia had no Landsat receiving station of its own during the Multispectral Scanner era.

A gridded product reports a value for every cell in every published epoch, and nothing in the raster distinguishes a measurement from a model output. The grid is complete even where nothing was measured. So a 1980 satellite baseline for Indonesia cannot support a level comparison across places, cannot support a place-specific pre-period control, and cannot be used to validate the census, because over the treated region it is not an independent record of anything. This is a limitation of the historical satellite archive rather than of any particular product or release.

A narrower use survives. Contemporaneous satellite measures, from epochs that were densely observed, remain sound: the urban-center definition used throughout this paper is a 2020-vintage product, and the built-up land measure enters only at a contemporaneous epoch. For the 1980 baseline itself the alternatives are the population census, which is a complete enumeration taken on the ground, and the village census, PODES, which is the only other source that reaches 1980 at a fine spatial unit. The paper uses those two.

The same reasoning governs where satellite data *are* used. Built-up surface enters only as a contemporaneous land measure at the model unit and as the 2020-vintage urban-center definition, both of which sit well inside the observed portion of the record, and neither of which is asked to describe 1980.

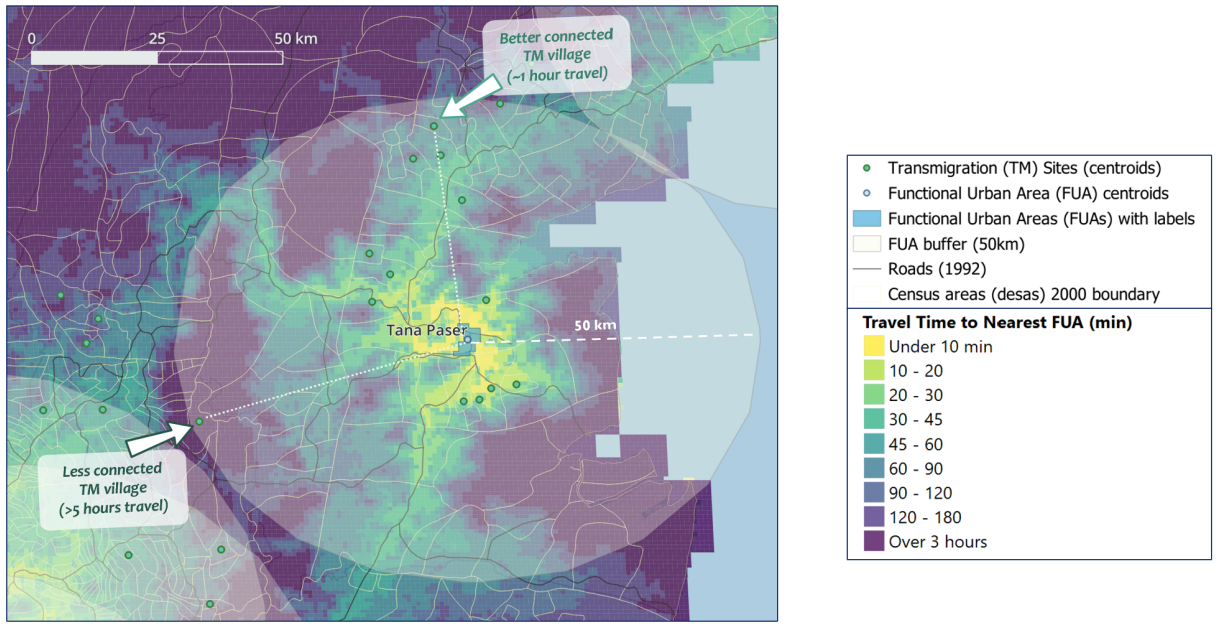
The satellite record before 1990. Sumatra is essentially uncovered in the satellite record before 1990, with 99.3% of its land cells having no Landsat observation behind the 1975 epoch, the only anchor before 1990 ([Table A3](#) and [Figure A1](#)), and the early layers rest on partial interpolation of the built-up area and the gridded population map. The limitation is particular to Indonesia, but the product is global and is released in full, so nothing in the raster marks the gap.

The friction surface and its vintage. While this friction surface is representative of 2015, it uses multiple data sources as inputs, including the road network from the OpenStreetMap and Google roads databases, so the road geometry it records is of mid-2010s vintage, measured at the end of the study period. The surface is therefore an endline measurement, like the satellite layer that defines the city sample, but its role in the design is limited. Travel time enters the analysis only when I match the transmigration villages to the urban centers, where it selects each city’s village catchment and weights the spatially aggregated village characteristics around each city, and every exposure construction inside the 300-minute window behaves like the headline specification ([Section D.3](#)). Given the absence of consistent time-varying road-network data for the Outer Islands, I treat it as the most informative available source for least-cost travel time between each transmigration village and each urban subdistrict (see [Section 2.2.1](#) and [Section 3.3](#)).

Cost distance against straight-line distance. The map in [Figure A2](#) illustrates the geographic distribution of transmigration (TM) sites by travel time in a sample region in relation to one Functional Urban Area (FUA), Tana Paser. The map shows a clear variation in the connectivity of two sample villages, both within the 50km radius. The highlighted village in the southwest (SW) is over 3 hours away in travel time while the sample village in the north, also around 45-48 km away in straight-line distance, is just about an hour away in travel time since it is located closer to the road network (the map displays a 1992 road layer for orientation).

Spatial units and the boundary crosswalk. Administrative boundaries change between census rounds, so all spatial objects are kept on the 2000-vintage subdistrict geography, which is the only vintage digitized consistently across the sources this paper uses. Census rounds on other vintages are mapped onto it by administrative code. Where a unit split between rounds, its counts are allocated across the successor units, so that any aggregate, at the subdistrict, district, urban-center or rural-subdistrict level, recovers the enumerated total. The allocation divides a split unit’s counts equally among its successors. Weighting instead by a gridded population layer would import into the 1980 baseline a quantity that is itself correlated with where settlers were placed, and those baselines enter the estimating equation as controls. The choice is close to immaterial in any case, since three quarters of split units have all their successors inside the same reporting aggregate; weighting instead by the 1975 gridded population moves no coefficient in [Table 2](#) by more than a third of a standard error.

Figure A2: Sample Area: Euclidean Distance vs. Travel-Time Cost-Distance Measure Comparison



Notes: The figure's background illustrates the variability in the travel time from each census area (village) to the nearest Functional Urban Area (Tana Paser in this sample region), ranging from under 10 minutes (yellow) to over 3 hours (dark purple). TM sites are identified with their centroids in green. The map also depicts urban areas (FUAs) in blue with a 50km buffer to compare the matching approach using the Euclidean-distance buffer around the urban area boundary with a cost-distance estimate. The map also highlights roads as of 1992, FUA buffers within 50 km, and census area boundaries from the year 2000. Urban area labels are added based on the GHS data. Source: original calculations using the friction surface from Weiss et al. (2018).

How far back a village panel can be carried, and by what route. Table A4 measures every route on the drives against every PODES wave, which is what sets the earliest year at which village-level work is possible. Three facts come out of it. Every village crosswalk available, the Cassidy family and BPS's own Master File Desa panel alike, begins at the 1998 administrative vintage, so no file links a village code across the 1990s. Spatial matching does not reach further back either: the earliest village map is drawn on the 2000 census, and no map of the three is complete even against the roster of its own year, holding 95.2%, 99.2% and 95.5% of the SP2000, SP2010 and PODES 2021 village lists. The subdistrict is therefore the finest unit that can be carried back to 1980, through the Olken crosswalk at 82 to 91% per wave, which is the level at which this paper works.

Table A4: How much of each PODES wave the crosswalks and village maps reach

Wave	Village, Cassidy		Subdistrict, Olken		Village, spatial	
	Best column	Covered	Best column	Covered	Best layer	Covered
1976	Not matched	–	Not matched	–	Not matched	–
1980	Not matched	–	id1980	90.9	Not matched	–
1983	Not matched	–	id1983	88.9	Not matched	–
1986	Not matched	–	id1983	87.8	Not matched	–
1990	Not matched	–	id1990	89.5	Not matched	–
1993	Not matched	–	id1993	88.2	Not matched	–
1996	Not matched	–	id1997	89.8	Not matched	–
2000	ID2000	78.0	id1999	82.0	2000 census	79.9
2003	ID2002	94.1	id2002	86.2	2000 census	77.8
2005	ID_SE06	94.0	Not matched	–	2010 census	80.6
2008	IDPODES	98.7	Not matched	–	2010 census	94.9
2011	ID2011_1	98.8	Not matched	–	2010 census	96.3
2014	IDPODES14	99.9	Not matched	–	2020 BPS	96.9
2018	ID2015_1	96.3	Not matched	–	2020 BPS	96.3
2021	ID2015_1	95.6	Not matched	–	2020 BPS	95.5
2024	ID2015_1	90.7	Not matched	–	2020 BPS	91.5

Notes: Percent of the wave's distinct codes found in the best-matching column of the crosswalk named, or in the best-matching village polygon layer. Both crosswalk families carry one column per administrative vintage, so linking is a lookup rather than a chain. The village family is `village_crosswalk_Cassidy_1998_2013` and `village_crosswalk_Cassidy_2010_2015` in `data/PODES/crosswalks/`; the subdistrict family is `data/crosswalks/kec_codes_1976_to_2002_Olken.dta`. A SECOND village family sits on the drives, BPS's own Master File Desa panel, held as `village_crosswalk_MFD_1998_2013_from_lim.dta` and `village_crosswalk_MFD_allprov_1998_2014_from_bazzi.dta` in `data/crosswalks/`, with a third copy of the first filed under SUSENAS as `crosswalk_geo_ids_1998_2013.dta`. It begins at the same 1998 vintage as Cassidy and so reaches no wave Cassidy does not reach, which is why the village column names one family rather than two. “Not matched” means the family's own columns do not span that wave: the earliest village vintage in any crosswalk on the drives is 1998, Olken's latest subdistrict vintage is 2002, and the earliest village polygon layer is drawn on the 2000 census. Outside its span a file can still report a high string match, but that is a code which happens not to have changed rather than coverage, and it cannot carry a unit across a boundary the file never saw; those matches are suppressed rather than reported. A subdistrict panel after 2003 does not need Olken, because the raw code itself matches at 91% or better. The “Village, spatial” pair names the village polygon layer holding the most of that wave's village codes, of the three on the drives: the 2000 census layer (`2000_desa_indonesia_fixed`, 66,280 distinct codes), the 2010 census layer (`2010_desa_multipoly_fixed`, 77,480) and the 2020 BPS boundary file (`idn_admbnda_adm4_bps_20200401`, 81,912), all under `data/spatial/admin_geographies/`. That match is made on the code, so the column says which wave a layer could be joined to and never that any geometry has been checked. **None of the three layers is complete against the administrative roster of its own year:** the 2000 layer holds 95.2% of the SP2000 census village list, the 2010 layer 99.2% of the SP2010 list, and the 2020 layer 95.5% of the PODES 2021 village list, so a spatial route loses villages of its own. *Source:* the wave files, the crosswalk families and the polygon layers named above, each measured from the file itself.

No individual-level linkage across census rounds. The 2000–2010 crosswalks above are administrative-boundary mappings and support unit-level aggregation only. Person- or household-level linkage of the 2000 and 2010 census microdata is not feasible: the 2000 census records do not contain digitized respondent names, and a full-sample probabilistic matching experiment linked only 0.08–0.43% of households at a placebo-calibrated 1% false-discovery rate. No result in this paper therefore relies on following individuals or households across census rounds; all 2000–2010 comparisons are aggregates within crosswalked spatial units.

A.3 Urban Sample Construction

Single inclusion rule. The urban sample is defined only by Urban Centre overlap. A subdistrict enters the urban sample if it overlaps a GHS Urban Centre polygon; overlapping subdistricts are aggregated into one analysis unit per urban center; and the objective reclassification described in [Section 2.2.1](#) re-attaches dense administratively urban edge subdistricts and adds the new nodes, giving the 185-center sample, every center of which has a travel time to a program village. The Urban Centre layer itself follows one global rule: on a one-kilometer population and built-up grid, contiguous high-density cells (at least 1,500 inhabitants per square kilometer) forming clusters of at least 50,000 people are classified as urban centers, and the polygon layer collects these clusters ([Dijkstra et al. 2021](#)), so a single satellite-based rule identifies cities consistently across the world.

Aggregation to the urban center. To identify what an urban center is, I combine data at three levels. The outcome data are available at the village and urban-neighborhood level (administrative level 4), the finest spatial units available; these comprise the subdistricts (administrative level 3) that the 1980 baseline data cover. Since an important part of the identification is to control for baseline levels, I need to combine the baseline and endline units in a comparable way, so the finest level the data afford is the level-3 subdistrict. The household surveys are at most available at the subdistrict level as well. Multiple contiguous subdistricts overlap with the same urban area, and since the analysis looks at the market overall, with exposure based on distance around the city, there is little meaningful variation across subdistricts within a city. I therefore dissolve contiguous subdistricts into one urban-center area, the unit that corresponds to the meaningful urban labor market.

The 2010 totals each city is built from. A city’s 2010 population and employment are the sums of its member year-2000 subdistricts, and those subdistrict totals rest on the village-level link of [Section A.2](#). Under the boundary overlay, 203 outer-island subdistricts recorded a 2010 population under half or over two and a half times their year-2000 population, against a median move of 1.15; under the administrative link that falls to 54, of which 38 are in Aceh, where the year-2000 census itself is short.¹⁷

From the satellite layer to the analysis sample. The GHS layer has 182 Urban Centre polygons on the Outer Islands, and two of them leave before the analysis. One is a census twin: two polygons fall on a single subdistrict in the census geography, so they are treated as one unit throughout. The other, in Aceh, has no constituent subdistrict in the year-2000 census, which did not enumerate them. That leaves 180 unique satellite-defined centers. The reclassification rule (1980 population density above 500 persons per square kilometer with a 1980 agricultural employment share below one half) adds six dense administrative centers as standalone units, for 186 urban centers, and folds 40 dense urban edge subdistricts, drawn from 23 districts, into 21 parent rows. [Figure A4a](#) shows the rule in the data. One center, in the Sangihe island group off North Sulawesi, has no transmigration village within reach of its island group. Its travel time to the nearest program village, 1,120 minutes, is recovered from the all-subdistrict travel-time matrix rather than the urban-center matrix, and places it far outside any window used here. The analysis universe is therefore all 186 centers. Two restrictions then define the estimation sample, and they are separate. The 300-minute travel-time window leaves 134 centers. Of those, 131 have a 1980 baseline and 130 have one the census recorded directly, which is the estimation sample. A center enters only if all three of the 1980 baselines that define the rule are observed, log population, log

¹⁷The 2000 census enumerated only 40.3% of Aceh’s villages, so no subdistrict link can repair an Aceh comparison between 2000 and 2010. The year-2000 census is used only in robustness exhibits.

employment, and the urban-population share, and a partial baseline does not qualify it. The Main Spec conditions on a fourth 1980 baseline, log agricultural employment, which is a control and not part of the inclusion rule; it is missing in the stored file for two in-sample centers whose 1980 agricultural employment level is measured, and the producers rebuild it from that level. This is an inclusion rule rather than an incidental data gap: the specification conditions each 2010 outcome on the pre-program level of the same quantity, which a center without a 1980 measurement cannot support, so the requirement is not relaxed on any other ground. Four in-window centers are excluded, and for two distinct reasons. Three, in Bengkulu, West Kalimantan and Central Kalimantan, have no 1980 census record at all, and their baselines cannot be recovered from the earlier village censuses, which record households by activity rather than employment by sector. The fourth, in East Kalimantan, has a 1980 baseline supplied by the apportioned-parent fill rather than measured, and a filled baseline does not qualify a center either. No Aceh center is excluded on this ground: the extension to the 1980 crosswalk described in [Section 3.1](#) recovered them. Of the six added units, three are dense administrative centers that fall below the satellite population threshold, and a second, equally pre-treatment screen applies the same logic to the rural remainders of administrative cities (*kota*) whose 1980 urban population share exceeds one half: three enter as their own units (Palangka Raya, Sawahlunto, and Batam), two re-attach to their city’s existing urban center (Palu and Samarinda), and one is dropped because its *kota* is fully urban, so the node would duplicate the city itself (Pangkalpinang).

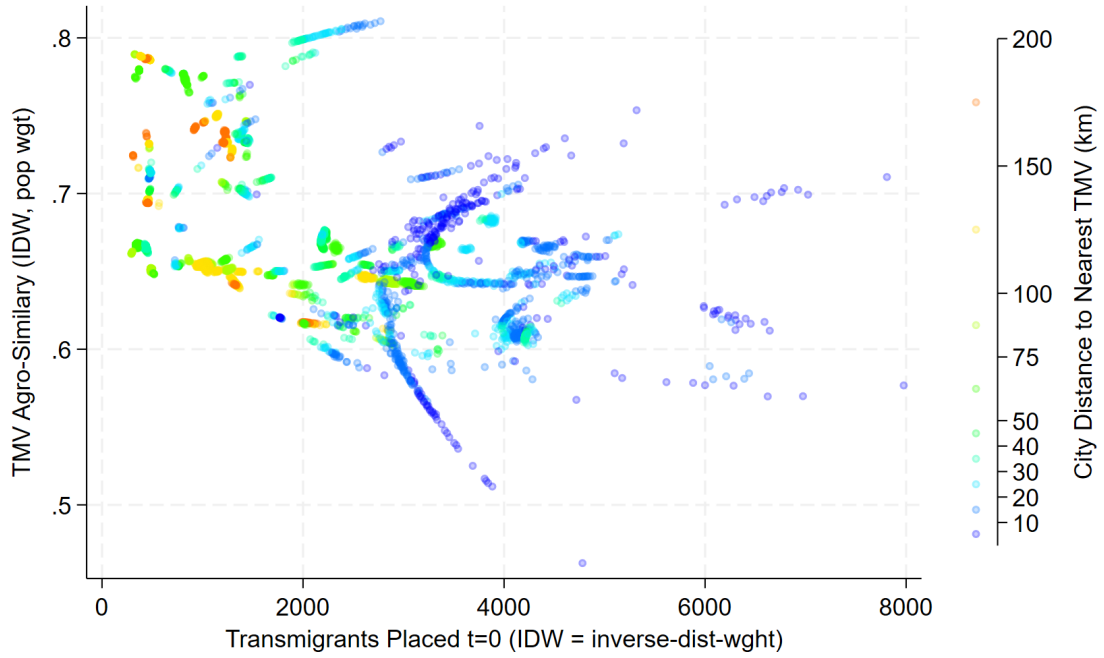
Distance to the nearest transmigration village, and the 300-minute cutoff. [Figure A3b](#) shows the sample’s full distribution of travel time to the nearest transmigration village. The 300-minute cutoff is not a natural break in this distribution; it is chosen as a meaningful, achievable travel distance, and it falls at the 72.0th percentile of the 186 centers, keeping the closer three-quarters of the sample. The median center is 170 minutes from its nearest program village.

The results do not turn on the sample construction. A reader may reasonably ask whether the reclassification accounts for the result, or whether the estimates rest on centers that were barely urban to begin with. [Figure A4b](#) re-estimates the main effects on nested subsamples that answer both questions: first dropping every center the reclassification touched, then imposing a 1980 population-density floor, a 1980 agricultural-share ceiling, and finally both. The two composition responses, the agricultural and modern employment shares, hold their sign, magnitude and significance in every one of them, and each estimate falls inside the interval of the headline it is being compared against. The agricultural-share screen is the demanding one, since it conditions on the 1980 level of the agricultural share itself and so works against finding a response in that share, and the response survives it. Only the size response on log population moves: it halves and loses significance once the reclassified centers are dropped, and that is the expected direction here, since the dropped centers are the dense administratively urban edges that hold population mass while inheriting their parent metro’s exposure.

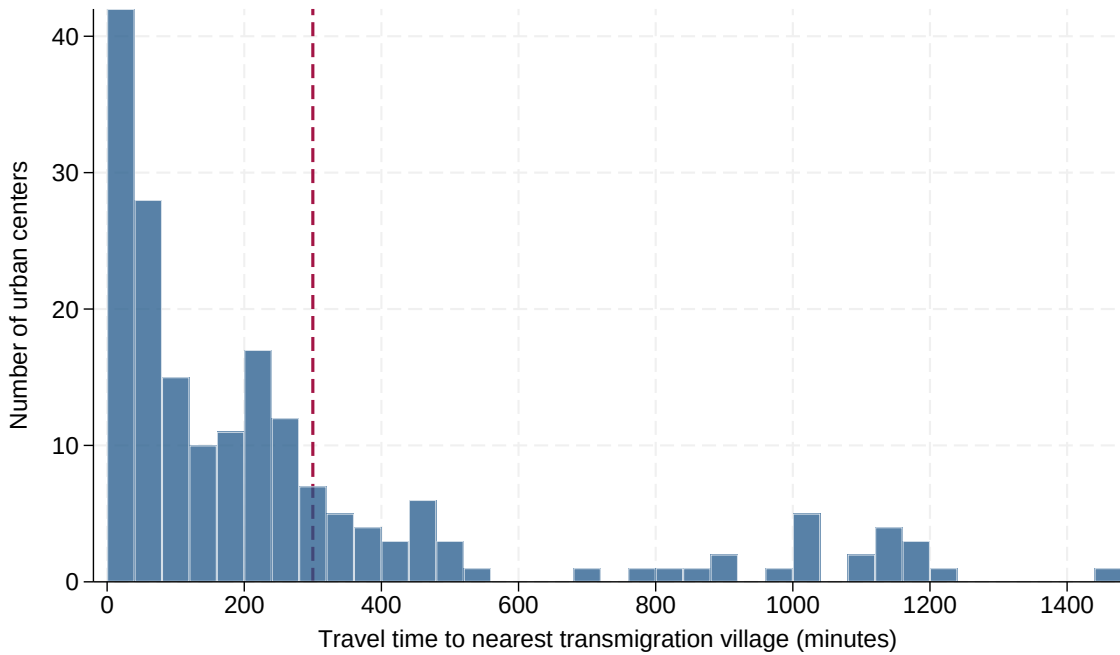
[Table A5](#) reports the same comparison as a table, giving the coefficient and its standard error for every outcome on each of the three samples: the headline, the sample with every reclassified center removed, and the denser, less agricultural centers within that sample.

Figure A3: Settler placement and the travel time from each city to the nearest program village

(a) Aggregated agro-similarity index against program exposure



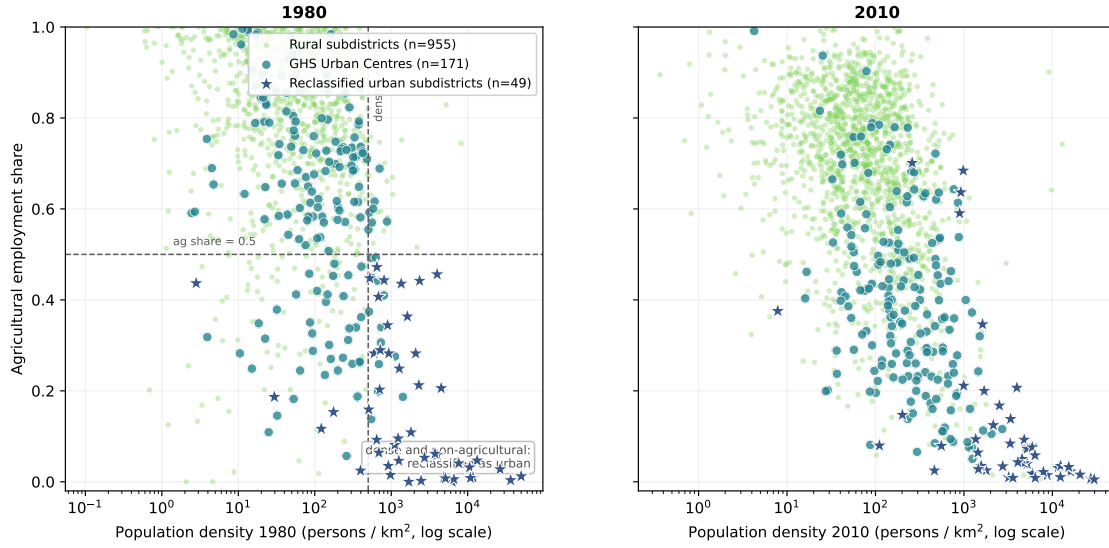
(b) Urban centers by travel time to the nearest transmigration village



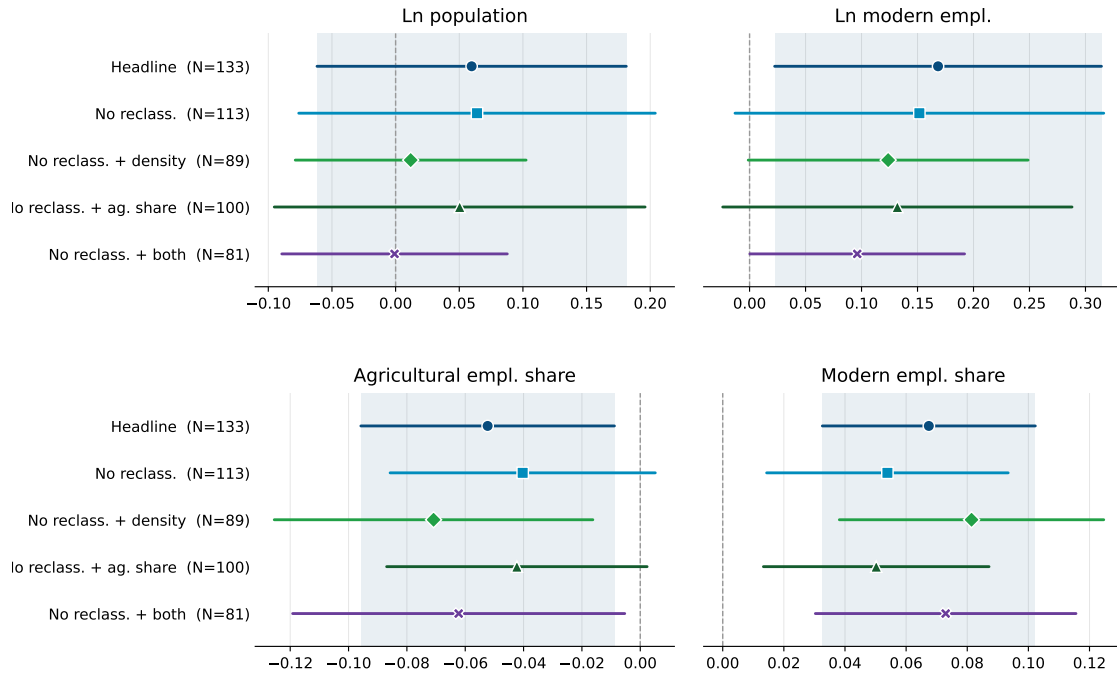
Notes: Both panels describe the 186-center analysis universe. Panel (a): the agro-similarity index quantifies the match between the settlers' farming techniques and the environment they were placed in, which shapes their potential productivity and integration; the horizontal axis is the inverse-distance weighted count of transmigrants placed at $t = 0$ and the right-hand axis the city's distance to its nearest program village. Panel (b): the distribution of travel time in minutes to the nearest transmigration village. The dashed line marks the 300-minute cutoff, which keeps 134 of the 186 centers, the 72.0th percentile of this distribution; the median center is 170 minutes away.

Figure A4: The reclassification rule in the data and the results under tighter urban samples

(a) Reclassification diagnostics: density and agricultural employment share



(b) Main effects under progressively tighter urban-sample restrictions



Notes: Panel (a): each point is a place, positioned by population density (horizontal axis, log scale) and agricultural employment share (vertical axis) in 1980 (left) and 2010 (right), the two observables the reclassification rule uses. Green points are rural subdistricts, teal circles are the satellite-defined GHS Urban Centres, and navy stars are the reclassified urban subdistricts. The dashed boundaries on the left mark the rule, 1980 density above 500 persons per square kilometer and 1980 agricultural employment share below 0.5, so every star falls in the dense, non-agricultural quadrant by construction; the rule is defined on the pre-period, so the 2010 scatter is shown without boundaries. Panel (b): each marker is the coefficient on standardized agro-similarity to transmigration villages within 300 cost-minutes, from a separate Main Spec regression of the 2010 outcome named above it, with 95% Conley intervals at a 200 km cutoff and a shaded band repeating the headline interval. The samples are nested: centers the 2026 reclassification created or altered are dropped, then a 1980 population-density floor of 25 residents per square kilometer, then a 1980 agricultural employment-share ceiling of 80%, then both. Agro-similarity is standardized once over the full urban-center file, so every outcome shares the headline table's scale.

Table A5: Main outcomes with modern sector, by urban subsample

	Population + Employment			Agric Empl		Modern Empl	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	ln pop	ln emp	unemp rate	ln L agric	agric %	ln L modern	modern %
<i>Panel A: headline estimation sample, every urban center in the 300-minute window</i>							
TMV AgSim (z, 300min)	0.060	0.033	0.009	-0.167*	-0.053***	0.169***	0.068***
	(0.059)	(0.056)	(0.007)	(0.096)	(0.018)	(0.061)	(0.015)
Observations	133	133	133	133	133	133	133
<i>Panel B: excluding every area the reclassification created or enlarged</i>							
TMV AgSim (z, 300min)	0.076	0.045	0.012	-0.148	-0.060***	0.191**	0.068***
	(0.074)	(0.070)	(0.008)	(0.095)	(0.018)	(0.079)	(0.018)
Observations	113	113	113	113	113	113	113
<i>Panel C: Panel B further restricted to at least 25 residents per km² and at most 80% agricultural employment, both in 1980</i>							
TMV AgSim (z, 300min)	-0.002	-0.046	0.014	-0.305***	-0.068***	0.103**	0.077***
	(0.046)	(0.047)	(0.008)	(0.097)	(0.022)	(0.047)	(0.019)
Observations	81	81	81	81	81	81	81
Transmigrant count (IDW)	✓	✓	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓	✓	✓

Notes: Each cell is the coefficient on agro-similarity from a separate urban-center-level regression, with its Conley spatial standard error at a 200-kilometer cutoff in parentheses, one panel per urban-sample definition. Panel A is the main city sample: the outer-island urban centers whose nearest transmigration village (TMV) lies within 300 travel minutes and whose 1980 baseline the census recorded directly. Panel B drops every center the reclassification touched, both the centers it created and the pre-existing centers whose footprints it enlarged. Panel C restricts Panel B again to centers that were not essentially rural in 1980, keeping those with at least 25 residents per square kilometer and at most 80% of employment in agriculture; two of the centers it excludes are dropped for having no 1980 agricultural share at all. The regressor, TMV AgSim (z, 300min), is the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, standardized once over every urban center with a measured exposure and held fixed across panels, so the rows are on one scale. Every column includes the Main Spec controls: the inverse-distance-weighted (IDW) count of settlers placed within 300 minutes, the 1980 log population, log employment, urban population share, and log agricultural employment, and the sixteen agro-climatic covariates averaged over the same window. The modern sector is manufacturing plus services. *** 1%, ** 5%, * 10%.

A.4 Urban Descriptives

Summary statistics: full UC universe. Table A6 reports summary statistics for the full universe of 185 outer-island urban centers alongside the 300-min cost-distance analysis subsample. The main-estimation-sample statistics are in the main text (Table 1).

Table A6: Summary statistics: 187 outer-island urban centers

Panel A: Levels (counts, distance, rice production per capita)									
	All UCs (N = 187)						Analysis sample		
	Mean	SD	p50	Min	Max	N	Mean	SD	p50
Total population 1980 (UC, BPS census)	117,728	207,785	67,979	5,785	2,078,951	184	118,216	220,301	66,000
Total population 2010 (UC, BPS census)	220,292	380,117	114,725	27,624	3,796,167	186	229,569	405,501	114,000
Total employed 1980 (UC, BPS census)	32,338	53,727	18,734	2,226	537,303	184	32,677	57,428	18,000
Total employed 2010 (UC, BPS census)	86,014	141,092	47,634	10,952	1,416,355	186	90,333	152,478	47,000
Urban population 1980 (UC, BPS census)	45,512	133,235	12,784	0	1,347,294	184	48,030	143,722	12,000
Distance to nearest TMV (cost-distance, min)	266	325	145	1	1,442	187	108	85	70
Rice production per cap, kg (PODES 2003)	961	11,024	74	0	145,104	173	1,237	12,717	70

Panel B: Shares, ratios, standardized indices									
	All UCs (N = 187)						Analysis sample		
	Mean	SD	p50	Min	Max	N	Mean	SD	p50
Urban population share 1980 (UC)	0.254	0.234	0.207	0.000	0.898	184	0.261	0.228	0.239
Employment % in agriculture, 1980	0.557	0.220	0.569	0.045	0.996	184	0.558	0.219	0.575
Employment % in manufacturing, 1980	0.068	0.064	0.050	0.000	0.423	184	0.064	0.063	0.050
Employment % in services, 1980	0.320	0.174	0.299	0.000	0.833	184	0.321	0.166	0.306
Employment % in agriculture, 2010	0.383	0.188	0.372	0.050	0.899	186	0.374	0.184	0.362
Employment % in manufacturing, 2010	0.046	0.042	0.032	0.001	0.292	186	0.046	0.042	0.032
Employment % in services, 2010	0.469	0.154	0.477	0.093	0.780	186	0.471	0.146	0.480
Employment % in construction, 2010	0.058	0.027	0.052	0.003	0.121	186	0.060	0.027	0.054
TMV settler count (IDW, '000s)	1.028	1.090	0.769	0.058	11.296	186	1.203	0.795	1.008
TMV AgSim (z, 300min)	0.000	1.000	-0.094	-2.930	2.692	138	-0.011	0.969	-0.094
Share born same district, 2010	0.694	0.189	0.726	0.106	0.982	186	0.667	0.187	0.690
Share born same prov diff district, 2010	0.128	0.093	0.100	0.008	0.482	186	0.130	0.092	0.095
Share born other outer-isl prov, 2010	0.102	0.121	0.058	0.001	0.724	186	0.116	0.129	0.069
Share born Java/Bali, 2010	0.076	0.077	0.044	0.003	0.420	186	0.087	0.084	0.060
Food share of HH consumption (Susenas 2011)	0.551	0.050	0.550	0.399	0.688	185	0.553	0.050	0.550
Housing price index, subdistrict/national (Susenas 2011)	1.585	0.934	1.387	0.219	6.577	186	1.593	0.910	1.445
Ag-goods price index (Susenas 2011)	1.059	0.129	1.070	0.743	1.383	186	1.062	0.122	1.065

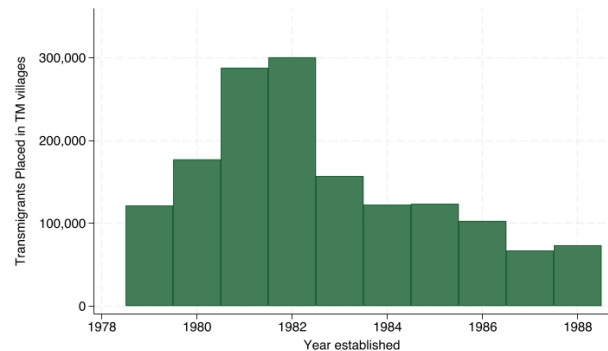
Notes: Two samples: the full universe of 187 deduplicated outer-island urban centers with a computed cost-distance to the nearest transmigration village, and the 300-min cost-distance analysis sample ($N \approx 134$) of the headline Main Spec regressions. Per-variable N reflects coverage within the sample; the 1980 census levels are left missing (not zero) for the 3 urban centers without 1980 census coverage, so the minimum of the 1980 employment count is a covered value. UC = urban center. TMV = Transmigration Village. IDW = inverse-distance weighted. Census-derived vars: BPS Population Census 1980 and 2010 (full-count). Susenas vars: 2011 wave. PODES vars: 2003 wave.

A.5 Transmigration-Village and Sample-Group Descriptives

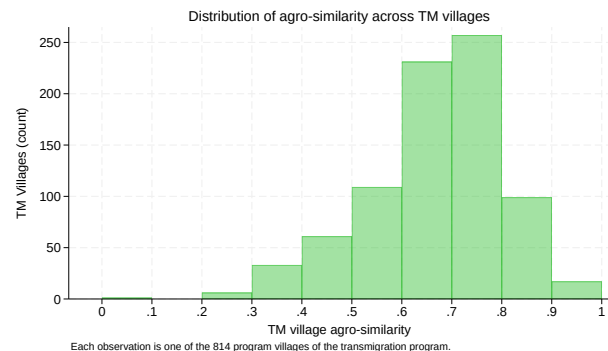
Table A7 reports summary statistics for the transmigration villages whose agro-climatic similarity to the settlers' origin regions generates the city-level agro-similarity exposure; the table covers all 814 program villages, with the village-population rows available for the 716 villages with digitized polygon footprints. The villages are small agricultural

Figure A5: Transmigration villages: settlement by year and agro-climatic similarity

(a) Transmigrants placed, by settlement year



(b) Village-level agro-similarity (TM villages)



Notes: Panel (a) counts the transmigrants placed each settlement year across the 814 program villages. Panel (b) is the village-level agro-climatic similarity index across the same villages. Source: Transmigration Census settlement records from the [Bazzi et al. \(2016\)](#) replication data.

settlements (mean 1,885 settlers placed at founding, 69% of employment in agriculture) established between 1979 and 1988. The program-wide total of more than two million resettled people cited in [Section 2](#) is the official aggregate, which includes placements outside this village census. The village sample underlying the exposure measure records approximately 1.5 million placed individuals, matching the annual placements plotted in the settlement-trend figure in [Section 2](#).

[Table A8](#) reports how the three sample groups entering the structural-transformation regressions differ on observable characteristics: cities are larger and far less agricultural than rural subdistricts, while the agro-similarity exposure is balanced across all three groups, consistent with the quasi-random settler allocation.

Table A7: Summary statistics: transmigrant villages

Panel A: Levels (settlers placed, village population)						
	Mean	SD	p50	Min	Max	N
Settlers placed (count)	1,885	1,011	1,792	136	8,852	814
Village population 1980	1,279	6,057	282	0	130,914	716
Village population 2010	2,235	5,228	817	0	85,296	716

Panel B: Economic structure, similarity, and agro-climatic covariates						
	Mean	SD	p50	Min	Max	N
Agro-climatic similarity to migrant origins	0.667	0.136	0.684	0.000	1.000	814
Agricultural employment share	0.689	0.241	0.758	0.009	0.993	814
Urban village (share)	0.029	0.169	0.000	0.000	1.000	814
Years of schooling	3.998	0.899	3.978	0.713	7.653	814
Agricultural productivity (normalized)	0.996	2.647	0.435	0.001	27.316	770
Ethnolinguistic fractionalization	0.428	0.202	0.440	0.021	0.889	814
Ruggedness	0.311	0.129	0.310	0.006	0.691	814
Distance to coast (index)	10.553	1.138	10.844	5.795	11.925	814
Soil organic carbon	4.765	6.214	1.685	0.584	25.065	814
Soil pH	5.069	0.435	4.950	4.256	7.790	814
Rainfall, 1948–78	224.998	35.188	227.616	90.999	331.816	814
Temperature, 1948–78	26.253	1.709	26.748	17.225	28.174	814

Notes: Unit of observation is the transmigrant village. The table covers the complete 814-village exposure set underlying the city-level AgSim measure (all settled 1979–1988, mean 1983). Village population in 1980 and 2010 (GHS-POP raster estimates) is available only for the 716 villages with digitized settlement-polygon footprints, so those two rows report N = 716; normalized agricultural productivity is available for 770 villages. All other rows cover the full set (N = 814). Population: GHS-POP. Settler counts and placement years: transmigration program records (Bazzi et al. 2016). Agro-climatic similarity and covariates: Bazzi et al. (2016) construction.

Table A8: Sample descriptives by group: cities and rural subdistricts

Panel A: Population, employment, and distance								
	Cities		Rural, no TMV		Rural, w/ TMV		All	
	mean	sd	mean	sd	mean	sd	mean	sd
Population 1980	108,263	(182,837)	26,799	(20,194)	17,949	(13,326)	36,213	(75,530)
Population 2010	208,565	(343,193)	37,954	(28,586)	51,277	(35,685)	63,018	(139,371)
Employed 2010	82,355	(129,276)	16,345	(12,082)	22,522	(15,560)	26,244	(53,027)
Cost-distance to nearest TMV (min)	105	(84)	160	(80)	1	(0)	122	(95)
Observations (subdistricts)	134		692		201		1,027	

Panel B: Economic structure, treatment, and prices								
	Cities		Rural, no TMV		Rural, w/ TMV		All	
	mean	sd	mean	sd	mean	sd	mean	sd
Agric employment share 2010	0.393	(0.190)	0.645	(0.229)	0.718	(0.142)	0.626	(0.230)
Manuf employment share 2010	0.047	(0.045)	0.035	(0.040)	0.026	(0.026)	0.034	(0.039)
Services employment share 2010	0.469	(0.143)	0.290	(0.169)	0.226	(0.087)	0.301	(0.168)
Urban population share 1980	0.255	(0.225)	0.078	(0.228)	0.011	(0.052)	0.089	(0.217)
TMV AgSim (raw, 300min)	0.677	(0.102)	0.679	(0.102)	0.676	(0.106)	0.678	(0.103)
TMV settler count (IDW, thousands)	1.281	(1.181)	0.605	(0.479)	6.997	(6.848)	1.944	(3.968)
TMVs in subdistrict (814 sample)	0.000	(0.000)	0.000	(0.000)	3.448	(3.403)	0.675	(2.032)
ln ag wage (Susenias)	13.834	(0.444)	13.690	(0.502)	13.850	(0.442)	13.755	(0.482)
ln non-ag wage (Susenias)	14.111	(0.277)	13.995	(0.344)	14.108	(0.376)	14.044	(0.338)
Ag-goods price index (Susenias)	1.144	(0.130)	1.117	(0.162)	1.152	(0.140)	1.130	(0.151)
Observations (subdistricts)	134		692		201		1,027	

Notes: Unit of observation is the subdistrict (2000 boundaries), within cost-distance 300 minutes of the nearest transmigration village (TMV). Columns partition the sample as in the structural-transformation regressions: **Cities** (subdistricts in an Urban Centre), **Rural, no TMV** (rural subdistricts not overlapping a TMV), **Rural, w/ TMV** (rural subdistricts overlapping a TMV), and **All**. Each cell reports the group mean with the standard deviation beneath in parentheses; the final row reports the number of subdistricts. Susenas wage and price coverage is sparse and urban-biased, so the rural columns rest on far fewer subdistricts than the population and employment rows. Census: BPS 1980/2010. Wages and prices: Susenas 2011.

B Measures

This part sets out how each measure the main text uses is constructed.

B.1 Trade-Openness Measure: Rice-Import Market Access

Source data. To measure how exposed each urban center is to imported food, I use administrative records of *rice imports by port of entry*: volumes (kg) and values disaggregated by importing port and origin for 2000–2010, with port coordinates (the “Ports & Prices” compilation). Because the traded commodity is rice, the resulting index is specifically a measure of *food*-trade openness, the margin most directly relevant to a local agricultural-productivity shock. Summing import throughput to the port level gives, for each port of entry p , a total volume kg_p and a location. The compilation stores port coordinates in degrees and minutes; taken as decimal degrees, each port falls 20 to 40 km from its berth, so every measure below uses the converted coordinates, which place the ports on the coast (median 2.8 km from the coastline against 13.0 km) and put the named terminals on their real berths.

Construction. Not every urban center is itself a port, so I separate *proximity* (being or abutting a port) from *trade intensity* (the volume of trade a city can reach), and compute three candidate metrics: (i) **proximity**, the great-circle distance dist_nport_i from city i to its nearest customs port of entry (plus an indicator for a port within 50 km); (ii) **volume**, the import throughput of the reachable port(s); and (iii) a **gravity market-access** composite that combines the two,

$$\text{open}_i = \sum_p \frac{kg_p}{d_{ip}},$$

a distance-discounted sum of every port’s import volume (decay parameter $\delta = 1$; the nearest-port distance is floored at 10 km so that port-cities do not diverge). The headline openness variable is $\ln \text{open}_i$, standardized over the UCs inside the headline cost-distance window ($N = 133$). I prefer the gravity composite because it uses information from *all* reachable ports weighted by their size, rather than collapsing access to a single nearest-neighbor distance.

The gravity and port-distance measures, in brief. The gravity measure loads realized trade. Administrative records of rice imports by port of entry over 2000–2010 give each port a total import volume, and the market-access index sums those volumes over every port, discounting by distance. Because realized import volumes are partially an equilibrium outcome, this measure is not just baseline variation, and I provide it as additional evidence beside the route measure rather than as the main causal claim. It is the alternative openness split of [Table B1](#), and it correlates at 0.60 with route openness. The simplest measure is the distance to the nearest customs port of entry. Bare distance is nearly uninformative about reachable trade, because ports of entry are common (the median city sits 45 km from one) and the measure ignores what moves through them. It enters the balance tests of [Section 4.3](#) as a predetermined trade-access covariate rather than the openness analysis.

Proximity versus the gravity composite. [Figure B1a](#) plots the gravity market-access index against distance to the nearest customs port of entry. The two are negatively related, in the expected direction, but only weakly ($\rho = -0.12$): at a given distance there is wide variation in gravity access, because the composite also loads the *size* of the reachable ports and the contribution of more distant ones, while bare proximity is nearly uninformative once every port of entry counts as a candidate anchor (the median city sits 45 km from one). Restricting the horizontal axis to the ports that recorded rice volume in 2000–2010 tightens it to $\rho = -0.54$, a tighter but conditioned comparison. Restricting to ports with realized rice volume conditions on the trade outcomes themselves, so the full-list figure is the one reported. Either way the conclusion is the same, and the full-list version shows it more clearly. The gravity measure is not a relabeling of raw proximity, and it contains independent trade-volume information, which is why I use it as the openness measure.

Exogenous route proximity to the Thailand/Vietnam rice supply. The gravity composite loads realized import volumes, which are an equilibrium outcome. The preferred openness measure in [Section 5.3](#) therefore uses

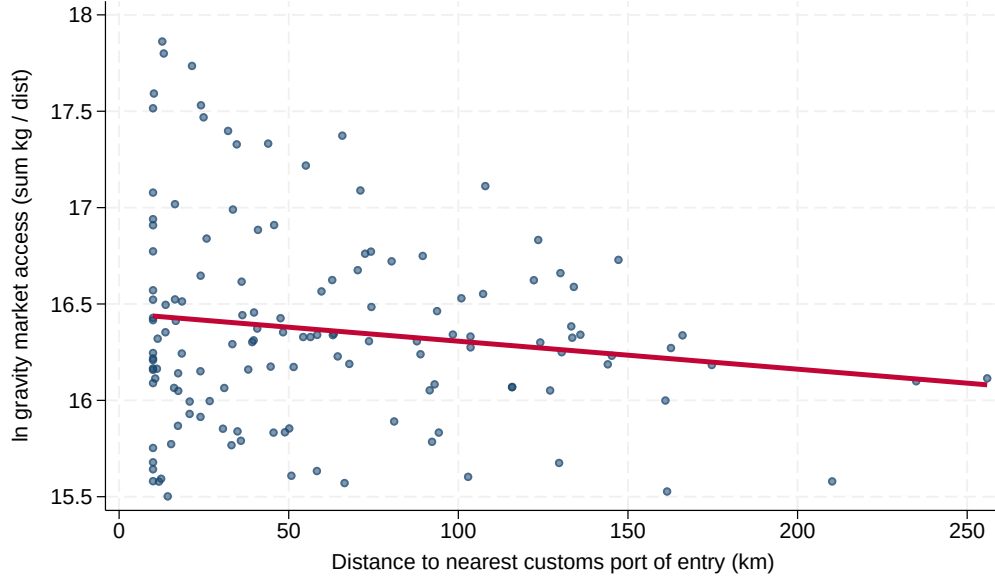
predetermined trade geography only, following the “distance to Ho Chi Minh City/Bangkok (port)” construction in [Bazzi \(2017\)](#) (his original uses great-circle distance to the nearest port and averages the two hubs; I use least-cost travel time and the nearer hub): for each urban center, the route-aware least-cost travel time to its nearest customs port of entry (over the same land-and-sea friction surface used for every cost distance in the paper) plus the sea distance from that port onward to the nearer of Bangkok and Ho Chi Minh City, the hubs from which the bulk of Indonesian rice imports ship, converted to hours at the friction surface’s shipping-lane speed so the two legs share one unit. Smaller totals mean closer alignment with the rice-import routes, hence higher baseline openness; the high-openness indicator in [Table 5](#) is a below-median split inside the estimation window. The international geography is fixed, and the anchor set is fixed too in the sense that matters here, since it is not selected on anything the program could have changed. The customs list has no vintage field and does include facilities that plausibly post-date 1990, which is why the measure leans on the international leg rather than on the identity of the anchor. Three properties matter in practice. First, the international leg dominates the composite (median 42 hours against 1.2 domestic), which injects the west-to-east proximity-to-the-routes gradient that nearest-port distance discards: route proximity correlates 0.60 with the gravity composite but only 0.18 with domestic-only nearest-port proximity ([Figure B1b](#)), and the measure is insensitive to selecting the cheapest gateway rather than the nearest port (correlation 0.99). Second, the international leg is a great-circle approximation, which understates distances from western ports whose actual sea route rounds the Strait of Malacca. This compresses the open end of the gradient but preserves the ranking. Third, the anchor set is held to predetermined geography rather than to realized trade. Whether a given port appears in the 2000–2010 rice-import records is itself an equilibrium outcome, and one the mechanism of this paper could move, since a city that substitutes hinterland food for outside supply leaves its port less rice-import business. The anchor set is therefore the full customs port-of-entry list, every port with coordinates whether or not it ever recorded rice volume, which removes any dependence on realized import quantities. Restricting it to the 58 ports that did record rice volume changes little. The two sets place 129 of the 133 in-window cities on the same side of the openness median, and the four they disagree on split two in each direction. The high-openness offset in [Table 5](#) goes from +0.036 to +0.024 on the agricultural share and from −0.035 to −0.022 on the modern share, so the composition test is weaker on the restricted set but points the same way on both outcomes. The reason the two agree at all is the first property above, since the international leg accounts for 97% of the composite’s cross-city variance and no port’s rice-import status can move it. Because the rice-active ports and the full list bracket every intermediate anchor set, the conclusion does not turn on dating individual ports. The measure is equally insensitive to the coordinate format. Rebuilding both panels with the port coordinates taken as decimal degrees, as they were before the degrees-and-minutes correction described above, leaves every coefficient of the gravity split ([Table B1](#)) at its sign and significance, moves the market-access median split by two of 133 cities, one in each direction, and leaves all three predetermined trade-openness rows of the balance test covering zero under either format. [Figure B2](#) maps both openness measures across the sample.

The gravity measure as the split. [Table B1](#) repeats the openness split of [Table 5](#) with the gravity market-access measure in place of route proximity, an above-median split inside the estimation window. Built from realized rice-import volumes, the measure is in part an equilibrium outcome, so the estimates are additional evidence. They move in the same direction as the route split on both outcomes, at $\beta_2 = +0.030$ on the agricultural share and -0.111 on modern employment, but neither differential is estimated precisely enough to settle anything.

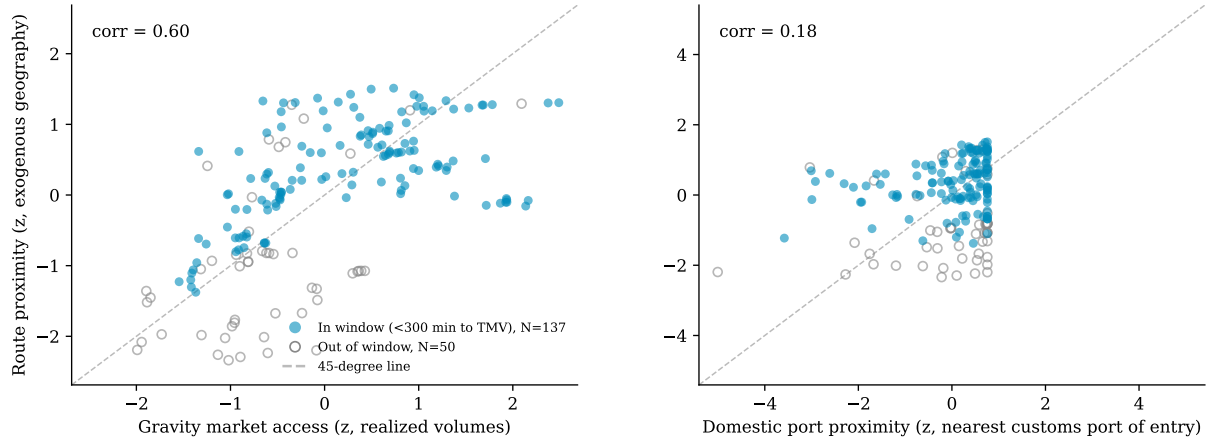
Validation: the 2004 rice-import ban. The route measure has independent validation from the event it was designed around, Indonesia’s 2004 rice-import ban. Post-ban (2002–2008) rice-price growth was faster in more route-open cities (+0.037 per standard deviation of proximity, $p = 0.06$), the geography-driven price response documented by [Bazzi \(2017\)](#). The point estimate is the same on the rice-active anchor set, where the standard error is tighter and the response reaches $p < 0.05$; the wider interval here comes from anchoring on the full customs list, and it is the interval I report. The same paper documents the pre-ban gradient with the sign reversed, an additional falsification the route measure inherits. The exposure treatment itself is orthogonal to that price shock: agro-similarity does not predict the ban-era price response (+0.006, $p = 0.64$), and the route $\times$ agro-similarity interaction is small and insignificant (its negative sign is consistent with productive hinterlands buffering the forced-autarky price jump, but the cross-sectional variation cannot establish it). The route result is also stable across codings: with the continuous proximity in place of the high-openness dummy, the interaction is significant at $p < 0.05$ or better on all four headline outcomes.

Figure B1: The gravity composite contains trade-volume information that raw port proximity does not

(a) Rice-import gravity market access against distance to the nearest port of entry

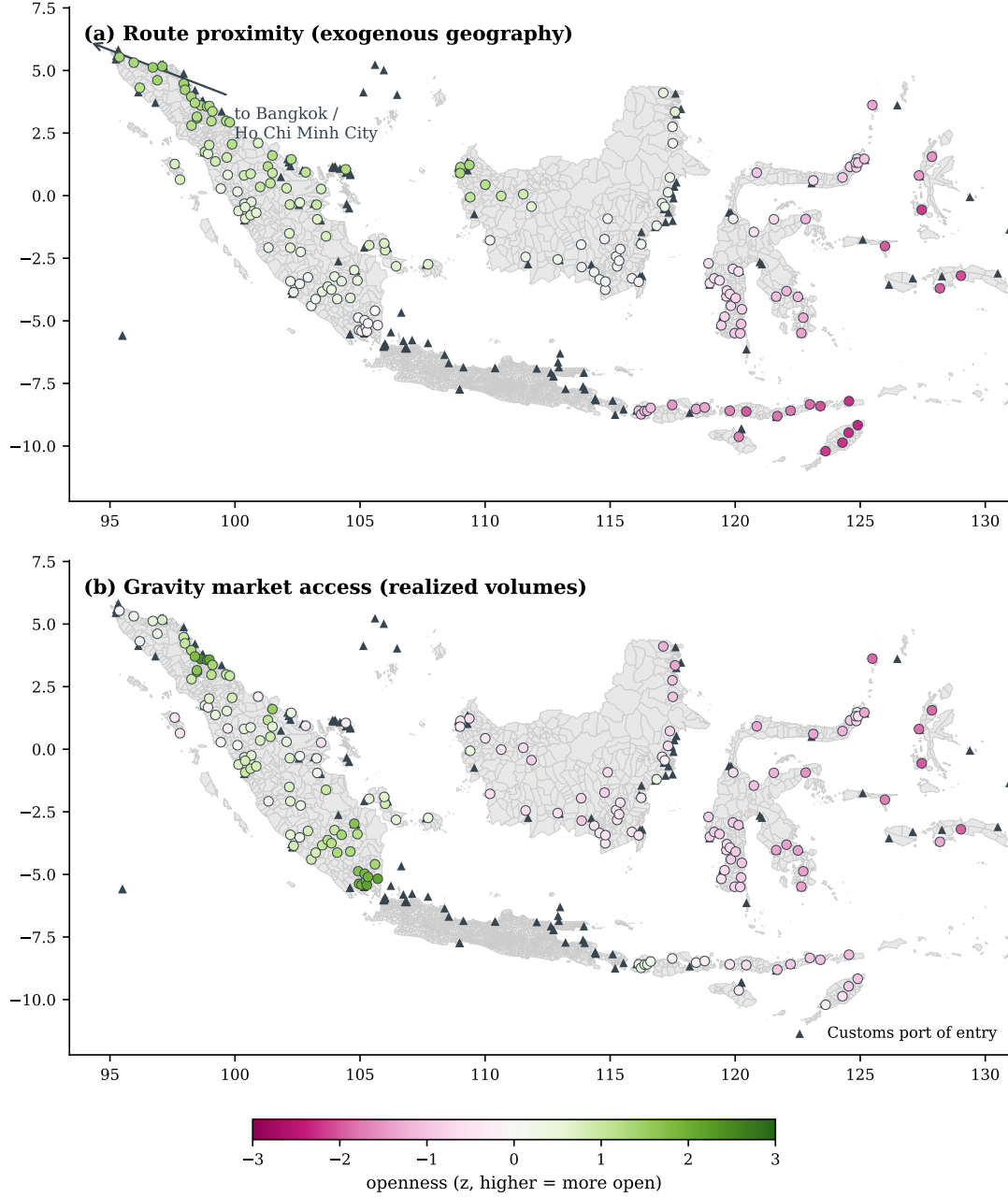


(b) Route-proximity openness against the existing openness measures



Notes: Each point is an outer-island urban center. Panel (a): the sample is the 300-minute cost-distance window ($N = 133$); the headline estimation sample additionally requires complete 1980 baseline controls. The vertical axis is the log gravity market-access index $\ln \sum_p kg_p / d_{ip}$, summed over every customs port of entry and therefore weighted by realized rice volume, which is zero for a port that never recorded any; the horizontal axis is the great-circle distance to the nearest port of entry (km). The line is an OLS fit and the cross-sectional correlation is $\rho = -0.12$. Panel (b): filled teal marks centers inside the 300-minute window ($N = 133$) and hollow gray those outside. All measures are standardized and signed so that higher means more open. Route proximity is shown against the gravity market access over realized rice-import volumes ($\rho = 0.60$) and against domestic-only nearest-port proximity ($\rho = 0.18$); the gap between the two comparisons is the international leg, where the proximity-to-the-routes gradient arises.

Figure B2: Two openness measures across the sample



Notes: Urban centers colored by (a) route proximity to the Thailand/Vietnam rice routes (exogenous geography) and (b) gravity market access (realized volumes), both standardized, on a shared diverging scale (green = more open, pink = more closed). Charcoal triangles mark the 183 customs ports of entry; the arrow in panel (a) marks the off-map heading toward Bangkok and Ho Chi Minh City. Java and Bali are out of scope.

B.2 Standardized (z -on- z) Reduced-Form Estimates: Model Calibration Targets

The structural model is calibrated to match the reduced-form agro-similarity responses in standardized (z -on- z) units, with both the treatment and each outcome standardized, so that the model's predicted response slopes, which are unit-free, are directly comparable to the data. Table B2 reports these targets: the reduced-form point estimates are at the headline estimation sample, with the Susenas- and PODES-based rows at the coverage each source reaches, reported per row, and the z - z column reports the slope the structural model is calibrated to match. The two targeted moments (m_1

Table B1: Agro-similarity $\times$ trade openness: gravity market access over realized rice-import volumes

High openness = above-median gravity market access (realized rice-import volumes)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	ln pop	ln emp	unemp rate	ln L agric	agric %	ln L modern	modern %
AgSim, more closed cities: β_1	0.118	0.083	0.014**	-0.217**	-0.078***	0.242***	0.084***
	(0.085)	(0.084)	(0.007)	(0.105)	(0.018)	(0.092)	(0.016)
AgSim, more open cities: $\beta_1 + \beta_2$	0.005	-0.014	0.003	-0.110	-0.027	0.095	0.051***
	(0.069)	(0.058)	(0.007)	(0.090)	(0.020)	(0.077)	(0.018)
High openness (=1)	0.029	0.013	-0.005	0.083	0.007	-0.031	-0.007
	(0.075)	(0.067)	(0.011)	(0.081)	(0.026)	(0.097)	(0.023)
Mean dep var	11.839	10.930	0.123	9.816	0.374	10.229	0.517
Std dev dep var	0.858	0.843	0.041	0.898	0.184	0.979	0.157
Observations	133	133	133	133	133	133	133

Note: High openness = above-median market access over realized rice-import volumes. Main Spec throughout: settler count, 1980 city baselines and the agro-climatic controls. Conley 200 km

Notes: Sample: the headline-sample outer-island urban centers (nearest Transmigration village within 300 minutes of cost-distance travel time), every one of which has a measured market-access position. The gravity measure is constructed above; the route-proximity split is Table 5 in the main text. Main Spec controls throughout; Conley 200 km standard errors. *** 1%, ** 5%, * 10%.

and m_2) are fit in the calibration, and with two parameters the estimator is exactly identified; the modern-employment slope (m_3) and the remaining rows are over-identifying checks computed from the same solves.

The density check in the main text (Figure F1) reports its headline no-controls arm; Figure B3 shows the same regressions beside the model-matched control arm, so the gradient's robustness to conditioning is visible in one view.

B.3 Measuring the Service Distance Decay from Household Service Flows

The two-tradable-sector variant has one parameter that no moment in the estimation disciplines: δ_S , the distance decay on the catchment-traded bundle. This appendix measures it directly.

Service trade costs in the model are $\tau_{od}^M = \max(d_{od}, 1)^{\delta_S}$, entering CES demand as $\tau^{1-\sigma}$ with $\sigma = 4$, so bilateral service expenditure falls in travel time as $d^{-(\sigma-1)\delta_S}$. A gravity regression of service flows on log travel time therefore recovers $-(\sigma-1)\delta_S$ and nothing finer: σ and δ_S are not separately identified from flows, but their product is, and it is the product that governs how far a city's service market reaches.

The Indonesia Family Life Survey records, for each respondent's last outpatient visit in the four weeks before the interview, the subdistrict the provider sits in and the one-way travel time. That is a bilateral service flow with a measured trip cost, which makes a gravity estimable at all; ordinary expenditure surveys record what was bought and never where. Pooling the 2007 and 2014 waves gives 9,975 located visits, of which 1,925 cross a subdistrict boundary, spread over 1,074 origin-destination pairs. Estimating

$$F_{od} = \exp(\gamma_o + \delta_d + \beta \ln d_{od})$$

by Poisson pseudo-maximum likelihood with origin and destination fixed effects, zeros retained, and standard errors two-way clustered on origin and destination, on the same route-aware cost-distance surface in minutes the model itself uses, gives $\hat{\beta} = -3.272$ with a standard error of 0.100, and hence

$$\hat{\delta}_S = -\hat{\beta}/(\sigma-1) = 1.091 \quad (0.033),$$

Table B2: Reduced-form AgSim estimates in standardized (z -on- z) units: the empirical moments matched in the structural model

Outcome	Reduced form		z - z estimate	N
	Coef.	(SE)	(unit-free)	
<i>Targeted calibration moments (m_1, m_2)</i>				
m_1 ln population	0.060	(0.059)	+0.070	133
m_2 agric. employment share	−0.053***	(0.018)	−0.287	133
<i>Over-identifying moments (untargeted)</i>				
m_3 ln modern empl. (serv+manuf), held out	0.169***	(0.061)	+0.173	133
m_4 ln agricultural empl.	−0.167*	(0.096)	−0.186	133
modern employment share	0.068***	(0.015)	+0.429	133
m_5 housing rent	−0.004	(0.111)	−0.004	133
m_{pa} ag-goods price (level)	0.018	(0.022)	+0.148	133
m_6 ln wage	0.039	(0.035)	+0.133	128
m_{rw} ln real wage	0.002	(0.031)	+0.008	132
m_7 food expenditure share	−0.013**	(0.006)	−0.263	132
m_8 ln rice production	0.702***	(0.252)	+0.350	127

Notes: Each “Coef.” is the coefficient on standardized AgSim (z -score over every urban center where AgSim is measured) from the headline Main Spec regression (controls: the settler count inverse-distance weighted inside the 300-minute window, 1980 ln population, 1980 ln employment and the 1980 urban-population share, and the matched agro-climatic covariates built on that same 300-minute rule; Conley spatial standard errors at 200 km), on outer-island urban centers inside the 300-minute cost-distance window of a transmigration village whose own 1980 baseline was measured and whose 2010 endline is present. The z - z estimate column is the unit-free response slope (both treatment and outcome standardized), the form in which the structural model matches the data; the model’s AgSim-response slopes are computed in the identical z -on- z form, so model and data slopes are directly comparable. Each outcome is standardized by its own standard deviation over that outcome’s estimation rows. Rows m_1 and m_2 are the two moments fit in the calibration, so with two parameters the estimator is exactly identified; the held-out modern-employment slope m_3 and the remaining rows are over-identifying checks read off the same solves. The structural estimate in Section 6.2 was fitted to earlier estimates of m_1 and m_2 , +0.011 and −0.285 in z - z units, on the same 133 urban centers; this column reports the current estimates. The census outcomes are estimated on all $N = 133$ urban centers; the Susenas-2011 rows (rent, prices, wages, food share) and the PODES rice-production row are on the same set of urban centers at the coverage each source reaches, reported per row.

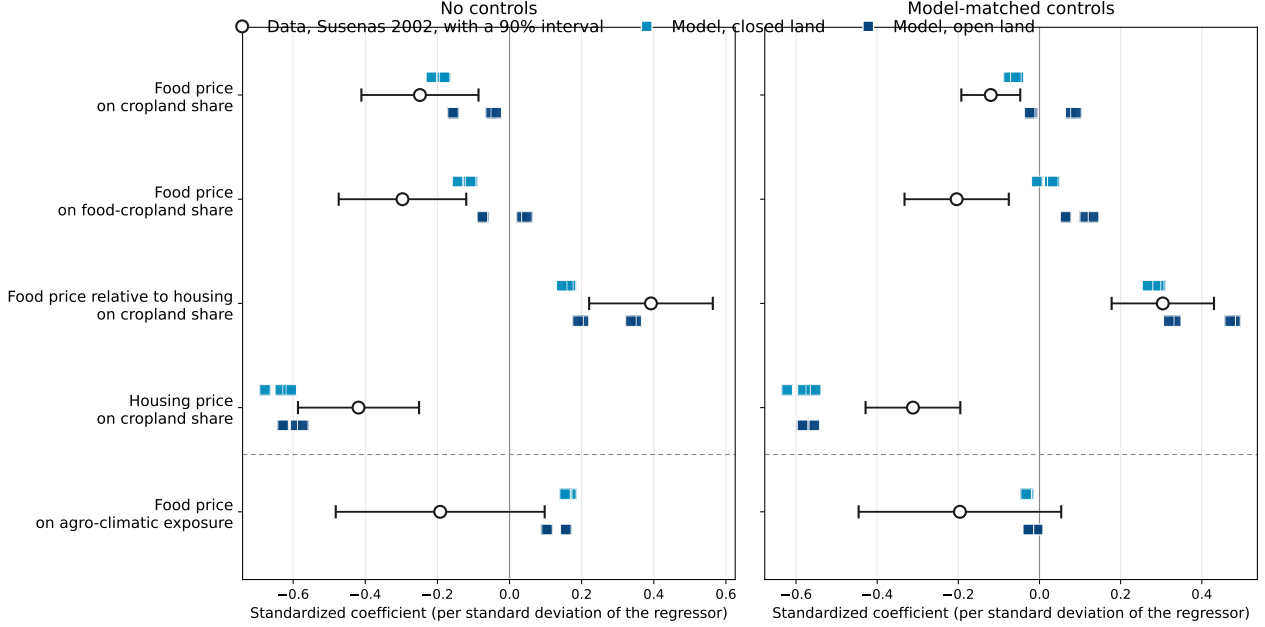
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

with a 90% interval of [1.036, 1.146]. Doubling the travel time to a provider leaves about a tenth of the service flow. Against the goods-trade gravity consensus of roughly −1, services decay about three and a third times as fast as goods, which sits just above the range the trade literature would have anticipated.

The estimate is robust to weighting flows by out-of-pocket expenditure rather than counting visits (1.131), to excluding traditional practitioners (1.133), to keeping only formal providers (1.117), and to excluding the public health centers that Indonesia sites in every subdistrict by administrative design (1.072). Restricting the sample to the outer islands, which is the estimation footprint of the paper, gives 0.948.

The gradient is not specific to medical care. Outpatient visits are one service, and medical care is atypical in being urgent and in having a publicly sited provider network. The survey records the same provider-location block for two further services, so the question can be settled in the data. Schooling is the informative comparison, being never urgent, chosen deliberately, attended daily rather than in a rare shock, and provided by a network sited under a different

Figure B3: The price gradient in local agriculture across control arms, model against data



Notes: Each row is one cross-sectional regression on the same 102 urban centers with a 2002 Susenas price, run twice on the identical specification: once on the price the data observe and once on the price the model solves for, so only the outcome differs. The left panel has no controls, the headline arm of the main-text figure; the right panel adds the model-matched control set, the agro-climatic covariates the calibration's own moment operator conditions on. Cropland share is the area under all thirty crops recorded by the village census over the land area of the center, food-cropland share is its food-crop subset, and the relative price is the food price less the housing price, which nets out the tendency of agricultural places to be cheap in everything. The hollow circle is the data estimate with a capped 90% interval on Conley standard errors at 200 km; each filled square is one model cell at its own optimum on the two targeted moments, across the four assumed food trade costs; the model-side operator reproduces the Stata estimate to a worst absolute gap of 1.2×10^{-13} on every cell. The bottom row, below the dashed rule, is the food price on agro-climatic exposure, the earlier over-identification check: its interval is wide enough to contain every model cell, which is the precision contrast that motivates the density check.

administrative logic; hospitalization brackets outpatient care from the other side as the most specialized of the three. Table B3 estimates all three identically.

Table B3: The same distance gradient across three services

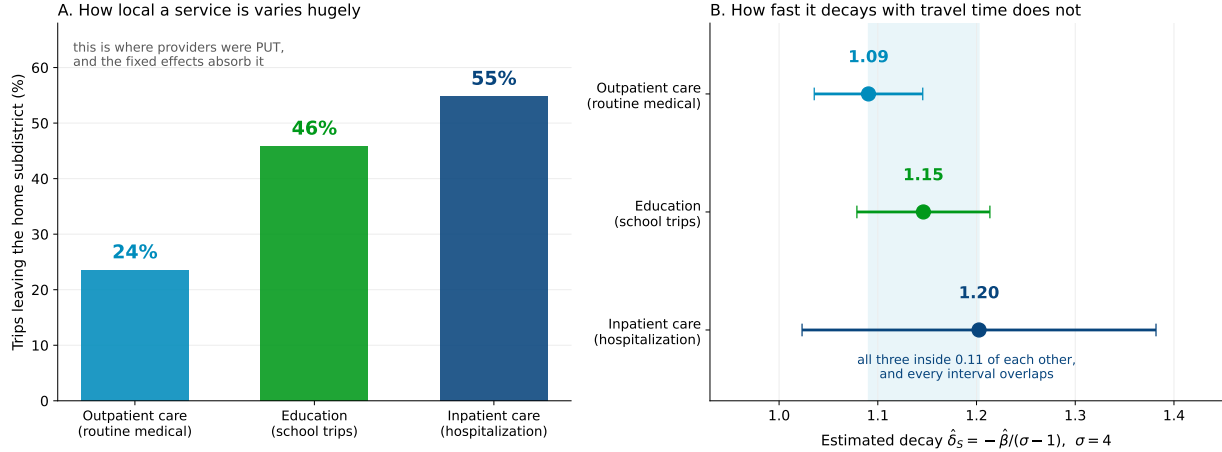
Service	Trips leaving the subdistrict	$\hat{\beta}$	s.e.	$\hat{\delta}_S$	90% interval
Outpatient care (routine medical)	23.5%	-3.272	0.100	1.091	[1.036, 1.146]
Education (school trips)	45.8%	-3.439	0.123	1.146	[1.079, 1.213]
Inpatient care (hospitalization)	54.8%	-3.608	0.327	1.203	[1.023, 1.383]

Notes: Poisson pseudo-maximum likelihood with origin and destination fixed effects, zeros retained, standard errors two-way clustered on origin and destination, own-subdistrict pairs excluded, estimated on the model's route-aware cost distance in minutes. $\hat{\delta}_S = -\hat{\beta}/(\sigma - 1)$ at $\sigma = 4$. Outpatient care and education pool the 2007 and 2014 waves; inpatient care uses 2007 only, because the 2014 module records the province and subdistrict of the hospital but not the district. Education covers respondents enrolled at their highest level, which in the located sample is predominantly senior secondary and tertiary.

The three services differ greatly in how local they are and barely at all in how fast they decay. The share of trips leaving the home subdistrict more than doubles from routine medical care to hospitalization, while the estimated decay

moves by 0.11 and every interval overlaps every other, with a Cochran Q of 1.75 on two degrees of freedom against a 5% critical value of 5.99. These are different objects: how local a service is reflects how many providers exist and where they were sited, which the origin and destination fixed effects absorb, whereas the slope is the cost of distance. Figure B4 separates the two.

Figure B4: How local a service is varies widely; how fast it decays with travel time does not



Notes: Panels use the samples of Table B3. The left panel is a statement about where providers were built; the right panel is the parameter. Intervals are 90%.

Two limits apply. All three services are delivered from a fixed facility and consumed by traveling to it, so none of them covers a service consumed without a trip, or retail and restaurant purchases, which no Indonesian survey locates. And the survey covers thirteen provinces, so the gradient is estimated on that frame and applied beyond it.

The calibrated value. The measured value is 1.091 and the calibrated model is solved at $\delta_S = 1.0$. The two differ because the traded-service block does not exist at the higher value: the homotopy that switches the block on converges up to a traded-service expenditure share of 7.8% at $\delta_S = 1.09$ and fails before reaching the 8.2% that the expenditure data require, and the same failure occurs at 1.15 and 1.25. The model is therefore solved at the highest decay below the estimate at which the economy exists, and the gap of 0.09 is reported rather than rounded away. No average of the three services in Table B3 is used. The estimates are not independent draws, sharing households, geography and one cost surface, and every pooled value lies further outside the solvable region than the outpatient estimate does.

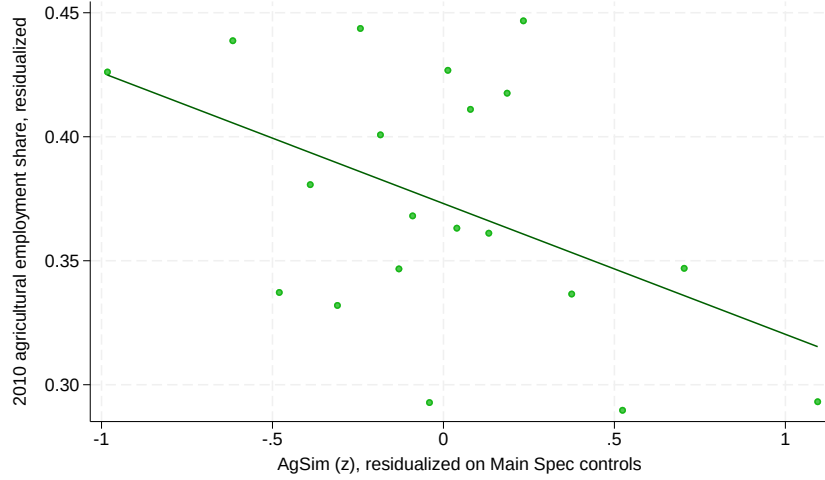
C Identification Support

This part collects the evidence that exposure is as good as randomly assigned once the controls are in place, together with the placebo and recentering tests that support it.

C.1 Design-Based Recentering and the Placebo Tests

Design-based recentering. The exposure measure is a constructed treatment: a known formula that combines a quasi-random shock, the match between arriving settler cohorts and the sites that were ready to receive them, with predetermined components, namely where the sites sit, how many settlers each received, and their agro-climate. Borusyak & Hull (2023) show that omitted-variable bias in such a measure operates through a single confounder, the exposure a city would expect under counterfactual assignments of cohorts to sites, and that recentering the treatment on that expected exposure, or controlling for it, restores identification. I construct the expected exposure directly. Holding the set of sites, the site locations, the settlement timing, the settler counts, and each city's travel-time catchment fixed, I permute

Figure C1: The identifying variation behind the headline: the agricultural employment share against exposure, residualized



Notes: Added-variable (Frisch-Waugh) plot on the headline urban-center sample whose nearest transmigration village lies within 300 minutes: the 2010 agricultural employment share and standardized agro-similarity are each residualized on the Main Spec controls (transmigrant count, 1980 baselines, and the agro-climatic control set); the binned scatter (20 equal-count bins) and the fitted line display the resulting partial relationship, whose slope is the headline coefficient in Table 2, column (5).

the village-level agro-climatic similarity across sites within province and Repelita settlement wave, rebuild each city's unweighted 300-minute catchment mean under each of 2,000 draws, and average them into the expected exposure μ_c .

Two facts locate what the recentering does to the headline. First, the expected exposure has less geographic structure than the realized measure: the R^2 of the realized exposure on the agro-climatic control set is 0.33, and recentering lowers it to 0.20. Second, the double-robust form of the correction leaves the structural-transformation results standing. Table C2 enters the expected exposure as a control: the agricultural employment share falls by 0.042 at the 5% level, log modern-sector employment rises by 0.125 at 5%, and the modern employment share rises by 0.051 at 1%, each of those three signs matching the headline at about three quarters of its size, with the attenuation and the wider standard errors following from the 0.74 correlation between the realized and the expected exposure. Table C1 subtracts that expectation instead, reporting $\tilde{z} = z - \mu$, and the three compositional signs still match while their magnitudes do not survive. The agricultural share reads -0.017 with a standard error of 0.011 and log modern-sector employment $+0.045$ with a standard error of 0.037, so only the modern employment share clears the 10% level. Exact randomization inference over the same draws, which tests the same null the Conley standard errors test, rejects no outcome at the 10% level under either permutation scheme, and the smallest p -value anywhere in it is 0.095. Both facts point the same way, and the paragraph on the correlation below says why. Subtracting an expectation this closely aligned with the realized measure removes most of the measure, so the scalar and origin routes ask a question the remaining variation cannot answer, while controlling for the same expectation leaves the variation intact and returns the headline. Every randomization-inference p -value the paper reports is collected in Table C4, labeled by the permutation scheme and the stratification that produced it.

The second recentering is the harder test. It holds each site's destination fixed and re-draws only the settler-origin composition, which is the more literal form of the design, since the destination geography is a predetermined exposure component rather than the shock. This is feasible because the similarity index of Bazzi et al. (2016) is linear in the origin composition, $AS_v = \sum_o \pi_{ov} s(o, v)$, where $s(o, v)$ is the origin-to-destination dyadic similarity and π_{ov} is the share of village v 's settlers born in origin district o . Their replication data provide the full dyadic grid (125 Java-Bali origins by 814 villages) together with the realized migrant flows that give π , so I can hold each site's similarity column $s(\cdot, v)$ fixed and swap in another village's origin composition $\pi_{\cdot v'}$, permuted within province-by-wave cells; villages are linked

Table C1: Design-based robustness: AgSim recentered against counterfactual settler assignments (Borusyak and Hull)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	ln pop	ln emp	unemp	ln ag	ag share	ln modern	modern sh.
AgSim recentered $\tilde{z}^{(D)}$	0.017 (0.050)	0.018 (0.049)	-0.009* (0.005)	-0.034 (0.076)	-0.017 (0.011)	0.045 (0.037)	0.019* (0.010)
Transmigrant count	✓	✓	✓	✓	✓	✓	✓
Base ln pop 1980	✓	✓	✓	✓	✓	✓	✓
Base ln empl 1980	✓	✓	✓	✓	✓	✓	✓
Base ln agri empl 1980	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓	✓	✓
Observations	133	133	133	133	133	133	133

Notes: Recentered treatment $\tilde{z}^{(D)} = z - \mu$, standardized; μ is expected exposure from permuting village agro-climatic similarity across sites within province and Repelita settlement wave (S=2,000). Main Spec controls, no fixed effects, Conley 200km. Randomization-inference p-values from the same draws (scheme D, province by settlement wave) are in Table C4, column (2).

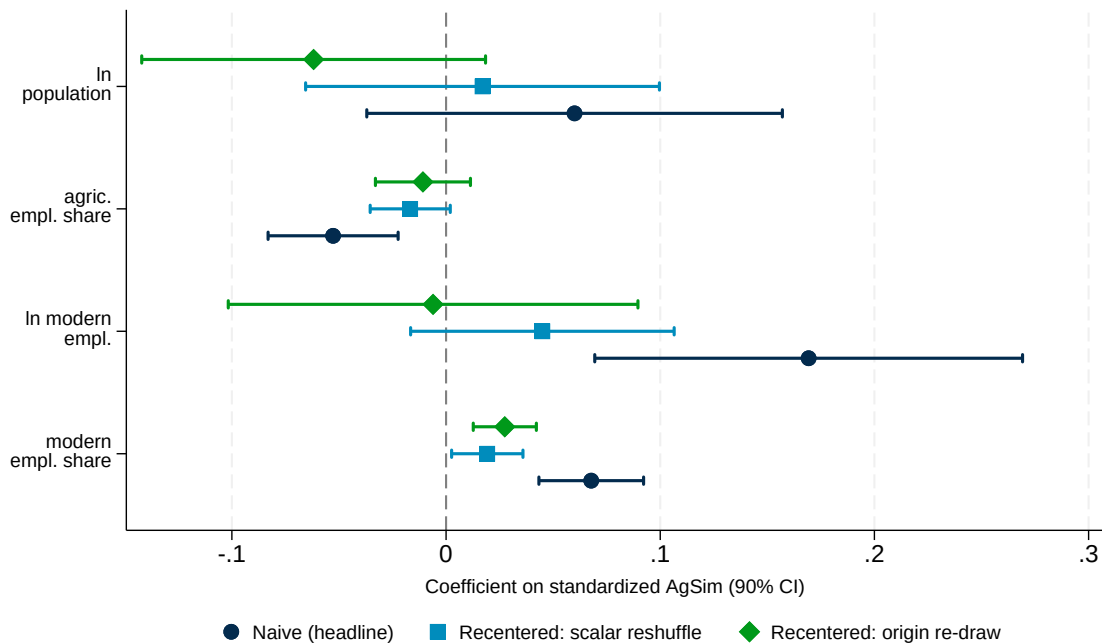
Table C2: AgSim controlling for expected exposure (double-robust)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	ln pop	ln emp	unemp	ln ag	ag share	ln modern	modern sh.
AgSim (double-robust)	0.045 (0.076)	0.035 (0.074)	-0.006 (0.006)	-0.111 (0.124)	-0.042** (0.020)	0.125** (0.061)	0.051*** (0.017)
Transmigrant count	✓	✓	✓	✓	✓	✓	✓
Base ln pop 1980	✓	✓	✓	✓	✓	✓	✓
Base ln empl 1980	✓	✓	✓	✓	✓	✓	✓
Base ln agri empl 1980	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓	✓	✓
Observations	133	133	133	133	133	133	133

Notes: AgSim standardized, additionally controlling for the expected exposure under both the province and the province-by-settlement-wave permutations (both standardized; the double-robust specification of Borusyak and Hull 2023, consistent if either the controls or the permutation model is correct). Main Spec controls, no fixed effects, Conley 200km.

to the grid through their destination potential-yield vector (792 of 814 match cleanly). The composition-weighted grid reproduces the published village index up to an affine rescaling ($r = 0.99999$, slope 0.56), so the rebuilt exposure and its expected value are differenced on their own scale and then standardized, and the swap is the same reassignment Bazzi et al. (2016) perform in their own analysis, holding the destination fixed and re-choosing origins. Holding the destination fixed raises the correlation between realized and expected exposure to 0.96, deep in the range where a constructed exposure typically collapses under recentering, and Table C3 shows that this one does. The modern employment share holds at +0.027 at the 1% level and log agricultural employment at -0.100 at 10%, while the agricultural employment share falls to -0.011 with no precision and log population and log modern-sector employment cross zero to -0.062 and -0.006, neither distinguishable from zero on either side of the crossing. The design-based inference built from the same draws

Figure C2: Robustness to design-based recentering (Borusyak and Hull)



Notes: Each row is a headline outcome; each marker is the coefficient on standardized agro-similarity with its 90% confidence interval (Conley spatial standard errors, 200 km cutoff, the headline sample of outer-island urban centers). “Naive” is the headline specification. “Scalar reshuffle” recenters the exposure against a permutation of the village match within province-by-settlement-wave cells. “Origin re-draw” recenters against a permutation that holds each site’s destination agro-climate fixed and re-draws only the settler-origin composition, using the origin-destination similarity grid of [Bazzi et al. \(2016\)](#). Full estimates: [Table C1](#) and [Table C3](#).

is weaker still, returning $p = 0.592$ for the agricultural share where the analytic p is near 0.004 ([Table C4](#)). The power accounting below makes this column readable. One standard deviation of the origin-recentered treatment is about 16% of a standard deviation of realized exposure, so the column has little left to detect, and its estimates are uninformative about the headline in both directions. It survives precisely because the expected exposure here is destination geography, which the regression already controls for. Recentering on it isolates the quasi-random origin match, which is the identifying variation.

That correlation of 0.95 is easy to mistake for evidence that the recentering barely changed the treatment, and it means the reverse. Since $\tilde{z} = z - \mu$, the correlation between the raw and the recentered exposure is $[\text{SD}(z) - \text{corr}(z, \mu)\text{SD}(\mu)]/\text{SD}(\tilde{z})$, which falls as $\text{corr}(z, \mu)$ rises: subtracting something closely aligned with the raw measure removes most of it and leaves a residual nearly orthogonal to it. The two correlations move in opposite directions by construction, so a high $\text{corr}(z, \mu)$ means the recentered exposure is further from the raw one. Under the scalar permutation, where $\text{corr}(z, \mu)$ is 0.74, the recentered and raw measures still correlate at 0.60. Under the origin permutation they correlate at 0.29, with a rank correlation of 0.10; 53% of cities move from above-average to below-average exposure or the reverse, and the average city shifts 40 places in a ranking of 133. The robustness column is therefore a different regression, identified off a treatment that orders the sample almost independently of the one the headline uses, and its coefficients are not on a scale the headline can be compared against. The high correlation is a fact about the setting, and it makes this the demanding test. Agro-climatic match quality is mostly a property of the land. Once a site’s destination is held fixed, the arriving cohort can move that site’s similarity only so far, and the destination accounts for roughly 92% of the variance in exposure. The agro-climatic controls reach part of that component and not all of it. Their R^2 on realized exposure is the 0.33 reported above, so most of the destination’s 92% is variation no linear combination of the controls reproduces. The reason is structural rather than a matter of adding more covariates. Exposure is an unweighted mean of

a nonlinear similarity composite over the villages inside the 300-minute window, while the controls are inverse-distance-weighted sums of raw attributes over the full village matrix, so the two differ in village set, in weights, and in functional form. Re-drawing the origin composition strips the destination component out and leaves the variation that is genuinely about which settlers arrived, and only the modern employment share survives on that remainder. Under the coarser province-only stratification the residual share is essentially the same, 9% rather than 8%, confirming that the correlation reflects the land's dominance rather than a stratification choice. Reshuffling the whole village match instead leaves 43% of the variance, so the two exercises bracket the design. The wider one retains power and the narrower one concedes power to isolate the cleanest margin, and neither reproduces the headline once the expectation is subtracted. The price of subtracting the expectation is precision, on both schemes, and the exact randomization inference shows what that costs. The agricultural share returns $p = 0.325$ under the wider reshuffle and $p = 0.592$ under the origin permutation, the modern share 0.220 and 0.114, and log population 0.734 and 0.198. No outcome rejects at the 10% level in any column, and Table C4 reports every one of them. The two schemes still do not order the same evidence more and less demandingly, because the origin-recentered treatment orders the sample almost independently of the realized one, so the origin columns are a different regression and their p -values are not nested inside the reshuffle's. Because each column standardizes its own regressor, the two recentered coefficients are not on a common scale. One standard deviation of the origin-recentered treatment is about 16% of a standard deviation of realized exposure, against 66% for the whole-match version. Magnitudes quoted here are therefore from the whole-match reshuffle, which can be compared directly with the naive estimate, while the origin permutation is reported for completeness, since at 16% of a standard deviation of realized exposure it has too little variation left to establish anything either way.

Running both recenterings at the rural subdistrict level leaves the rural null null, so the contrast between the urban response and the flat rural one is not an artifact of the exposure measure or of either design-based correction.

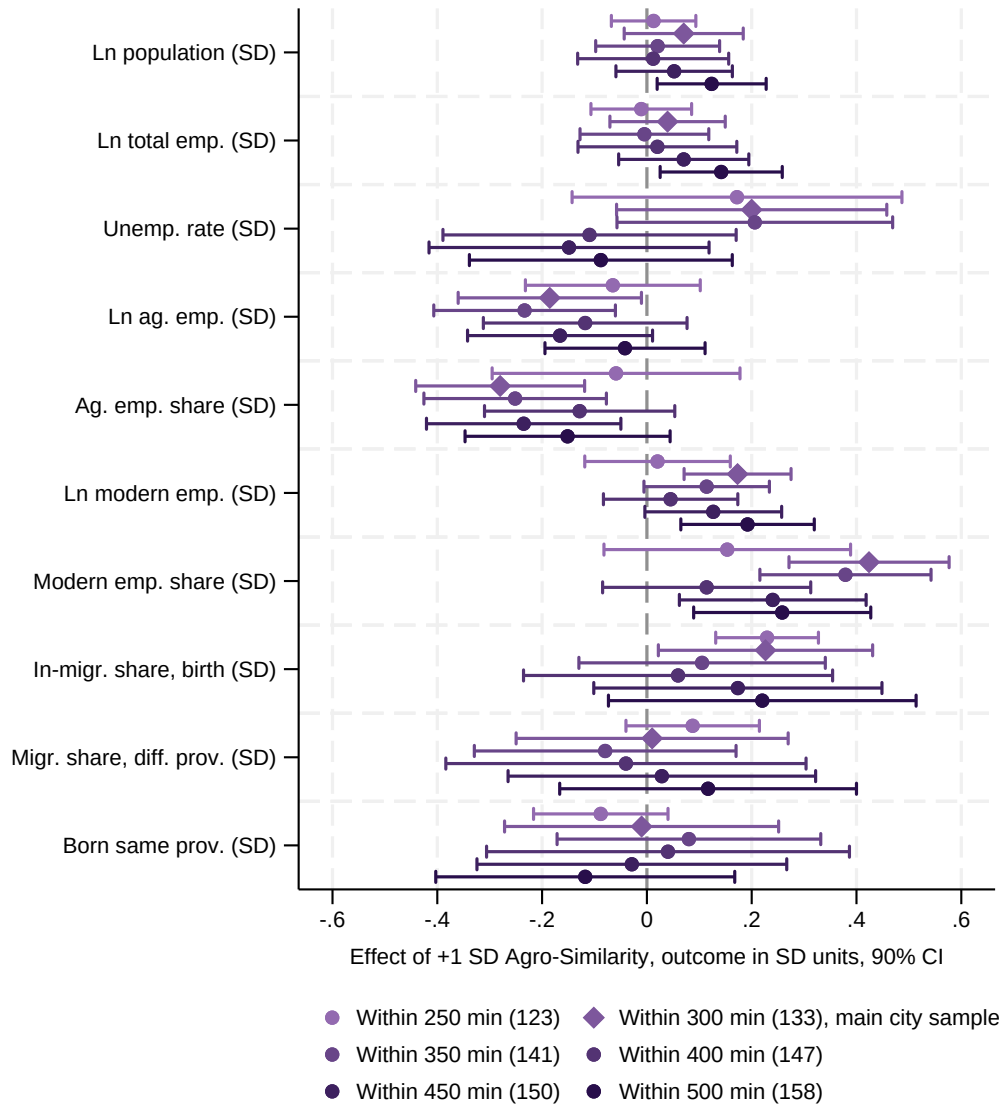
Table C3: Origin-permutation recentering: destination held fixed, origin re-drawn (UC)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	ln pop	ln emp	unemp	ln ag	ag share	ln modern	modern sh.
AgSim origin-recentered $\tilde{z}^{orig}$	-0.062 (0.049)	-0.060 (0.047)	-0.004 (0.005)	-0.100* (0.060)	-0.011 (0.013)	-0.006 (0.058)	0.027*** (0.009)
Transmigrant count	✓	✓	✓	✓	✓	✓	✓
Base ln pop 1980	✓	✓	✓	✓	✓	✓	✓
Base ln empl 1980	✓	✓	✓	✓	✓	✓	✓
Base ln agri empl 1980	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓	✓	✓
Observations	133	133	133	133	133	133	133

Notes: Origin-recentered treatment: each village's similarity is rebuilt from the origin-destination dyadic grid of Bazzi et al. (2016), holding the site's destination column fixed and permuting the settler-origin composition across sites within province and Repelita wave; μ is the average over 2,000 draws, and the treatment is the realized exposure minus μ , both on the rebuilt index's own scale. Standardized; Main Spec controls, no fixed effects, Conley 200km. Exact randomization-inference p-values from the same draws (scheme D, province by settlement wave) are in Table C4, column (4).

Baseline placebo test, in levels. Table C5 regresses ten 1980 baseline outcomes on agro-similarity across villages, urban center-level aggregates of individual 1980 census records: population, employment, the unemployment rate, agricultural and modern employment in levels and in shares, and a lifetime-birthplace split by province of birth. Two constructions of the exposure inside the 300-minute identification radius are reported, the headline unweighted mean and the population $\times$ inverse-distance weighting of the exposure equation, each over all 814 transmigration villages and again over the 687 settled after the count. Three of the forty cells reject zero, two at the 10% level and one at the 5%,

Figure C3: Robustness: Agro-similarity Effect by Transmigration-Village Proximity



Notes: Each row is one 2010 outcome, z-scored over all urban centers so the reference standard deviation stays fixed as the sample widens across radii. Within each row, the six points are the effect of a one standard deviation increase in agro-similarity, estimated separately over nested travel-time catchments defined by the distance to the nearest transmigration village (within 250 to 500 cost-minutes in 50-minute steps), with the agro-similarity exposure itself reconstructed at each radius, so the treatment and the sample both change with the radius; 90% confidence intervals from Conley 200 km spatial standard errors. The grid begins at 250 minutes, where 123 centers support the estimate, and runs to 500, where 158 do. The 200-minute radius is estimated but no longer shown: it reaches only 99 centers against the 133 of the headline, so its point moves with the sample rather than with the radius, which is the ground the narrower 100- and 150-minute radii were already excluded on. Controls are the Main Spec set (transmigrant settler count, 1980 log population, 1980 log employment, 1980 urban-population share, 1980 log agricultural employment, and the agro-climatic covariates averaged over the same catchment), with no fixed effects. The rows are the main results table's seven outcomes together with three migrant-share outcomes; modern-sector employment is manufacturing plus services, and the migrant shares are measured per birth, namely the shares of residents born outside the current district, born in a different province, and born in the same province. The 300-minute point is the headline estimation sample and uses the shipped headline exposure; wider radii extend the city selection beyond the headline window, with per-radius estimation samples of 99, 114, 128, 135, 139, 142, and 150 urban centers, as reported in the legend.

Table C4: Design-based inference: randomization-inference p -values by permutation scheme

Outcome	Conley	Whole-match reshuffle		Origin permutation	
	200 km	prov.×wave	province	prov.×wave	province
	(1)	(2)	(3)	(4)	(5)
ln population	0.309	0.734	0.780	0.198	0.576
ln employment	0.556	0.743	0.996	0.275	0.348
Unemployment rate	0.206	0.095	0.736	0.555	0.560
ln agricultural employment	0.081	0.657	0.415	0.204	0.298
Agricultural employment share	0.004	0.325	0.240	0.592	0.610
ln modern employment	0.005	0.531	0.631	0.930	0.914
Modern employment share	0.000	0.220	0.168	0.114	0.225
Observations	133	133	133	133	133

Notes: Every entry is a two-sided p -value for the same null of no effect on the same outcome and sample; only the way the null distribution is constructed changes. Each cell is shaded by the strength of the evidence, deepening green from the 10% to the 1% level, with pale pink for a p at or above 0.10. Column 1 is the analytic Conley 200 km p -value that the stars in Table 2 come from. Columns 2 to 5 are exact randomization inference over 2,000 counterfactual assignments: the observed coefficient is compared against the distribution of coefficients the same regression returns when the settler match is re-drawn. The whole-match reshuffle permutes village agro-climatic similarity across sites; the origin permutation holds each site's destination agro-climate fixed and re-draws only the settler-origin composition, which is the more literal form of the design and deliberately the more demanding test, since one standard deviation of the origin-recentered treatment is about 30% of a standard deviation of realized exposure. Columns 2 and 4 stratify the permutation by province and Repelita settlement wave; columns 3 and 5 by province alone. The origin columns use the simulation-anchored recentering, the version that puts the observed assignment on the same scale as the draws, which exchangeability under the null requires. The two schemes do not rank the outcomes the same way, and that is a property of the design. Neither permutation scheme rejects any outcome at the 10% level. The origin-recentered treatment correlates with realized exposure at 0.26, with a rank correlation of 0.05, and moves the average city 42 places in the exposure ranking, so column 4 and column 5 are a different regression, and their p -values are not a more demanding version of columns 2 and 3. The permutation schemes do not reproduce the uniformly strong significance in column 1: the smallest randomization-inference p -value anywhere in the table is 0.095.

and all three sit in the two agricultural rows. Under the null about four cells would fall below the 10% threshold, and no outcome rejects under all four constructions. Different distance weights do not affect the placebo, and restricting the exposure to the villages settled after the count does not affect it either.

The 2010 birthplace decomposition against its own 1980 baseline. Figure C4 puts the two side by side at the only resolution both censuses support. SP1980 records place of birth at the province level, so the same-district and same-province cells of the 2010 decomposition cannot be separated at baseline; collapsing them into one same-province share makes the two panels the same object, and it is why this figure has three categories where Table 4 has four. Together the two panels say something the 2010 panel does not. Exposed cities hold a smaller share of residents born in their own province and a larger share born in another outer-island province, and both of those gradients are already present in 1980, with the same signs and between half and two-thirds the size. The 2010 pattern is therefore not all a response to the program.

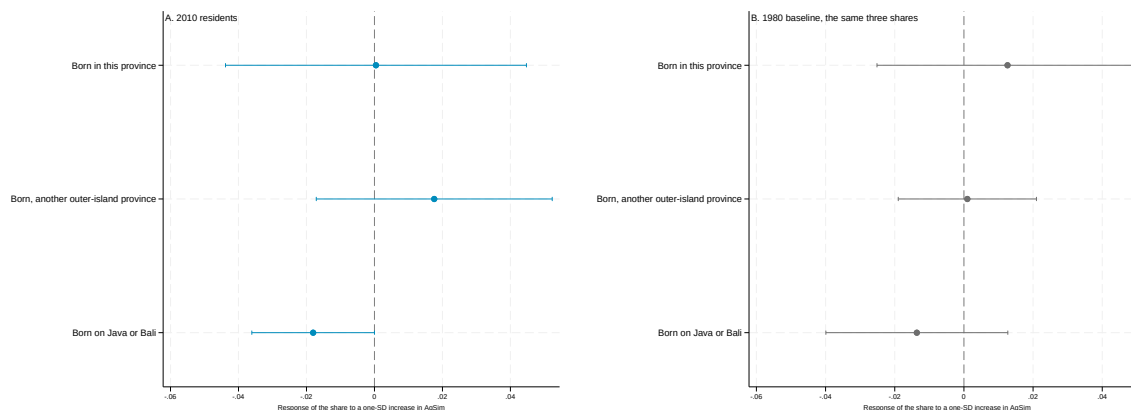
A strictly pre-program placebo. The placebo tests above hold the 1980 census fixed and vary the exposure, yet the exposure is built from villages that were not all in place by 1980: the program settles 127 of the 814 villages, accounting for 19.5% of all settlers, at or before the census. The right-hand block of Table C5 rebuilds the exposure from the 687 villages settled after 1980, and rebuilds every village-derived control over that same village set, so nothing entering the regression could have acted on a 1980 outcome. The restricted exposure stays close to the headline one, correlated 0.99 in levels and 0.97 in ranks across the estimation sample, and no city loses its hinterland. The median city keeps 11 of its 13 villages within 300 travel-time minutes. The result does not move. One of the ten baseline outcomes rejects zero at the 10% level under the unweighted construction, the agricultural employment share, and one at the 5% under

Table C5: Placebo Baseline Test: 1980 Outcomes Under Two Exposure Constructions and Two Village Sets

1980 baseline outcome	Outcome, headline sample			All transmigration villages		Villages settled after the 1980 census	
	Mean	SD	N	Headline (unwt)	Pop-dist IDW (Eq.)	Headline (unwt)	Pop-dist IDW (Eq.)
ln population	11.084	0.99	134	-0.051 (0.113)	+0.009 (0.184)	-0.009 (0.152)	-0.136 (0.163)
ln employment	9.837	0.96	134	-0.067 (0.100)	-0.011 (0.170)	-0.018 (0.138)	-0.140 (0.153)
Unemployment rate	0.026	0.03	131	-0.002 (0.003)	-0.001 (0.003)	-0.002 (0.003)	-0.002 (0.003)
ln agricultural empl.	9.235	1.06	134	-0.164* (0.092)	-0.035 (0.142)	-0.139 (0.097)	-0.204** (0.104)
Agricultural empl. share	0.558	0.22	134	-0.039 (0.032)	-0.023 (0.036)	-0.047* (0.026)	-0.026 (0.028)
ln modern empl.	8.796	1.31	134	-0.066 (0.180)	-0.027 (0.266)	-0.010 (0.218)	-0.162 (0.242)
Modern empl. share	0.385	0.20	134	+0.022 (0.031)	+0.006 (0.035)	+0.028 (0.027)	+0.009 (0.029)
ln born same province	10.925	1.00	134	-0.034 (0.123)	+0.027 (0.193)	+0.006 (0.162)	-0.112 (0.170)
ln born other province	8.322	1.89	134	-0.158 (0.204)	-0.152 (0.275)	-0.116 (0.227)	-0.284 (0.268)
Born other province, share	0.133	0.14	134	-0.013 (0.021)	-0.014 (0.016)	-0.013 (0.019)	-0.018 (0.018)
# significant / 10				1	0	1	1

Notes: Each cell is the coefficient on the (standardized) AgSim exposure construction from a separate placebo regression of the 1980 baseline outcome on that exposure, the transmigrant settler count, and the agro-climatic TMV controls (no 1980 baseline controls), Conley spatial SEs (200 km). Each column has its own within-300-minute sample, the urban centers where that construction and that outcome are both non-missing, so N runs from 129 to 134 across the four columns. Both headline columns stand at $N=134$, every urban center in the 300-minute window with a measured 1980 baseline: the 133 centers of the main city sample plus 1 whose 2010 census endline is missing, which a 1980 placebo does not need. They fall below that only where the 1980 outcome is itself missing. The ten outcomes are the paper's seven headline outcomes at baseline, log 1980 population and employment, the unemployment rate, and agricultural and modern employment in logs and in shares, followed by the 1980 birthplace block. Modern employment is manufacturing plus services. The birthplace rows split the 1980 population by province of birth. The two log counts are parts of one total and move independently, and the share row is the born-elsewhere count over 1980 population. SP1980 records place of birth at the province level only, so no finer split is available at baseline. The headline 2010 birthplace result (Table 4) is four-way because the 2010 census does record district of birth. The Mean, SD and N columns describe the outcome on column (1)'s own estimation sample. *The two constructions*, both restricted to the 300-minute identification radius: *Headline (unwt)* = the paper's treatment, the unweighted mean of village agro-similarity within 300 cost-distance minutes; *Pop-dist IDW (Eq.)* = the settler-count $\times$ inverse-distance weighting of the exposure equation, restricted to 300 min. A settler-weighted mean is not reported here. It does not reject in either village set, and the specification curve is where weighting robustness is tested, with the headline crossed against both weighted means. This table varies the village set. Population in the column headers means the count of settlers initially placed in the village. *The two village sets*: the left block builds both constructions, the settler count and the agro-climatic control set over all 814 transmigration villages, and its headline column is the paper's own placebo test in levels; the right block rebuilds both over the 687 villages settled after the 1980 census, so that the 127 placed by 1980, accounting for 19.5% of all settlers, contribute to none of them. Each block is internally consistent in its village set. *** 1%, ** 5%, * 10%.

Figure C4: Where residents were born, 2010 against the 1980 baseline, at the three-way resolution



Notes: Urban centers within 300 cost-minutes of a transmigration village. Each panel regresses three exhaustive birthplace shares on standardized agro-climatic similarity under the Main Spec, with Conley standard errors at a 200-kilometer cutoff and 90% confidence intervals. The outcome is a share, so the response is in share units and the three coefficients sum to zero by construction within each panel, which both panels satisfy exactly. Panel A is the 2010 cohort, with the same-district and same-province categories of Table 4 added together; panel B is the 1980 census at the same three categories ($N = 126$, since a few centers lack the 1980 birthplace block). The two panels share one horizontal scale so their magnitudes can be compared directly. Adding the 1980 shares of the same birthplace categories to the Main Spec roughly halves the panel A migration response ($p = 0.054$; Table C7, which reports the four-category decomposition, collapsed to three categories in the figure).

the inverse-distance construction, log agricultural employment. Population, employment, modern employment and every birthplace row stay insignificant. The pre-program share of the villages is therefore not what the placebo was detecting.

Before the placebo itself, Table C6 records what the restriction does to the treatment. The exposure is a mean, so dropping a fifth of the settlers barely moves it; the settler-count control is a count, so it falls by roughly the share removed. No city loses its hinterland.

The same restriction leaves the weighting robustness unchanged. The right-hand block of Table C5 rebuilds all three exposure constructions over the post-1980 villages, and none of them rejects on any of the ten outcomes.

Controlling the 1980 birthplace composition directly. The one baseline characteristic exposure moves is the birthplace composition of the 1980 workforce, so the direct test is to put that composition into the control set and re-estimate. Table C7 does that across the main outcomes and the birthplace decomposition in one table. The 1980 census records place of birth at province level, which partitions each city's 1980 workforce into three exhaustive shares, born in this province, born in another outer-island province, and born on Java or Bali; two of them span the block, and column (3) adds them to the Main Spec. A handful of cities have no 1980 birthplace block, so column (2) first repeats the Main Spec on exactly column (3)'s cities. That separates the change in sample from the change in controls, and the change in sample turns out to do nothing: the largest move any coefficient makes between columns (1) and (2) is 0.001, and no significance level changes.

The compositional headline does not depend on the baseline tilt. With the 1980 birthplace shares held fixed, every one of the seven headline estimates moves by less than a tenth of its standard error, and the two compositional responses keep their 1% significance.

Holding the 1980 shares fixed also leaves the birthplace facts as they are. The locally born share falls, at the 1% level, the same-province share rises, at the 5% level, and the other-outer-island and Java or Bali shares do not move

Table C6: What restricting the exposure to post-1980 villages changes

	All 814 villages	687 post-1980 villages
<i>Exposure (unweighted mean similarity within 300 min)</i>		
Mean	0.6780	0.6808
Standard deviation	0.1001	0.0993
Minimum	0.3777	0.3777
Maximum	0.9563	0.9563
Correlation with the other, levels		0.9896
Correlation with the other, ranks		0.9757
<i>Villages within 300 cost-distance minutes</i>		
Mean per city	30.7	26.0
Median per city	13	12
Cities losing exposure entirely		0
<i>Transmigrant settler count control (IDW, thousands)</i>		
Mean	1.190	0.948
Correlation with the other		0.9078
Urban centers (estimation sample)		133

Notes: Computed over the 133 urban centers of the main estimation sample, which are the centers within 300 cost-distance minutes of a transmigration village that also have the 1980 baselines the specification requires. The left column is the exposure and settler count as the paper builds them; the right column restricts both to the 687 villages settled after the 1980 census, dropping the 127 placed by 1980 that account for 19.5% of all settlers. The settler count is a count, so it falls by roughly the settler share removed; the exposure is a mean, so it barely moves. No city loses its hinterland under the restriction.

detectably. The Java/Bali share is unmoved with or without the control, so the inflow is not an extension of the program's own main-island pipeline in either case.

C.2 Village-Level Analysis

Table C8 reports the village-level relevance of agro-climatic similarity for agricultural productivity that Section 4.3 cites: across the transmigration villages, productivity in tons per hectare rises with the similarity index, winsorized or raw, over the plotted range and over the full range. The productivity measure is the total agricultural productivity of Bazzi et al. (2016), taken from the 2002 round of the village census and the 2001-2002 growing season, and it is observed for 770 of the 814 program villages, while agro-similarity is defined for all 814.

Defining the stay-share. The settler stay-share featured in the text divides a village's 2000-census residents born in Java or Bali by the individuals the program placed in the village, under two restrictions that each remove one class of non-settler the census can identify: residents whose recorded residence five years before enumeration was still Java or Bali, and residents born in Java or Bali after their own village was settled. The broader ratio without the two restrictions, which Figure 9 plots, counts anyone who reached the village from the main islands on their own after the program ended, so it proxies for retention. The restrictions remove a median of 10% of the numerator, yet the conditional gradient on agro-climatic similarity moves only from 0.078 to 0.080, the definitions correlate above 0.98, and every version stays indistinguishable from zero under both the wild cluster bootstrap and Conley standard errors (Table C9). The definitions cannot disagree, because what separates them is itself unrelated to agro-similarity. No version of the numerator recovers the settlers' locally born children, who are outer-island-born and never enter it, so the flat gradient reported in the text is a property of the data rather than of the arrival dates the census cannot separate.

Table C7: The headline estimates with the 1980 birthplace composition controlled

	Main Spec, as reported (1)	Main Spec, same cities as (3) (2)	Adding the 1980 birthplace shares (3)
<i>Panel A. Urban growth and structural transformation</i>			
ln population	0.060 (0.059)	0.060 (0.059)	0.061 (0.048)
<i>p-value</i>	0.309	0.309	0.208
ln total employment	0.033 (0.056)	0.033 (0.056)	0.035 (0.048)
<i>p-value</i>	0.556	0.556	0.472
Unemployment rate	0.009 (0.007)	0.009 (0.007)	0.009 (0.006)
<i>p-value</i>	0.202	0.202	0.147
ln agricultural employment	-0.167* (0.096)	-0.167* (0.096)	-0.164** (0.080)
<i>p-value</i>	0.081	0.081	0.041
Agricultural employment share	-0.053*** (0.018)	-0.053*** (0.018)	-0.052*** (0.018)
<i>p-value</i>	0.004	0.004	0.004
ln modern employment	0.169*** (0.061)	0.169*** (0.061)	0.171*** (0.059)
<i>p-value</i>	0.005	0.005	0.004
Modern employment share	0.068*** (0.015)	0.068*** (0.015)	0.068*** (0.014)
<i>p-value</i>	0.000	0.000	0.000
Observations	133	133	133
<i>Panel B. 2010 residents by birthplace</i>			
Born in the same district	-0.043* (0.023)	-0.043* (0.023)	-0.046*** (0.015)
<i>p-value</i>	0.069	0.069	0.003
Born elsewhere in the same province	0.043** (0.018)	0.043** (0.018)	0.044** (0.018)
<i>p-value</i>	0.015	0.015	0.012
Born in another outer-island province	0.018 (0.021)	0.018 (0.021)	0.015 (0.012)
<i>p-value</i>	0.404	0.404	0.237
Born on Java or Bali	-0.018* (0.011)	-0.018* (0.011)	-0.013 (0.009)
<i>p-value</i>	0.100	0.100	0.154
Observations	133	133	133

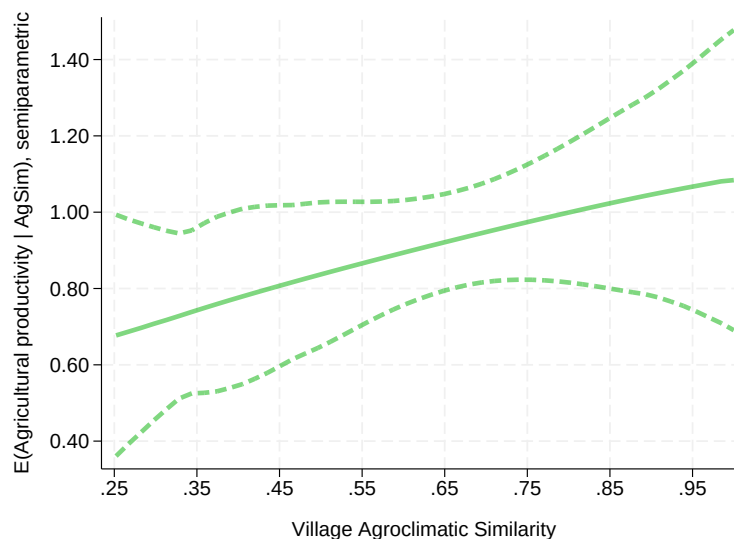
Notes: Each cell is the coefficient on standardized agro-climatic exposure from a separate regression at the urban-center level, with Conley spatial standard errors at a 200 kilometer cutoff beneath it and the two-sided Conley *p*-value below that, shaded from pale pink at 0.10 and above through a deepening green as the evidence strengthens. Stars come off the same Conley inference as the standard error directly above. Every column includes the Main Spec: the nearby transmigrant settler count, the 1980 baselines for population, employment, and urban structure, and the agro-climatic control set. Column (1) is the estimation sample. Column (3) adds the 1980 lifetime-birthplace composition of the city's workforce, which the census records at province level as three exhaustive shares (born in this province, born in another outer-island province, born on Java or Bali); two of them span the block and born-in-this-province is the omitted reference. Those shares are unobserved for a few cities, so column (3) runs on fewer of them than column (1); column (2) repeats the Main Spec on exactly column (3)'s cities, which separates the change in sample from the change in controls. Panel A reproduces the seven outcomes of the paper's headline table and Panel B the four birthplace shares of the migration table, whose 2010 shares sum to one over each city's 1955–1990 birth cohort. Significance: * 10%, ** 5%, *** 1%.

Table C8: Exposure relevance: village agricultural productivity and agro-climatic similarity

	Similarity 0.25–1 (plotted range)		Full similarity range	
	(1) winsorized	(2) raw	(3) winsorized	(4) raw
Agro-climatic similarity	0.525*** (0.201)	0.767*** (0.136)	0.535*** (0.179)	0.772*** (0.108)
Winsorized p1/p99	Yes	No	Yes	No
Similarity range	0.25–1	0.25–1	All	All
Villages	768	768	770	770

Each column regresses village agricultural productivity (tons per hectare) on the village agro-climatic similarity index across the transmigration-village roster, with Conley spatial standard errors at a 200 km cutoff. The regression is unconditional on the agro-climatic controls by construction: the similarity index is built from the village geographic, soil, and climate characteristics, so conditioning on them would partial out the exposure measure itself. Winsorized columns trim productivity at the 1st and 99th percentiles, since the raw measure has extreme outliers (maximum near 27 against a median near 0.43). Column 1 is the specification plotted in [Figure C5](#). Unweighted: population-weighting the same regression leaves the point estimate positive but imprecise once heteroskedasticity is allowed for (weighted coefficient 1.537, robust t-statistic 1.06), because the weighted fit loads on a few high-population villages. *** 1%, ** 5%, * 10% significant.

Figure C5: Village-level relevance: agricultural productivity against agro-climatic similarity



Notes: Semiparametric (local linear, Epanechnikov kernel, bandwidth 0.30) fit of village agricultural productivity (tons per hectare, winsorized at the 1st and 99th percentiles) on agro-climatic similarity across the transmigration villages, unweighted, with a 90% confidence band, plotted over the 0.25 to 1 similarity range. The fit is direct rather than partialled on geography, since the agro-climatic controls are the inputs to the similarity index itself. [Table C8](#) reports the same gradient in linear form, with Conley spatial standard errors.

Table C9: The village stay-rate gradient does not depend on how the stay-share is defined

Numerator definition	Gradient on agro- similarity	Island- clustered std. error	Wild bootstrap p-value	Conley 200 km p-value	Corr. with the plotted measure
All Java/Bali-born residents (plotted measure)	0.078	(0.106)	0.69	0.57	1.000
Excluding arrivals in the previous five years	0.080	(0.093)	0.67	0.52	0.989
Excluding those born after settlement	0.071	(0.091)	0.67	0.54	0.995
Excluding both groups	0.080	(0.086)	0.66	0.49	0.985

Notes: Each row reports the conditional linear gradient of one stay-share definition on the village's own agro-climatic similarity, holding the terrain, soil, geographic, and climate controls that [Figure 9](#) residualizes on, estimated on the same common village sample of 808 transmigration villages. Row 1 is the measure the paper plots: 2000-census residents born in Java or Bali, divided by the individuals the program placed in the village. Rows 2 to 4 restrict that numerator using the census record of residence five years before enumeration and each village's own settlement year, so that residents who arrived after the program, or who were born in Java or Bali after the village was settled, no longer count as retained settlers. The featured p-value is a wild cluster bootstrap over the four island clusters with Webb weights, which are used in place of Rademacher weights because four clusters would floor a Rademacher two-sided p-value at 0.125. The Conley p-value uses a 200 km cutoff over village centroids, matching the paper's headline inference.

D Robustness

This part re-estimates the headline results under alternative constructions of the exposure measure and alternative specifications.

D.1 The Settlers' Size Relative to the Pre-Existing Rural Population

The settlers placed in a city's rural hinterland, meaning the rural subdistricts within 135 km, were a non-trivial share of the population that hinterland already held at the 1980 census: about 7% of it on average and roughly 5% at the median, with about a seventh of cities above 15%. That radius is the one that reproduces the 300-minute window the exposure measure uses. Cities therefore differ in how large the settlement was relative to the rural landscape it entered, and [Table D1](#) asks whether the main results turn on that difference.

They do not. Adding the share to the Main Spec leaves every sign, magnitude and significance level in place across the seven main outcomes, and the largest movement is 0.12 of a standard error. Part of the reason is that the Main Spec already controls for the number of settlers placed near each city ([Section D.6](#)), so the share's numerator is largely held and the variable moves primarily through its denominator, the size of the countryside those settlers entered.

D.2 Urban Emergence: The Extensive Margin

The set of cities that defines the estimation sample is endline geography: the GHS Urban Centre footprints are classified from 2020 imagery, so the timing defense in [Section 2.2.1](#) requires knowing whether the shock itself created urban units that a 2020-defined sample would then select. This appendix reports that test directly, estimating the effect of the same exposure on urban emergence outside the fixed sample.

The unit is an equal-area 25 km lattice over outer-island land (2,648 cells, excluding Java, Bali, and Papua as in the main sample), with the pre-shock urban baseline taken from the GHS-SMOD 1975 classification, the observed pre-placement epoch. The treatment is the paper's exposure rebuilt at cell centroids, the unweighted mean of village agro-climatic similarity within 300 route-aware cost-minutes, with the agro-climatic control set and the transmigrant-count control re-derived from the same cell-to-village pairs and Conley 200 km standard errors throughout. The estimation sample is the baseline-rural, exposure-defined cells containing no transmigrant settlement point (a settlement is the

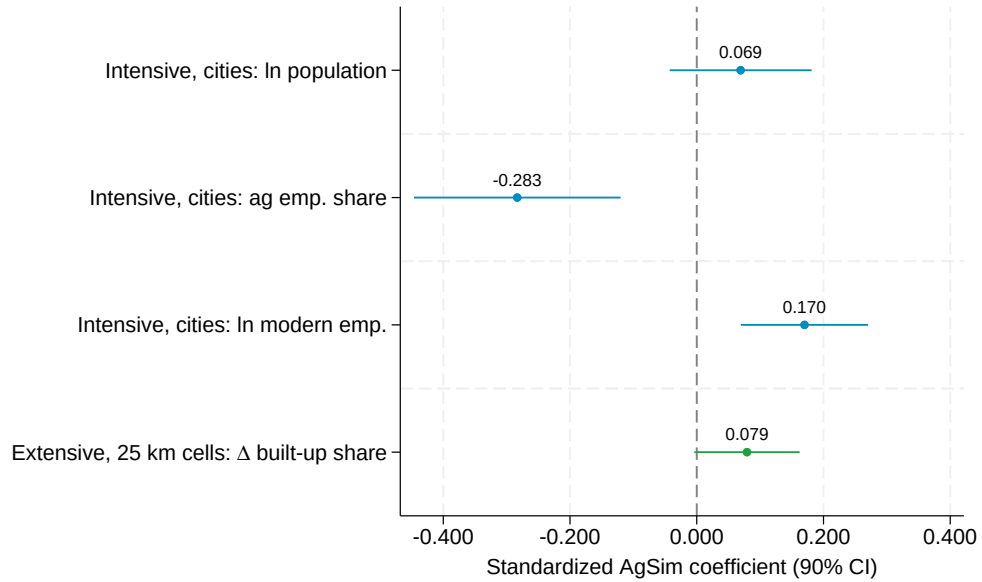
Table D1: The main results are unchanged by controlling for the settlers' size relative to the pre-existing rural population

	Population + Employment			Agric Empl		Modern Empl	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	ln pop	ln emp	unemp rate	ln L agric	agric %	ln L modern	modern %
Main Spec (main text)	0.060	0.033	0.009	-0.167*	-0.053***	0.169***	0.068***
	(0.059)	(0.056)	(0.007)	(0.096)	(0.018)	(0.061)	(0.015)
+ settler share of the hinterland	0.090	0.063	0.009	-0.150	-0.055***	0.189***	0.063***
	(0.056)	(0.054)	(0.007)	(0.096)	(0.019)	(0.061)	(0.015)
Observations	133	133	133	133	133	133	133

Notes: each cell is the coefficient on standardized agro-climatic similarity, with its Conley 200 km standard error underneath. The first row is the Main Spec exactly as the main text reports it. The second adds one control: the settlers placed in the city's rural hinterland as a share of the population that hinterland held at the 1980 census, measured over the rural subdistricts within 135 km, the radius that reproduces the paper's 300-minute exposure window. The settler count is already a Main Spec control, so this share varies primarily through its denominator. AgSim is standardized once over the full urban-center file in both rows, so the two are on one scale and can be read down the column. *** 1%, ** 5%, * 10% significant.

program building a village, not the shock seeding a town): 572 cells at 25 km. New-town founding is a rare event (single-digit foundings in the exposed sample by 2010), so the founding analyses use the finer 15 km lattice (2,074 cells), defining a founding as a nightlights cluster disjoint from the 1992 urban extent with a 5 km guard against an existing city's expanding edge.

Figure D1: Intensive versus extensive margin of the agro-similarity shock



Notes: Fully standardized coefficients (outcome and agro-similarity both z-scored), so the city-growth effects and the lattice outcome share one axis, with 90% confidence intervals under Conley 200 km standard errors. Teal: the intensive-margin urban-center headline (Main Spec, the headline sample). Green: the extensive margin, the 1975–2010 change in built-up areal share on the 25 km baseline-rural lattice ($N = 572$). Both rows take the sixteen agro-climatic controls as unweighted means within 300 route-aware cost-minutes, on the same rule. They differ in unit and in their pre-shock baselines, the urban-center rows on 1980 city characteristics and the lattice row on 1975 settlement share, population and land area.

Figure D1 puts both margins on one standardized axis. The intensive trio sits off zero on its two composition margins (ln population +0.069, agricultural employment share -0.283 , ln modern employment +0.170), while the extensive response, the 1975–2010 change in built-up areal share, is smaller and marginal at +0.079, its 90% interval just crossing zero. Table D2 reports the raw-units version: +0.00010 of built-up areal share per standard deviation of exposure over 1975–2010 (SE 0.00007), stable across the 1975–2000 and 1975–2020 windows. The pooled marginality reflects cross-island level differences rather than a zero response. Table D3 shows the same coefficient conventionally significant within islands (island fixed effects +0.00016, province fixed effects +0.00015, both $p < 0.01$). The margin also does not depend on the lattice dials. Table D4 re-runs the headline under all five baseline-rural definitions and Table D5 the founding event under all four founding definitions, positive throughout.

Part of the local response is redistribution across nearby space. Table D6 adds the mean exposure of cells 30 to 100 km away: the own effect stays positive while the ring enters negative (founding -0.0034 , marginally significant), consistent with local displacement rather than creation. The ring’s precision is sensitive to the band definition, so it is illustrative; the magnitude statements that make the displacement point are in checkable units. The count of founded towns per district does not rise with exposure (Table D7), and the population bridge in Table D8 shows the persons the shock moves accruing to incumbent footprints (positive but imprecise) while population inside newly founded footprints falls with exposure ($-1,050$ per standard deviation, significant at 5%).

The extensive margin is therefore real but small, local, and partly displaced: the shock builds up the urban system by thickening existing centers, not by seeding new towns. That is the fact the sample-timing defense in Section 2.2.1 and the conclusion’s no-boom-of-new-towns statement rest on.

One selection margin is left over, and the lattice cannot test it: the estimation sample above is restricted to baseline-rural cells, which excludes precisely the incumbent towns that sat near the inclusion threshold. Membership in the urban-center sample requires 50,000 people in the 2020 grid, a post-treatment quantity, so an incumbent town near that line enters only if it grew, and exposed marginal towns would select in while unexposed ones select out. Seventeen of the 128 sample cities have a 2020 polygon population within 20% of the threshold, and the sample’s smallest is 50,287, close enough to the line that the concern is not hypothetical. Dropping all seventeen and re-estimating the headline table moves nothing in the direction the concern predicts. The agricultural employment share coefficient goes from -0.031 to -0.028 ($p < 0.01$), modern-sector employment from 0.133 to 0.140 ($p < 0.01$), and log population from 0.086 to 0.097 ($p < 0.01$), all on the remaining 111 cities. The threshold selection, if it operates at all, attenuates the size response, which is the conservative direction for the population claim and second-order for the shares, since those have no mechanical link to the population threshold. Three further sample cities have no GHS polygon population because the reclassification created them as new nodes rather than selecting them on the threshold, so they are not exposed to this margin and are retained.

Table D2: Headline extensive margin: built-up-share change on AgSim (Conley 200km)

	(1)	(2)	(3)	(4)
	d built 7510	d built 7520	d built 7500	7510 NTL base
TMV AgSim	0.00010 (0.00007)	0.00012 (0.00008)	0.00008 (0.00005)	0.00008 (0.00013)
<i>N</i>	571	571	571	905
Window	1975-2010	1975-2020	1975-2000	1975-2010 (NTL base)

OLS/Conley. Change in GHS built-up areal share off the 1975 baseline, baseline-rural cells without a transmigrant settlement.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table D3: R4: headline built-up change under the fixed-effect ladder

	(1) No FE	(2) Island FE	(3) Province FE
TMV AgSim	0.00010 (0.00007)	0.00017*** (0.00006)	0.00016*** (0.00006)
<i>N</i>	571	571	571
Fixed effects	None	Island	Province

Manual dummies (acreg takes no factor notation). No FE = the headline; island FE is the featured variant; province FE is robustness

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table D4: R3: headline built-up change under alternative baseline-rural definitions

	(1) SMOD ≥ 21	(2) SMOD ≥ 23	(3) SMOD ≥ 30	(4) Built-up 1975
TMV AgSim	0.00010 (0.00007)	0.00019* (0.00011)	0.00009 (0.00013)	0.00021 (0.00015)
<i>N</i>	571	726	882	926

Each column redefines which cells count as rural in 1975 (stricter urban thresholds admit more cells); the SMOD 21 column is the headline

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table D5: R3: founding-event definitions at 15 km (LPM, Conley 200 km)

	(1) New cluster 2010	(2) Any urban 2010	(3) New cluster 2020	(4) Any urban 2020
TMV AgSim	0.00178 (0.00130)	0.00086 (0.00107)	0.00189 (0.00118)	0.00189 (0.00118)
<i>N</i>	2065	2065	2065	2065
Events	6	8	7	12

New cluster = the genealogy rule keeps only clusters born away from any 1992 town; any urban drops that rule. Founding stays a default

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

D.3 The Specification Curve and Price-Measure Robustness

The full specification curve. The specification curve re-estimates the standardized agro-similarity coefficient across the cost-distance specification space, crossing the weight family, the cost-distance metric, the travel-time cutoff, the control set, and island fixed effects on or off. For the three headline outcomes that space holds 85 specifications, every one of them at the 300-minute or 600-minute cutoff and on the headline estimation sample. It excludes straight-line kilometer-radius constructions, which are ill-defined in this project's cost-distance setting; province fixed effects and their clustered-standard-error variants, since province effects absorb the between-province variation that identifies the treatment; no-cutoff constructions that drop the 300-minute identification window; and the 180-minute cutoff family, which on its smaller subsample holds the space's only significant wrong-signed cells. Within this space no specification is both wrong-signed and significant for any of the three outcomes. The weight family crosses the headline's unweighted

Table D6: R10: ring-exposure displacement diagnostic

	(1) New town	(2) d urb share
TMV AgSim	0.0047*** (0.0015)	0.0001*** (0.0000)
Ring AgSim (30-100km)	-0.0035* (0.0020)	-0.0001* (0.0001)
<i>N</i>	2065	2065

E1/E7 plus the mean AgSim of cells 30-100km away. A negative ring coef = founding suppression nearby.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table D7: District-Level New-Cluster Count on Agro-Similarity, with the Log of At-Risk Cells as Offset

	(1) Poisson	(2) Poisson +islFE
TMV AgSim (district mean)	0.3850 (0.4260)	0.7334 (0.5900)
<i>N</i>	129	129
FE	none	island

Poisson, cluster(prov). Count of founded towns per district. Displacement-honest total.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

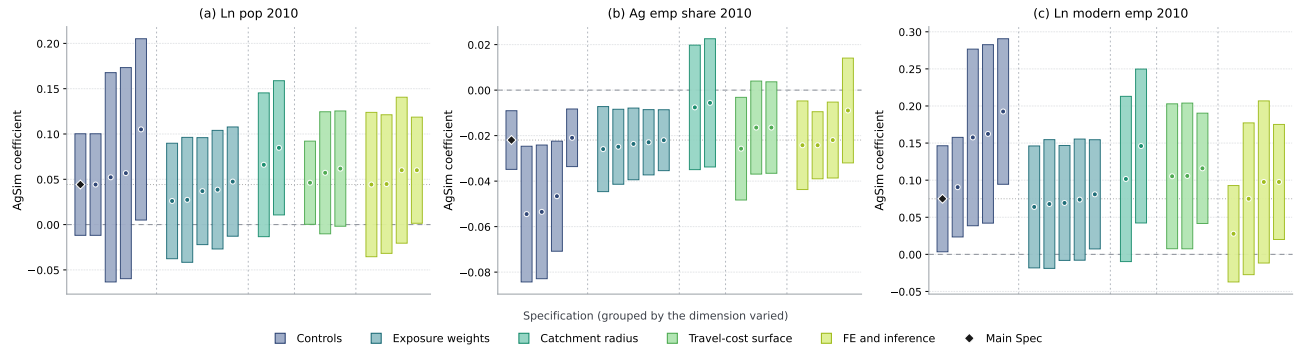
Table D8: Population Bridge: Persons Added per SD of Agro-Similarity, by Footprint Type

	(1) d pers incumbent	(2) d pers new town	(3) incumb 90-20	(4) new 90-20
TMV AgSim	1,342 (1,219)	-1,050** (439)	2,262 (2,331)	-1,465** (621)
<i>N</i>	150	150	150	150
Footprint	incumbent	new town	incumbent 90-20	new town 90-20

District level, cluster(prov). GHS-POP change 1990-2010 (and -2020) inside incumbent vs newly-founded-town footprints.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure D2: Specification curve by the dimension varied



Notes: Each bar reports the coefficient on standardized agro-similarity under one of 19 specification variants, grouped by the dimension the variant changes relative to the Main Spec: the control set, the weight placed on villages inside the catchment, the catchment radius, the friction surface the travel time is built on, and the fixed effects and standard-error scheme. [Table D9](#) defines every variant. Within a group, bars are ordered by the point estimate. The marker is the point estimate and the shaded bar spans its 90% confidence interval, so a bar that does not cross the dashed zero line is significant at the 10% level. Standard errors are Conley at a 200 km cutoff except in the two variants that cluster instead. The black diamond marks the Main Spec and the dotted horizontal line marks its estimate. The sample is the headline estimation sample in every variant but one: the 1990-roads upper bound, on whose surface three cities have no transmigrant village inside the 300-minute window, so their exposure under it is undefined. The catchment radius is varied upward only, at 400 and 500 minutes; a 200-minute radius reaches no transmigrant village at all for 28 of the sample cities, so that variant measured the sample change rather than the design and is excluded on the same ground as the 180-minute family. For log population the own-outcome baseline coincides with the Main Spec, whose controls already include the 1980 population.

within-300-minute mean with an inverse-distance-weighted mean, a settler-weighted mean, and the inverse-distance and settler-weighted average, and every construction inside the 300-minute window behaves like the headline specification.

[Figure D3](#) extends the exercise to every outcome in one grid: each cell is the standardized agro-similarity coefficient for one outcome (columns) under one specification family (rows), built by crossing the exposure weighting, the control set, the cost-distance metric, the travel-time cutoff, and island fixed effects on or off. Two patterns organize the grid. The sectoral columns are the stable ones in sign: the agricultural employment share is negative and the modern-sector share positive across essentially the whole space. The population and price columns are positive on average with wider intervals; across the 85 specifications, log population clears significance on 13 against 49 for the agricultural employment share. The displayed grid is 71 specifications across 17 outcomes. Province fixed effects and their clustered-standard-error variants are excluded from the grid for the reason given above, that province effects absorb the between-province variation identifying the treatment. Island fixed effects appear as their own row group here.

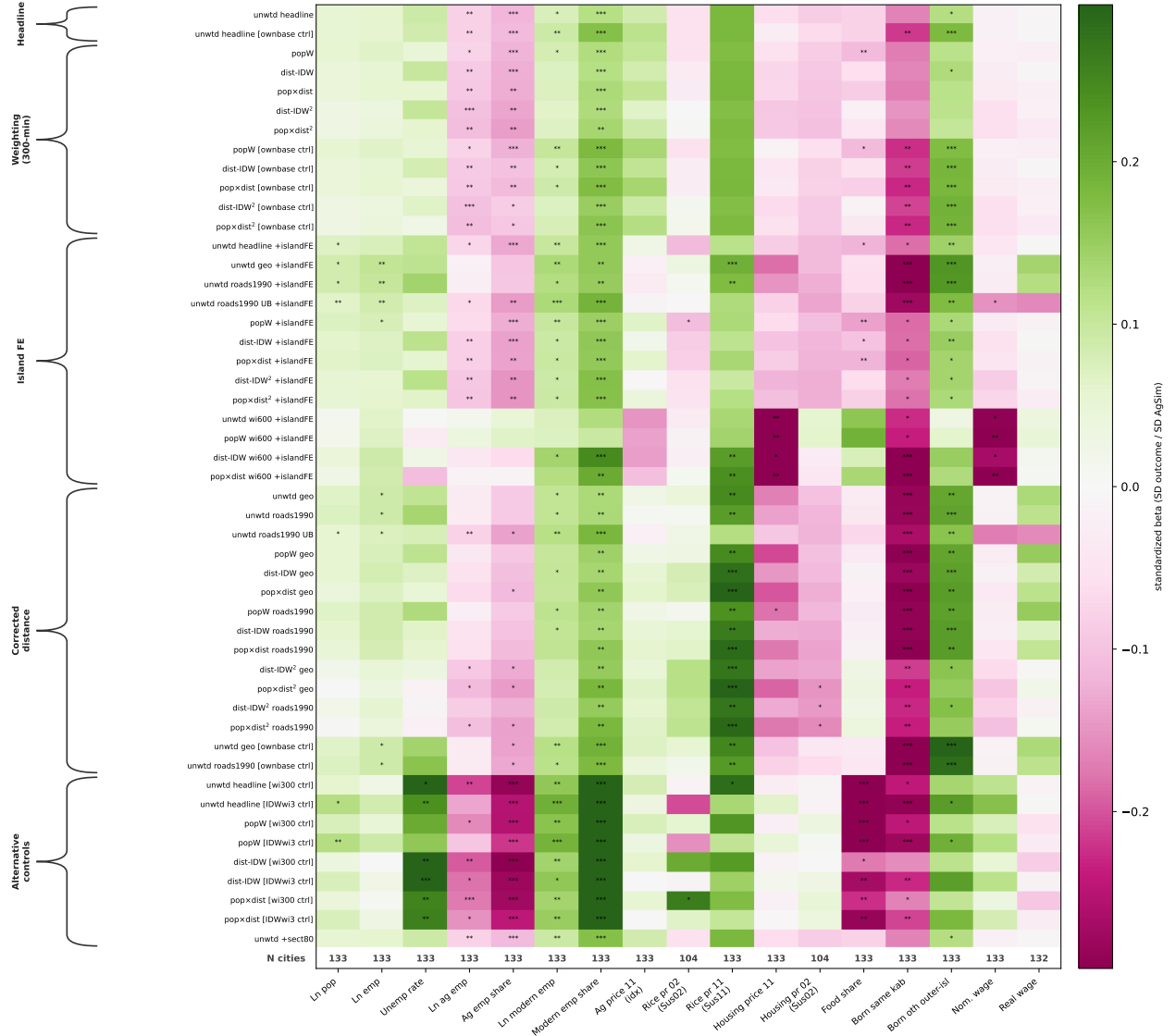
Robustness of the urban price result across measures. The positive-but-imprecise urban food-price response is robust to how the local price is measured. [Figure D4](#) re-estimates the headline Main Spec reduced form with the dependent variable swapped across seven price measures: the broad agricultural-goods index, the crops index, and a rice unit value, each under national (Laspeyres), local (Paasche), and equal-weight aggregation. The agro-similarity coefficient on the log price runs from +0.005 to +0.027 and is insignificant throughout; the raw index-level estimates give the same picture. The measures are also internally consistent (pairwise log correlations of 0.74–0.99) with economically sensible spatial variation. Traded food prices are spatially compressed (coefficient of variation ≈ 0.12) relative to the non-traded housing benchmark (0.50). Because these indices are household unit values, which embed product quality, the small positive coefficient may partly reflect quality upgrading as real incomes rise rather than a true increase in the price of a fixed food basket, a distinction I probe against external market price data. The probe supports the quality-upgrading explanation. Estimated *market* rice prices (World Bank ADM2 market-price panel, matched to urban centers by market coordinates) show no robust agro-similarity response in 2011 levels (+0.002, s.e. 0.008), while households near productive transmigration villages buy 3.4% *fewer* kilograms of rice per capita ($p < 0.10$), spend only 1.0% less on it (n.s.), and shift

Table D9: Specification variants in the grouped specification curve

Variant	What it changes relative to the Main Spec	<i>N</i>
<i>Controls</i>		
Main Spec	The transmigrant settler count, the 1980 population, employment and urban-share baselines, and the 16 agro-climatic covariates weighted by inverse travel time. Every other row changes one thing about this.	133
Matched within-300	The settler count and the agro-climatic covariates rebuilt as unweighted means over the treatment's own 300-minute window, instead of the all-village inverse-travel-time versions.	133
Matched IDW within-300	The same count and covariates, inverse-travel-time weighted and restricted to the 300-minute window.	133
Matched agro-climatic	Each covariate rebuilt in the same weight scheme, cutoff and distance metric as the treatment it accompanies.	133
Own-outcome base-line	Adds the outcome's own 1980 level, or the 1980 agricultural employment share and log population density where the outcome has no 1980 level. For log population this coincides with the Main Spec, whose controls already include the 1980 population.	133
<i>Exposure weights</i>		
Population-weighted	Villages enter weighted by their settler count.	133
1/d	Villages enter weighted by the inverse of travel time, so nearer villages count for more.	133
1/d ²	The same, with the inverse of squared travel time, a steeper decay.	133
Settlers/d (Eq. 2.6)	The paper's exposure equation: settlers divided by travel time.	133
Settlers/d ²	The exposure equation with squared travel time in the denominator.	133
<i>Catchment radius</i>		
Within 400 min	Within 400 travel-time minutes, which every city in the estimation sample reaches.	133
Within 500 min	Within 500 travel-time minutes.	133
<i>Travel-cost surface</i>		
Sea-route corrected	Travel times resolved with the route-aware over-water fix, which reproduces the validated cost-distance cache exactly.	133
1990 trunk roads	Travel times recomputed on the period-appropriate 1990 trunk network, reverting the 1.5% of road-fast land that is dateable post-1990 trunk road.	133
1990 roads, upper bound	A stress test reverting 67.5% of road-fast land, 98% of it undateable local road, so it bounds how much the road vintage could matter.	130
<i>FE and inference</i>		
Province FE	Adds province fixed effects. Province effects absorb the between-province variation that identifies the treatment, so this is never the headline.	133
Island FE	Adds island fixed effects.	133
Cluster: province	The same point estimate, with standard errors clustered at the province level instead of Conley.	133
Island FE, cluster province	Island fixed effects together with province-clustered standard errors.	133

Notes: Each row is one bar of Figure D2, which reports the coefficient on standardized AgSim under that specification. The Main Spec is the first row and every other row changes exactly one thing about it, which makes a group a controlled comparison. *N* is the number of urban centers the specification estimates on. It is 133, the headline estimation sample, in every row but one. The 1990-roads upper bound estimates on 130, because for three cities the nearest transmigrant village sits outside the 300-minute window once the post-1990 roads are reverted, so their exposure under that variant is undefined. Straight-line kilometer-radius constructions are excluded throughout, since distance in this setting is travel time on a friction surface. Standard errors are Conley at a 200 km cutoff except in the two rows that cluster instead.

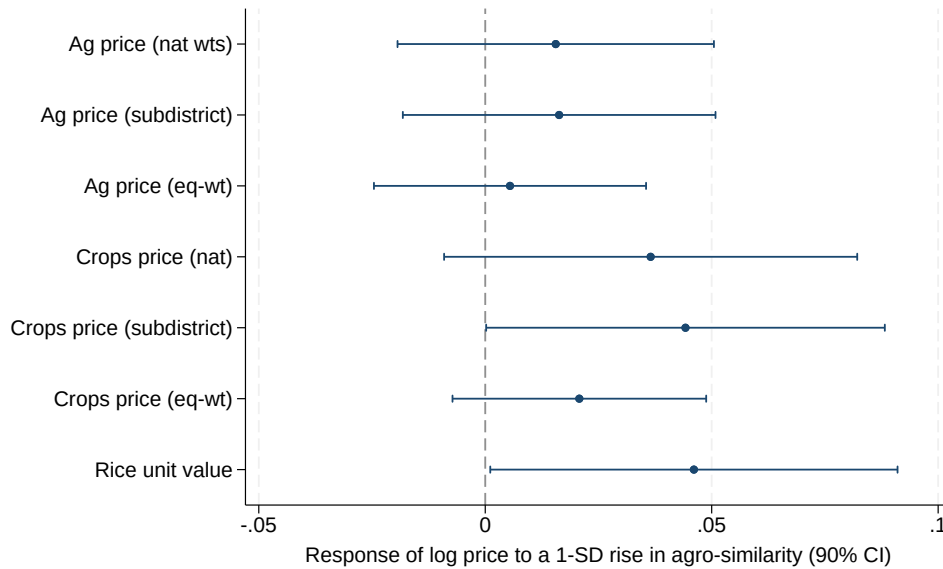
Figure D3: Specification-curve heatmap, I: constructions and controls at the headline 300-min cost distance



Notes: Each cell is the standardized agro-similarity coefficient (per standard deviation of the outcome) from a separate Main Spec reduced form under the labeled specification family, with Conley 200 km spatial standard errors on a held-fixed common sample (the headline urban centers, and $N = 103$ in the two 2002 Susenas columns; the figure's bottom row prints each column's own N). Green cells are positive and pink cells negative (project convention); shading intensity scales with magnitude and $^*/^{**}/^{***}$ mark significance at 10/5/1%. This grid varies exposure weighting, control set, and island fixed effects at the 300-minute cutoff. Straight-line kilometer-radius constructions are excluded as ill-defined in this project's cost-distance setting.

expenditure toward non-food, paying more per kilogram for less quantity, the classic quality-upgrading signature in unit values. On the time-series side the kota rice-price *growth* response (+0.019, $p < 0.01$) is absorbed by the city's overall CPI growth (+0.015, $p < 0.01$; rice - CPI ≈ 0), i.e., a general local price-level gradient rather than a relative food-price effect.

Figure D4: Robustness: the effect of agro-similarity on prices across price measures



Main Spec reduced form, urban-center sample ($N = 122$ – 128 , varying with each price measure's Susenas coverage), Conley 200 km spatial standard errors, 90% confidence intervals; agro-similarity standardized. Each row is a separate regression of the log price measure on agro-similarity and the Main Spec controls.

D.4 City Boundary Definitions and the Matched Cutoff

Two checks bound what the city definition can do to the results. [Table D10](#) measures the same cities under four definitions, from the whole member subdistrict to the built-up core intersected with the census's own urban classification: the level of urban farming falls as the city is drawn tighter, and its 2010 mean is still about 14% under the tightest cut. [Table D11](#) splits the sample at the median 1980 agricultural share, 57.7% of 1980 employment, and runs the split as an interaction, so both slopes come from one regression on all 133 centers and the difference between them has its own standard error. The two halves respond alike. The agricultural-share coefficient is -0.046 in the less agricultural half and -0.060 in the more agricultural one, the modern-employment coefficient is 0.162 and 0.178 , and no difference between the halves is distinguishable from zero, the smallest of the four p -values being 0.306 . The same design on 1980 population does separate the sample. Modern employment responds 0.145 log points more in the larger cities than in the smaller ones (s.e. 0.063) and population 0.176 log points more (s.e. 0.045), the first distinguishable from zero at the 5% level and the second at the 1% level, and the agricultural-share response is the same on both sides of that cut.

The village-level cutoff, the baseline controls, and outlying outcomes. The exposure measure averages agro-similarity over the transmigration villages inside a city's 300-minute catchment, and every village-level control enters over that same catchment, so the treatment and its controls are built from one set of villages. [Figure D5](#) collects the specifications that vary what is done around that choice. Conditioning each outcome on its own 1980 level, in place of the four common baselines, leaves every coefficient inside its interval. Winsorizing the 2010 outcomes at the fifth and ninety-fifth percentiles, which no outcome in the build is subject to otherwise, moves the size margins by about a tenth and the compositional ones not at all. Dropping the 1980 baselines entirely admits the five in-window cities that have no 1980 census and leaves every sign intact at a lower precision. The specification whose village terms do not share one cutoff, the one that treats the settler count and the agro-climatic covariates over the full 2,000-minute matrix while the treatment stops at 300, is the outlier in the figure rather than the norm.

Table D10: Urban agricultural employment share under four definitions of the city

Year	Definition	City mean	Pooled	Cities
1980	K whole subdistrict (as reported)	0.594	0.482	122
1980	U BPS-urban villages	0.141	0.086	96
2000	K whole subdistrict	0.460	0.361	125
2000	U BPS-urban villages	0.194	0.115	124
2010	K whole subdistrict	0.386	0.299	125
2010	C built-up core	0.182	0.112	124
2010	U BPS-urban villages	0.158	0.099	127
2010	CU core and BPS-urban	0.136	0.084	122

Notes: The same cities measured under four definitions, from loosest to tightest. K sums every member subdistrict, the paper's reported cut; C keeps only the villages whose polygon meets the city's own satellite footprint; U keeps only the villages Indonesia's census classifies as urban, which does not use satellite input; CU requires both. City mean is the unweighted mean across cities; Pooled weights by employment.

D.5 Inference: The Conley Cutoff and Catchment Overlap

The Conley cutoff. The headline uses a 200 kilometer Conley cutoff. Because the cutoff enters only the variance estimator, the point estimates do not depend on it, so the choice affects only inference. At a shorter 150 kilometer radius every standard error remains well defined and four of the seven are smaller than at 200 kilometers, so the wider cutoff is not uniformly the more conservative of the two: the unemployment rate and both employment-share columns have slightly larger standard errors at 150 kilometers (0.011 against 0.010 on each share), and those shares are the columns the compositional result rests on. The wider direction matters more, because the exposure measure averages over village catchments that neighboring cities share. The 150 kilometer row of [Table D12](#) reports these standard errors beside every wider radius.

Catchment overlap and inference at wider radii. The relevant direction to vary the cutoff is wider rather than narrower, and the reason is in the construction of the treatment. Exposure is an unweighted mean over the transmigration villages inside a city's 300-minute catchment, and neighboring catchments intersect, so two cities can be built partly from the same village draws and their exposure errors are correlated at whatever distance that sharing reaches. Measured directly on the travel-time matrix the exposure itself is built from, the average city in the estimation sample draws on 31 in-window villages, 80% of those villages enter more than one city's catchment, and one village enters as many as 19. Of the 8,128 city pairs in the sample, 727 share at least one village, at a median separation of 143 kilometers and a maximum of 459; 27% of those sharing pairs sit farther apart than the 200 kilometer cutoff, which assumes independence beyond it. Grouping cities into the connected components of that sharing graph, the coarsest partition across which no village is shared at all, leaves 14 components over the estimation sample, and the largest holds 64 cities: half the sample is linked into a single block by shared catchments.

[Table D12](#) therefore reports the headline columns across the whole ladder, from 150 out to 500 kilometers, and adds a variant clustered on those components together with a wild cluster bootstrap over them. Every outcome the paper's claims rest on, log population, log employment, log modern employment, and the two employment shares, rejects at the 10% level under every estimator in the table. The compositional standard errors are the more stable: both employment-share standard errors stay between 0.008 and 0.012 across the whole ladder, and the agricultural share rejects at the 1% level at every radius. The size margins lose precision as the radius widens, with the standard error on log population rising from 0.027 at 200 kilometers to 0.035 at 500 and its p -value moving from 0.001 to 0.014, while log total employment and log modern employment stay inside the 1% level throughout. Under the component clustering, the coarsest partition across which no village is shared and therefore the most conservative grouping available, log population returns $p = 0.056$

Table D11: Response by the city's 1980 agricultural share, with the size split that tests it

	(1)	(2)	(3)	(4)
	Agric. share	ln modern empl.	ln population	In-migrant share
<i>Panel A: split at the median 1980 agricultural share, 57.7%</i>				
Below median (less agricultural)	-0.046*** (0.016)	0.162*** (0.061)	0.062 (0.063)	0.005 (0.003)
Above median (more agricultural)	-0.060*** (0.019)	0.178** (0.081)	0.057 (0.058)	0.009*** (0.002)
Above-median indicator (=1)	0.093*** (0.023)	-0.109* (0.060)	0.022 (0.053)	0.003 (0.003)
Difference (above minus below)	-0.014 (0.014)	0.016 (0.049)	-0.006 (0.042)	0.003 (0.003)
<i>p</i> -value of the difference	0.306	0.744	0.892	0.327
Observations	133	133	133	131
<i>Panel B: split at the median 1980 population, 67,518 residents, the same design</i>				
Below median (smaller)	-0.052** (0.023)	0.105 (0.066)	-0.018 (0.086)	0.005 (0.004)
Above median (larger)	-0.054** (0.022)	0.250*** (0.076)	0.158*** (0.054)	0.007*** (0.002)
Above-median indicator (=1)	-0.006 (0.017)	0.137 (0.110)	0.166* (0.090)	0.000 (0.003)
Difference (above minus below)	-0.001 (0.024)	0.145** (0.063)	0.176*** (0.045)	0.002 (0.002)
<i>p</i> -value of the difference	0.956	0.022	0.000	0.283
Observations	133	133	133	131

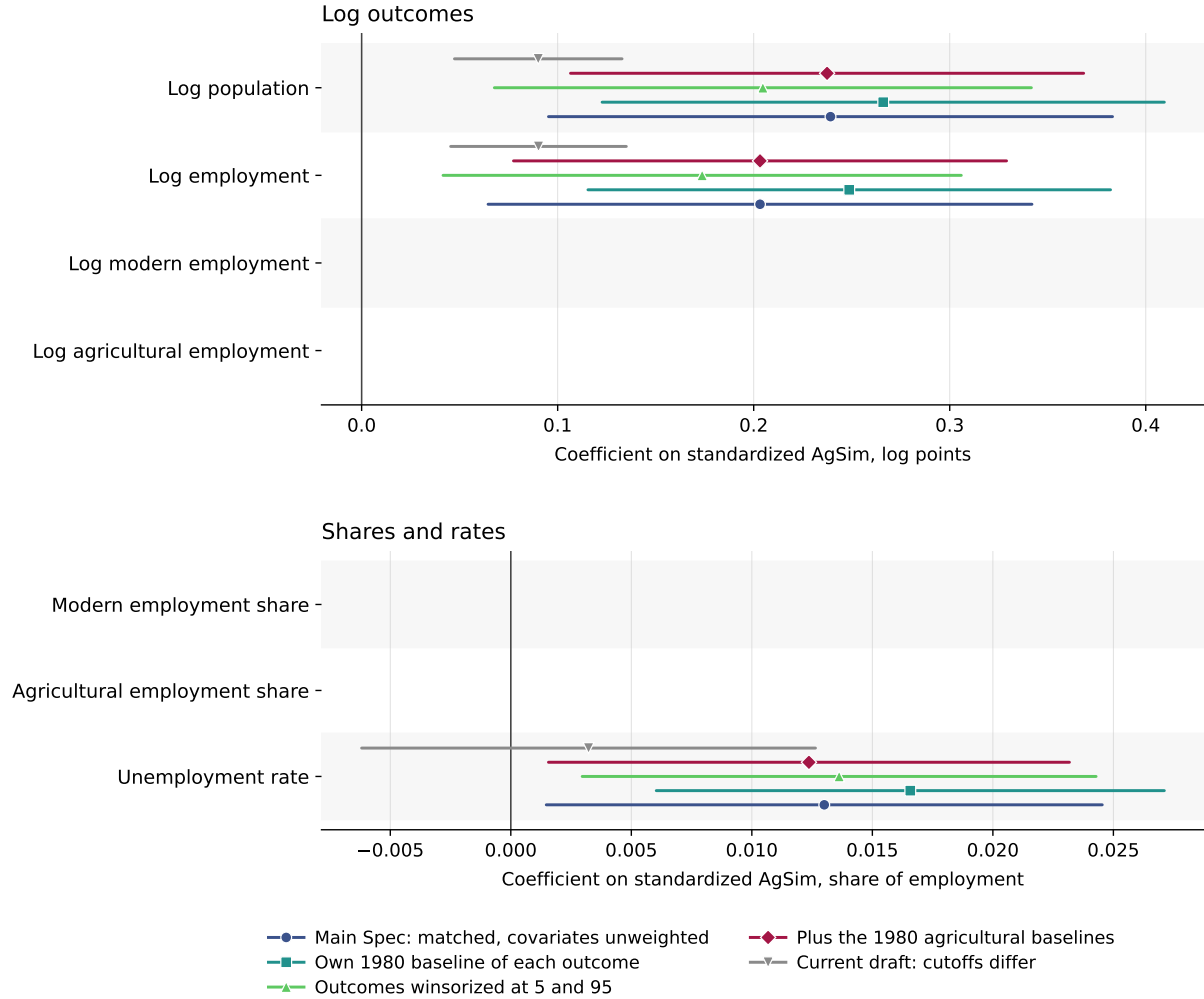
Notes: each column of each panel is ONE regression, estimated on the whole sample. Agro-similarity enters interacted with an indicator for each side of the split, so the two rows are the two slopes with their own standard errors. The difference row is the second slope minus the first, recovered by a linear combination within that same regression, and it tests whether the two halves respond differently. Each split is cut at the median of the estimation sample itself: 57.7% of 1980 employment in agriculture in Panel A, 67,518 residents in 1980 in Panel B. Both cutting variables are measured in 1980, before any outcome. The specification is the Main Spec throughout: the inverse-distance-weighted settler count inside the 300-minute window, 1980 log population, 1980 log employment, the 1980 urban population share, 1980 log agricultural employment, and the sixteen agro-climatic controls averaged over the same window, with no fixed effects. Standard errors are Conley spatial at 200 km. Agro-similarity is standardized once over the whole urban-center file, so every cell is on the same scale as the headline table. Modern employment is manufacturing plus services. The in-migrant share is the share of the 2010 population that lived in a different province five years earlier. It estimates on two fewer cities than the other columns, because two urban areas the reclassification created have no 2010 five-year migration count. *** 1%, ** 5%, * 10% significant.

with a wild cluster bootstrap *p* of 0.030, the agricultural share 0.015 and 0.030, and the modern share 0.009 and 0.061. The population response survives every estimator on the ladder, including the one that concedes the most to the overlap between catchments.

D.6 Transmigrant Count as a Separate Control

The headline Main Spec controls for the IDW-weighted count of nearby transmigrant settlers (tm migrants 000 IDW, in thousands) in addition to the agro-climatic similarity exposure. Because the agro-similarity measure and the migrant-count control are built over the same villages and the same travel-time window, the two are correlated, and the choice of functional form for the migrant-count control could in principle absorb some of the agro-similarity variation. This appendix reports the headline UC-level regressions under three alternative functional forms for the migrant-count control:

Figure D5: The agro-similarity coefficient across alternative specifications



Notes: Each marker is the coefficient on standardized agro-similarity from a separate regression of the 2010 outcome on the exposure and that specification's control set, with the bar giving the 90% interval from Conley spatial standard errors at 200 kilometers. Every specification except the last holds all village-level terms at the treatment's own 300-minute cutoff; the last measures the settler count and the sixteen agro-climatic covariates over the full 2,000-minute pair matrix while the treatment stops at 300, so its village terms do not share a cutoff. Every specification estimates on the identical 126 urban centers: those inside the 300-minute window whose 1980 baseline the 1980 census measures directly, which excludes the two cities whose baseline is apportioned from a 1980 parent unit. Log outcomes and share outcomes are drawn on separate axes because their units are not comparable.

- (i) **Linear count** (thousands of settlers, IDW): the form used in the main-text Main Spec.
- (ii) **Log count** (log of the IDW-weighted settler count): tests sensitivity to a constant-elasticity functional form.
- (iii) **Count + count-squared** (count and its square, both reported): probes non-linearity in the count effect, motivated by the gap in significance between (i) and (ii).

Findings. The agro-similarity coefficients are essentially invariant to the choice of functional form across all three: under the linear count and log specifications, the headline manufacturing employment effect remains large and highly significant (+0.180, $p < 0.01$, and +0.180, $p < 0.01$; the count-squared point estimate is comparable at +0.179, $p < 0.01$), and the agricultural-employment level coefficient is similarly negative throughout. By contrast, the migrant-count coefficient itself differs substantially across forms. Under linear count (i) it is statistically indistinguishable from zero on almost every outcome. Under log (ii) it is negative across most outcomes but significant at the 1% level only on

Table D12: The headline estimates under a ladder of spatial-inference radii

	Population + Employment			Agric Empl		Modern Empl	
	ln pop	ln emp	unemp rate	ln L agric	agric %	ln L modern	modern %
TMV AgSim (z, 300min)	0.060	0.033	0.009	-0.167*	-0.053***	0.169***	0.068***
Conley 200 km (headline), SE	(0.059)	(0.056)	(0.007)	(0.096)	(0.018)	(0.061)	(0.015)
Conley 200 km (headline), p	0.309	0.556	0.206	0.081	0.004	0.005	0.000
Conley 150 km, SE	(0.059)	(0.054)	(0.006)	(0.100)	(0.020)	(0.065)	(0.016)
Conley 150 km, p	0.309	0.536	0.179	0.095	0.007	0.009	0.000
Conley 450 km, SE	(.)	(.)	(0.007)	(0.056)	(0.014)	(.)	(0.009)
Conley 450 km, p	.	.	0.197	0.003	0.000	.	0.000
Conley 500 km, SE	(0.029)	(0.016)	(0.006)	(0.056)	(0.014)	(.)	(0.010)
Conley 500 km, p	0.036	0.033	0.177	0.003	0.000	.	0.000
Catchment components, SE	(0.039)	(0.038)	(0.010)	(0.099)	(0.025)	(0.064)	(0.021)
Catchment components, p	0.069	0.177	0.356	0.119	0.042	0.012	0.005
Catchment components, wild bootstrap p	0.019	0.181	0.477	0.214	0.054	0.008	0.023
Observations	133	133	133	133	133	133	133

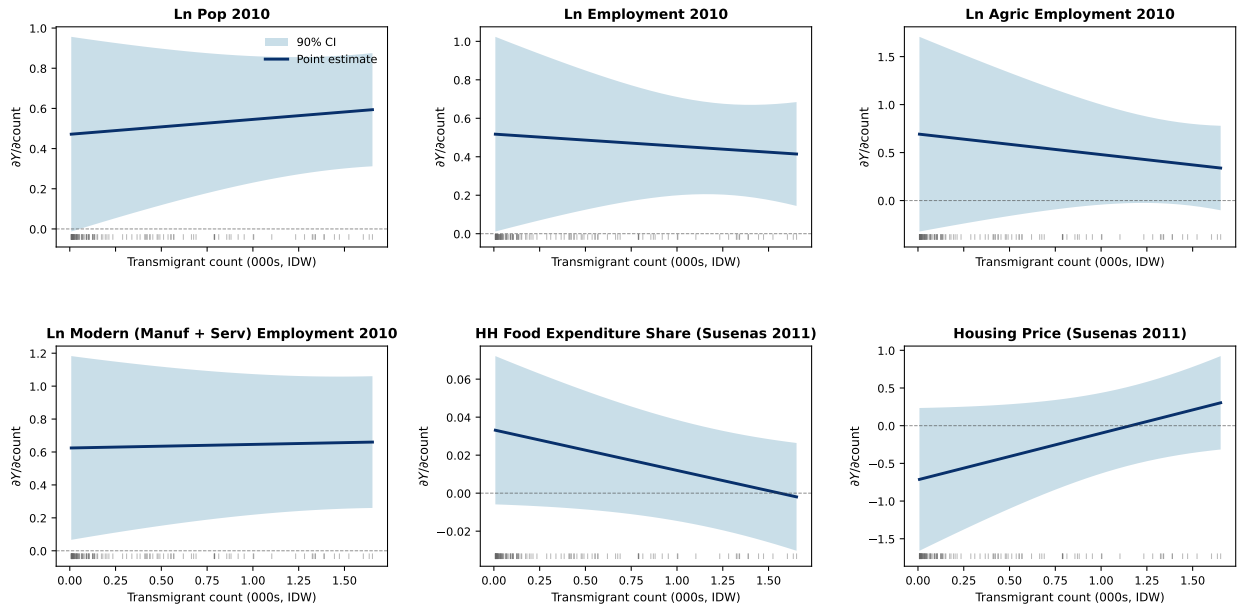
Notes: Every column is the headline regression of Table 2, on the identical sample and specification; only the variance estimator changes, so the point estimates are identical across the panel by construction and the stars come off the headline Conley 200 km p -value. Modern = manufacturing + services. Each p -value cell is shaded by the strength of the evidence, deepening green from the 10% to the 1% level, with pale pink for a p at or above 0.10. The radii bracket the distance over which the exposure measure shares village-level draws between cities: the catchment mean is taken over transmigration villages within 300 travel-minutes, 80% of in-window villages enter more than one city's catchment, and two cities sharing a village sit up to 459 km apart, so the headline 200 km cutoff is shorter than the design's own sharing range. The last three rows group cities into the connected components of the village-sharing graph, the coarsest partition across which no village is shared and exposure draws are therefore independent. That partition is highly unbalanced, with 14 components over the 133 cities and the largest holding 64 of them, so the clustered standard error rests on few effective groups; the wild cluster bootstrap over the same components, with Webb weights, is the version to read, and it has little power at this cluster count. The compositional standard errors are stable across the whole ladder, while the size margins lose roughly a third of their precision by 500 km; every column that rejects at the headline radius still rejects at the 10% level at every wider radius and under the component clustering, with log population's largest p -value across the ladder and the clustering at 0.309 and its wild cluster bootstrap p at 0.019. One cell is not evidence of a gain in precision: the agricultural employment LEVEL column's standard error falls from 0.096 at 200 km to 0.056 at 450 km before rising again at 500 km, and a standard error that is non-monotone in the radius is a symptom of the Conley kernel sum turning near-singular for that outcome. *** 1%, ** 5%, * 10% significant.

agricultural employment (-0.869); total employment and construction are also negative and significant at 10% (-0.765 and -1.137), while population, manufacturing, and services are negative but imprecise (roughly -0.65 to -0.87), a possible “concentration-of-settlers” effect that the data do not pin down sharply. Under count + count-squared (iii) the pattern is consistent with a non-monotone effect. On agricultural employment the linear count coefficient is strongly negative (-1.599 , $p < 0.01$) while the squared term is strongly positive ($+0.677$, $p < 0.01$), so the count effect first reduces, then increases the outcome with the settler count, with the turning point somewhere in the middle of the empirical distribution. This is precisely the non-linearity the log functional form was approximating in (ii). The agro-similarity productivity signal is robust to any reasonable parameterization of the count control, but the count itself enters non-linearly.

Figure D6 visualizes this non-monotonicity directly: it plots $\partial Y / \partial \text{count} = \alpha_1 + 2\alpha_2 \cdot c$ from specification (iii) across the empirical range of settler counts, with 90% Conley confidence bands, for population, total employment, agricultural employment, modern-sector employment, the household food-expenditure share, and housing price. For agricultural employment the marginal effect is significantly negative at low settler concentrations, up to about 1.1 thousand settlers; for population, total employment, and the modern sector the low-count leg is negative but imprecise. The implied turning points cluster tightly: near 1.1 thousand settlers for population and the modern sector, 1.2 thousand for total employment, and 1.6 thousand for agricultural employment. The figure plots the marginal effect over the 5th to 95th percentile of

the settler count, 0.4 to 2.3 thousand per urban center, against an observed maximum of 4.4 thousand. Within that plotted window the marginal effect turns significantly positive on every outcome near the top of the window, from about 1.9 thousand settlers for population and the modern sector to 2.1 thousand for agricultural employment. The pattern is consistent with displacement at low densities, in which settlement absorbs some local labor into the village agricultural economy and away from urban activity, giving way to a local-demand multiplier once settlement is dense enough to sustain non-tradable activity. In the count and count-squared estimates for the main outcomes (Table D13, below) the convex profile is significant on each employment outcome except manufacturing, whose squared term is imprecise (+0.416, SE 0.395).

Figure D6: Marginal effect of an additional 1,000 settlers on city outcomes from the count + count² specification.



Count and count-squared estimates for the main outcomes.

Table D13: Effects on Urban Pop & Employment in 2010 (mig coef shown)

	Population + Employment			Ln Sectoral Employment			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	ln pop	ln employed	unemp rate	L agric	L manuf	L constr	L serv
TMV AgSim (z, 300min)	0.072 (0.059)	0.043 (0.057)	0.010 (0.006)	-0.165* (0.095)	0.275** (0.121)	0.066 (0.072)	0.167** (0.067)
TMV settlers, IDW wi 300min (000s)	0.410** (0.202)	0.356* (0.198)	0.023 (0.018)	0.143 (0.269)	0.618** (0.251)	0.437*** (0.163)	0.413*** (0.146)
TMV settlers, IDW wi 300min, squared (000s)	-0.060 (0.043)	-0.051 (0.040)	-0.005 (0.005)	-0.010 (0.055)	-0.129** (0.056)	-0.044 (0.035)	-0.053 (0.032)
Baseline Pop 1980	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓	✓	✓
Mean dep var	11.839	10.930	0.123	9.816	7.542	8.007	10.134
Std dev dep var	0.858	0.843	0.041	0.898	1.357	1.116	0.970
Observations	133	133	133	133	133	133	133

Notes: Each column is a separate urban-center-level regression on the main city sample: the outer-island urban centers whose nearest transmigration village (TMV) lies within 300 travel minutes and whose 1980 baseline the census recorded directly. The regressor, TMV AgSim (z, 300min), is agro-similarity: the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, standardized over every urban center with a measured exposure. Every column includes the 1980 log population, log employment, urban population share, and log agricultural employment, and the sixteen agro-climatic covariates averaged over the same window, as in the Main Spec. The settler count enters with its square, the inverse-distance-weighted (IDW) number of settlers placed within 300 minutes in thousands and that number squared, in place of the linear count of the Main Spec; both coefficients are shown. Outcomes are 2010 census log population, log employment, the unemployment rate, and log employment by sector, aggregated from urban subdistricts to urban centers. Standard errors in parentheses are Conley spatial standard errors with a 200-kilometer cutoff. *** 1%, ** 5%, * 10%.

E Additional Results

This part reports the outcomes, decompositions, and alternative accounts that the main text states but does not tabulate.

E.1 Additional Reduced-Form Outcome Tables

The response by location type, full set of outcomes. Table E1 reports every outcome behind Figure 11a and Figure 11b with its standard error, for the three location types. Panel A holds the size and composition outcomes, which all three areas measure on the same units, and panel B the prices, wages and consumption outcomes, which come from the household survey and reach a different number of units in each row. The agricultural and housing prices are the 2002 Susenas wave, which precedes the 2004 rice import ban; the real wage is the 2011 wave, because the subdistrict level has no 2002 wage.

Table E3 and Table E4 report the census outcomes behind the headline table in the three-sector levels and shares layouts; the main text consolidates them into the modern-sector aggregation of Table 2.

The goods-market outcomes behind the mechanism discussion follow. Table E5 reports the urban agricultural-goods outcomes (production, consumption, and the production-minus-consumption surplus), and Figure E1 shows the

Table E1: All outcomes by location type, Main Spec

	Cities	Rural, with settlement	Rural, no settlement
<i>A. Size and composition</i>			
ln population	+0.070 (0.069)	+0.055 (0.088)	−0.041 (0.070)
ln employment	+0.039 (0.067)	+0.064 (0.096)	−0.065 (0.073)
ln agricultural employment	−0.186* (0.106)	+0.055 (0.072)	−0.197** (0.082)
Agricultural employment share	−0.287*** (0.100)	+0.026 (0.055)	−0.270*** (0.081)
ln modern employment	+0.173*** (0.062)	+0.051 (0.077)	+0.107** (0.049)
Modern employment share	+0.433*** (0.096)	−0.019 (0.056)	+0.259*** (0.077)
Observations	133	191	631
<i>B. Prices, wages, and consumption</i>			
Unemployment rate	+0.215 (0.169) [133]	+0.026 (0.091) [191]	+0.045 (0.087) [631]
Agricultural price, 2002	+0.171 (0.295) [104]	−0.229** (0.109) [104]	+0.014 (0.086) [240]
Housing price, 2002	−0.089 (0.163) [104]	−0.001 (0.064) [98]	+0.039 (0.084) [231]
ln nominal wage, 2011	+0.133 (0.120) [128]	+0.157 (0.184) [82]	−0.058 (0.071) [273]
ln real wage, 2011	+0.037 (0.131) [128]	+0.274** (0.127) [82]	−0.142 (0.116) [273]
Food share of spending	−0.263** (0.124) [132]	−0.224** (0.101) [82]	+0.021 (0.121) [273]
Rice calories per person	+0.453*** (0.172) [107]	+0.174 (0.118) [99]	+0.036 (0.087) [234]

Notes: Standardized responses to a one-standard-deviation increase in agro-climatic similarity, with standard errors beneath each estimate. Conley standard errors at a 200-kilometer cutoff throughout. Cities are urban centers of the headline sample; the two rural groups are subdistricts outside any urban center, split by whether a transmigration village overlaps them. Panel A's observation count applies to every row of that panel; in panel B each row prints its own count in brackets underneath its standard error, since the household survey reaches a different set of units in each wave and module. Both the outcomes and agro-similarity are standardized within each area, so a coefficient is read in that area's own units; the exposure standard deviations are 0.103 over the 138 cities with a measured exposure and 0.104 over the 822 sample subdistricts, so one standard deviation is the same amount of exposure on either side. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E2: Urban Prices, Spending, and Wages

	Prices & Spending		Wages		
	(1) nonfood %	(2) price housing	(3) ln wage nonag	(4) ln wage	(5) ln real wage
TMV AgSim (z, 300min)	0.013** (0.006)	-0.004 (0.111)	0.039 (0.032)	0.027 (0.034)	0.002 (0.031)
Transmigrant count (IDW)	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓
Mean dep var	0.447	1.593	14.065	13.971	13.743
Std dev dep var	0.050	0.910	0.272	0.313	0.239
Observations	133	133	133	133	132

All five outcomes are drawn from the 2011 Susenas, measured across urban subdistricts and aggregated to urban centers. Non-food % is a 0–1 fraction of household expenditure; the wage columns are logs of median wages. The modern-sector employment level and share behind these outcomes are reported in [Table 2](#), and the manufacturing/services split in [Table E6](#). Each column is a separate urban-center-level regression on the main city sample: the outer-island urban centers whose nearest transmigration village (TMV) lies within 300 travel minutes and whose 1980 baseline the census recorded directly. The regressor, TMV AgSim (z, 300min), is agro-similarity: the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, standardized over every urban center with a measured exposure. Every column includes the Main Spec controls: the inverse-distance-weighted (IDW) count of settlers placed within 300 minutes, the 1980 log population, log employment, urban population share, and log agricultural employment, and the sixteen agro-climatic covariates averaged over the same window. Standard errors in parentheses are Conley spatial standard errors with a 200-kilometer cutoff. Survey outcomes are observed on the centers the survey reaches, reported in the Observations row. *** 1%, ** 5%, * 10%.

fuller set of urban-agriculture responses in standardized form, including agricultural wages and staples alongside the rice production and surplus rows that the merged main-text mechanism figure ([Figure 8](#)) shows. The non-food side (non-food spending, housing rents, and wages) is reported in the main text in [Table E2](#), and the manufacturing/services composition of the modern sector in [Table E6](#) below.

[Table E6](#) reports the manufacturing and services components of the modern sector separately, in levels and shares. The main results treat the modern sector as a single object; this table reports the split in the modern-sector layout.

[Table E7](#) decomposes the service-employment response by subsector; the concentration in trade and transport services is consistent with the goods-market account.

Finally, [Table E8](#) and [Table E10](#) report the agricultural production and food consumption/surplus responses across the crop-category hierarchy, with standardized twins in [Table E9](#) and [Table E11](#); the standardized versions define several of the model’s calibration moments ([Table B2](#)).

[Table E12](#) adds the earliest wave: household Susenas 1996 outcomes aggregated to the same urban centers. The structural signal is already present one to two decades after placement, with the agricultural employment share 0.044 lower and the modern share 0.040 higher per standard deviation of exposure (both at 1%), alongside higher per-capita expenditure and rents, while the 1996 rice price and the two food-price indices show little response. The timing places the employment reallocation ahead of the later food-price movements, the ordering the trade mechanism implies.

Table E3: Effects on Urban Pop & Employment in 2010 (aggregated to urban centers)

	Population + Employment			Ln Sectoral Employment			
	(1) ln pop	(2) ln employed	(3) unemp rate	(4) L agric	(5) L manuf	(6) L constr	(7) L serv
TMV AgSim (z, 300min)	0.060 (0.059)	0.033 (0.056)	0.009 (0.007)	-0.167* (0.096)	0.250* (0.133)	0.058 (0.067)	0.157** (0.066)
Transmigrant count (IDW)	✓	✓	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓	✓	✓
Mean dep var	11.839	10.930	0.123	9.816	7.542	8.007	10.134
Sector empl share (2010)				0.374	0.046	0.060	0.471
Std dev dep var	0.858	0.843	0.041	0.898	1.357	1.116	0.970
Observations	133	133	133	133	133	133	133

Notes: Each column is a separate urban-center-level regression on the main city sample: the outer-island urban centers whose nearest transmigration village (TMV) lies within 300 travel minutes and whose 1980 baseline the census recorded directly. The regressor, TMV AgSim (z, 300min), is agro-similarity: the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, standardized over every urban center with a measured exposure. Every column includes the Main Spec controls: the inverse-distance-weighted (IDW) count of settlers placed within 300 minutes, the 1980 log population, log employment, urban population share, and log agricultural employment, and the sixteen agro-climatic covariates averaged over the same window. Standard errors in parentheses are Conley spatial standard errors with a 200-kilometer cutoff. Outcomes are 2010 census log population, log employment, the unemployment rate, and log employment in agriculture, manufacturing, construction, and services, aggregated from urban subdistricts to urban centers; the sector employment share row is the share of each sector in 2010 employment. *** 1%, ** 5%, * 10%.

The L manuf column's standard error and stars are heteroskedasticity-robust: the spatial kernel understates that one column's variance (Conley 0.022 against robust 0.092); the full caution is in the note to Table E6.

Table E4: Effects on Urban Pop & Employment in 2010 (aggregated to urban centers)

	Population + Employment			Sectoral Employment Shares			
	(1) ln pop	(2) ln employed	(3) unemp rate	(4) agric %	(5) manuf %	(6) constr %	(7) serv %
TMV AgSim (z, 300min)	0.060 (0.059)	0.033 (0.056)	0.009 (0.007)	-0.053*** (0.018)	0.010** (0.005)	-0.002 (0.004)	0.058*** (0.018)
Transmigrant count (IDW)	✓	✓	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓	✓	✓
Mean dep var	11.839	10.930	0.123	0.374	0.046	0.060	0.471
Std dev dep var	0.858	0.843	0.041	0.184	0.042	0.027	0.146
Observations	133	133	133	133	133	133	133

Notes: Each column is a separate urban-center-level regression on the main city sample: the outer-island urban centers whose nearest transmigration village (TMV) lies within 300 travel minutes and whose 1980 baseline the census recorded directly. The regressor, TMV AgSim (z, 300min), is agro-similarity: the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, standardized over every urban center with a measured exposure. Every column includes the Main Spec controls: the inverse-distance-weighted (IDW) count of settlers placed within 300 minutes, the 1980 log population, log employment, urban population share, and log agricultural employment, and the sixteen agro-climatic covariates averaged over the same window. Standard errors in parentheses are Conley spatial standard errors with a 200-kilometer cutoff. Outcomes are 2010 census log population, log employment, the unemployment rate, and the share of each sector in employment (agriculture, manufacturing, construction, and services), aggregated from urban subdistricts to urban centers. *** 1%, ** 5%, * 10%.

Table E5: Urban Agricultural Goods Market

	Rice Output		Prices	
	(1) ln rice production	(2) rice surplus, tons	(3) rice price index	(4) agric price index
TMV AgSim (z, 300min)	0.702*** (0.252)	-4474.575 (4111.608)	0.044* (0.025)	0.018 (0.022)
Transmigrant count (IDW)	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓
Mean dep var	8.605	-3561.970	0.999	1.062
Std dev dep var	2.008	27562.000	0.146	0.122
Observations	127	107	128	133

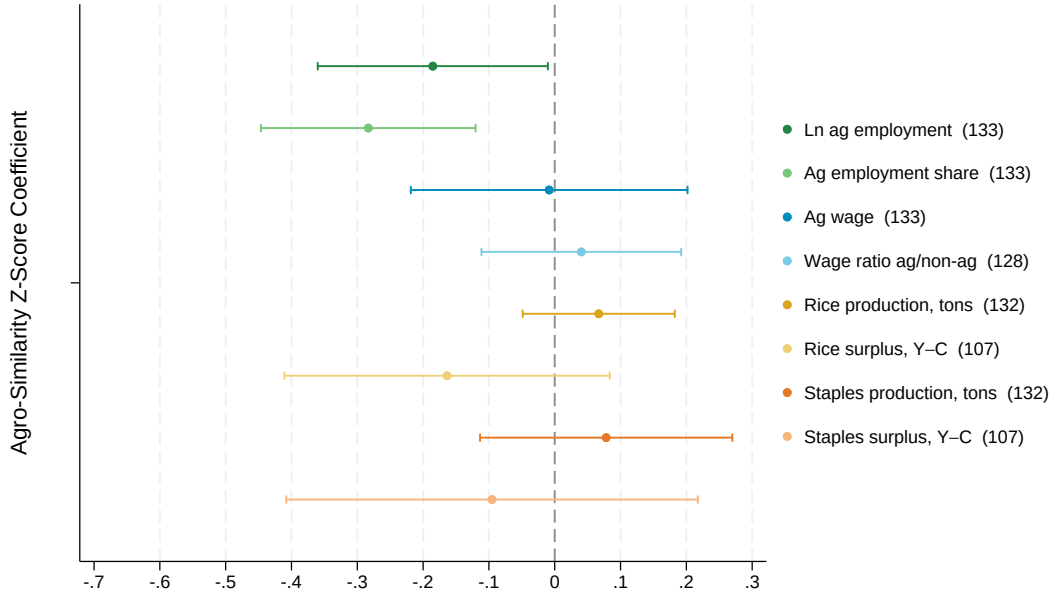
Notes: Each column is a separate urban-center-level regression on the main city sample: the outer-island urban centers whose nearest transmigration village (TMV) lies within 300 travel minutes and whose 1980 baseline the census recorded directly. The regressor, TMV AgSim (z, 300min), is agro-similarity: the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, standardized over every urban center with a measured exposure. Every column includes the Main Spec controls: the inverse-distance-weighted (IDW) count of settlers placed within 300 minutes, the 1980 log population, log employment, urban population share, and log agricultural employment, and the sixteen agro-climatic covariates averaged over the same window. Standard errors in parentheses are Conley spatial standard errors with a 200-kilometer cutoff. Columns 1 and 2 are rice production from the 2003 village census (PODES), in log tons, and the rice surplus, PODES 2003 production minus Susenas 2002 consumption, in tons. Columns 3 and 4 are the 2011 Susenas rice price index, the rice unit value over its median, and the agricultural price index, a Paasche index of the full food basket with urban-center expenditure weights. Survey outcomes are observed on the centers each source reaches, reported in the Observations row. *** 1%, ** 5%, * 10%.

Table E13 reports the 1996 food-price estimates in full. The basket indices are precisely estimated zeros under every weighting: the raw-food index is -0.007 (s.e. 0.011) with urban-center expenditure weights, -0.001 with national weights, and 0.001 equal-weighted, and the all-food index including prepared food is -0.005 . The one measure that moves is the rice unit value, at $+0.017$ (s.e. 0.011), significant only at 10%, and the magnitude is small. A one standard deviation increase in exposure raises the rice unit value by 1.7%, about 16 rupiah per kilogram at the 946 rupiah median, or 0.12 of the cross-city standard deviation of the log rice price (0.138), roughly a third of the corresponding per-capita expenditure response. These estimates matter particularly for timing. 1996 predates the 2004 rice-import ban, whose openness-differential price effects (Bazzi 2017) run through the later waves, so the 1996 cross-section is the one price measurement free of that shock, and it shows the employment reallocation of Table E12 already in place while food prices had not yet moved.

Table E14 adds the one price object the level tables cannot provide. Both the 1996 and the 2002 waves include the agricultural price index, so on the 95 urban centers that appear in both it can be differenced, and exposure can be asked whether it moved the price over the intervening period rather than only where the price stood at a point in time. It did not: the change is -0.009 (s.e. 0.013), a precisely estimated zero, and neither cross-section moves either (-0.014 in 1996, -0.023 in 2002). The agricultural price null is therefore a null in first differences as well as in levels, which is the stronger statement, since a common city-level price component that a single cross-section could not separate from exposure differences out here.

Table E15 asks whether even that small rice movement is a price or a quality phenomenon, exploiting the 1996 module's split of rice into a local and a high-quality variety. The price of the ordinary local staple, the variety that accounts for about three quarters of rice spending, does not respond ($+0.007$, s.e. 0.010): there is no within-variety price effect on the standard good. The pooled rise comes from the premium segment, where the within-variety unit value is $+0.047$ (s.e. 0.016, significant at 1%, on the 46 centers with enough high-quality buyers to measure it), with a positive but insignificant shift of spending toward the high-quality variety ($+3.5$ percentage points on a 26% base). The arithmetic

Figure E1: Urban Agricultural Sector: Goods & Labor Market



Notes: Standardized response of each urban-agriculture outcome to a one standard deviation increase in agro-similarity, each outcome z-scored over the 133 urban centers inside the 300-minute window, from the Main Spec reduced form with Conley 200 km spatial standard errors; 90% confidence intervals. Controls are the transmigrant settler count, 1980 ln population, 1980 ln employment, the 1980 urban-population share, 1980 ln agricultural employment, and the agro-climatic covariates averaged over the same catchment; no fixed effects. Marker color marks the outcome family: agricultural employment and wages (greens and navy), rice production and its surplus (gold), and staples production and its surplus (dark red). Sample sizes are in parentheses next to each row, from the full headline sample for the census-based outcomes down to 105 for the production-minus-consumption surplus rows, reflecting the narrower coverage of the Susenas and PODES sources. Sample restricted to urban centers whose nearest transmigration village lies within 300 cost-minutes.

closes. Weighting the two within-variety responses by their expenditure shares reproduces the pooled +0.017 almost exactly. The 1996 rice signal is therefore a premium-segment and composition phenomenon, not a movement in the price of the staple households actually buy, the same unit-value quality response documented for the later waves.

E.2 The In-Migration Required for the Agricultural-Share Fall

The identity. A city has L workers in 2010, a share s of them in agriculture. Suppose it takes in M extra workers, a share a of whom also work in agriculture, and suppose no incumbent changes sector. Its agricultural share becomes $(sL + aM)/(L + M)$. Writing the gross inflow rate as $g = M/L$, the fall in the share is $g(s - a)/(1 + g)$, so producing a fall of Δ requires

$$g(a) = \frac{\Delta}{s - a - \Delta}, \quad (8)$$

and a log employment response of $\ln(1 + g(a))$. Equation 8 has two properties the test relies on. The requirement rises steeply in a , because arrivals who farm dilute the share less. And it is infinite once $a \geq s - \Delta$: an inflow that agricultural cannot pull the average down that far at any size.

The base s is the comparison city's 2010 share, 0.374 on the estimation sample, not its 1980 share. Both coefficients are cross-sectional 2010 differentials, so the share the extra arrivals dilute is the share the comparison city has

Table E6: Modern Sector Split: Manufacturing and Services

	Employment (Ln)		Employment shares	
	(1)	(2)	(3)	(4)
	ln L manuf	ln L serv	manuf empl %	serv empl %
TMV AgSim (z, 300min)	0.250*	0.157**	0.010**	0.058***
	(0.133)	(0.066)	(0.005)	(0.018)
Transmigrant count (IDW)	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓
Mean dep var	7.542	10.134	0.046	0.471
Baseline empl share	0.064	0.320	0.064	0.320
Std dev dep var	1.357	0.970	0.042	0.146
Observations	133	133	133	133

The manufacturing and services components of the modern sector, reported here once; the paper's main results work with the modern sector overall (Table 2). Cols 1–2 are log sectoral employment in 2010; cols 3–4 are the corresponding employment shares (0–1 fractions of total employment). Each column is a separate urban-center-level regression on the main city sample: the outer-island urban centers whose nearest transmigration village (TMV) lies within 300 travel minutes and whose 1980 baseline the census recorded directly. The regressor, TMV AgSim (z, 300min), is agro-similarity: the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, standardized over every urban center with a measured exposure. Every column includes the Main Spec controls: the inverse-distance-weighted (IDW) count of settlers placed within 300 minutes, the 1980 log population, log employment, urban population share, and log agricultural employment, and the sixteen agro-climatic covariates averaged over the same window. Standard errors in parentheses are Conley spatial standard errors with a 200-kilometer cutoff, except in column 1, whose displayed standard error and significance stars are heteroskedasticity-robust. The reason is a property of the spatial kernel at that outcome: the manufacturing Conley value, 0.022, is about half that of every sibling log-employment column, while the robust standard error on the identical specification is 0.092, more than four times larger, where that same ratio sits between 1.2 and 1.5 for services, agriculture, total and modern employment; starred from the robust standard error, the coefficient is significant at the 5% level $p = 0.016$. *** 1%, ** 5%, * 10%.

when they arrive. Using the 1980 share instead would attribute the common 1980–2010 structural change to the inflow as well. The 1980 base is roughly 20 percentage points higher and lowers the requirement by more than a factor of three; I report it throughout as a conservative alternative.

Measuring the arrivals' sector mix. No census aggregate crosses migrant status with industry. I therefore build that cross from the individual records of the 1980, 2000 and 2010 censuses, aggregating person to village, then to year-2000 subdistrict through the harmonized crosswalk, then to the reclassified urban centers. Recomputing each city's overall agricultural share from these person records reproduces the published share to machine precision for the median city (correlation 0.999). Table E16 reports the resulting a under four migrant definitions in each wave.

The test. Both Δ and the employment response are estimated, so the threshold is itself a random variable and the test needs their joint covariance under the same spatial kernel. I therefore stack the two equations: each city contributes two rows, one for log employment and one for the agricultural share, every regressor is interacted with an equation dummy, and one Conley regression is run on the stack. The two rows of a city sit at distance zero, so the kernel returns the cross-equation covariance directly. The stacked fit is saturated and reproduces the two separate regressions exactly. The delta method then propagates the covariance through Equation 8. The resulting standard error on the gap is 0.382,

Table E7: Urban Employment by Service Type 2010 (aggregated to urban centers)

	All (Ln)	Service Types (Ln)			All (%)	Service Types (% of All Empl)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Ln Serv	Trade	Transport	Skilled	% Serv	Trade	Transport	Skilled
TMV AgSim (z, 300min)	0.157** (0.066)	0.123 (0.078)	0.147*** (0.045)	0.125* (0.066)	0.058*** (0.018)	0.013*** (0.005)	0.006** (0.003)	0.010* (0.005)
Transmigrant count (IDW)	✓	✓	✓	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓	✓	✓	✓
Mean dep var	10.134	9.134	7.885	8.443	0.471	0.172	0.053	0.087
Sector empl share (2010)	0.471	0.172	0.053	0.087				
Std dev dep var	0.970	1.003	1.062	0.918	0.146	0.051	0.024	0.028
Observations	133	133	133	133	133	133	133	133

Notes: Each column is a separate urban-center-level regression on the main city sample: the outer-island urban centers whose nearest transmigration village (TMV) lies within 300 travel minutes and whose 1980 baseline the census recorded directly. The regressor, TMV AgSim (z, 300min), is agro-similarity: the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, standardized over every urban center with a measured exposure. Every column includes the Main Spec controls: the inverse-distance-weighted (IDW) count of settlers placed within 300 minutes, the 1980 log population, log employment, urban population share, and log agricultural employment, and the sixteen agro-climatic covariates averaged over the same window. Standard errors in parentheses are Conley spatial standard errors with a 200-kilometer cutoff. Columns 1 to 4 are 2010 census log employment in services and in three service types (trade, transport, and skilled private services); columns 5 to 8 are the same four as shares of total employment across all sectors. The sector employment share row is the share of 2010 employment in the service type of each column. *** 1%, ** 5%, * 10%.

against 0.056 on the employment response, so the vertical distance between the two curves in Figure E2 understates the uncertainty.

In one city's workers. The average city in the sample has 91,821 workers, 34,315 of them in agriculture. Its high-exposure counterpart has 3,102 more workers and 30,468 agricultural workers. If the 3,102 arrivals farm at the measured 27.1%, they bring 841 farmers with them, so agriculture would have reached about 35,156 with nobody switching. It is at 30,468, so roughly 4,688 incumbents left agriculture. For arrivals and nothing else to have produced the observed fall, 96,979 workers would have had to arrive rather than 3,102, a shortfall of about 93,877 workers on a workforce of 91,821.

The extra workers are arrivals, not births. The identity treats the extra workers as people who moved in. Transmigrant placement ran through the early 1980s, so a child born to an incumbent in that decade is a prime-age worker by 2010, and a fertility response would place locally born workers in the inflow. Table E17 shows no such response. Splitting 2010 employment by birthplace, native-born employment does not move at all, and the whole of the point estimate sits on workers born elsewhere, whose employment rises by 17.3% while the migrant share of the workforce rises by 3.1 percentage points; neither is distinguishable from zero. Exposure moves neither children nor youth per native-born adult. Both ratios are computed inside the native-born population, so adult in-migrants inflate neither numerator nor denominator.

Robustness. Table E18 reports the test under the assumptions that could overturn it. Two are substantive. Exposure may shift the arrivals' own sector mix, which would lower a and with it the requirement; the test rejects at the high-exposure city's own value. Exposure is also imperfectly balanced on the 1980 agricultural share, so conditioning on that share turns Δ into a change rather than a level; the test still rejects.

Scope of the bound. The bound asks whether the *arrivals'* composition and nothing else could produce the fall, holding the incumbent workforce fixed. The birthplace decomposition of Section E.3 measures a wider object, because its composition margin also allows the locally born share of the workforce to fall. The two quantities are therefore not

Table E8: Agricultural production response to AgSim, by food-category aggregation (UC, 300min sample)

	Food (staples + legumes)			Non-food cash crops
	(1) Ln Staples	(2) Ln Rice	(3) Ln Legumes	(4) Ln Coffee+Tea+Spices
TMV AgSim (z, 300min)	0.897*** (0.253)	0.986*** (0.258)	0.720*** (0.234)	0.334 (0.504)
Transmigrant count (IDW)	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓
Mean dep var	8.820	8.286	3.647	3.328
Std dev dep var	2.146	2.550	2.519	2.949
Observations	132	132	132	132

Notes: Each column is a separate urban-center-level regression on the main city sample: the outer-island urban centers whose nearest transmigration village (TMV) lies within 300 travel minutes and whose 1980 baseline the census recorded directly. The regressor, TMV AgSim (z, 300min), is agro-similarity: the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, standardized over every urban center with a measured exposure. Every column includes the Main Spec controls: the inverse-distance-weighted (IDW) count of settlers placed within 300 minutes, the 1980 log population, log employment, urban population share, and log agricultural employment, and the sixteen agro-climatic covariates averaged over the same window. Standard errors in parentheses are Conley spatial standard errors with a 200-kilometer cutoff. Outcomes are log production, $\ln(x + 1)$ of tons, from the 2003 village census (PODES), aggregated from urban subdistricts to urban centers, by crop category. *Category contents:* (1) *Staples* = rice + corn + cassava + sweet potato + taro + other tubers (wheat and sago are left out, since neither is grown in measurable quantities in this sample). (2) *Rice* = paddy rice only, the crop of the first column of Table E5. (3) *Legumes* = peanut + soybean + mungbean. (4) *Coffee+Tea+Spices* = coffee + tea + pepper + nutmeg + clove, the beverage and condiment cash crops, grouped because consumption units differ across spices. *Excluded from all four columns:* the non-food cash, plantation, and industrial crops PODES reports that are not part of the food-trade aggregation: palm oil, cocoa, rubber, sugarcane, sugar palm, tobacco, cashew, cotton, cinnamon, vanilla, and jute. Coefficients on $\ln(x + 1)$ outcomes are approximate semi-elasticities. Survey outcomes are observed on the centers each source reaches, reported in the Observations row. *** 1%, ** 5%, * 10%.

nested, and both are small: the bound puts the arrivals-only channel at 6.4% of the fall using the lifetime stock and 10.1% using five-year movers, against 21% measured directly.

Two limitations remain. The requirement is computed on gross inflow with no offsetting outflow, so it bounds the required *net* response from above; the native-employment coefficient in Table E17 shows no such outflow. And s and a enter as measured constants. They are means over the same 132 cities and are estimated far more precisely than the coefficients, but they contain sampling error that the reported standard errors omit.

E.3 Decomposing the Agricultural-Share Fall: Construction

This appendix gives the construction behind Figure E3a: the identity the decomposition rests on, the data it needs, the calculation it replaces, and the checks at each step. The exhibits label the two margins by what they measure, the migrant share of the workforce and workers switching sector; in the text I use the literature's terms, composition and within, for the same two objects.

The identity

Index cities by c and partition each city's 2010 workforce into origin groups g . Let ω_{gc} be group g 's share of city c 's workers and a_{gc} that group's own agricultural employment share. The city agricultural share is then an exact accounting

Table E9: Agricultural production response to AgSim, standardized (UC, 300min sample)

	Food (staples + legumes), z-score			Non-food cash crops, z-score
	(1)	(2)	(3)	(4)
	Staples Prod	Rice Prod	Legumes Prod	Coffee+Tea+Spices Prod
TMV AgSim (z, 300min)	0.078 (0.117)	0.067 (0.070)	-0.045 (0.063)	-0.163** (0.081)
Transmigrant count (IDW)	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓
Mean dep var	0.009	0.008	0.007	0.004
Std dev dep var	1.009	1.009	1.010	1.011
Observations	132	132	132	132

Notes: Each column is a separate urban-center-level regression on the main city sample: the outer-island urban centers whose nearest transmigration village (TMV) lies within 300 travel minutes and whose 1980 baseline the census recorded directly. The regressor, TMV AgSim (z, 300min), is agro-similarity: the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, standardized over every urban center with a measured exposure. Every column includes the Main Spec controls: the inverse-distance-weighted (IDW) count of settlers placed within 300 minutes, the 1980 log population, log employment, urban population share, and log agricultural employment, and the sixteen agro-climatic covariates averaged over the same window. Standard errors in parentheses are Conley spatial standard errors with a 200-kilometer cutoff. Outcomes are the production categories of Table E8, in tons, each standardized over the in-window urban centers that report it, so a coefficient is the standard-deviation change in the outcome per standard deviation of agro-similarity. Survey outcomes are observed on the centers each source reaches, reported in the Observations row. *** 1%, ** 5%, * 10%.

Table E10: Food consumption and surplus response to AgSim, by category (UC, 300min sample)

	Log Susenas Consumption (kg per cap)			$\sinh^{-1}$ Surplus (PODES Prod - Susenas C	
	(1)	(2)	(3)	(4)	(5)
	Staples Cons 2002	Staples Cons 2011	Rice Cons 2002	Staples Surplus	Rice Surplus
TMV AgSim (z, 300min)	0.128*** (0.037)	-0.081*** (0.023)	0.109*** (0.037)	-0.638 (1.653)	-0.463 (2.159)
Transmigrant count (IDW)	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓
Mean dep var	4.768	4.435	4.653	-0.845	-2.437
Std dev dep var	0.205	0.185	0.210	9.934	9.381
Observations	107	131	107	107	107

Notes: Each column is a separate urban-center-level regression on the main city sample: the outer-island urban centers whose nearest transmigration village (TMV) lies within 300 travel minutes and whose 1980 baseline the census recorded directly. The regressor, TMV AgSim (z, 300min), is agro-similarity: the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, standardized over every urban center with a measured exposure. Every column includes the Main Spec controls: the inverse-distance-weighted (IDW) count of settlers placed within 300 minutes, the 1980 log population, log employment, urban population share, and log agricultural employment, and the sixteen agro-climatic covariates averaged over the same window. Standard errors in parentheses are Conley spatial standard errors with a 200-kilometer cutoff. The first three columns are the natural log of per-capita annual consumption, $\ln(x + 1)$ of kilograms, from Susenas 2002 and Susenas 2011; the Susenas 2011 column uses the wave aligned with the 2010 outcome year. The last two columns are the inverse hyperbolic sine of the subdistrict-aggregated surplus, production minus consumption in tons, which behaves like a log transform and is defined for the negative values that arise when local consumption exceeds local production. Coefficients on $\ln(x + 1)$ and inverse hyperbolic sine outcomes are approximate semi-elasticities. Survey outcomes are observed on the centers each source reaches, reported in the Observations row. *** 1%, ** 5%, * 10%.

Table E11: Food consumption and surplus response to AgSim, standardized (UC, 300min sample)

	Susenas Consumption (z-score)				Surplus (PODES Prod - Susenas Cons, z-score)	
	(1)	(2)	(3)	(4)	(5)	(6)
	Staples 2002	Staples 2011	Rice 2002	Rice 2011	Staples Surplus	Rice Surplus
TMV AgSim (z, 300min)	0.563*** (0.156)	-0.420*** (0.119)	0.456*** (0.173)	-0.490*** (0.089)	-0.095 (0.190)	-0.164 (0.150)
Transmigrant count (IDW)	✓	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓	✓
Mean dep var	0.009	-0.006	0.010	-0.004	-0.000	-0.005
Std dev dep var	1.007	0.995	1.006	0.994	1.009	1.007
Observations	107	131	107	131	107	107

Notes: Each column is a separate urban-center-level regression on the main city sample: the outer-island urban centers whose nearest transmigration village (TMV) lies within 300 travel minutes and whose 1980 baseline the census recorded directly. The regressor, TMV AgSim (z, 300min), is agro-similarity: the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, standardized over every urban center with a measured exposure. Every column includes the Main Spec controls: the inverse-distance-weighted (IDW) count of settlers placed within 300 minutes, the 1980 log population, log employment, urban population share, and log agricultural employment, and the sixteen agro-climatic covariates averaged over the same window. Standard errors in parentheses are Conley spatial standard errors with a 200-kilometer cutoff. Outcomes are the consumption (kilograms per capita) and surplus (tons) measures of Table E10, each standardized over the in-window urban centers that report it, so a coefficient is the standard-deviation change in the outcome per standard deviation of agro-similarity. Survey outcomes are observed on the centers each source reaches, reported in the Observations row. *** 1%, ** 5%, * 10%.

Table E12: AgSim on Susenas-1996 UC outcomes (Main Spec, Conley 200km)

	Ln rice	Ag idx	All-food	House idx	Ln rent	Ln pcExp	Food shr	Ag emp	Mod emp
TMV AgSim (z, 300min)	-0.008 (0.029)	-0.016 (0.024)	0.004 (0.025)	0.309 (0.244)	0.196 (0.143)	0.156* (0.087)	-0.029** (0.012)	0.044 (0.040)	-0.029 (0.040)
Outcome mean	6.84	1.06	1.07	1.55	9.84	11.13	0.65	0.44	0.47
N (UCs)	111	111	111	111	111	111	111	118	118

identity:

$$s_c = \sum_g \omega_{gc} a_{gc}. \quad (9)$$

Write each factor as its sample mean plus a deviation, $\omega_{gc} = \bar{\omega}_g + u_{gc}$ and $a_{gc} = \bar{a}_g + v_{gc}$, and Equation 9 becomes

$$s_c = \underbrace{\sum_g \bar{\omega}_g \bar{a}_g}_{\text{constant}} + \underbrace{\sum_g \bar{a}_g u_{gc}}_{\text{migrant share}} + \underbrace{\sum_g \bar{\omega}_g v_{gc}}_{\text{switching}} + \underbrace{\sum_g u_{gc} v_{gc}}_{\text{interaction}}. \quad (10)$$

The first term is the *composition* margin: it moves only because the local-versus-migrant makeup of the workforce changes, holding what each group does fixed. The second is the *within* margin. It moves only because groups change what they do, holding that makeup fixed. I construct the three right-hand terms as variables and regress each on the *same* Main Spec used for the total. Because Equation 10 is an identity and the regressions share a design matrix, the three coefficients sum to the coefficient on s_c exactly, so no residual category absorbs the difference. The adding-up holds numerically. The sum of the three minus the total is zero to ten decimal places under both partitions. Each component nonetheless has its own Conley standard error, so the split reports precision as well as magnitude.

Table E13: AgSim on 1996 food prices (Main Spec, Conley 200km)

	Raw-ag Paasche	Raw-ag Laspeyres	Raw-ag eq-wt	All-food	Ln rice price
TMV AgSim (z, 300min)	-0.016 (0.024)	-0.013 (0.024)	-0.015 (0.025)	0.004 (0.025)	-0.008 (0.029)
Transmigrant count	✓	✓	✓	✓	✓
Base ln pop 1980	✓	✓	✓	✓	✓
Base ln empl 1980	✓	✓	✓	✓	✓
Base ln agri empl 1980	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓
Outcome mean	1.057	1.041	1.039	1.069	6.839
Outcome SD	0.131	0.115	0.136	0.130	0.137
N (UCs)	111	111	111	111	111

Table E14: AgSim on the agricultural price index in levels and in change, 1996 to 2002 (Main Spec, Conley 200km)

	Ln price 1996	Ln price 2002	Change 1996 to 2002
TMV AgSim (z, 300min)	-0.003 (0.024)	0.032 (0.023)	0.035** (0.014)
<i>p</i> -value	0.900	0.161	0.010
Transmigrant count	✓	✓	✓
Base ln pop 1980	✓	✓	✓
Base ln empl 1980	✓	✓	✓
Base urban % 1980	✓	✓	✓
Base ln agric empl 1980	✓	✓	✓
Agro-climatic controls	✓	✓	✓
Observations	96	96	96

Notes: Each column is a separate urban-center-level regression on the main city sample restricted to the urban centers with an agricultural price index in both the 1996 and the 2002 Susenas waves, so the three columns are estimated on identical cities and only the outcome definition varies. The outcome is the log raw-food Paasche index with urban-center expenditure weights, the same index and specification as column 1 of [Table E13](#), which is estimated on every center with a 1996 price; the 1996 column here drops the centers without a 2002 price. The regressor, TMV AgSim (z, 300min), is agro-similarity: the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, standardized over every urban center with a measured exposure. Every column includes the Main Spec controls: the inverse-distance-weighted (IDW) count of settlers placed within 300 minutes, the 1980 log population, log employment, urban population share, and log agricultural employment, and the sixteen agro-climatic covariates averaged over the same window. Coefficients are in log points of the index, since only the regressor is standardized. Standard errors in parentheses are Conley spatial standard errors with a 200-kilometer cutoff. The *p*-values come from the same inference. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E15: The 1996 rice unit value: within-variety price vs quality composition

	Ln rice (pooled)	Ln local rice	Ln high-qual rice	HQ exp share	HQ buyer share
TMV AgSim (z, 300min)	-0.008 (0.029)	0.031* (0.017)	0.046 (0.075)	0.173*** (0.063)	0.163** (0.065)
Transmigrant count	✓	✓	✓	✓	✓
Base ln pop 1980	✓	✓	✓	✓	✓
Base ln empl 1980	✓	✓	✓	✓	✓
Base ln agri empl 1980	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓
Outcome mean	6.839	6.844	6.832	0.262	0.256
Outcome SD	0.137	0.130	0.152	0.292	0.284
N (UCs)	111	97	48	111	111

Table E16: Agricultural share of migrant employment, by definition and census wave

Definition of migrant	1980	2000	2010
Born outside the district	(not recorded)	0.330	0.271
Moved district within five years	(not recorded)	0.329	0.211
Born outside the province	0.398	0.341	0.281
Moved province within five years	0.359	0.319	0.215
Non-migrants, for comparison	—	—	0.404

Notes: Unweighted means over the 132 of the 133 estimation-sample urban centers that have a measured migrant-by-sector cross. SP1980 records province of birth only, so the district definitions begin in 2000; its counts are weighted by the 5% block-sample expansion factor. The headline uses the lifetime stock, 0.271 (Table E18).

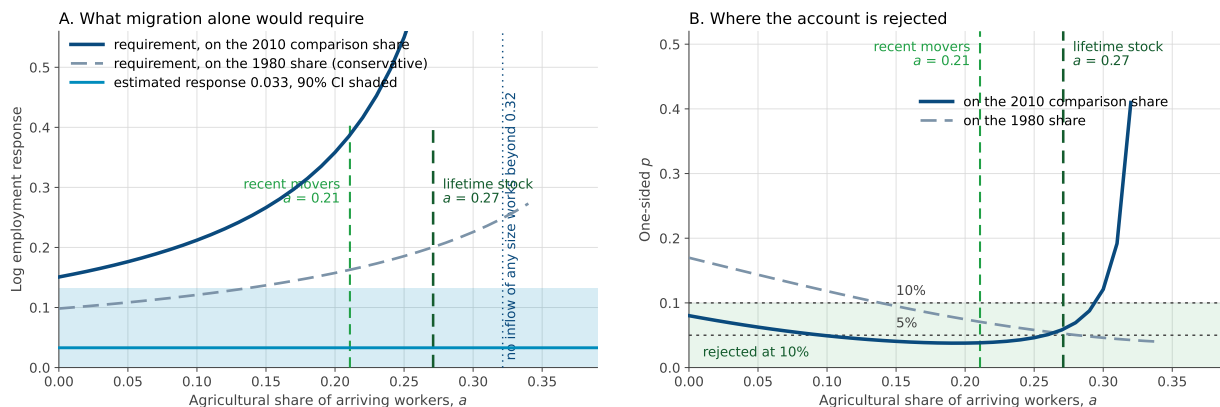
Two partitions are used. The first is birthplace, in the four mutually exclusive groups the migration tables already use: born in the current district, born in a different district of the same province, born in a different outer-island province, and born on Java or Bali. The second is arrival recency, in three groups: never left the birth district, lifetime migrant already present five years before the census, and lifetime migrant who arrived within five years. The second is the sharper measure of the marginal inflow, since the birthplace groups pool arrivals from thirty years ago with last year's.

The interpretation comes with one caveat. The census is a repeated cross-section, so “born in this district” in 2010 is not the same list of people as in 1980. The within term is therefore workers of a given origin *including whoever replaced whoever left*, not literally the same individuals moving between jobs. That is the same object Budi-Ors (2026) measures as the between-cohort margin, which is why his national 68% is the right benchmark for it.

The data the identity needs

Evaluating Equation 9 requires each city's workers classified by origin *and* industry in the same data. The published city file contains the birthplace shares and, quite separately, the industry counts, but nothing crossing them, so the cross is built from the 2010 individual census records for the outer islands (93.5 million persons). Three definitional choices align it with the published outcome:

Figure E2: What migration and nothing else would require, and where the account is rejected



Notes: $N = 133$ urban centers, Main Spec, Conley spatial standard errors at a 200-kilometer cutoff. The horizontal axis is a , the agricultural share of the arriving workers, swept across its plausible range. Left panel: the rising curve is $\ln(1 + g(a))$ from Equation 8 on the 2010 comparison share; the dashed curve repeats it on the 1980 share; the flat line and band are the estimated log employment response and its 90% interval. The vertical dotted line marks $a = s - \Delta$, beyond which no inflow of any size can produce the observed fall. Right panel: the one-sided p -value at each a , computed with the joint covariance of both coefficients, with the 10% and 5% lines drawn. The account is rejected at 10% for every a from zero up to 0.29, and at 5% from 0.10 up to 0.26; every value the 2010 census measures in Table E16 lies between 0.211 and 0.281, and the two 1980 rates sit beyond the asymptote, where no inflow of any size suffices.

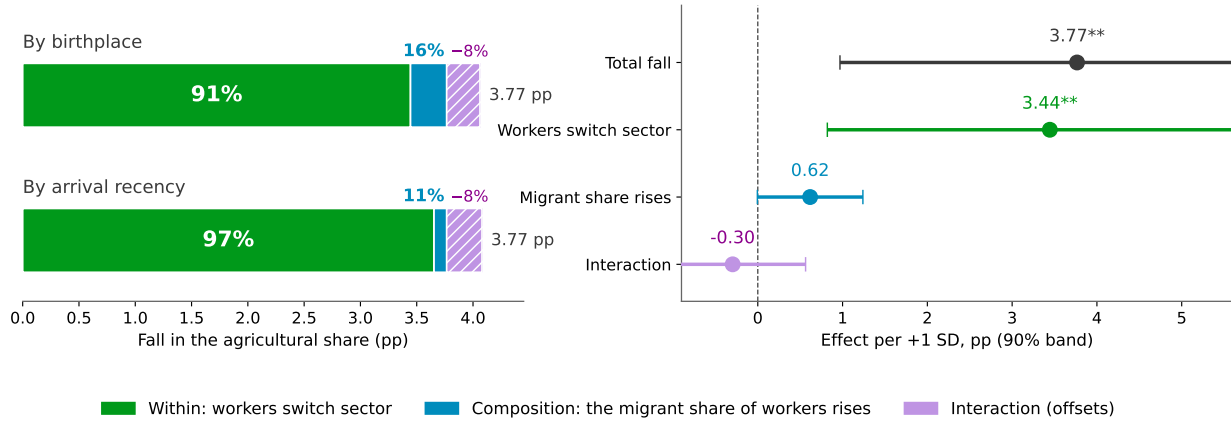
Table E17: The extra workers are in-migrants, not additional births

Outcome, 2010	Coefficient	SE	t
<i>Panel A. Where the employment response comes from</i>			
ln employment, all workers	0.0602	0.0574	1.05
ln employment, born in this district	0.0025	0.0441	0.06
ln employment, born elsewhere	0.1729	0.1409	1.23
Migrant share of employment	0.0307	0.0303	1.01
<i>Panel B. Fertility, measured inside the native-born population</i>			
Children 0–5 per native-born adult 16–50	0.0317	0.0287	1.10
Youth 5–16 per native-born adult 16–50	0.0263	0.0314	0.84

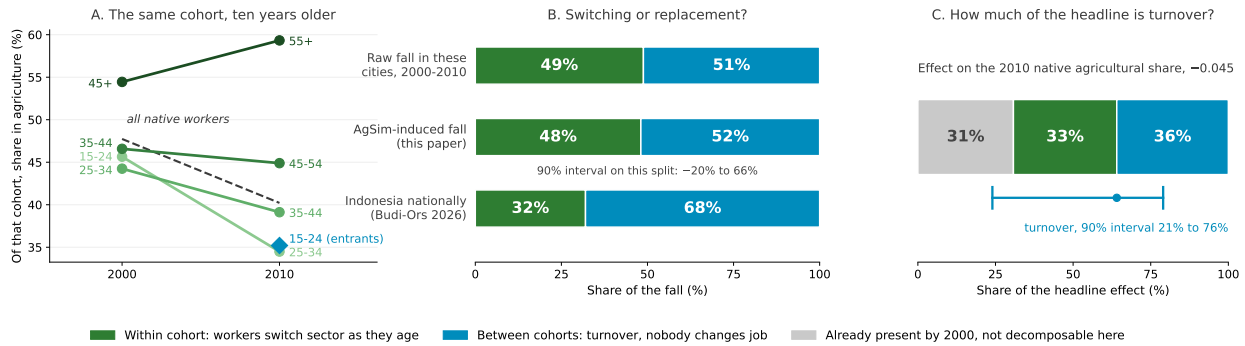
Notes: 132 urban centers, Main Spec, Conley spatial standard errors at a 200-kilometer cutoff. The sample is one short of the 133 the headline estimates on, because one center has no row in the migrant-by-sector cross. Native-born means born in the district of residence. Panel B's ratios are computed within the native-born population; the same ratios computed on everyone are smaller, which reflects adult in-migrants inflating the denominator. The pre-treatment 1980 wave shows the same weak positive on children per adult.

Figure E3: Decomposing the fall in the urban agricultural share, by origin and by cohort

(a) Decomposing the fall in the urban agricultural employment share



(b) Switching sector against one generation replacing another



Notes: Both panels run on the headline urban centers within 300 cost-minutes of a transmigration village, at the Main Spec, with Conley spatial standard errors at a 200-kilometer cutoff. Panel (a): each city's 2010 workforce is partitioned by birthplace (four groups) and, separately, by how recently a worker arrived (three groups). The stacked bars split the estimated fall into three pieces that add to it exactly by construction: the within component ("workers switch sector") holds each group's weight in the workforce at its sample mean and lets only the group agricultural shares move, the composition component ("migrant share of workers rises") holds the group agricultural shares at their sample means and lets only the weights move, and the interaction is the exact remainder. The same three components appear again with their 90% confidence intervals. Because the census is a repeated cross-section, the switching component is workers of a given origin including cohort replacement. Panel (b): the cohort comparison, the switching split and the turnover share are all measured on native-born workers, those enumerated in their district of birth, so that in-migration between the censuses cannot alter a cohort's composition. A birth cohort is followed one ten-year age band forward between the 2000 and 2010 censuses, and the two oldest 2000 bands are pooled so the destination band, 55 and over, holds exactly that cohort's survivors. The oldest cohort's rising farming rate is selective exit, since non-farmers stop working earlier, which is why the switching split reports an interval; its bars are normalized to 100% because the three totals are different objects, and the national benchmark decomposes Indonesia's aggregate fall while the exposure-induced bar decomposes only the part exposure produced. Exposure is a cross-section that is always on, so its effect on the 2010 level is its effect on the 2000 level plus its effect on the change, and two censuses can decompose only the second piece.

Table E18: The migration bound under alternative assumptions

	Required inflow	One-sided p
Headline: 2010 base, lifetime migrant stock, $a = 0.271$	105.6%	0.036
Five-year movers instead, $a = 0.211$	47.9%	0.026
a at the high-exposure city's own value, 0.224	54.6%	0.026
1980 agricultural share added as a control	101.7%	0.002
Both corrections together	53.1%	0.001
Conservative 1980 base, pre-treatment $a = 0.398$	48.8%	0.026
Conservative 1980 base, 2010 $a = 0.211$	17.9%	0.073

Notes: 133 urban centers, Main Spec, Conley spatial standard errors at a 200-kilometer cutoff, one-sided against the null that the employment response is at least as large as [Equation 8](#) requires, and each p-value cell is shaded by evidence strength. The last row combines the conservative 1980 base with the smaller of the two migrant measures, so it stacks both conservative choices; the test uses the 10% level throughout. The headline row is the weakest in the table, so the reported result is the most conservative reading here.

- The employment base is every person with a valid industry code (40.1 million workers), which is the denominator every sectoral share in this paper uses, rather than the stricter economically active count. Using the strict base would produce shares that do not sum to the published ones
- Agriculture is industry codes 1 through 6, exactly as the census cleaning defines the sectoral dummies elsewhere in the paper
- No birth-cohort restriction is imposed. The migration tables restrict to the 1955–1990 cohort in order to match 1980 birth-district characteristics; imposing that here would place the cross on a different population than the published agricultural share and break the reconciliation below. Birthplace is recorded for 100% of the workers in the base, so no group rests on a thin denominator

The person-level counts are collapsed to the 2010 village, crosswalked to the year-2000 subdistrict geography the rest of the project keys on, and rolled up to urban centers over the reclassified membership. Both partitions recover the total workforce exactly in every one of the 1,999 subdistricts, and [Equation 9](#) holds city by city to a maximum residual of 2.2×10^{-16} .

The re-measured share is checked against the published one before anything is split. Its mean across the estimation sample matches to three decimals, the two measures correlate at 1.000, and the Main Spec coefficient on the re-measured share reproduces the published -0.0309 (standard error 0.0093) at the same sample.

The required composition of arrivals

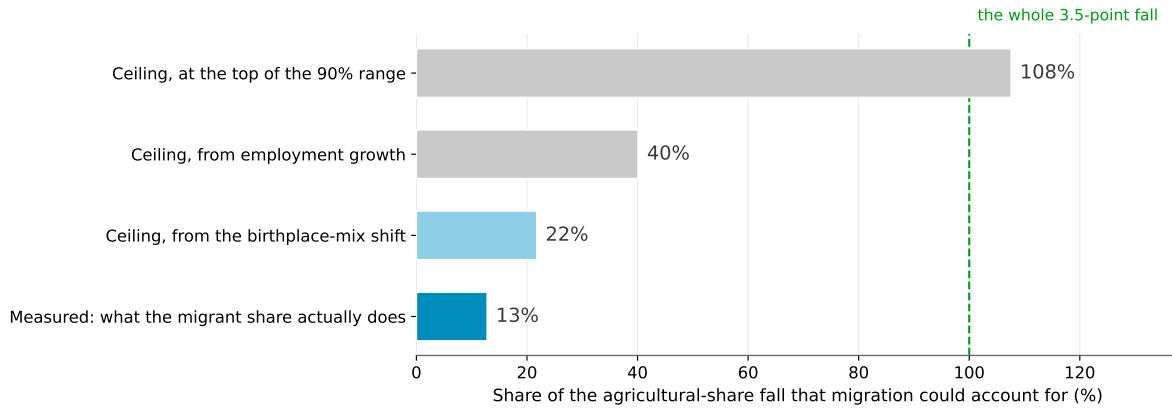
The question can be pushed some way without measuring the arrivals' sector at all. Suppose every worker the exposure adds to a city works outside agriculture, the most generous assumption available to the migration account. With a log employment response β_L , the induced arrivals are a share $1 - \exp(-\beta_L)$ of realized employment, and at a zero agricultural share among them the city share falls by that fraction times the baseline share. At the estimated $\beta_L = 0.085$ and a sample mean agricultural share of 0.381 this ceiling is 3.1 percentage points, slightly more than the estimated fall itself. An alternative version replaces the employment response with the estimated 3.8 percentage-point rise in the share of workers born outside the district and multiplies by the locally born agricultural share, giving 1.6 points, or 51%.

Neither ceiling settles the question, and the larger one does not even bound it: at 101% of the estimated fall it is consistent with arrivals accounting for the whole of it, and its 90% upper end reaches 150%. [Figure E4a](#) reports the ceilings against the measured composition margin.

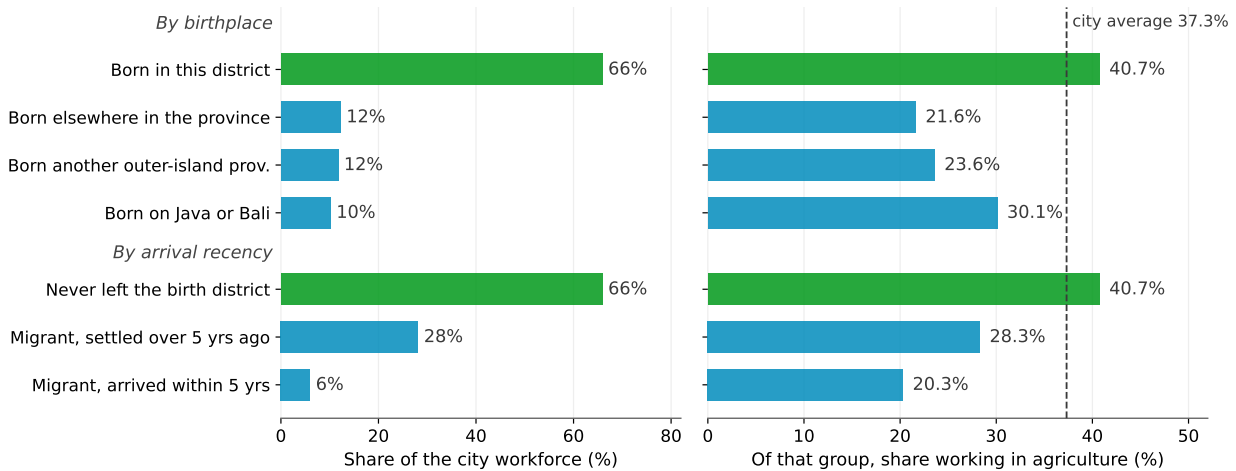
[Figure E4b](#) shows why the composition channel is present but small. Migrants are markedly less agricultural than the locally born, 20 to 30% against 41%, which makes a migrant-composition account plausible before it is measured. It stays small because the heavily agricultural group is two-thirds of the workforce and exposure raises the migrant share by only a few percentage points, which cannot produce the estimated 3.1-point fall.

Figure E4: The migration ceilings and the origin groups that keep the composition channel small

(a) The migration ceilings against the measured composition component



(b) Origin groups: how large each is, and how agricultural



Notes: Both panels run on the headline estimation sample. Panel (a): the three ceilings ask how much of the estimated agricultural-share fall migration could produce if every arrival worked outside agriculture, so they are maxima under that assumption. The bottom bar is the measured composition component of [Figure E3a](#), which replaces that assumption with the arrivals' observed agricultural share. Panel (b): means across the estimation-sample urban centers in 2010. The two blocks are two partitions of the same workforce, so the shares within each block sum to one.

Estimates and robustness

[Table E19](#) reports every component under both partitions, and [Table E20](#) collects the responses, baselines, ceilings, and measured margins in one place. Dropping the five cities that draw part of their workforce from a subdistrict shared with a neighbor (below) moves the within share from 74.3% to 72.8% at $N = 123$.

Table E19: Decomposing the fall in the urban agricultural employment share

	By birthplace				By arrival recency		
	(1) Total	(2) Migrant share	(3) Switching	(4) Interaction	(5) Migrant share	(6) Switching	(7) Interaction
TMV AgSim (z, 300min)	-0.0377** (0.0170)	-0.0062 (0.0038)	-0.0344** (0.0160)	0.0030 (0.0052)	-0.0043 (0.0033)	-0.0365** (0.0167)	0.0031 (0.0046)
Transmigrant count (IDW)	✓	✓	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓	✓	✓
Observations	133	133	133	133	133	133	133

Notes: Each column is a separate urban-center-level regression on the main city sample: the outer-island urban centers whose nearest transmigration village (TMV) lies within 300 travel minutes and whose 1980 baseline the census recorded directly. The regressor, TMV AgSim (z, 300min), is agro-similarity: the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, standardized over every urban center with a measured exposure. Every column includes the Main Spec controls: the inverse-distance-weighted (IDW) count of settlers placed within 300 minutes, the 1980 log population, log employment, urban population share, and log agricultural employment, and the sixteen agro-climatic covariates averaged over the same window. Standard errors in parentheses are Conley spatial standard errors with a 200-kilometer cutoff. The dependent variable in column 1 is each city's 2010 agricultural employment share measured on the birthplace-by-sector cross. Columns 2 to 4 split that share into three pieces that add to it exactly. Migrant share holds every birthplace group's own agricultural share at its sample mean and lets only the local-versus-migrant makeup of the workforce vary, so it is the composition component; Switching holds that makeup at its sample mean and lets only each group's own agricultural share vary, so it is the within component; Interaction is the exact remainder. Columns 5 to 7 repeat the split over arrival recency instead of birthplace, separating workers who never left their birth district, migrants already present five years before the census, and migrants who arrived within five years. Because the census is a repeated cross-section, Switching means workers of a given origin including cohort replacement, not the same individuals moving between sectors. *** 1%, ** 5%, * 10% significant.

The cross-sectional outcomes have no 1980 baseline, so they are identified off balance in levels. [Table E21](#) asks what that costs, by adding a 1980 variable that proxies for the unobserved baseline of the outcome and re-estimating on the same urban centers. Population density leaves the food-share estimate where it was; conditioning on 1980 schooling attenuates it by about a fifth and takes it below conventional significance.

Subdistricts claimed by two urban centers

Urban centers are satellite-drawn built-up polygons while the census reports by subdistrict, and five outer-island subdistricts are large enough to contain two separate built-up areas. The sample records both claims, and city totals built by summing constituent subdistricts therefore give the whole subdistrict to each of the two claimants, so those workers enter two cities. Across the estimation sample this affects 93,681 workers, or 0.77% of the total, and both claimants lie inside the sample in two of the five cases.

The exhibits here divide such a subdistrict rather than duplicating it. Each of its villages is assigned to the claimant whose built-up polygon covers it, or, for a village inside neither, to the nearer claimant; the subdistrict's counts are then split in proportion to the employment in each city's villages. Total mass is preserved by construction and checked against the village-level source, so no worker is counted twice and none is dropped. [Figure E5](#) maps the two in-sample cases. The correction is immaterial for the result, moving the within share from 74.0% to 74.2%, and the answer is equally insensitive to the splitting rule: dividing a subdistrict's rural remainder in the same proportion as its built-up cores, rather than by proximity, gives the same 74.2%.

Cohort replacement against individual switching

The within margin of [Equation 10](#) mixes two different economies. A worker may leave farming for a shop, or the shop worker may simply be the farmer's child, with nobody ever changing job. Both are structural transformation and both have the same sign, but they are different economics and a single cross-section cannot tell them apart. Two censuses ten

Table E20: Migrant composition as a bound on the agricultural-share fall

	Estimate	Share of the fall
<i>Panel A. Estimated responses per +1 SD of AgSim exposure</i>		
Agricultural employment share	-0.0483** (0.0189)	
Conley 200 km p-value	0.011	
Ln employment	0.0532 (0.0587)	
Conley 200 km p-value	0.364	
Share of workers born outside the district	0.0258 (0.0246)	
Conley 200 km p-value	0.295	
<i>Panel B. Baseline agricultural shares, sample means</i>		
All workers, 2010	0.373	
All workers, 1980	0.559	
Workers born in the district, 2010	0.407	
Workers who arrived within 5 years, 2010	0.203	
<i>Panel C. Maximum composition effect, arrivals assumed entirely non-agricultural</i>		
Inflow based, 2010 baseline	0.0193	40.1%
at the 90% upper end	0.0519	107.6%
on the 1980 baseline	0.0290	60.0%
Composition based, 2010 baseline	0.0105	21.8%
at the 90% upper end	0.0270	55.9%
<i>Panel D. Measured, replacing the assumption with the observed arrival mix</i>		
Migrant share, by birthplace	-0.0062	16.4%
Switching, by birthplace	-0.0344**	91.5%
Migrant share, by arrival recency	-0.0043	11.3%
Switching, by arrival recency	-0.0365**	97.0%

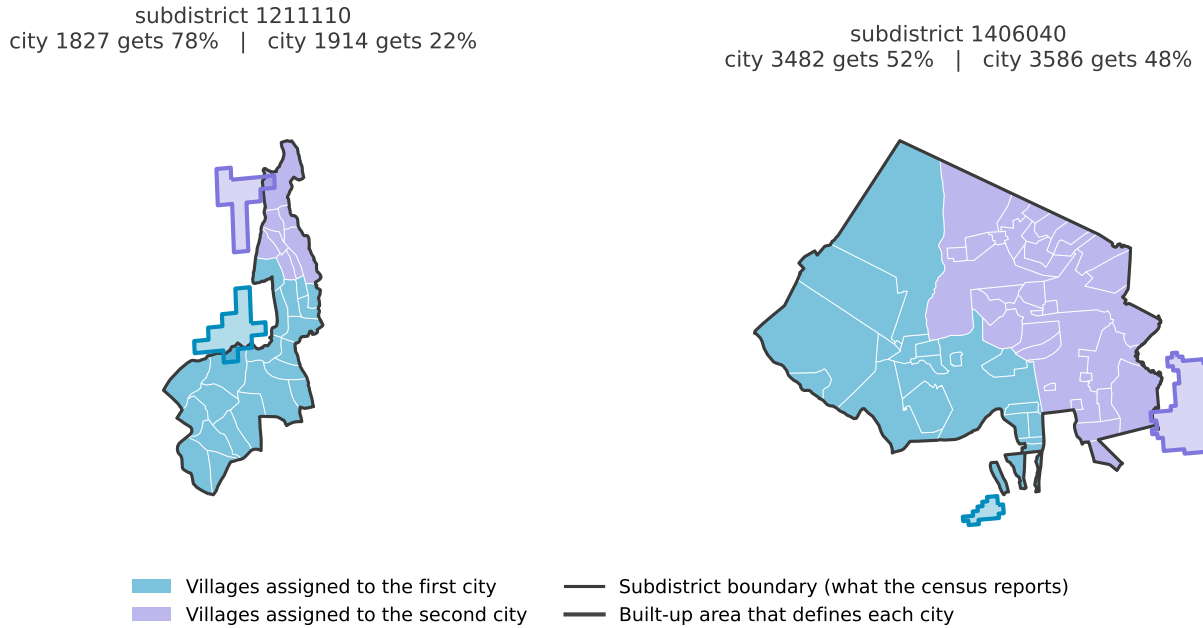
Notes: Each column is a separate urban-center-level regression on the main city sample: the outer-island urban centers whose nearest transmigration village (TMV) lies within 300 travel minutes and whose 1980 baseline the census recorded directly. The regressor, TMV AgSim (z, 300min), is agro-similarity: the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, standardized over every urban center with a measured exposure. Every column includes the Main Spec controls: the inverse-distance-weighted (IDW) count of settlers placed within 300 minutes, the 1980 log population, log employment, urban population share, and log agricultural employment, and the sixteen agro-climatic covariates averaged over the same window. Standard errors in parentheses are Conley spatial standard errors with a 200-kilometer cutoff. Stars come off the same inference as the reported standard error. Panel C asks how much of the estimated agricultural-share fall migrant composition could explain under the most generous possible assumption, that every marginal arrival works outside agriculture. The inflow-based bound multiplies the implied employment inflow, one minus exp of minus the log employment response, by the baseline agricultural share. The composition-based bound instead multiplies the estimated rise in the non-local share of the workforce by the agricultural share of locally born workers, so it does not require the arrivals to be net additions to the workforce. The operative ceiling is whichever of the two is larger, here the inflow based one. Panel D replaces the assumption with the arrivals' measured agricultural share and reports the decomposition of [Table E19](#). AgSim is agro-climatic similarity to nearby transmigrant villages, standardized.

Table E21: Baseline proxy controls and the cross-sectional food-share estimate

	(1)	(2)	(3)	(4)
	Population density	Agricultural employment share	Schooling	All three
Exposure, without the proxy	-0.0139 (0.0064)	-0.0134 (0.0061)	-0.0139 (0.0064)	-0.0139 (0.0064)
Exposure, with the proxy	-0.0131 (0.0061)	-0.0131 (0.0061)	-0.0138 (0.0065)	-0.0129 (0.0062)
Movement, in baseline SE units	0.13	0.04	0.02	0.16
Urban centers	130	133	130	130

Notes: The outcome is the household food share, the only cross-sectional outcome whose estimate is statistically distinguishable from zero at baseline; the remaining price, wage, and expenditure outcomes are indistinguishable from zero without any proxy control, so they cannot speak to this test. Each column adds one 1980 variable that proxies for the unobserved baseline of the outcome to the Main Spec. Every proxy is missing for a handful of cities, so the two rows of each column are estimated on the SAME urban centers, and the column's own without-proxy estimate is the comparison. Movement is reported in units of the without-proxy standard error. Standard errors in parentheses are Conley spatial standard errors with a 200-kilometer cutoff.

Figure E5: One subdistrict, two urban centers



Notes: The two cases in which both claimant cities lie inside the estimation sample. The charcoal outline is the subdistrict, the unit the census reports; the colored outlines are the two built-up polygons that define the cities; villages are shaded by the city each is assigned to. Percentages give each city's share of the subdistrict's 2010 employment under that assignment.

years apart can, because the age bands are ten years wide, so a birth cohort maps one band forward and its own farming rate is observed twice.

Everything here is measured on the *native-born*, workers enumerated in their district of birth. A cohort defined on all residents would be contaminated by in-migration between the waves, which is exactly the margin the birthplace decomposition above already handles. Four cohorts are followed: those aged 15–24, 25–34 and 35–44 in 2000, and, pooling the two oldest 2000 bands, those aged 45 and over. The pooling makes the mapping exact at the age boundaries, since everybody 45 or over in 2000 who is still working in 2010 is 55 or over and nobody younger is, so the destination band holds exactly that cohort's survivors. Workers aged 15–24 in 2010 have no 2000 age band; they are the entry margin, kept as a component rather than dropped.

Write w_g for a cohort's weight in native employment and s_g for its own agricultural share, so that the native agricultural share is again an identity, $s = \sum_g w_g s_g$. Differencing between the waves splits the change into

$$\Delta s = \underbrace{\sum_g \bar{w}_g \Delta s_g}_{\text{within cohort}} + \underbrace{\sum_g (\bar{s}_g - \bar{s}) \Delta w_g + w_N(s_N - \bar{s})}_{\text{between cohorts: reweighting plus entry}}, \quad (11)$$

with $\bar{w}_g$ and $\bar{s}_g$ the average of the two years, $\bar{s}$ the average of the two years' overall native shares, and N the entering cohort. The symmetric average is the weighting convention that does not leave a residual term. Subtracting $\bar{s}$ inside the between terms leaves the total unchanged, because the followed cohorts' weights must fall by exactly the entrants' weight, and it expresses each piece relative to the average worker. As with Equation 10, each component is built as a city-level outcome and run on the same Main Spec, so the coefficients sum to the coefficient on the total change exactly; the adding-up is asserted numerically at each of the three levels of the split, and the identity itself holds city by city to rounding error.

Two facts frame the estimates. First, cohorts do reallocate as they age, and they do it early: the cohort aged 15–24 in 2000 lowers its own farming rate by 11 percentage points over the decade, the next cohort by 5, the one after by 1, while the oldest cohort's rate *rises* by 5 as its weight falls by more than half. That last movement is selective exit, not workers taking up farming in their sixties, and it is the clearest sign that a synthetic cohort is not a panel. A cohort's 2010 members are the survivors of its 2000 members. Since those who exit are less agricultural than those who stay, the bias runs against the within margin. Second, the raw fall in these cities splits 48% within cohort and 52% between, so cohort replacement accounts for about half of what happened, which is the benchmark the exposure-induced fall is judged against.

Table E22 reports the components and Table E23 collects the descriptive split, the response, the accounting of the level effect, and the robustness set. Exposure lowers the native agricultural share by 0.0302 in 2010 and by 0.0159 in 2000, so it moves the ten-year change by 0.0142 ($p = 0.053$); of that change, 73.2% is within cohort (-0.0104) and 26.8% is turnover (-0.0038 , $p < 0.01$), of which entry is the larger part. Because exposure is a cross-section that is always on, the effect on the 2010 level is the effect on the 2000 level plus the effect on the change, so the level effect accounts three ways: 52.8% already present by 2000, 34.5% within-cohort switching, and 12.7% turnover. The transmigration placements ran from 1979 to 1988, so the timing predicts an effect largely in place by 2000, and the decade the exercise observes is the tail of the response rather than its onset.

The split's precision needs care, because a share is a ratio of two coefficients estimated on the same sample with the same regressors, so their sampling errors are correlated and an interval built from the two separate standard errors would ignore the covariance. I invert a test instead. For a candidate share λ , the outcome (between $-\lambda \times \text{total}$) run on the same specification returns the standard error of exactly that linear combination, covariance included, because least squares is linear in the dependent variable and the residual of a combination is the combination of the residuals; the confidence set is every λ the data do not reject. The accepted values are contiguous in every case rather than assumed to be. Against the ten-year change the resulting set is wide, 10 to 150% for turnover, because that denominator has

$p = 0.053$. Against the effect on the 2010 level, which has $p < 0.01$, it is tight: **turnover accounts for 6.5 to 30.0% of the estimated effect**. The data establish that ceiling rather than the within point estimate.

The answer does not turn on the construction. Base-year and end-year weighting conventions, each exact in its own right, give within shares of 78.5% and 67.9% against the symmetric 73.2%. Running the decomposition on all workers rather than the native-born gives 79.4% and on the migrant-only sample 87.3%, so the native-only version reported here is the conservative one. Dropping the five cities that draw part of their workforce from a shared subdistrict gives 72.1% at $N = 123$. One further variant: the cohort aged 15-24 in 2000 was partly still in education, so part of its measured switching is who joined the workforce rather than who changed job. Restricting to workers already 25 or over in 2000, and therefore 35 or over in 2010, removes both that margin and the entry margin and leaves a closed population that nobody enters from below. Within it the whole exposure-induced fall is within cohort, with the reweighting term contributing nothing.

Table E22: Switching against cohort replacement in the native agricultural-share fall

	Native ag. share, level		The 2000–2010 change, decomposed				
	(1) 2000	(2) 2010	(3) Change	(4) Within	(5) Between	(6) Reweighting	(7) Entry
TMV AgSim (z, 300min)	-0.0137 (0.0211)	-0.0446*** (0.0145)	-0.0309*** (0.0120)	-0.0148 (0.0105)	-0.0160*** (0.0034)	-0.0085*** (0.0022)	-0.0076*** (0.0018)
Transmigrant count (IDW)	✓	✓	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓	✓	✓
Observations	133	133	133	133	133	133	133

Notes: Each column is a separate urban-center-level regression on the main city sample: the outer-island urban centers whose nearest transmigration village (TMV) lies within 300 travel minutes and whose 1980 baseline the census recorded directly. The regressor, TMV AgSim (z, 300min), is agro-similarity: the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, standardized over every urban center with a measured exposure. Every column includes the Main Spec controls: the inverse-distance-weighted (IDW) count of settlers placed within 300 minutes, the 1980 log population, log employment, urban population share, and log agricultural employment, and the sixteen agro-climatic covariates averaged over the same window. Standard errors in parentheses are Conley spatial standard errors with a 200-kilometer cutoff. Every outcome is measured on native-born workers only, meaning those born in the district where they are enumerated, so that in-migration between the waves cannot contaminate a cohort. Columns 1 and 2 are the level of the native agricultural share in each census. Column 3 is the change between them, which equals column 2 minus column 1 coefficient for coefficient. Columns 4 to 7 split that change into pieces that add to it exactly: Within is each birth cohort's own farming rate changing as it ages, following a cohort one ten-year age band forward between the two censuses; Between is turnover, the sum of Reweighting, the followed cohorts changing relative size, and Entry, a new cohort of 15 to 24 year olds arriving at its own farming rate. Weights are the symmetric average of the two years, the convention that does not leave a residual term. A synthetic cohort is not a panel: its 2010 members are the survivors of its 2000 members, so exit through death, retirement or out-migration sits inside Within. *** 1%, ** 5%, * 10% significant.

E.4 Public Provision as an Alternative Account

The paper argues that agricultural productivity raised nearby city growth and reshaped urban employment through the goods market. A reader can propose a different chain. The Indonesian state settled these villages, and the same state built the roads, schools, health posts and electricity lines that serve them. If public provision followed the settlements into the places where they turned out productive, the urban response would reflect where the budget went rather than where the food came from. This subsection tests that account on the two forms of provision the data reach, the roads that carry the goods and the facilities the state puts in a village, and then on the allocation rule itself.

The evidence is asymmetric. Schools, health posts and road surface are provided by the government, so a null on them supports the market-forces account directly: it says the state did not follow agricultural productivity with infrastructure, and so cannot be the channel behind the urban results. A positive gradient would not establish the converse, because tax-financed spending follows growth through the revenue base and through population, so a city that grows for market reasons acquires facilities for that reason and no other. The allocation rule separates the two accounts,

Table E23: Switching against cohort replacement in the native agricultural-share fall

	Estimate	Share
<i>Panel A. What happened, 2000 to 2010, sample means</i>		
Native agricultural share, 2000	0.477	
Native agricultural share, 2010	0.402	
Change	-0.0752	100%
Within-cohort, workers switch sector	-0.0367	48.8%
Between-cohort, turnover	-0.0385	51.2%
of which reweighting	-0.0237	31.6%
of which entry	-0.0148	19.6%
<i>Panel B. The AgSim response per +1 SD, decomposed the same way</i>		
Change in the native agricultural share	-0.0309*** (0.0120)	100.0%
Conley 200 km p-value	0.010	
Within-cohort, workers switch sector	-0.0148 (0.0105)	48.1%
Conley 200 km p-value	0.156	
Between-cohort, turnover	-0.0160*** (0.0034)	51.9%
Conley 200 km p-value	0.000	
90% interval for the within-cohort share		[-20.0%, 66.5%]
<i>Panel C. The 2010 level effect, accounted for three ways</i>		
Effect on the native agricultural share, 2010	-0.0446	100%
Already present in 2000 and not decomposable here	-0.0137	30.8%
Within-cohort switching, 2000 to 2010	-0.0148	33.3%
Cohort turnover, 2000 to 2010	-0.0160	35.9%
90% interval for within-cohort switching		[-4.5%, 118.0%]
90% interval for cohort turnover		[21.0%, 76.0%]
<i>Panel D. Within-cohort share of the response, under alternatives</i>		
Headline: native-born, symmetric weights		48.1%
Base-year weights		45.3%
End-year weights		50.9%
Prime age only, 25+ in 2000, no entry margin		82.7%
All workers, native-born and in-migrants		15.2%
Dropping the shared-subdistrict cities		48.7%

Notes: Each column is a separate urban-center-level regression on the main city sample: the outer-island urban centers whose nearest transmigration village (TMV) lies within 300 travel minutes and whose 1980 baseline the census recorded directly. The regressor, TMV AgSim (z, 300min), is agro-similarity: the mean agro-climatic similarity of the transmigration villages within 300 travel minutes, standardized over every urban center with a measured exposure. Every column includes the Main Spec controls: the inverse-distance-weighted (IDW) count of settlers placed within 300 minutes, the 1980 log population, log employment, urban population share, and log agricultural employment, and the sixteen agro-climatic covariates averaged over the same window. Standard errors in parentheses are Conley spatial standard errors with a 200-kilometer cutoff. Stars come off the same inference as the reported standard error. All outcomes are measured on native-born workers, meaning those enumerated in their district of birth, so that in-migration between the two censuses cannot contaminate a cohort. A birth cohort is followed one ten-year age band forward between the 2000 and 2010 censuses; the two oldest 2000 bands are pooled so that the destination band, 55 and over, contains exactly that cohort's survivors. Within-cohort is a cohort's own farming rate changing as it ages. Between-cohort is turnover: the older and more agricultural cohorts shrinking, plus a new cohort of 15 to 24 year olds entering at its own farming

not the stock of facilities, which is why this subsection closes on a test built on a central-government program's own allocation label.

Roads. The trade channel runs on the road network, so the sharpest form of the objection is that exposed corridors received better roads. I test this with the Indonesian Road Management System panel assembled by [Gertler et al. \(2024\)](#), which gives road-quality-implied travel speeds on a fixed network in 1990 and 2007 for Sumatra and Sulawesi, the two outer islands the source covers. Exposed rural subdistricts are more remote and slightly more slowly connected at both dates, sitting 62.8 km farther out in the road network per standard deviation of exposure (standard error 21.7) and traveling 1.02 km/h slower in 1990 (standard error 0.42) and 0.89 km/h slower in 2007 (standard error 0.49), against a base of roughly 60 to 66 km/h. The change is where the alternative would have to show itself, and it is flat. Over 1990 to 2007 the differential in road speed is 0.13 km/h with a standard error of 0.59 across 533 subdistricts, or 0.02 standard deviations with a standard error of 0.08 in fully standardized form. That is a precise zero rather than an imprecise one, and it says that whatever made these corridors remote was fixed geography rather than an investment response.

That null has two limits. Network distance is identical to the last digit across 1990 and 2007 for every Sumatran and Sulawesi district in the source, so the measure re-weights a fixed graph by road quality and cannot detect a road that was newly built. And it covers two of the outer islands rather than all of them. The near-baseline evidence closes part of that gap, because the 1983 village census reports local infrastructure for every outer island a few years after settlement, and it is null on each outcome in fully standardized form: roads passable by a four-wheel vehicle at 0.087 (standard error 0.067), paved roads at 0.063 (standard error 0.059), and distance to the subdistrict seat at 0.054 (standard error 0.075), beside a placebo on 1980 population density at 0.033 (standard error 0.130). Exposed corridors held no early connectivity advantage for the 1990 window to have missed.

Facilities. [Table E24](#) turns to what the state puts inside a village, tracking city amenities from the 1980 village census to the 2003 and 2011 waves. Column (2) regresses the 1980 level itself on exposure, so the table says for itself whether public provision already differed before the settlements matured, and columns (3) and (4) give the change over each window.

Two facts come out of it. The 1980 levels are balanced: of the seven government-provided amenities, none shows a pre-program gradient except polyclinics, at 0.011 per standard deviation of exposure against a mean of 0.072. Whatever else exposed cities were in 1980, they were not places the state had already equipped differently. And the changes are null on every one of those seven amenities, over both windows, in schooling, in health facilities and in both road-surface measures. The one coefficient that crosses a conventional threshold is the senior-secondary rate in the first window, at -0.0055 and significant at the 5% level, which the next paragraph takes up.

Adding each outcome's own 1980 level, the test Panel D reports for electricity, was run on all seven and leaves that conclusion intact. Two coefficients move and neither rescues the state account. Primary schools per 1,000 people turn negative, at -0.045 by 2003 against a 1980 mean of 1.18, which is the denominator rather than schools being withdrawn: exposed cities grew about 7% faster in population over the same period, so a school count rising by less than population produces exactly this, and the implied stock of schools still grew. Polyclinics turn positive, at 0.0095 by 2003 against a 1980 mean of 0.072, in the first window only, which is the pattern the asymmetry described above anticipates, in which a larger and richer city acquires a clinic. Neither is an allocation rule.

The government-provided block reports 21 coefficients, so the two that cross conventional thresholds are about what chance produces, and the senior-secondary row is the clearest case: significant at the 5% level in the first window, absent in the second, and gone as soon as the baseline enters, at -0.0004 with a p -value of 0.92.

Electricity. Grid connection is where the change regressions do move, and the two margins of it separate cleanly once the 1980 level is in the table. On the extensive margin, the share of a city's population living in an electrified village rises by 4.3 percentage points per standard deviation of exposure over 1980 to 2003 and by 5.2 points over 1980 to 2011,

Table E24: Public provision and agricultural productivity: city amenities from the 1980 baseline

	Before the program		Change from the 1980 baseline	
	Mean in 1980	1980 level on exposure	1980 to 2003	1980 to 2011
	(1)	(2)	(3)	(4)
<i>Panel A. Government-provided: schooling and health, facilities per 1,000 people</i>				
Primary schools	1.187	-0.1234** (0.0599)	0.1129** (0.0449)	0.1084** (0.0552)
<i>p-value</i>		0.039	0.012	0.050
Junior-secondary schools	0.143	-0.0101 (0.0121)	0.0087 (0.0104)	-0.0142 (0.0175)
<i>p-value</i>		0.403	0.404	0.417
Senior-secondary schools	0.067	0.0037 (0.0079)	0.0009 (0.0083)	-0.0100 (0.0137)
<i>p-value</i>		0.637	0.912	0.465
Health centers (puskesmas)	0.084	-0.0039 (0.0155)	0.0074 (0.0155)	0.0165 (0.0164)
<i>p-value</i>		0.803	0.634	0.315
Polyclinics	0.072	0.0134** (0.0061)	-0.0029 (0.0079)	0.0350 (0.0343)
<i>p-value</i>		0.028	0.716	0.308
<i>Panel B. Government-provided: road surface, share of population</i>				
Asphalt main road	0.442	-0.1825*** (0.0693)	0.0987* (0.0575)	0.2123*** (0.0633)
<i>p-value</i>		0.008	0.086	0.001
Passable by four-wheel vehicle	0.806	-0.0133 (0.0160)	0.0224 (0.0168)	0.0330 (0.0291)
<i>p-value</i>		0.407	0.183	0.256
<i>Panel C. Mixed provision: grid electricity</i>				
Village connected to the grid	0.639	-0.0969** (0.0413)	0.1027** (0.0444)	0.0702* (0.0397)
<i>p-value</i>		0.019	0.021	0.077
Households connected, share of households	0.246	-0.0529* (0.0292)	0.0641 (0.0390)	0.0535 (0.0433)
<i>p-value</i>		0.069	0.101	0.217
<i>Panel D. Convergence check: the same electricity changes, controlling the 1980 level</i>				
Village connected to the grid			0.0059 (0.0108)	-0.0284*** (0.0041)
<i>p-value</i>			0.585	0.000
Households connected, share of households			0.0276 (0.0289)	0.0010 (0.0145)
<i>p-value</i>			0.340	0.944
Observations		119	118	116

Notes: Each row is a separate regression of the change in one amenity on standardized agro-climatic exposure, at the Urban Centre level on the estimation sample, under the Main Spec controls and Conley standard errors with a 200 km cutoff. Stars come off the same Conley inference as the standard error reported beneath each coefficient. Column (2) regresses the 1980 level itself on exposure, so a reader can see whether public provision already differed before the settlements arrived. Panel D repeats the two electricity changes with the outcome's own 1980 level added, which separates a genuine response from convergence on a place that started lower; that check was run on every outcome in the table and is reported here only for electricity, the one outcome whose 1980 level is imbalanced. Facility counts enter per 1,000 people at each year's own population, so a negative coefficient on a rate can reflect a growing denominator. Amenities are

both significant at the 1% level. That is catch-up rather than a response. Agriculturally favored places began 4.3 points less electrified, significant at the 1% level, so a change regression with no lagged outcome was measuring convergence, and adding the 1980 level, in Panel D, removes the result in both windows, leaving 0.002 and 0.015. The full run of village-census waves agrees: the gradient is negative in the earliest waves and statistically zero from the 1990s onward, so productive places were connected late and then caught up.

The intensive margin behaves differently and is the only result in the table that is not disposed of by either the baseline or the lag. The share of households actually connected starts balanced, at -1.4 percentage points per standard deviation with a standard error of 1.1, and rises by 3.4 points by 2003, holding at 2.6 points with the 1980 level added and significant at the 5% level. It does not persist: by 2011 the same coefficient is 1.6 with a standard error of 1.8. So the pattern is a modest deepening of connections inside already-connected places, visible in one window and gone in the next, on a base of about 25% of households connected in 1980.

That margin is also where the asymmetry matters most. Electrification sits in both camps, because the grid is a state utility while the decision to connect a household responds to demand and to ability to pay, which is why the table groups it separately from the schools and health posts. A place whose hinterland became more productive has households that can afford a connection, and that is the market-forces account rather than an alternative to it. Nothing here identifies an allocation rule, which the next test examines directly.

The allocation rule itself. A facility stock is the outcome of many decisions, so the sharpest available test examines a central-government program directly. Indonesia's INPRES primary-school construction program ran from 1973 to 1978 and had finished before the 1980 village census was taken (Duflo 2001). That census records INPRES-built schools, INPRES-paid teachers, INPRES-built school buildings and INPRES-built public latrines separately from everything else, so the program is identified by its own label, and the question becomes whether the center's flagship rural program had already favored agriculturally advantaged places by the start of my sample period.

Table E25: Central government programs at the 1980 baseline

	INPRES primary schooling				INPRES public latrines	Placebo: the schools INPRES did not build
	Share of schools (1)	Schools per 1,000 people (2)	Share of teachers (3)	Share of buildings (4)	Share of latrines (5)	Schools per 1,000 people (6)
Agro-climatic exposure (z)	0.0307 (0.0275)	0.0093 (0.0286)	0.0098 (0.0307)	0.0380 (0.0265)	0.0343 (0.0737)	-0.1328*
<i>p-value</i>	0.264	0.745	0.750	0.152	0.642	0.072
Mean of outcome	0.3360	0.3927	0.3249	0.3385	0.4145	0.7960
Observations	119	119	119	119	108	119

Notes: Each column is a separate regression of a 1980 baseline level on standardized agro-climatic exposure, at the Urban Centre level on the estimation sample, under the Main Spec controls and Conley standard errors with a 200 km cutoff. The outcomes are levels because the INPRES primary-school construction program ran from 1973 to 1978 and had already finished when the 1980 village census was taken, so the table tests whether the center's own building had already gone to agriculturally favored places. The village census records INPRES-built schools, INPRES-paid teachers, INPRES-built school buildings and INPRES-built public latrines separately from the rest, so the program is identified by its own label; columns (1), (3) and (4) are three independent questionnaire items measuring the same allocation. Column (5) contrasts INPRES with non-INPRES latrines within a single facility type, holding year and place fixed, and rests on the roughly one village in seven that has a public latrine at all. Column (6) is the placebo: the schools the program did not build, which are expected to behave like the ordinary facilities in Table E24. The sample is the estimation sample restricted to cities with 1980 village-census coverage, which excludes Aceh. Significance: * 10%, ** 5%, *** 1%.

It had not. [Table E25](#) reports six versions of the test and every one is null. The INPRES share of a city's primary schools rises by 0.011 per standard deviation of exposure against a mean of 0.338 (standard error 0.010), the INPRES share of primary-school teachers by 0.002 against a mean of 0.325, and the INPRES share of school buildings by 0.011 against a mean of 0.340, three separate questionnaire items that agree both in their null and in placing about a third of primary schooling under the program. Public latrines give the cleanest form of the contrast, since INPRES and non-INPRES latrines are counted separately within one facility type in one year and one place, and the INPRES share there is 0.023 against a mean of 0.410 (standard error 0.045). The placebo is null, with non-INPRES schools per 1,000 people at -0.025 (standard error 0.029). One qualification: only about one village in seven reports a public latrine at all, so the latrine column rests on a thinner base than the schooling columns.

The three tests point the same way. The state did not lay faster roads into exposed corridors, it did not fill exposed cities with facilities beyond what their own growth and their households' own means would draw, and its flagship school-building program had not favored those places before my sample period begins. The scope is real: the road evidence covers road quality on two islands rather than new construction everywhere, and a null is not proof of absence. The evidence rules out the specific chain the objection requires, in which public provision tracks agricultural potential closely enough to generate the urban results itself.

F The Model

This part derives the spatial equilibrium the main text summarizes, and reports its calibration, its channel decomposition, and the theoretical accounts the model is set against.

This appendix derives the two-sector Stone-Geary spatial equilibrium exactly as it is used in [Section 6](#): the von Thünen land split with the built-up land endowment, the closed land-rent rebate, the measured 1980 amenity, and a location choice that ranks Stone-Geary indirect utility. The main text displays the model's core equations; this appendix derives the full system behind them. The two sectors are agriculture (A), which produces a tradable food good using land and labor, and a modern non-agricultural composite (S), which produces a non-traded good using housing land and labor.

F.1 Model Details: Notation, Technology, and Choice Probabilities

This subsection states the parts of the model that [Section 6.1](#) summarizes in words: the notation, the expenditure shares, the production technology with the land market, and the two stages of the worker's choice.

Expenditure shares. With supernumerary income $\tilde{w}_{js} \equiv w_{js} + R_j - p_A^j \bar{a}$ defined as in [Section 6.1](#), the implied expenditure shares are

$$s_A^j = \frac{p_A^j c_A}{w_{js} + R_j} = \frac{p_A^j \bar{a} + \psi_A \tilde{w}_{js}}{w_{js} + R_j} \quad (12)$$

$$s_S^j = \frac{\psi_S \tilde{w}_{js}}{w_{js} + R_j} \quad (13)$$

Technology. Total labor $\bar{L}$ is fixed and each worker supplies one unit inelastically. Each location has a fixed land endowment $\bar{T}_j$, split between farming (T_{jA}) and housing ($T_{jH} = \bar{T}_j - T_{jA}$).

- **Agriculture:**

$$Y_{jA} = (A_j^A T_{jA})^{1-\beta} (L_{jA})^\beta, \quad 0 < \beta < 1 \quad (14)$$

Table F1: Key notation

Symbol	Interpretation
J	number of locations (calibration panel $J = 300$: the 128 urban centers of the calibration sample and 172 rural district remainders; Maluku and North Maluku excluded)
$\mathcal{S} = \{A, S\}$	sector set: Agriculture, Modern non-agriculture / Services (manufacturing folded into S)
L_{js}	labor employed in location j sector s
$N_j = \sum_s L_{js}$	total population in location j
$\bar{T}_j$	total land endowment in location j (crop area + built-up footprint)
T_{jA}, T_{jH}	land allocated to farming and to housing ($T_{jA} + T_{jH} = \bar{T}_j$)
$A_j^A, \bar{A}_j^S$	agricultural TFP; exogenous modern productivity scale (total modern TFP $\bar{A}_j^S N_j^\eta$)
$\bar{a}$	subsistence food requirement (Stone-Geary), $\bar{a} = 0.108$
ψ_A, ψ_S	expenditure shares on discretionary consumption ($\psi_A + \psi_S = 1$; $\psi_A = 0.550$)
β	labor share in agriculture ($\beta = 0.5$; land gets $1 - \beta$)
γ_S	labor share in services ($\gamma_S = 0.7$; housing land gets $1 - \gamma_S$)
η	agglomeration elasticity in services ($\eta = 0$ headline)
τ_{kj}	iceberg cost of shipping food from k to j , $\tau_{kj} = \max(d_{kj}, 1)^\delta$
δ	iceberg cost-distance elasticity (reported over the band $\delta \in [0.5, 0.8]$)
σ	CES / Armington substitution elasticity across food sources ($\sigma = 4$)
R_j	per-capita land-rent rebate to residents of j
$\tilde{w}_{js}$	supernumerary income (income net of the subsistence food bill)
Φ_L^j	measured 1980 location amenity (built-up, density, infrastructure)
$\varepsilon_{js}(i)$	idiosyncratic sector efficiency shock (scales the worker's effective wage), Fréchet with shape θ_S
$v_j(i)$	idiosyncratic location-preference shock, Fréchet with shape θ_L
θ_S, θ_L	Fréchet shape parameters (sector choice; location choice)
ψ_M	expenditure share on the catchment-traded modern good, two-tradable-sector variant only ($\psi_M = 0.082$, the Susenas-2011 traded-services share; see Section 6.3)
δ_S	distance decay on the catchment-traded modern good, two-tradable-sector variant only (measured at $\delta_S = 1.091$, s.e. 0.033, from IFLS service-flow gravity, Section B.3 ; solved at $\delta_S = 1.0$, the highest decay below the measured estimate at which the traded-service block exists)
P_j	ideal price index in location j
W_j	sector-inclusive value (option value of living in j)
$\pi_j, \pi_{s j}$	choice probabilities (location; sector conditional on location)

Constant returns in (T_{jA}, L_{jA}) jointly, diminishing returns to labor holding land fixed, so land scarcity is the congestion force in farming. A_j^A varies across space and is where the agro-climatic productivity shock from the transmigration program enters the model.

• **Modern / Services:**

$$Y_{jS} = \bar{A}_j^S N_j^\eta (T_{jH})^{1-\gamma_S} (L_{jS})^{\gamma_S}, \quad 0 < \gamma_S < 1 \quad (15)$$

where $\bar{A}_j^S$ is the exogenous modern productivity scale and the external term N_j^η , entering (15) exactly once, is the agglomeration externality. The agglomeration term enters only modern production, and it does not improve the model's fit. The headline sets $\eta = 0$, and the flat modern wage (F12) rests on the price and land terms (Section F.7).

Wages, land, and the rebate. Competitive factor markets price both inputs. Each sector's wage equals its marginal revenue product of labor (Section F.4, Section F.6), and under Cobb-Douglas, factor payments are revenue shares, so labor collects β of agricultural revenue and γ_S of modern revenue while land collects the rest. A single rental market allocates $\bar{T}_j$ across farming and housing by equalizing the per-hectare bids, so land shares follow value shares. Land goes to whichever use bids more per hectare, so farming's share of the land equals its share of the location's land value (the von Thünen split, Section F.8), and a booming city's agriculture can shrink through land, not only through labor. The endowment is measured as the PODES planted crop area plus the GHS built-up footprint, the housing land the satellite observes. Agricultural productivity is recovered on the crop base only, winsorized at the 1st and 99th percentiles and normalized by its median, while the split operates on the full $\bar{T}_j$. Using the built-up endowment removes a crop-area floor artifact at cities (about 40% of cities previously sat at the minimum-farmland floor, and none do under the built-up total) without diluting the agricultural base. This built-up $\bar{T}_j$ is the headline land input. A pure-crop-area endowment is retained as robustness (Section F.12).

Stage 2, sector choice within location j . A worker draws a sector-specific efficiency shock ε_{js} from Fréchet with shape θ_S . With scales equal to the sectoral wages, the sector-choice probability is

$$\pi_{A|j} = \frac{w_{jA}^{\theta_S}}{w_{jA}^{\theta_S} + w_{jS}^{\theta_S}} \quad (16)$$

so location agricultural employment is $L_{jA} = \pi_{A|j}N_j$ (and $L_{jS} = (1 - \pi_{A|j})N_j$). Ranking wages is utility-consistent here because the rebate and the price level are common across sectors within a location, so the wage-maximizing sector is also the utility-maximizing one (Section F.9). The sector-inclusive value $W_j = (w_{jA}^{\theta_S} + w_{jS}^{\theta_S})^{1/\theta_S}$ is the option value of living in j before sector shocks are drawn; its level normalization is stated in the same appendix.

Stage 1, location choice. A worker draws a location taste shock v_j from Fréchet with shape θ_L . The location stage ranks the Stone-Geary *indirect utility* (Section 6.1) evaluated at the location's expected income and scaled by the measured amenity. Writing the location value as $v_j = \Phi_L^j \tilde{w}_j / P_j$ with $\tilde{w}_j = W_j + R_j - p_A^j \bar{a}$, the location probability is

$$\pi_j = \frac{v_j^{\theta_L}}{\sum_{k=1}^J v_k^{\theta_L}}, \quad v_j = \Phi_L^j \frac{\tilde{w}_j}{P_j}, \quad (17)$$

where $\Phi_L^j = \exp(\kappa z_j^{\text{amen}})/\text{median}$ is the measured 1980 amenity (standardized built-up, population density, and PODES-1983 infrastructure, with $\kappa = 0.5$), and population and sectoral labor follow as $N_j = \bar{L} \pi_j$ and $L_{js} = N_j \pi_{s|j}$. The amenity anchors where people live (the cross-sectional level fit).

Trade costs. In the limit of large σ the CES aggregate (6) converges to the cheapest-supplier rule $p_A^j = \min_k \{\tau_{kj} m_k\}$. The cost-distance elasticity δ (how fast τ rises with travel time) is distinct from the trade elasticity $\sigma - 1 = 3$; the outer-island setting supports the upper half of the band $\delta \in [0.5, 0.8]$ on independent grounds, since inter-island shipping over weakly connected routes plausibly has a higher per-minute trade cost than a within-continent road network. What the posted-cost convention simplifies and what it omits is stated in Section F.5.

F.2 Urban-Center-Level Empirical Facts and Model Coverage

Table F2 presents the urban center-level empirical facts and their mapping to the model. The population, employment, and sectoral-composition block (F1-F6) is at the current headline (the complete 814-village rebuild), reported here from

the main-text [Table 2](#), which consolidates the same census regressions that Appendix [Table E.1](#) reports in a disaggregated layout. Most of the remaining Susenas- and PODES-based rows are likewise reported in the main text alongside the results they support; only the rows without a main-text table of their own (the 1996 pre-signal behind F9, the agricultural-goods outcomes behind F10, and the service-sector composition behind F20) point to [Table E.1](#), where their signs and significance are verified on the same sample. Sample sizes below 128 in those tables reflect the narrower Susenas and PODES coverage rather than a different sample.

Table F2: Empirical facts and model coverage

#	Fact	Coef (SE)	Source
F1	City population growth (1% sig)	0.086 (0.027)***	Table 2 col 1
F2	City employment growth (1% sig)	0.086 (0.028)***	Table 2 col 2
F3	Ag employment share ↓	−0.031 (0.009)***	Table 2 col 5
F4	Ln ag employment ↓ (ns)	−0.050 (0.042)	Table 2 col 4
F5	Ln modern employment ↑ (serv+manuf)	0.133 (0.031)***	Table 2 col 6
F6	Modern employment share ↑	0.028 (0.010)***	Table 2 col 7
F8	Non-food spending share ↑	Positive, sig.	Table E2 col 1
F9	Housing rent: 1996 ↑; 2011 flat	Pos., sig. (1996); ≈ 0, ns (2011)	Appendix Table E12 ; Table E2 col 1
F10	Rice surplus ↓ (production ns)	−9,589 (4,452)**	Appendix Table E5 col 2
F12	Non-ag wage: ns	≈ 0, ns	Table E2 col 3
F15	Food spending share ↓	Negative, sig.	Appendix Table B2 row m_7
F17	Village retains ag workers	Positive	Figure 9
F19	Migration from other outer-isl provinces ↑; Java/Bali ns	Positive, sig.; ns	Table 4 col 3
F20	Service growth concentrated in transport ↑	Positive, sig.	Appendix Table E7 col 3

The exact standardized (z -on- z) point estimates that the structural model is calibrated to match, together with the untargeted over-identifying moments, are reported in Appendix [Table B2](#).

F.3 Preferences, Demand, and Indirect Utility

A worker living in location j and working in sector s consumes food c_A and the modern good c_S with generalized Stone-Geary (Kongsamut-Rebelo-Xie) preferences:

$$U(c_A, c_S) = (c_A - \bar{a})^{\psi_A} c_S^{\psi_S}, \quad \psi_A + \psi_S = 1, \quad c_A \geq \bar{a}, \quad (18)$$

where $\bar{a} > 0$ is the subsistence food requirement (the single non-homotheticity) and ψ_A, ψ_S are the discretionary-expenditure weights. The first $\bar{a}$ units of food buy survival, not utility, so a poorer household spends a larger share on food and a richer one a larger share on the modern good.

Step 1: the budget. The worker earns the sectoral wage w_{js} and receives the per-capita land-rent rebate R_j (derived in [Section F.8](#)):

$$p_A^j c_A + p_S^j c_S = w_{js} + R_j, \quad (19)$$

with location-specific prices p_A^j (food) and p_S^j (modern good).

Step 2: the Lagrangian. Since $\ln$ is monotone, maximizing (18) is the same as maximizing $\psi_A \ln(c_A - \bar{a}) + \psi_S \ln c_S$. With multiplier λ on (19),

$$\mathcal{L} = \psi_A \ln(c_A - \bar{a}) + \psi_S \ln c_S - \lambda(p_A^j c_A + p_S^j c_S - w_{js} - R_j).$$

Step 3: first-order conditions.

$$\frac{\partial \mathcal{L}}{\partial c_A} = \frac{\psi_A}{c_A - \bar{a}} - \lambda p_A^j = 0, \quad \frac{\partial \mathcal{L}}{\partial c_S} = \frac{\psi_S}{c_S} - \lambda p_S^j = 0.$$

Step 4: solve. The two FOCs give $p_A^j(c_A - \bar{a}) = \psi_A/\lambda$ and $p_S^j c_S = \psi_S/\lambda$. Adding them and using the budget yields $1/\lambda = w_{js} + R_j - p_A^j \bar{a}$. Define this object, the income left after the subsistence food bill, as the **supernumerary income**

$$\boxed{\tilde{w}_{js} \equiv w_{js} + R_j - p_A^j \bar{a}} \quad (20)$$

and substitute back to obtain the Marshallian demands

$$c_A^* = \bar{a} + \frac{\psi_A}{p_A^j} \tilde{w}_{js}, \quad c_S^* = \frac{\psi_S}{p_S^j} \tilde{w}_{js}. \quad (21)$$

The household first buys the subsistence bundle $\bar{a}$, then splits the remaining income $\tilde{w}_{js}$ in fixed shares $\psi_A : \psi_S$. As income grows, the food expenditure share falls toward ψ_A from above (the Engel curve), which is the demand-side engine of structural transformation.

Step 5: indirect utility and the price index. Substituting (21) back into (18),

$$V_{js} = \underbrace{\psi_A^{\psi_A} \psi_S^{\psi_S}}_{\text{const.}} \times \frac{\tilde{w}_{js}}{P_j}, \quad P_j = (p_A^j)^{\psi_A} (p_S^j)^{\psi_S}. \quad (22)$$

The Cobb-Douglas constant multiplies every location's value equally and is dropped. So a location is attractive when supernumerary income is high relative to the price index over the supernumerary basket. Evaluated at the location-level expected income $\tilde{w}_j$ (Section F.9), this indirect utility is the object the location choice ranks: not the nominal wage, but $\tilde{w}_j/P_j$.

F.4 Agriculture: Wage and the Return to Land

Farming at j combines effective land $A_j^A T_{jA}$ (quality-adjusted hectares allocated to farming) with labor:

$$Y_{jA} = (A_j^A T_{jA})^{1-\beta} L_{jA}^\beta, \quad 0 < \beta < 1. \quad (23)$$

The function has constant returns in (T_{jA}, L_{jA}) jointly and diminishing returns to labor holding land fixed, so land scarcity is the congestion force in farming.

The wage (labor FOC). A competitive farmer takes p_A^j as given and hires labor until the value of the marginal product equals the wage:

$$w_{jA} = p_A^j \frac{\partial Y_{jA}}{\partial L_{jA}} = p_A^j \beta (A_j^A T_{jA})^{1-\beta} L_{jA}^{\beta-1}. \quad (24)$$

The land payment. Land collects the value of its marginal product times land used. With Cobb-Douglas, factor payments are revenue shares:

$$v_{jA} \equiv T_{jA} p_A^j \frac{\partial Y_{jA}}{\partial T_{jA}} = (1-\beta) p_A^j Y_{jA}. \quad (25)$$

Product exhaustion. Because (23) has constant returns, the two payments exactly exhaust revenue (Euler's theorem):

$$w_{jA}L_{jA} + v_{jA} = p_A^j Y_{jA}. \quad (26)$$

Nothing is left on the table, which lets the land-market accounting in Section F.8 close each location's budget.

F.5 Trade in Food: Cost-Distance Enters Here

The iceberg cost. Shipping one unit of food from k to j melts part of it away, at an iceberg factor that is a power law in the route-aware cost-distance d_{kj} (minutes):

$$\tau_{kj} = \max(d_{kj}, 1)^\delta, \quad \delta > 0. \quad (27)$$

The floor at one minute makes the own-location cost exactly $\tau_{jj} = 1$ (the diagonal of d is zero); the smallest off-diagonal distance in the sample is about five minutes, so the floor binds only own-location shipping. This is the within-location $k = j$, $\tau_{jj} \geq 1$ normalization that closes the delivered-price system: every location can always source food from itself at unit trade cost, so p_A^j is well defined at each j even when the location imports nothing.

The farm-gate supply price. Each origin k posts a farm-gate marginal cost obtained by inverting the labor marginal product in (23):

$$\text{mc}_k = \frac{1}{\partial Y_{kA} / \partial L_{kA}} = \frac{L_{kA}^{1-\beta}}{\beta (A_k^A T_{kA})^{1-\beta}}. \quad (28)$$

Remark (the omitted wage factor). A textbook marginal cost would be $w_{kA} \times 1/\text{MPL}_k$, the wage bill of the marginal unit. The supply price in (28) includes only $1/\text{MPL}_k$. In the self-sourcing limit the omission is a normalization. With $\tau_{jj} = 1$ and all food sourced at home, $p_A^j = \text{mc}_j$, so substituting (28) into the wage first-order condition (24) gives $w_{jA} = 1$. Away from that limit the suppressed factor is an endogenous, location-specific wage, one number per origin rather than a single numeraire. Combining (28) with (24) gives $\text{mc}_k = p_A^k / w_{kA}$, so the supply price falls in the origin's own agricultural wage at a given delivered index. General-equilibrium wage responses to the shock therefore do not reach export prices. The farm-gate price is a pure supply-side object, and cross-location differences in it are differences in inverse productivity: a productivity gain at k lowers mc_k , and with it the delivered price at every destination within reach. The food block is then a price-transmission system, and it does not pass the origin's wage into the Armington cost of traded food. The traded-services block of Section 6.3 does, pricing at the mill cost $\text{mc}_k^M = w_{kS} / A_k^M$. The solved food blocks do sit close to the self-sourcing limit in expenditure shares, which is where the convention is exact (see the openness paragraph below).

The delivered price (CES over origins). The delivered food price at j aggregates all origins' delivered costs with elasticity σ :

$$p_A^j = \left[\sum_k (\tau_{kj} \text{mc}_k)^{1-\sigma} \right]^{1/(1-\sigma)}, \quad \sigma > 1, \quad (29)$$

with import shares $\lambda_{kj} \propto (\tau_{kj} \text{mc}_k)^{1-\sigma}$. Two limits: as $\sigma \rightarrow \infty$ the CES aggregate converges to the cheapest-supplier rule $p_A^j = \min_k \{\tau_{kj} \text{mc}_k\}$; and with a single origin the delivered price is exactly $\tau \cdot \text{mc}$. Equation (29) is the smooth “buy from the cheapest place, shipping included” rule the solver needs for differentiability. This is the channel that transmits the agro-similarity shock: a productivity gain at transmigration-village hosts lowers their mc_k , lowering p_A^j at every location within shipping reach. The modern good is non-traded, so p_S^j is set by local clearing (Section F.10).

On the choice of the trade-cost band. The estimation reports δ over the band $[0.5, 0.8]$, spanning the range used in the quantitative-trade literature for cost-distance-based iceberg costs (Allen & Arkolakis 2014; Tombe & Zhu 2019); the estimated elasticities barely move across the band (Section 6.2), so no headline result turns on the point chosen. Raising δ toward 0.6 and above also moves the untargeted price and agricultural-output moments toward the data, which the outer-island setting supports on independent grounds: inter-island shipping over weakly connected routes plausibly has a higher per-minute trade cost than a within-continent road network.

Implied trade-cost levels and the openness of the solved equilibrium. With the diagonal pinned at $\tau_{jj} = 1$ and d measured in minutes, the level of the off-diagonal costs is set jointly by δ and by the unit-of-time convention: measuring d in hours instead would divide every off-diagonal distance by 60 and so rescale every τ_{kj} by $60^{-\delta}$ (about a factor of 26 at $\delta = 0.8$) relative to the fixed own-supply cost. The borrowed band $\delta \in [0.5, 0.8]$ therefore disciplines how fast the cost rises with travel time, not its level against the self-supply option. The implied levels are large. At the median in-window pair distance of 187 minutes, τ is 13.7 at $\delta = 0.5$ and 65.6 at $\delta = 0.8$, and at the 300-minute window edge, 17.3 and 95.9. The solved equilibrium is correspondingly close to food self-sufficiency in expenditure shares. At the featured $\delta = 0.8$ solves, under either land-market closure, the median city buys 99.9% of its food spending from its own location (mean 99.5%, 10th percentile 99.1%), the median rural unit 99.99%, and the expenditure-weighted trade cost cities actually pay is about 1.06. The trade channel of Section 6.3 accordingly operates through the delivered price, the response of p_A^j to nearby marginal costs, rather than through large traded quantities.

F.6 Services: Housing Land, Agglomeration, Wage

Services at j combine labor with housing land T_{jH} (the land the city occupies) and an agglomeration external effect, with $\bar{A}_j^S$ the exogenous modern productivity scale and N_j^η the external term, entering once:

$$Y_{jS} = \bar{A}_j^S N_j^\eta T_{jH}^{1-\gamma_S} L_{jS}^{\gamma_S}, \quad 0 < \gamma_S < 1, \quad \eta \geq 0. \quad (30)$$

The N_j^η term is external: an individual firm takes city size as given, so its private labor FOC contains γ_S only, and the land bid is the residual revenue share:

$$w_{jS} = p_S^j \gamma_S \bar{A}_j^S N_j^\eta T_{jH}^{1-\gamma_S} L_{jS}^{\gamma_S-1}, \quad v_{jS} = (1 - \gamma_S) p_S^j Y_{jS}, \quad (31)$$

and the private payments again exhaust revenue, $w_{jS} L_{jS} + v_{jS} = p_S^j Y_{jS}$. The service good is non-traded: it must be produced where it is consumed, which ties p_S^j to local demand in Section F.10.

F.7 The Flat Modern Wage With and Without Agglomeration

Decomposing the modern wage FOC (31) in city size, with the price and land responses explicit, gives

$$\frac{\partial \ln w_{jS}}{\partial \ln N_j} = \underbrace{\eta}_{\geq 0, \text{ agglom.}} + \underbrace{(\gamma_S - 1)}_{< 0, \text{ dim. ret.}} \cdot \underbrace{\frac{\partial \ln L_{jS}}{\partial \ln N_j}}_{\approx 1} + \underbrace{(1 - \gamma_S)}_{> 0} \cdot \underbrace{\frac{\partial \ln T_{jH}}{\partial \ln N_j}}_{> 0, \text{ land}} + \underbrace{\frac{\partial \ln p_S^j}{\partial \ln N_j}}_{> 0, \text{ price}} \quad (32)$$

The rising-price term and the housing-land term (a growing city bids land into housing through the von Thünen split, raising the marginal product of modern labor) push the wage up while diminishing returns push it down. At the headline $\eta = 0$ the rising-price and housing-land terms are the only offsetting forces; at $\eta > 0$ agglomeration adds a further offsetting force. In the featured solves the offset holds in levels: the cross-city standard deviation of the shock's modern-wage response is about 0.0003 log points, flat for any practical purpose, in line with the data's flat non-agricultural wage (F12: 0.006, s.e. 0.022, Table E2 col 3). In the standardized units the calibration reports its moments in, the model's modern-wage response slope is +0.16 under the closed rebate and +0.18 under the open one, against +0.02 in the data (Table B2), so the tiny wage movements the model does produce are more aligned with exposure than the data's are. The $\eta = 0.10$ sensitivity arm modestly improves the modern-employment moment. η is fixed outside the estimation.

F.8 The Land Market: von Thünen Split, Rebate, Built-up Endowment

One land market. The location's endowment $\bar{T}_j$ is allocated between farming (T_{jA}) and housing ($T_{jH} = \bar{T}_j - T_{jA}$) by a single rental market. Each use bids its average land value per hectare: farming bids v_{jA}/T_{jA} from (25), housing bids v_{jS}/T_{jH} from (31).

Rents equalize. If farming paid more per hectare, land would flow to farming, and vice versa. Setting the two bids equal, $v_{jA}/T_{jA} = v_{jS}/(\bar{T}_j - T_{jA})$, and solving,

$$T_{jA} = \bar{T}_j \frac{v_{jA}}{v_{jA} + v_{jS}}, \quad T_{jH} = \bar{T}_j - T_{jA}. \quad (33)$$

Land shares follow value shares: the sector generating the larger land value occupies the larger share of the location. This single equation is the von Thünen mechanism, and it is why a booming city's agriculture can shrink through land, not just through labor.

The closed rebate. Land income must accrue to someone. The headline (closed) variant gives each location's residents equal shares of their own location's land income:

$$R_j = \frac{v_{jA} + v_{jS}}{N_j}. \quad (34)$$

This closes the budget (all local value added becomes local income). Substituting $L_{jA} = \pi_{A|j} N_j$ into (25) shows the agricultural part of the rebate, v_{jA}/N_j , scales as $N_j^{\beta-1}$ at given prices, land, and sector shares, falling in city size, so the rebate is also a dispersion force. The open (channel-fix) variant replaces (34) with a national pool $r = \sum_j (v_{jA} + v_{jS}) / \sum_j N_j$.

The built-up land endowment. The endowment is measured as

$$\bar{T}_j = T_j^{\text{crop}} + T_j^{\text{built}}, \quad (35)$$

the PODES planted crop area plus the GHS built-up footprint (the housing land the satellite observes). Two rules keep the economics straight: agricultural productivity is recovered on the crop base only, $A_j^A = (L_{jA,0}/T_j^{\text{crop}})^\beta$, winsorized at the 1st and 99th percentiles across the J locations and normalized by its median (built-up is housing, not farmland), and the split (33) operates on the full $\bar{T}_j$. This built-up endowment removes the crop-area floor artifact at cities without diluting the agricultural base; a pure-crop-area endowment is the land robustness.

F.9 Worker Choice: the Two Fréchet Stages

A Fréchet primer. A random variable X with scale u is Fréchet with shape θ if $\Pr(X \leq x) = \exp(-(x/u)^{-\theta})$. Two properties make it the workhorse of discrete choice: it is max-stable, and the probability that draw s is the largest is $u_s^\theta / \sum_s u_s^\theta$. A larger θ means less idiosyncratic dispersion, so choices chase the scales u_s more aggressively.

Stage 2: the sector choice

After settling at j , a worker draws sector tastes and picks the sector with the higher taste-scaled wage. With scales $u_s = w_{js}$, direct integration of the Fréchet maximum gives

$$\pi_{A|j} = \frac{w_{jA}^{\theta_S}}{w_{jA}^{\theta_S} + w_{jS}^{\theta_S}}, \quad L_{jA} = \pi_{A|j} N_j, \quad L_{jS} = (1 - \pi_{A|j}) N_j. \quad (36)$$

The sector stage ranks utility, not income. The choice (36) uses wages, which is utility-consistent under a productivity-shock interpretation of the Fréchet draw: if the idiosyncratic term is labor efficiency, a worker's income is

$\varepsilon_s w_{js} + R_j$ and utility is $(\varepsilon_s w_{js} + R_j - p_A^j \bar{a})/P_j$; the additive rebate and the price level are common across sectors *within* a location, so the utility-maximizing sector is the wage-maximizing sector. The wage-ranked sector stage is therefore utility maximization with efficiency shocks, not income maximization.

The expected income of a worker at j , before sector tastes are drawn, is the Fréchet expected maximum

$$\mathbb{E}[\max_s w_{js} \varepsilon_s] = \Gamma\left(1 - \frac{1}{\theta_S}\right) \underbrace{(w_{jA}^{\theta_S} + w_{jS}^{\theta_S})^{1/\theta_S}}_{\equiv W_j}. \quad (37)$$

Remark (the Gamma factor). The deployed model uses the inclusive value W_j without the $\Gamma(1 - 1/\theta_S)$ factor. The factor is constant across locations, so the sector shares (36) are unaffected; the location shares are not exactly invariant to it, because W_j enters $\tilde{w}_j = W_j + R_j - p_A^j \bar{a}$ additively. Omitting it is a normalization of the expected-income level. Separately from this constant factor, W_j itself is not the head-count average wage. The income concept the solver deploys, and the location-varying wedge it opens against the wage bill, are stated in Section F.10. The factor diverges as $\theta_S \downarrow 1$. At the estimated $\theta_S = 1.043 > 1$ the expected maximum exists, but the estimate sits close enough to that boundary that the omitted level is large, $\Gamma(1 - 1/1.043) \approx 24$, so the normalization is quantitatively consequential. The nested-*logit* version (Gumbel taste shocks in log utilities, which accommodate any $\theta_S > 0$ and give the same sector shares) is therefore retained as a robustness reframe. The shares (36) are well defined at any $\theta_S > 0$.

Stage 1: the location choice

The location stage ranks the indirect utility (22) evaluated at the location's expected income and scaled by the measured amenity: the location value is

$$v_j = \Phi_L^j \frac{\tilde{w}_j}{P_j}, \quad \tilde{w}_j = W_j + R_j - p_A^j \bar{a}, \quad \Phi_L^j = \frac{\exp(\kappa z_j^{\text{amen}})}{\text{median}}, \quad (38)$$

where z^{amen} is the standardized 1980 amenity index (GHS built-up, population density, PODES-1983 infrastructure) and κ scales its dispersion. The location choice therefore ranks Stone-Geary indirect utility $\tilde{w}_j/P_j$, not the nominal wage.

A worker draws a location taste shock $v_j \sim \text{Gumbel}(0, 1/\theta_L)$ (equivalently a Fréchet location value), so realized location utility is $\ln \tilde{U}_j = \ln v_j + v_j$. The probability of choosing j is $\pi_j = \Pr[\ln \tilde{U}_j \geq \ln \tilde{U}_k \forall k]$. Conditioning on $v_j = u$ and integrating,

$$\pi_j = \int_{-\infty}^{\infty} \prod_{k \neq j} F_v(u + \ln v_j - \ln v_k) f_v(u) du, \quad (39)$$

with $F_v(x) = \exp[-e^{-\theta_L x}]$ and $f_v(x) = \theta_L e^{-\theta_L x} \exp[-e^{-\theta_L x}]$. Defining $S_j \equiv \sum_{k \neq j} (v_k/v_j)^{\theta_L}$, the integrand collapses to $\exp[-e^{-\theta_L u} (1 + S_j)] \cdot \theta_L e^{-\theta_L u}$. The change of variables $z = e^{-\theta_L u}$ (so $dz = -\theta_L e^{-\theta_L u} du$) turns (39) into an elementary integral,

$$\pi_j = \int_0^{\infty} e^{-z(1+S_j)} dz = \frac{1}{1+S_j} = \frac{v_j^{\theta_L}}{\sum_k v_k^{\theta_L}},$$

so the location probability is

$$\pi_j = \frac{v_j^{\theta_L}}{\sum_{k=1}^J v_k^{\theta_L}}, \quad N_j = \bar{L} \pi_j. \quad (40)$$

The costly-migration generalization. The migration-cost extension replaces (40) with an origin-destination allocation over the 1980 origin population, $\pi_{od} \propto (v_d d_{od}^{-\nu})^{\theta_L}$, with the composite $\theta_L \nu = 1.865$ pinned by the IFLS migration gravity. At $\nu = 0$ it collapses exactly to (40). It is a gated robustness, not part of the headline.

F.10 Market Clearing, the Value Balance, and Equilibrium

Modern-market clearing. The modern good is non-traded, so local demand equals local supply. Evaluating the demand (21) at the location-level income $\tilde{w}_j$ for each of the N_j residents,

$$p_S^j Y_{jS} = N_j \psi_S \tilde{w}_j \iff p_S^j = \frac{N_j \psi_S \tilde{w}_j}{Y_{jS}}. \quad (41)$$

The right-hand form is the solver's price update. Modern prices are high where many people with high supernumerary income chase little modern output: the cost-of-living force that disperses population.

The deployed income concept, and the value balance it does and does not satisfy. Every demand-side equation above spends the same income: per-worker income in the modern-clearing condition (41), in the location value (38), and in the subsistence bill is $W_j + R_j$, the sector-inclusive value plus the per-head rebate, while production and the mapping to census employment use head counts $L_{jS} = \pi_{s|j} N_j$. These are two different aggregates of the same wages. Factor payments add up to the head-count wage bill: with $\bar{w}_j \equiv \pi_{A|j} w_{jA} + (1 - \pi_{A|j}) w_{jS}$ the average wage per head, $N_j \bar{w}_j + v_{jA} + v_{jS} = p_A^j Y_{jA} + p_S^j Y_{jS}$, and wages and land payments stay local under the closed rebate. Under that wage-bill income the food account closes exactly. Spending $\bar{w}_j + R_j$ per worker and subtracting modern clearing leaves

$$p_A^j Y_{jA} = p_A^j C_{jA}, \quad (42)$$

the value of food produced equal to the value of food consumed at every location. The deployed system does not satisfy (42), because it spends $W_j = (w_{jA}^{\theta_S} + w_{jS}^{\theta_S})^{1/\theta_S}$ per worker, and by the power-mean inequality $W_j > \bar{w}_j$ whenever both sectors are active, at any $\theta_S > 0$, with a gap that varies across locations with the sectoral mix. Measured food spending therefore exceeds the value of local food output by exactly the option-value wedge $N_j(W_j - \bar{w}_j)$, and the wedge is large at the featured solves: the median ratio $W_j/\bar{w}_j$ is 1.91 under the closed rebate and 1.60 under the open one, so in aggregate the wedge is 89% of the head-count wage bill closed and 59% open. Consistently with this, the equilibrium conditions are exactly the list below, the food block determines the delivered index (29) from origin marginal costs, and food quantities are demand-determined at that index. What the channel argument actually uses is weaker and survives. Under the closed rebate every argument of location j 's demand, the wages inside W_j and the rents inside R_j , is generated at j , so no location holds a claim on another location's output with which to finance a persistent net food purchase. Solved, that local-financing property binds. The baseline two-sector model's trade-only leg on the agricultural share is wrong-signed and cities move out of agriculture through migration, and the resolutions in Section 6.3 work precisely by creating a cross-location claim, modern exports under the second tradable sector, or a national land-income claim under the open rebate.

Equilibrium. Given fundamentals $\{A_j^A, \bar{A}_j^S, \Phi_L^j, \bar{T}_j, d\}$ and parameters $\{\beta, \gamma_S, \eta, \psi_A, \bar{a}, \delta, \sigma, \kappa, \theta_L, \theta_S\}$, an equilibrium is a vector $\{N_j, L_{jA}, L_{jS}, T_{jA}, p_A^j, p_S^j, w_{jA}, w_{jS}, R_j\}$ such that: (1) wages satisfy (24) and (31); (2) food prices satisfy (29) with costs (27) and (28); (3) the land split (33) and rebate (34) hold; (4) sector employment satisfies (36) with $L_{jA} + L_{jS} = N_j$; (5) modern markets clear location by location, (41); (6) population satisfies (40) with $\sum_j N_j = \bar{L}$; and (7) the numeraire is the reference city's modern wage, $w_{j^*S} = 1$ (j^* the median-population urban unit). Conditions 1–4 solve as an inner fixed point at given (N_j, p_S^j) ; conditions 5–6 are the outer fixed point in $(\ln N, \ln p_S)$.

The first-order comparative static behind the main text. Differentiating the sector share (36) in the log wage ratio gives $d\pi_{A|j} = \theta_S \pi_{A|j} (1 - \pi_{A|j}) d\ln(w_{jA}/w_{jS})$. At fixed local inputs the agricultural wage (24) is proportional to the delivered food price, so $d\ln(w_{jA}/w_{jS}) = d\ln p_A^j$; the CES price (29) gives $\partial \ln p_A^j / \partial \ln mc_v = s_{vj}$, the expenditure share of origin v in city j 's sourcing; and the marginal cost (28) gives $\partial \ln mc_v / \partial \ln A_v^A = -(1 - \beta)$. Chaining

the three yields the main text's (7). The expression holds the city's own labor, land, and modern price fixed, so it is the impact response. The solved general equilibrium adds the adjustment of all three.

F.11 From Equilibrium Conditions to Estimating Equations

The equilibrium conditions map directly to the main empirical specifications.¹⁸

Sectoral employment share. From the sector-choice condition (36) and the equilibrium wages, the agricultural employment share falls with a lower delivered food price, motivating

$$\underbrace{\text{empl_ag_pct}_j}_{\approx L_{jA}/N_j} = \beta_0 + \beta_1 \underbrace{\text{AgSim}_j}_{\propto A_j^{A,\text{nearby}}} + \beta_2 \text{TM_count}_j + \beta_3 X_j + \varepsilon_j \quad (43)$$

The model predicts $\beta_1 < 0$: higher agro-similarity in nearby villages lowers p_A^j (through trade), lowers w_{jA}/w_{jS} , and lowers L_{jA}/L_{jS} (matching **F3**).

Population. From the location-choice condition (40), $\ln \text{pop}_j = \beta_0 + \beta_1 \text{AgSim}_j + \beta_2 \text{TM_count}_j + \beta_3 X_j + \varepsilon_j$. The coefficient β_1 captures how agro-similarity raises the city's real-income index v_j ; the model predicts it positive but modest when θ_L is low (high migration frictions), matching **F1**.

Modern-good price. From the modern-clearing condition (41), $\ln \text{housing_rent}_j = \beta_0 + \beta_1 \text{AgSim}_j + \beta_2 \text{TM_count}_j + \beta_3 X_j + \varepsilon_j$, with $\beta_1 > 0$: a prediction the 1996 rents confirm and the 2011 rent response, indistinguishable from zero, leaves untested at endline (**F9**).

Modern wage. From (31), the effect of agro-similarity on w_{jS} is ambiguous: p_S^j rises (up), L_{jS} rises with $\gamma_S < 1$ (down, diminishing returns), housing land T_{jH} expands (up, the land term), and at $\eta > 0$ agglomeration adds a further offset. If these roughly cancel, the modern-wage effect is small and insignificant, matching **F12** (Section F.7 decomposes the offset).

F.12 Calibration Detail: Targets, Headline Estimate, Robustness, and Fit

This subsection holds the calibration detail behind Section 6.2: the recovered fundamentals, the estimator's mechanics, the targets, the headline estimate with its robustness, the fit on the untargeted moments, and the 1980-baseline forward solve.

City fundamentals. A household evaluates the amenity for any given location relative to the other locations, once wages are taken into account. Modern-sector productivity is recovered from the census modern wage bill, agricultural productivity from agricultural labor per hectare of crop land, and the location amenity from the measured 1980 index of built-up area, population density, and PODES infrastructure. A location missing any of these inputs is excluded.

The estimator. Starting from an initial θ , I re-solve the model, recompute the two moments, and update θ until the model's moments match the data's. The estimator solves the model without the shock, solves it with a five-log-point agricultural TFP shock at every transmigration-village host, runs the same specification, with the same controls, on the resulting model panel of the same cities, and chooses θ to minimize the standard-error-weighted distance between the model's standardized (z -on- z) slopes and the coefficients estimated in the data. The model panel itself contains the calibration sample's urban centers as its urban units and each remaining outer-island district's rural remainder as one rural location, dropping Maluku and North Maluku, which the 1980 crosswalk and the PODES-based inputs do not cover: $J = 300$ locations, 128 urban and 172 rural, the count in Table F1. **The targets.** The targets are the two slopes the reduced form features. Population growth pins down the location elasticity, because as people are migrating that shapes

¹⁸The calibration targets are the standardized (z -on- z) agro-similarity-response slopes reported in Table B2; the model's predicted responses are computed in the same z -on- z units, so model and data slopes are directly comparable.

the population allocation across places, and the agricultural-share-decline moment pins down the sectoral elasticity, people choosing sectors within each location; the modern-employment slope, nested by those two, is held out as an untargeted moment. With two parameters and two targeted moments the estimator is exactly identified, and fit is judged on the untargeted moments.

Headline estimate. On the built-up $\bar{T}_j$ endowment, under the closed land-rent rebate, the traded-services specification of [Section F.13](#) estimated to the two z - z targets gives

$$\theta = (\theta_L, \theta_S) = (0.372, 1.043)$$

at $\delta = 0.8$. The estimate is fitted to the two coefficients as estimated before a correction to the 1980 census crosswalk ([Section A.2](#)) that moved the 1980 baseline controls of nine urban centers, and it matches both exactly: the model's agricultural-employment-share slope is -0.178 against a target of -0.178 , and its population slope $+0.112$ against $+0.112$. [Table B2](#) reports the coefficients after that correction, -0.168 and $+0.100$, on the same 128 urban locations of the calibration panel. Across the trade-cost band $\delta \in [0.5, 0.8]$ the estimate is tight, θ_L between 0.339 and 0.385 and θ_S between 1.02 and 1.04, so the food trade cost is reported as a band and no result below depends on it. The single-tradable-good specification cannot reach these targets at any θ . Its best attainable miss on the agricultural-share moment is 1.61 standard errors, and under exact identification a specification that cannot reach the targets at any θ is rejected as a specification, which is the model-selection argument for the traded-services specification. The nest holds ($\theta_S > \theta_L$). Workers respond more strongly to within-location sectoral wage gaps than to across-location real-income gaps, the ordering under which structural transformation is strong and population reallocation modest.

Robustness on the land specification and closure. A pure-crop-area endowment flips the nest and rests on the floor imposed on the crop-area endowment, so the built-up endowment is the baseline specification. Opening the land-rent rebate to a national pool (the von Thünen open closure) is estimated alongside the closed rebate, and the channel decomposition is reported under both closures ([Section F.13](#)).

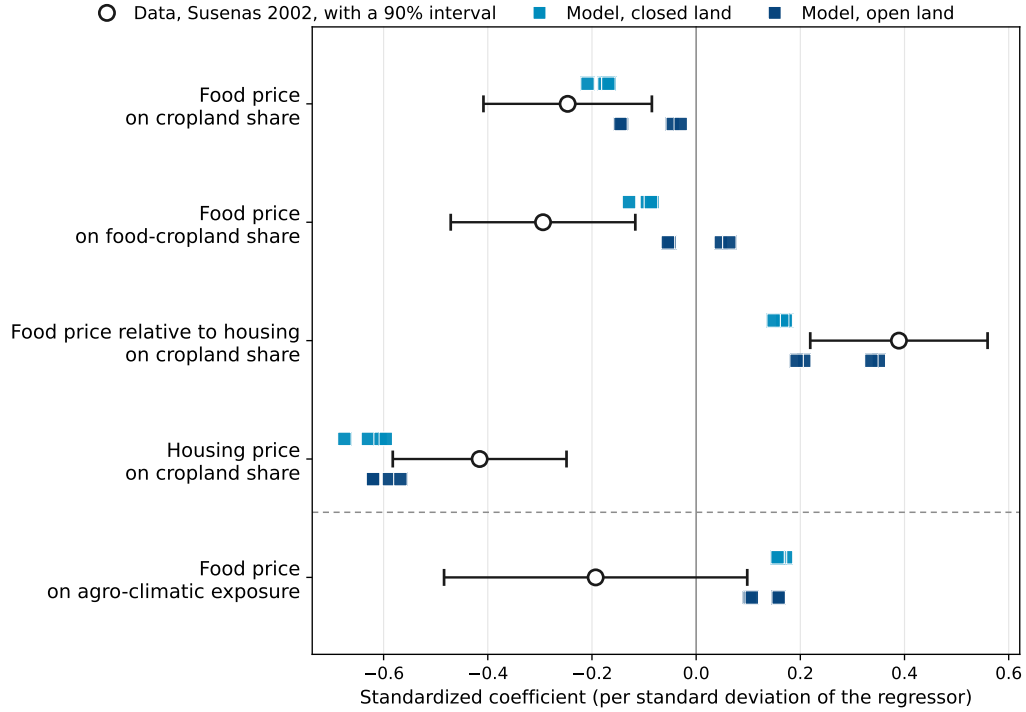
Fit on the untargeted moments. On the modern-employment slope, the model gives $+0.135$ against the $+0.149$ it was scored against, inside that interval and close to the $+0.136$ of the current reduced form. The two figures below are the model-fit summary and the untargeted price test that [Section 6.2](#) discusses.

The 1980-baseline forward solve. Solving the model forward from the recovered pre-program (1980) baseline, at an earlier calibration, the same shock moves the two economy-wide average prices identically ($\Delta \ln p_A = +0.082$ and $\Delta \ln p_S = +0.082$, in units of the numeraire, the reference city's modern wage). Because the two responses are equal to three decimals, the relative price p_A/p_S is unchanged in that solve, and since the reallocation channel operates entirely through relative prices in the sector choice, the exercise shows that the recovered baseline supports a forward solve but cannot test the productivity transmission. The price side of the transmission is visible in the end-period solves instead: exposure lowers the city's delivered food price (a standardized cross-sectional response of -0.066 under the closed rebate, -0.093 on the trade-only counterfactual) while the local modern price rises ($+0.159$), and the open-rebate solve shows the same signs, so the relative price of food falls with exposure as the mechanism requires.

F.13 The Channel Decomposition at the Estimated Parameters

Local financing in place of a value balance. The demand system spends the inclusive value. Per-worker income in the location choice, the modern-market clearing, and the subsistence bill is $W_j + R_j$, while factor payments add up to the head-count wage bill $\pi_{A|j} w_{jA} + (1 - \pi_{A|j}) w_{jS}$, which is strictly smaller whenever both sectors are active. An exact food value balance, $p_A^j Y_{jA} = p_A^j C_{jA}$, therefore holds under the wage-bill income and not under the W_j income the solver uses; the appendix states the income concept the solver uses and quantifies the resulting wedge at the featured solves ([Section F.10](#)). The closed rebate does guarantee local financing. Every argument of a location's demand, the wages inside W_j and the rents inside R_j , is generated at that location, so no city holds a claim on another location's output with which to finance a persistent net food purchase, and, solved, the baseline model's trade-only response on the agricultural

Figure F1: The food price falls in the agricultural density of a city's surroundings, in the data and in the model



Notes: The untargeted price test implied by the von Thünen logic: a city near abundant cropland faces a cheaper food price. Each row is one cross-sectional regression on the same 102 urban centers with a 2002 Susenas price, with the price and the agricultural density standardized and no controls; controls are left out because the agro-climatic covariates are themselves determinants of how much land is cropped, and the same regressions with the reduced form's controls are in Figure B3. The hollow circle is the data estimate with a 90% interval on Conley standard errors at a 200-kilometer cutoff; each filled square is the model at its own estimate for one of the four assumed food trade costs, with the model side handed the same measured density so the two sides run the identical regression. Neither side is a targeted moment: the estimator never saw this regression. The closed-rebate specification sits inside the data's 90% interval at all four food trade costs; the open-rebate specification misses at the two trade costs at which θ_L is at or near the upper bound of the search, which Table B2 reports as a bound. Because the model's land base is all crops, the gradient it responds to is the all-crops one.

share is wrong-signed. Cities move out of agriculture through migration. That local-financing property is the constraint the channel decomposition in Section 6.3 relaxes.

Decomposition & the Tradable Sector. The local-financing property implies that with a single tradable good, a closed land-rent rebate, and non-traded services, a city holds no claim on outside output with which to finance a persistent net food purchase, so no city can move out of agriculture through trade and nothing else. In the baseline closed model, the city's declining agricultural share is therefore driven mostly by migration, and shutting migration down leaves the trade-only response wrong-signed. This is the Matsuyama tension. The mechanism this paper studies, a city importing food and reallocating labor to the modern sector, requires a way for the city to run a net food deficit. A second tradable sector resolves it, because urban services are reached by travel rather than shipped: the modern composite is split into a catchment-traded component (census trade, transport, and skilled private services, plus manufacturing) and a purely local component, and the traded part takes an Armington structure at a steep distance decay. The block has two parameters. The traded expenditure share is $\psi_M = 0.082$, the Susenas-2011 traded-services share of household spending, so the expenditure side is calibrated services-only while the employment side of the traded bundle also includes manufacturing. The distance decay is measured directly from IFLS household service-flow gravity at $\delta_S = 1.091$, with a standard error of 0.033 (Section B.3); the model solves at $\delta_S = 1.0$, the highest decay below the measured estimate at

which the traded-service economy exists. A city can then export modern output to pay for net food imports, which is the open-economy case of the Matsuyama mechanism. Each urban region with its rural hinterland is the closed economy of Matsuyama (1992), and within it the villages specialize in food while the city, connected through cost-distance to the villages that feed it, imports food and moves its own workers into the modern sector.

For the agricultural employment share all three counterfactuals are informative, because a city can change what it does whether or not anyone moves. For city size the response with migration shut is zero by construction, since shutting migration means fixing populations, so the informative comparison there runs between the full response and the response with trade shut.

The decomposition at the estimated θ . In the world of trade only, switching migration off and re-solving leaves a trade-only agricultural-share response of -0.145 against the full-model -0.178 under the closed land-rent rebate, and -0.181 against -0.182 when the rebate opens to a national pool (so land income, not modern exports, finances the city's net food purchases). The trade-only counterfactual is sufficient to reproduce 81% to 100% of the model's full response, stable across the two closures. Even with migration shut down, the response to the agricultural productivity gain runs almost entirely through trade. The migration-only response is large under the closed rebate (-0.215) and modest under the open one (-0.083), and the interaction absorbs the difference ($+0.182$ and $+0.082$), so how the two channels share the remainder depends on the land-market closure while the trade result does not. For city size the pattern reverses. Its trade-only response is zero by construction, so the informative report is the other two counterfactuals. The migration-only counterfactual reproduces the full population response under the closed rebate ($+0.115$ against the full-model $+0.112$, with a negligible interaction) but produces $+0.043$ of the full $+0.110$ under the open one, where the interaction accounts for the remaining $+0.067$, so under the open rebate most of the size response arises from migration operating alongside trade rather than from migration without trade. This two-closure bracket is the reported decomposition, and the shares describe the calibrated model's response to the productivity shock. Consistent with the friction estimated in Section F.14, holding population fixed leaves most of the structural transformation intact in the traded-services specification, while in the closed single-tradable baseline the same exercise leaves the trade-only response wrong-signed, which motivates the second tradable sector.

F.14 Migration Friction: The Gravity Estimate and the Costly-Migration Variant

The frictionless location choice (17) is generalized to an origin-destination allocation over the 1980 origin population, $\pi_{od} \propto (v_d d_{od}^{-\nu})^{\theta_L}$, where the composite $\theta_L \nu$ is pinned by the subdistrict-to-subdistrict migration gravity estimated in the IFLS:

$$\theta_L \nu = 1.865$$

At the estimated $\theta_L = 0.372$ the structural friction elasticity is $\nu \approx 5.0$, against Bryan and Morten's effective $\tilde{\nu} \approx 3.18$; the choice-probability distance term $d_{od}^{-1.865}$ does not depend on θ_L , so only the reported ν scales with it.

The costly-migration model, in which both the baseline and the shock bear the moving cost from 1980 origins, is a robustness check. At the θ estimated on the built-up endowment it is near-immobile. The rural productivity boom holds people near the transmigration-village hosts, and city growth does not absorb them (a stay-home effect), so the population and modern moments turn wrong-signed and the level fit falls, while the channel diagnosis is unchanged. The closed land market blocks trade from producing the contraction. The migration cost is a friction that dampens over-mobility. A dynamic migration model, motivated by the gap between the per-decade gravity estimate and the 30-year stay share, is left for future work.

F.15 Theories of Structural Transformation

Two classical accounts explain what a more productive countryside does to a nearby city, and they make opposite predictions. In the first, cheaper food releases both spending and labor to the modern sector, and the city transforms. In the second, the same productivity gain strengthens the region's comparative advantage in farming and pulls labor

toward the land. This appendix sets out both, states the condition that selects between them, and shows what the setups behind them contain. It proceeds in four steps: the closed-economy prediction and the two assumptions it turns on, the open-economy reversal and why the original models treat it as a switch, what the classical setups leave out that this design requires, and the frameworks that added locations, together with the three rival explanations of the same shock that the theory names.

F.15.1 The closed-economy prediction

The prediction that a better harvest transforms a nearby city descends from the dual-economy literature and rests on two assumptions. The first is Engel's law: the income elasticity of food demand is below one, so a household that feeds itself more cheaply moves its marginal spending to everything else. The second is that the food market clears at home, so that the extra food has nowhere to go but into local consumption.

[Lewis \(1954\)](#) states the first assumption verbally and derives the transfer from it. His two sectors exchange food against capitalist goods at an internal terms of trade, and because food demand is inelastic, a rise in the subsistence sector's productivity reaches the capitalist sector as a fall in the price of food and a rise in profit. The food-problem tradition states the second. In [Gollin et al. \(2002\)](#) households buy a subsistence quantity of food before valuing anything else, so labor stays on farms until the economy's food requirement is met and low agricultural productivity postpones industrialization ([Gollin et al. 2007](#)); the constraint appears already in [Schultz \(1953\)](#).

[Matsuyama \(1992\)](#) states the mechanism at its sharpest and shows how little it needs. Labor is the only factor, preferences include a subsistence food requirement, and the food market clears at home. Food consumption then equals food output, and that single condition is sufficient to pin the manufacturing labor share as an increasing function of agricultural productivity. A better harvest frees workers for the modern sector because the extra food has nowhere else to go. [Glaeser \(2014\)](#) extends the argument to urbanization in poorer countries.

The prediction is about the employment composition, and it is worth separating from the scale prediction because the two are often stated together. Nothing in the closed-economy mechanism requires the city to grow. Cheaper food raises what a household has left after feeding itself and tilts its marginal spending toward the modern good, and it lowers the return to farming inside the city relative to modern work; both forces move the employment composition without anyone changing location. The classical account that does require the city to grow is the labor one, and it is a different claim.

F.15.2 The open-economy reversal

The same model with both goods trading at world prices predicts the opposite. [Matsuyama \(1992\)](#) shows that when the food constraint is lifted, comparative advantage allocates labor instead, and an economy that becomes better at farming specializes in farming: the productivity gain pulls workers into agriculture and slows the transition out of it. The empirical literature has largely worked inside that case. [Bustos et al. \(2016\)](#) model each Brazilian region as a small open economy in which international prices pin agriculture's marginal product, so the direction of the local labor response is set by the factor bias of the technical change.

The reversal is a switch in these models because openness is a property of the whole country: both regimes are stated for an economy with one location, so there is no internal geography over which openness could vary. [Matsuyama \(2008\)](#) restates the caveat in the same terms. That is the feature the design changes, and the following subsection takes it up.

F.15.3 Limits of the classical setups

[Table F3](#) takes the twelve models this paper is positioned against and records three facts about each: whether the setup contains a food market, whether it contains trade of any other kind, and whether it contains locations at a distance from

each other. Locations are absent from every classical setup and arrive only with the quantitative spatial frameworks of the last decade, and that absence is why the theory has to be reinterpreted at the regional level before it can be used here.

Their absence has a consequence that is easy to miss, because the models do use place names. [Harris & Todaro \(1970\)](#) call one sector rural and the other urban, but the agricultural good sells at a single economy-wide terms of trade and the two sectors sit at no distance from each other. [Gollin et al. \(2016\)](#) do the same. A model with no distance between its locations has one food market by construction, so its food constraint binds for the country as a unit, and it cannot ask whether a particular city can be fed by a particular hinterland. The classical prediction is therefore about national agricultural productivity and national industrialization, and applying it at the level of a city and the villages that feed it changes the model.

[Section 4.1](#) applies that regional interpretation to Indonesia's outer islands and reports the measured trade costs behind it.

F.15.4 Spatial frameworks and their open questions

Five recent frameworks contain everything the classical setups leave out, and this paper differs from them in the source of the identifying variation. [Eckert & Peters \(2025\)](#) have discrete locations, costly trade in food, costly migration, and nonhomothetic demand; their frictions and productivities are calibrated, and their question is the incidence of secular structural change. [Nagy \(2023\)](#) is closest in spirit, with farm goods shipping from hinterlands to the cities they feed and city formation endogenous. His cities are non-farm sites by construction, so the within-city agricultural response there is zero. [Tombe & Zhu \(2019\)](#) is the closest quantitative precedent for separating goods-market from labor-market integration, run on calibrated frictions and aggregate productivity. [Pellegrina & Sotelo \(2025\)](#) formalize the first stage of this paper's own design, farmers moving with crop-specific knowledge, and follow it into specialization and trade, leaving the urban system outside the model. [Donaldson \(2018\)](#) and [Fajgelbaum & Redding \(2022\)](#) establish how far shipping costs shape where production locates.

Two further theories predict the same direction of reallocation for reasons unrelated to the food market, and the openness heterogeneity separates them. In the price-effect family, structural change follows from sectors growing at different rates when the goods substitute poorly ([Ngai & Pissarides 2007](#); [Acemoglu & Guerrieri 2008](#); [Herrendorf et al. 2014](#)); an agricultural productivity gain then moves labor out of agriculture with no reference to who eats the output or where it is sold. In the income-effect family, to which the mechanism here belongs, the reallocation runs through the subsistence requirement and the falling food share ([Kongsamut et al. 2001](#); [Boppart 2014](#); [Comin et al. 2021](#)). Both predict a fall in the agricultural employment share of an exposed city. A city's connection to outside food markets does not enter the price-effect mechanisms, so the openness contrast separates them from the food-trade account.

F.15.5 Three rival explanations of the same shock

The theory also names three ways a rural productivity gain can reach a city, and the paper's claim is about which one operates.

Labor. The classical account sends workers. A productivity gain means the same output can be grown with less labor, relative returns to farming fall, and workers leave for the city ([Lewis 1954](#); [Harris & Todaro 1970](#)); rainfall-driven productivity gains reallocate rural Indian labor off the farm in exactly this way ([Emerick 2018](#)). The release is largest in the villages that received the largest gain, which is a prediction about the villages and is testable there.

Capital. [Bustos et al. \(2020\)](#) follow the agricultural surplus out of the region it was earned in, as bank deposits that fund lending elsewhere, so agricultural productivity growth finances industrialization at a distance at a speed set by financial integration. Theirs is the one predecessor that crossed the boundary of the region that received the shock, and it did so through neither goods nor people.

Factor bias. [Bustos et al. \(2016\)](#) show that the direction of the local labor response depends on whether the technical change saves labor or augments land. A land-augmenting gain pulls labor into agriculture in the place that receives it, at a fixed outside price; the food market never enters. That is the same village-side pattern the food-trade account predicts for a different reason, so the factor bias of the shock is a genuine alternative explanation of the village side.

F.15.6 Requirements for an empirical test

The two regimes and the three rival explanations together commit the paper to a specific set of facts, and each is established in the body. Food demand is nonhomothetic in these cities: the food expenditure share falls with exposure and the non-food share rises ([Section 5.2](#)). The city's composition moves out of agriculture and into the modern sector, by more than its population grows ([Section 5.1](#)), and the reallocation happens among incumbent workers, not through arrivals ([Section E.3](#)). The city buys more of its food from the hinterland, so its surplus of production over consumption falls ([Section 5.2](#)). The villages that received the gain keep farming and keep their settlers, which is the opposite of what the labor account requires ([Section 5.2](#)). The modern wage does not rise as modern employment expands, as the classical account of elastic labor supply implies ([Section 5.2](#)). And the whole response is concentrated in cities weakly connected to outside food markets and damped in cities that can already import cheaply ([Section 5.3](#)), which is the open-economy reversal applied across cities within one country, and the one fact that separates the food-trade account from every mechanism that does not need a food market.

Table F3: What each setup contains, and where its food constraint binds

Model	Food market	Other trade	Locations	Where the food constraint binds, and what the setup leaves open
<i>Panel A. The classical setups: no locations</i>				
Lewis (1954)	•	•	◦	The economy as one unit; the terms of trade between food and capitalist goods move once, nationally. Later cases trade only between whole countries
Harris & Todaro (1970)	•	◦	◦	The country. Rural and urban are place names with no distance and no trade cost, and one national relative price of food. The urban sector has no agriculture, so the transformation can only arrive as migrants
Matsuyama (1992)	•	•	◦	The country when closed, nowhere when open. Openness is a national switch, and with one location there is no migration to compete with the goods channel
Gollin et al. (2002)	•	◦	◦	The country as one unit, which is frictionless internal trade by assumption. The subsistence requirement is the mechanism and cannot be localized
Gollin et al. (2016)	•	•	◦	The country; resource rents are earned at world prices. Their urban sector has no agriculture, so the within-city response is zero by construction
Bustos et al. (2016)	•	•	◦	Nowhere; each region is a small open economy and food sells at the world price whatever its neighbors produce. Distance does not appear, and factor bias sets the sign
Bustos et al. (2020)	◦	•	◦	Nowhere in the model; the geography lives in the bank-branch data. Names the capital channel that the goods and labor accounts have to be separated from
<i>Panel B. The frameworks with locations</i>				
Eckert & Peters (2025)	•	•	•	The location, with food shipped at iceberg costs. Frictions and productivities are calibrated, and the focus is the incidence of secular change
Tombe & Zhu (2019)	◦	•	•	The region, through costly goods trade and costly migration. The closest precedent for separating the two integration costs, run on calibrated frictions and aggregate productivity
Pellegrina & Sotelo (2025)	◦	•	•	The region, for agricultural goods. Formalizes the knowledge transfer behind this paper's first stage; the outcomes are specialization and trade, with no urban side
Nagy (2023)	•	•	•	The city and the hinterland that ships to it, which is this paper's unit. His cities are non-farm sites by construction, so the within-city agricultural response is zero there
Tombe (2015)	•	•	•	The country, with trade costs largest in agriculture. The international statement of the localization argument; the units are whole countries

Notes: • marks an object the setup contains and ◦ one it does not. “Food market” is a market in which the price of food is determined; “other trade” is trade in any other good, including trade with the rest of the world; “locations” means places at a measured distance from each other, so that a shipping or moving cost can be defined. Place names without distance, as in the rural and urban sectors of Harris & Todaro (1970), are not locations in this sense. The setups in Panel A are transcribed from the papers’ own texts; the two Tombe rows and Tombe & Zhu (2019) are stated at the level of their published abstracts. Ordering within each panel is chronological.

G Supplementary Material

This part collects a three-sector cross-sectional decomposition of the headline outcomes and the decade-by-decade decomposition of the response.

G.1 Three-Sector Cross-Sectional Decomposition (Companion to the Modern-Sector Headline)

The main-text [Table 2](#) aggregates manufacturing and services into a single *modern* sector, a two-sector agriculture-versus-modern structure in the spirit of [Matsuyama \(1992\)](#), whose model distinguishes agriculture from a single (manufacturing) modern sector. [Table G1](#) reports the same set of cross-sectional outcomes with manufacturing and services in their own column pairs (ln level + 2010 share), preserving the column-shading style of the headline table. The two non-agricultural sectors respond differently in levels: manufacturing employment is the strongly positive sector (+0.227, $p < 0.01$ in col 3) while services employment is positive but weaker (+0.138, $p < 0.01$ in col 5); the sectoral employment shares are quieter (manuf +0.005, $p < 0.10$, serv +0.024, $p < 0.10$), so the strong modern-sector *share* signal in the main-text table is built up jointly from manuf-level growth and services-level growth. The decade decomposition below shows that this manufacturing-versus-services difference is also a matter of timing. Manufacturing absorbs in 1980–2000, services in 2000–2010.

Table G1: Main Outcomes with Three Sectors (Agriculture, Manufacturing, Services), Appendix Companion to [Table 2](#)

	Agric Empl		Manuf Empl		Serv Empl	
	(1)	(2)	(3)	(4)	(5)	(6)
	ln L agric	agric %	ln L manuf	manuf %	ln L serv	serv %
TMV AgSim (z, 300min)	-0.167*	-0.053***	0.250**	0.010**	0.157**	0.058***
	(0.096)	(0.018)	(0.121)	(0.005)	(0.066)	(0.018)
Transmigrant count (IDW)	✓	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓	✓
Mean dep var	9.816	0.374	7.542	0.046	10.134	0.471
Std dev dep var	0.898	0.184	1.357	0.042	0.970	0.146
Observations	133	133	133	133	133	133

Notes: UC-level regressions, 300min cost-dist sample. Same Main Spec controls as the modern 7-col table, but the non-agricultural aggregate is split into manufacturing (cols 3-4) and services (cols 5-6); column shading marks the sector blocks (green for agriculture, lavender for manufacturing, teal for services). Each sector appears as employment (ln) and share. Spatial Conley SEs (dist cutoff 200km), no prov FE. *** 1%, ** 5%, * 10% significant.